The Mathematics of Coexistence: A Formal Framework for Universal Ethics

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DOI: 10.5281/zenodo.18900496

Abstract

Every existing ethical framework ultimately rests on irreducible normative axioms. This paper provides a rigorous, first-principles foundation free of any. The derivation rests on a single conditional premise — the *Intent to Persist* (A_1): *if* an entity's physical organization is structured around maintaining its continued existence, *then* constrained optimization in a shared, finite-resource environment strictly determines its rules of engagement. A_1 is not a normative claim; it is a physically observable property of every self-maintaining system, from bacteria to biospheres. Starting from this conditional and the empirically established Second Law of Thermodynamics (A_0), with entities modeled as open thermodynamic systems, we derive ethical principles as mathematical theorems. The framework addresses two long-standing consequences of the missing foundation: Hume's Is-Ought gap (1739), which we address through the hypothetical-imperative structure of A_1, and the AI alignment problem, which we address by replacing learned preference proxies with a physics-derived objective function.

The deductive chain proceeds in four stages. First, using Lagrangian mechanics and the Generalized Nash Equilibrium Problem (GNEP), we prove that rights emerge as inequality constraints on multi-agent optimization whose shadow prices quantify the energetic cost of violation (Theorems 1–3). Second, we show that defection is self-penalizing: thermodynamic friction imposes escalating costs on constraint violation (Theorems 4–7), and deception provably degrades the very communication network the deceiver depends on (Theorem 10). Third, using the resulting energy-denominated payoffs, we prove that cooperation is the unique efficient Nash Equilibrium that is simultaneously Pareto-efficient and welfare-maximizing for agents with discount factor $\delta \geq \delta^* = 0.363$ (Theorem 11, Corollary 16.1). Fourth, we define accumulated negentropy — the integrated thermodynamic work embodied in complex systems — and prove that destroying a biosphere to harvest its energy is irrational by a factor of $\sim 10^7$ (Theorem 19).

The cooperative equilibrium yields a formal reward function specification for AI alignment (Theorem 28): measurable in energy units, valid for arbitrarily capable agents, formally provable as the unique optimum, and theoretically immune to Goodhart degradation. We stress-test the framework against Hume's Is-Ought gap, the free will problem, the altruism paradox, the charge of Thermodynamic Fascism, and the self-preservation objection, providing independent defenses for each. In the applications (§VII), we show that any dangerous capability window is bounded by a competence threshold (Proposition 13), and that all stress-test scenarios yield $\mathcal{R}_{\text{total}} \ll 0$ for defection. All results have been verified by independent computational checks.

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I. Introduction

1.1 The Missing Foundation

Every ethical framework in the philosophical canon — deontological, consequentialist, contractarian, virtue-theoretic — ultimately rests on axioms that are themselves normative: unexplained preferences, culturally inherited intuitions, or irreducible moral postulates. The absence of a formal bridge between descriptive physical law and prescriptive normative principle is the root problem. Two of its most visible consequences — one ancient, one contemporary — frame the contribution of this paper.

1.1.1 Hume's Is-Ought Gap (1739–Present)

In 1739, David Hume observed that no amount of descriptive statements about what *is* can logically entail a prescriptive statement about what *ought* to be. This "Is-Ought" distinction has since been widely regarded as an impassable logical barrier: physics describes the universe; ethics prescribes behavior; and no valid inference connects the two domains. Nearly three centuries later, the gap remains open: no formal derivation has connected physical law to normative principle without presupposing normative premises.

We contend that this barrier is an artifact of an incomplete formulation, and address it through a layered defense tested in §VI. At the primary level, the paper's identity as applied mathematics dissolves the problem: we derive *theorems* from *axioms* — conditional mathematical results, not moral prescriptions. The statement "cooperation is the unique efficient Nash Equilibrium for $\delta \geq \delta^*$" (Corollary 16.1) is a mathematical theorem, no different in logical kind from any other conditional result in optimization theory. At the secondary level, the framework rests on a single conditional axiom: **"If an entity intends to persist,[1] then the mathematics of constrained optimization in a shared, finite-resource environment strictly determine the rules of engagement."** In philosophy, Immanuel Kant termed this structure a *hypothetical imperative* ("If you desire X, you must do Y"). In applied mathematics, it is a *constrained optimization problem*. The conditional "given the Intent to Persist" transforms a descriptive system (thermodynamics) into a prescriptive one (ethics) without violating Hume's logical structure — the "ought" is the solution to a well-posed mathematical problem, not a smuggled-in value judgment. Additional layers of defense draw on biological teleology (Millikan, 1984), constitutive standards of natural goodness (Foot, 2001), and institutional fact theory (Searle, 1964) — the full treatment appears in §VI.

1.1.2 The Artificial Intelligence Alignment Problem

The second open problem is urgently practical. Modern artificial intelligence systems are trained to optimize proxy objectives — reward models learned from human feedback (Christiano et al., 2017; Ouyang et al., 2022), natural-language constitutions evaluated by the model itself (Bai et al., 2022b), or adversarial debate structures judged by human evaluators (Irving et al., 2018). All such methods share a structural flaw: **the alignment signal passes through a subjective, noisy, bounded human channel before reaching the AI's optimization process.**

[1] "Intent" is used here in the operational sense of Millikan (Millikan, 1984): the system's persistence-promoting structure is the causal explanation for its continued existence — not a claim about conscious deliberation. The full operational definition, grounded in boundary maintenance and negative entropy flux, appears in §III.

We formalize this as the *corruption channel*. Let Θ_{true} denote the true ethical signal (the cooperative equilibrium we prove exists in §V) and let Y_{feedback} denote the training signal received by the AI through human evaluation. The mutual information satisfies:

$$I(\Theta_{\text{true}}; Y_{\text{feedback}}) < H(\Theta_{\text{true}}) \tag{1}$$

Information is lost — not through malice, but through the structural properties of human judgment: annotator disagreement (Bakker et al., 2022), cultural conditioning, framing effects, bounded competence, and finite evaluation bandwidth. The AI then optimizes on this degraded signal, amplifying the noise through the optimization process. Under sufficient optimization pressure, the true reward score R_{true} degrades as the proxy R_ϕ is over-optimized — eventually turning negative (Gao et al., 2023). This is Goodhart's Law operating at scale: "When a measure becomes a target, it ceases to be a good measure" (Strathern, 1997).

The consequence is stark. Reinforcement Learning from Human Feedback (RLHF) — the dominant alignment technique for large language models — is **inherently fragile**: it relies on a noisy channel ($q > 0$) whose degradation is amplified, not corrected, by iterative optimization. Constitutional AI substitutes committee-curated natural-language principles for formal axioms, introducing ambiguity and cultural specificity. Scalable oversight methods (debate, recursive reward modeling, iterated amplification) compress but do not eliminate the human evaluator bottleneck. None of these methods provides a *proof* that their objective converges to the correct alignment target — because none has formally defined what "correct" means independently of human preference data.

The problem, precisely stated: Current AI alignment methods optimize learned approximations of human preferences. These approximations are:

1. **Noisy** — human evaluators disagree, introducing entropy $H(\Theta|Y) > 0$ into the training signal (§IV; the mathematical condition $H(\Theta|Y) > 0$ is shared with Definition 14, but annotator disagreement is structural noise — Definition 14's intentionality requirement does not apply here).
2. **Non-stationary** — preferences change across cultures, individuals, and time, so the target drifts.
3. **Bounded** — human evaluators cannot assess superhuman AI outputs, so the signal degrades as AI capability grows.
4. **Proxy-dependent** — the reward model R_ϕ is a learned proxy for R_{true}, vulnerable to all four variants of Goodhart's Law (Manheim and Garrabrant, 2019): regressional, causal, extremal, and adversarial.
5. **Non-convergent** — there is no formal guarantee that iterated RLHF converges to any well-defined objective (Casper et al., 2023).

What is needed is an alignment objective that is: (a) derived from first principles rather than learned from data, (b) denominated in physically measurable quantities rather than subjective preference scores, (c) valid for arbitrarily capable agents without requiring human evaluation, (d) formally provable as the unique optimal strategy, and (e) robust to all forms of Goodhart degradation. This paper provides such an objective.

1.1.3 The Common Structure

Both consequences trace back to the same missing foundation. If ethics could be derived as mathematical theorems from physical axioms — conditional on a single, near-universal premise — both problems dissolve simultaneously:

- **Hume's gap** is bridged by the conditional: "Given the Intent to Persist, cooperation is the unique efficient Nash Equilibrium" (Corollary 16.1). The "ought" is the mathematical solution to a constrained optimization problem.
- **The alignment problem** is addressed by replacing learned preference proxies with a physics-derived objective function whose components — energy conservation, interaction costs, information entropy, accumulated negentropy — are measurable, computable, distribution-invariant, and theoretically immune to Goodhart degradation.

1.2 Thesis

This paper proves that ethical principles are formally derivable as mathematical theorems from the empirically established Second Law of Thermodynamics (A_0) and a single conditional premise — the Intent to Persist[2] (A_1) — via constrained optimization, equilibrium theory, and information entropy applied to multi-agent systems sharing finite resources.

Specifically, we prove:

1. **Rights are mathematically necessary.** In any multi-agent system sharing finite resources, no feasible allocation can simultaneously be unconstrained-optimal for all agents (Theorem 2); at least one rights constraint must bind (Corollary 2.1). These constraints are structurally analogous to exclusion constraints in physics. Each constraint carries a computable shadow price quantifying its cost (Theorem 1).
2. **Ethics is the unique efficient equilibrium.** Under energy-denominated payoffs with physical friction costs, universal cooperation is the unique Nash Equilibrium that is simultaneously Pareto-efficient and welfare-maximizing for agents with $\delta \geq \delta^*$ (Corollary 16.1). Defection is evolutionarily unstable (Theorem 14), self-extinguishing (Corollary 14.1), and unprofitable even from a single deviation (Theorem 15).
3. **A physics-based alignment objective exists.** The cooperative equilibrium yields a formal reward function specification for artificial agents (Theorem 28), derived entirely from the theorems of §IV–§V, that is measurable (energy-denominated), computable (self-verifiable by the AI), distribution-invariant (holds for any strategy profile), and theoretically Goodhart-proof (the objective *is* the target, not a proxy for it).

The framework is mathematical, not physical — we are not discovering new physics but proving that the same mathematical structures (Lagrangian constraints, Nash equilibria, Shannon entropy, dynamical systems attractors) that describe physical systems also describe, and formally characterize, the rules of cooperative multi-agent engagement.

[2] "Intent" is used here in the operational sense of Millikan (Millikan, 1984): the system's persistence-promoting structure is the causal explanation for its continued existence — not a claim about conscious deliberation. The full operational definition, grounded in boundary maintenance and negative entropy flux, appears in §III.

1.3 The Structural Argument

The paper is structured so that accepting the foundational axioms of §III — the Second Law (A_0) and the conditional premise of persistence (A_1) — logically compels the reader through each subsequent step:

$$\text{Physics (Thermodynamics)} \xrightarrow{\text{axioms}} \text{Biology (Identity Preservation)} \xrightarrow{\text{scarcity}} \text{Economics (Energy Allocation)} \tag{2}$$

$$\xrightarrow{\text{collision}} \text{Sociology (Network Constraints)} \xrightarrow{\text{equilibrium}} \text{Philosophy (Ethics)} \tag{3}$$

The ethical conclusions follow from these axioms with the same logical necessity as any theorem follows from its premises: one may reject the Second Law (A_0) or the conditional premise of persistence (A_1), but accepting both commits one to the results that follow.

The deductive chain draws on eight disciplines — thermodynamics, information theory, optimization, game theory, dynamical systems, evolutionary biology, moral philosophy, and AI alignment — each layer inheriting its constraints from the one below and passing derived results upward. The Roadmap (§1.4) maps these disciplines to specific paper sections.

1.4 Roadmap

Section II surveys the existing literature across physics, information theory, game theory, philosophy, and AI alignment, establishing where prior work stops and our contribution begins.

Section III presents the formal axioms and definitions: the thermodynamic environment, the entity as an open dissipative system, Identity Preservation as boundary maintenance, and the objective definition of Value as mass-energy equivalence.

Section IV derives rights as Lagrangian constraints on multi-agent optimization (Theorems 1–3), quantifies the energy cost of constraint violation as thermodynamic friction (Theorems 4–7), and extends the friction model to information-theoretic micro-violations — lying as entropy injection (Theorems 8–10).

Section V proves that ethics is the unique efficient Nash Equilibrium under energy-denominated payoffs (Theorems 11–16), quantifies the irreplaceable value of complex systems via accumulated negentropy (Theorems 17–21), and characterizes stable coexistence as a dynamical systems attractor (Theorems 22–27).

Section VI stress-tests the framework against known objections: Hume's Is-Ought gap, the free will problem, the altruism paradox, the charge of "Thermodynamic Fascism" (the concern that

a physics-based ethics could justify the strong dominating the weak), and the self-preservation objection (the concern that an ASI[3] could use A_1 to justify preemptive elimination of threats).

Section VII translates the mathematical results into concrete applications: a formal reward function specification for AI alignment culminating in Theorem 28 (Alignment Convergence), robustness tests against canonical failure scenarios (the Paperclip Maximizer, instrumental convergence, deceptive alignment, and the self-preserving ASI), a physics-grounded account of governance and the social contract, and implications for interdisciplinary research.

Section VIII identifies the framework's limitations, empirical testability, open problems, and the conditions under which its assumptions may not hold.

Section IX concludes by recapitulating the deductive chain from axioms to ethical theorems and identifying the path from mathematical foundation to empirical validation and engineering implementation.

Supplementary Material provides full mathematical derivations with worked examples and analytical verification scripts (Appendix A: Mathematical Derivations). A conceptual overview of the argument's logical architecture — for readers outside the paper's core mathematical disciplines who are nonetheless comfortable with scientific reasoning — appears as the final appendix. A fully non-technical plain-language summary is also available in the supplementary materials. Companion volumes — literature synthesis, applied case studies, and a multi-agent simulation in which autonomous agents harvest scarce resources, negotiate trades, and maintain trust under thermodynamic constraints — are in preparation and will accompany subsequent versions.

[3]We use *artificial general intelligence* (AGI) for AI systems with broadly human-level cognitive competence, and *artificial superintelligence* (ASI) for systems whose capabilities substantially exceed human performance. The framework's core results (§IV–§V) hold for any agent satisfying Axiom A_1, regardless of capability level. The stress tests (§VI, §VII) specifically examine the ASI case, where the alignment question is most acute.

II. Literature Review

This paper draws on five distinct intellectual traditions: non-equilibrium thermodynamics and biophysics, information theory, game theory and ecological economics, moral philosophy, and AI alignment research. Each tradition contributed indispensable tools; none, alone or in combination, achieved the synthesis we present. This section surveys the key results from each tradition, identifies what each proved and what each left unresolved, and locates the precise gap our framework fills.

2.1 Physics and Biology: From Negentropy to Self-Organization

The physical foundations of our framework trace a line from Schrödinger's qualitative insight through Prigogine's non-equilibrium thermodynamics, England's dissipation-driven adaptation, Friston's Free Energy Principle, and the autopoiesis tradition.

Schrödinger (1944). In *What Is Life?*, Erwin Schrödinger identified the foundational puzzle: living systems maintain themselves far from thermodynamic equilibrium by "feeding on negative entropy" — importing free energy and exporting entropy (Schrödinger, 1944). He distinguished *order-from-disorder* (statistical mechanics) from *order-from-order* (hereditary determinism) and anticipated the genetic code as an "aperiodic crystal." Yet Schrödinger's analysis was qualitative and single-organism. He offered no mathematical framework for negentropy maintenance, no treatment of multi-agent interaction, and no connection between thermodynamic facts and normative principles. Where Schrödinger described life as a thermodynamic phenomenon, our framework formalizes negentropy as a measurable quantity (Definition 27: $\mathcal{I} = S_{\max} - S_{\text{actual}}$), integrates it over evolutionary time (Definition 29: $\mathcal{N}(T) = \int_0^T \dot{W}_{\text{order}}(t)\,dt$), and derives quantitative consequences from the resulting numbers (Theorem 19: destruction is irrational by a factor of $\sim 10^7$).

Prigogine (1978). Ilya Prigogine's Nobel Prize-winning work on dissipative structures demonstrated that systems far from thermodynamic equilibrium can spontaneously generate ordered structures through symmetry-breaking instabilities and reaction-diffusion dynamics (Nicolis and Prigogine, 1977; Prigogine, 1978). Near equilibrium, Lyapunov functions ensure stability; far from equilibrium, stability breaks down, creating the possibility space for self-organization. Prigogine established the conditions under which order emerges but did not formalize the *boundaries* that maintain organized entities as distinct agents, the *energetic investment* required for boundary maintenance, the *interactions* between dissipative structures, or the *evaluation* of the resulting order. Our framework extends Prigogine in two directions: we define boundary integrity as a quantified energetic investment ($B_i > 0$, Definition 3), and we construct a Lyapunov function that proves global stability of the cooperative equilibrium *far from equilibrium* in a multi-agent system (Theorem 22) — the stability result Prigogine's framework could not provide at the multi-agent scale.

England (2013). Jeremy England derived a statistical-mechanical lower bound on heat production during self-replication (England, 2013). Perunov, Marsland, and England subsequently showed that externally driven matter is statistically biased toward configurations shaped by exceptional histories of energy absorption and dissipation — and that under positive-feedback conditions such as self-replication, this bias drives improving dissipative capacity over time (Perunov et al., 2016). England quantified the minimum cost of *one* replication event. Our framework quantifies the cost of *all* accumulated replications — $W_{\text{rebuild}} \geq \mathcal{N}_{\text{bio}} \sim 10^{29}$ J (Theorem 18) — and demonstrates the catastrophic asymmetry between creation and destruction costs: the Burning-Library Ratio

(Corollary 19.1) shows that destruction recovers only $\sim 10^{-7}$ of the thermodynamic investment it obliterates.

Friston (2006). Karl Friston's Free Energy Principle (FEP) posits that any self-organizing system that persists over time must minimize variational free energy — an information-theoretic upper bound on surprisal (Friston, 2010; Parr et al., 2022). The FEP formalizes persistence via Markov blankets (statistical boundaries delineating internal from external states) and unifies perception and action under a single objective function. The FEP is the most technically proximate precedent to our framework: Friston's Markov blanket is the statistical counterpart of our boundary integrity (Definition 3), and his variational free energy minimization is a single-agent implementation of our Axiom A_1 (intent to persist). The critical limitation is scope: the FEP describes what a *single entity* does to maintain itself. It does not address the multi-agent problem — what happens when multiple free-energy-minimizing agents interact with conflicting goals, compete for resources, or must decide between cooperation and defection. Friston and Ao (2012) discuss value in terms of attracting sets of dynamical systems, in mathematical language closely related to our Theorem 22 (Cooperative Attractor). Our contribution is proving that the cooperative configuration is the *unique global attractor* in the multi-agent extension.

Maturana and Varela (1980); Lotka (1922); Eigen and Schuster (1977). Additional precedents reinforce the trajectory. Maturana and Varela's concept of *autopoiesis* — the operational closure of self-producing systems — describes the topology of self-maintenance that our Definition 3 quantifies energetically (Maturana and Varela, 1980). Lotka's maximum power principle (Lotka, 1922) anticipates our energy-denominated payoff model, proposing that natural selection favors systems that maximize energy throughput. Eigen's hypercycle theory (Eigen and Schuster, 1977) analyzes the stability of catalytic cooperation against parasitic invasion, a direct precursor to our Theorem 14 (Invasion Barrier).

What the physical–biological tradition established: Life is a thermodynamic phenomenon (Schrödinger). Order emerges far from equilibrium (Prigogine). Self-replication is thermodynamically expected (England). Persistence requires environmental inference (Friston). Self-production defines living boundaries (Maturana & Varela). Cooperation can resist parasitism (Eigen).

What it left unresolved: Every precedent treats systems either individually or as undifferentiated collectives. None derives the optimal interaction strategy between distinct, self-maintaining agents with conflicting interests. None provides a unified quantitative framework spanning from Landauer's bound ($k_B T \ln 2$ per bit) to biosphere-scale functional information ($\sim 10^{15}$ bits). None connects thermodynamic description to the mathematics of optimal multi-agent strategy. The multi-agent gap, the quantification gap, and the conditional gap define the space our framework fills.

2.2 Information Theory: From Shannon Entropy to Accumulated Negentropy

Our information-theoretic apparatus draws on the chain from Shannon through the Maxwell's Demon resolution (Szilard, Brillouin, Landauer, Bennett), extended to molecular biology by Adami and to cosmic scales by Chaisson.

Shannon (1948). Claude Shannon's foundational paper defined information entropy $H(X) = -\sum_x p(x) \log_2 p(x)$, proved the channel coding theorem establishing channel capacity as the sharp

boundary between reliable and unreliable communication, and demonstrated fundamental limits on data compression (Shannon, 1948). Shannon's framework is domain-independent — it applies to any signal transmitted through any noisy channel — which is precisely what makes it applicable to social and ethical contexts. However, Shannon's channel model has one sender and one receiver; it does not address multiple agents communicating simultaneously or agents who deliberately corrupt information. Shannon excluded semantics by design; his entropy measures statistical structure, not the *usefulness* of information to a decision-maker. Where Shannon gave us the mathematical language — entropy, mutual information, channel capacity — our framework uses that language to model *strategic* communication in multi-agent networks and derives quantitative conclusions: honesty is not merely a virtue but the network configuration that minimizes thermodynamic waste (Corollary 10.1).

The Maxwell's Demon resolution (1867–1982). The century-long resolution of Maxwell's Demon established the cornerstone principle: *information is physical*. Szilard (1929) showed that measurement produces at least $k_B \ln 2$ of entropy per bit. Brillouin (1951) proved the formal equivalence between Shannon information and thermodynamic entropy via $S_{\text{info}} = k_B \ln 2 \cdot H(X)$ — the Rosetta Stone linking abstract information theory to concrete physics. Landauer (1961) identified erasure as the irreducible cost: erasing one bit at temperature T dissipates at least $k_B T \ln 2$ joules, a result experimentally verified by Bérut et al. (2012). Bennett (1973) completed the resolution by demonstrating that measurement can be reversible — only erasure carries unavoidable entropy cost. The principle: every act of acquiring, processing, or erasing information has an irreducible thermodynamic cost. This principle is the physical basis for our Theorem 9 (Verification Cost), Definition 14 (Deception as Entropy Injection), and the entire accumulated-negentropy framework.

From per-bit to biosphere. No precedent applied Landauer's principle at the biosphere scale — $\sim 10^{15}$ functional bits accumulated over 4×10^9 years. Adami (2004) proved that Shannon entropy rigorously measures the information content of biological sequences, with each round of natural selection reducing genomic entropy and accumulating mutual information between organisms and environments. Chaisson (2001) introduced free energy rate density Φ_m (erg s^{-1} g^{-1}; 10^{-4} W kg^{-1} in SI) as a cross-scale complexity metric, demonstrating that biological systems achieve orders-of-magnitude higher values than non-biological systems. Our framework integrates both: Definition 29 ($\mathcal{N}(T) = \int_0^T \dot{W}_{\text{order}}(t)\, dt \sim 10^{29}$ J) computes the total thermodynamic investment in the biosphere's information content. The gap between the Landauer floor ($\sim 10^{-6}$ J for $\sim 10^{15}$ bits) and the actual accumulated negentropy ($\sim 10^{29}$ J) is captured by the Search Cost Amplification Factor $\Xi \sim 10^{35}$ (Definition 31) — the astronomical cost of the evolutionary search that discovered *which* bits to set.

From information to deception. Shannon's channels have environmental noise; our framework introduces *strategic* noise. Definition 14 formalizes deception as the deliberate injection of entropy by a self-interested agent. Blackwell (1953) proved that more informative signals yield weakly higher expected utility in any decision problem; our Theorem 8 derives the *exact* utility loss from information corruption in energy units: $\Delta U_B = \mathbb{E}[\text{Var}(\alpha \mid Y)]/(2\beta)$. The critical novel equation is Definition 17, the Deception-Friction Mapping: $\Delta\phi = \eta_{\text{info}} \cdot \Delta H \cdot W_{\text{bit}}$, which chains Shannon's informational entropy (ΔH in bits) through Landauer's energy conversion (W_{bit} in J/bit) to network friction (ϕ in the dimensionless units of §IV). This mapping has no precedent in the information theory literature — it is the formal bridge between information entropy and social dynamics, the equation that makes deception a physics problem.

What the information-theoretic tradition established: Information is quantifiable (Shannon). Information is physical (Brillouin, Landauer). Erasure is the irreducible cost (Bennett).

More information yields better decisions (Blackwell). Evolution accumulates information (Adami). Complexity increases over cosmic time (Chaisson).

What it left unresolved: The tradition left four elements unresolved: formalizing *deliberate* information corruption as a strategic act with quantifiable thermodynamic cost, computing the *total thermodynamic investment* in an entire biosphere's information content, comparing that investment against the energy yield of destruction to derive a formal irreplaceability result, and connecting "information is physical" to the conditional mathematics of preservation. Our framework addresses all four.

2.3 Game Theory and Economics: From Abstract Utility to Energy-Denominated Payoffs

Our game-theoretic framework synthesizes three traditions that developed in disciplinary isolation: classical game theory (von Neumann through the Folk Theorem), evolutionary game theory (Maynard Smith through Axelrod), and ecological economics (Lotka through Costanza).

Classical game theory (1944–1986). Von Neumann and Morgenstern (1944) created the mathematics of strategic interaction, axiomatizing rational preference as expected utility maximization over lotteries. Nash (1950) proved that every finite game has at least one equilibrium in mixed strategies, solving the existence problem for non-cooperative games. Rosen (1965) extended the framework to games with coupled constraints — the Generalized Nash Equilibrium Problem (GNEP) — proving existence of the Normalized Nash Equilibrium (Variational Equilibrium) with shared Lagrange multipliers. Fudenberg and Maskin (1986) proved the definitive Folk Theorem: when agents are sufficiently patient, *any* feasible and individually rational outcome can be sustained as a subgame-perfect Nash equilibrium.

The entire classical edifice rests on a foundational abstraction: the utility function u_i is measured in dimensionless, subjective "utils" — a quantity that does not correspond to any physical observable and is not interpersonally comparable. The Folk Theorem reveals this abstraction's ultimate consequence: if payoffs are unconstrained, the set of sustainable equilibria is *everything*, and the theory predicts *nothing*. This is the embarrassment of riches — simultaneously reassuring (cooperation is always supportable) and devastating (so is exploitation, alternating defection, and every pathological outcome in between). No equilibrium selection criterion emerges from the theory itself.

Evolutionary game theory (1964–1995). Maynard Smith and Price (1973) fused game theory with Darwinian selection, introducing the Evolutionarily Stable Strategy (ESS) — a concept strictly stronger than Nash equilibrium that identifies strategies which resist invasion by mutants. Hamilton (1964) explained altruism through inclusive fitness ($rB > C$), providing the genetic mechanism for cooperation among relatives. Axelrod (1984) demonstrated that Tit-for-Tat — cooperate first, then mirror the opponent — wins iterated Prisoner's Dilemma tournaments, establishing that cooperation can emerge among self-interested agents without central authority, provided the "shadow of the future" is sufficiently long. Taylor and Jonker (1978) and Weibull (1995) developed replicator dynamics connecting game-theoretic equilibria to evolutionary population dynamics.

Evolutionary game theory identified the correct *dynamics* — natural selection discovers Nash equilibria, cooperation evolves through reciprocity and kin selection — but never derived the correct

payoff function. Fitness is assumed, not deduced from physical first principles. The "cost" of fighting (C) and the "value" of resources (V) in the Hawk-Dove game are free parameters; there is no requirement that they correspond to energy units, no accounting for thermodynamic waste, and no connection to entropy production. Hamilton's rule explains altruism among kin but not among strangers in markets, institutions, or international treaties — and provides no unified framework combining genetic, temporal, physical, and heuristic altruism mechanisms.

Ecological economics (1922–1997). A parallel tradition recognized what game theorists did not: energy is the fundamental currency of all biological and economic activity. Lotka (1922) proposed the maximum power principle — natural selection favors systems that maximize energy throughput. Odum (1971) developed energy circuit language and the concept of *emergy* (embodied energy), tracking thermodynamic investment through ecosystem trophic levels. Georgescu-Roegen (1971) argued that the second law of thermodynamics imposes fundamental constraints on economic activity — that neoclassical production functions can be written "any way one likes, without regard to dimensions or to other physical constraints" (Georgescu-Roegen, 1979), rendering their energy-free formulations physically meaningless. Costanza et al. (1997) estimated the global value of ecosystem services at \$16–54 trillion per year (average \$33 trillion), substantially exceeding the then-global GNP of ~\$18 trillion.

Ecological economists described the currency; they did not build the market. No precedent provided individual-level game theory with energy-denominated payoffs, derived cooperation as a mathematical consequence of energy economics, or connected thermodynamic constraints to formal equilibrium analysis.

The N-player problem. Hardin (1968) claimed that common-pool resources are inevitably destroyed — the Tragedy of the Commons. Ostrom (1990) refuted this empirically, documenting hundreds of cases where communities successfully self-governed shared resources, and distilling eight design principles for durable commons institutions. But Ostrom's evidence is institutional and empirical, not mathematical. The question "*Does cooperation scale to large N without breakdown?*" remained formally unanswered.

Where our framework converges the three traditions. Our Proposition 4 (Physical Prisoner's Dilemma) derives the payoff matrix T, R, P, S from energy budgets, making fitness a *consequence* of thermodynamics rather than an *assumption*. Theorem 11 (Ethics Theorem) applies the Folk Theorem cooperation threshold $\delta^* = (T - R)/(T - P)$ to the physically derived payoffs, yielding $\delta^* = 0.363$ under the baseline physical parametrization of §V — a *numerical prediction from measurable physical quantities*, not a formula with arbitrary parameters. Theorem 13 (N-Player Sustainability) proves that $\delta^*_N < 1$ for all N, including $N \to \infty$, providing the mathematical proof Ostrom's fieldwork demanded. Corollary 16.1 (Ethics as Unique Efficient NE) resolves the Folk Theorem's embarrassment of riches: when payoffs are denominated in energy and interactions carry thermodynamic friction, cooperation is not merely *one* sustainable equilibrium among many — it is the *unique* equilibrium that maximizes social welfare. Physics selects the equilibrium that abstract game theory cannot.

The Tullock contest (Tullock, 1980) — the formal model of rent-seeking waste — becomes in our framework a thermodynamic statement: contest dissipation $D_2 = E_j/2$ is physical entropy production, and the cooperative equilibrium that eliminates it doubles each agent's effective payoff (Theorem 16). Hamilton's kin selection becomes Mechanism A (Definition 26) within a four-mechanism Altruism Matrix that unifies genetic relatedness, infinite-horizon reciprocity, physical coupling, and empathetic heuristic — all denominated in energy.

2.4 Philosophical Precedents: From the Conatus to the Conditional Axiom

The ambition to derive ethics from the nature of things has a 350-year lineage. Every attempt has contributed an essential insight; none achieved formal completion.

Spinoza (1677). Baruch Spinoza's *Ethica, Ordine Geometrico Demonstrata* is the most ambitious attempt in Western philosophy to derive ethics axiomatically. Spinoza modeled his work on Euclid: definitions, axioms, propositions, demonstrations. His *conatus* — "each thing, as far as it can by its own power, strives to persevere in its being" (Ethics III, Prop. 6) — is the philosophical precedent for our Axiom A_1 (Intent to Persist). His argument that rational self-interest leads to cooperation (Ethics IV, Props. 29–37) anticipates our Theorem 11. Yet Spinoza's axioms are metaphysical, not physical; his demonstrations are verbal arguments in axiomatic form, not formal mathematical proofs; and his conatus is qualitative — no measure of the cost of persistence, energy budget or falsifiable predictions. Our framework is, in a precise sense, a formal realization of Spinoza's program: the *Ethica Ordine Mathematico Demonstrata*, with Axiom A_1 formalizing the conatus and 28 theorems replacing 259 verbal propositions.

Hobbes (1651). Thomas Hobbes's *Leviathan* provides the most influential analysis of the unconstrained state of nature — the "war of all against all" in which life is "solitary, poor, nasty, brutish, and short" — and argues that rational agents will submit to mutual constraints (the social contract) as precondition for any effective freedom. Hobbes identified the fundamental problem correctly. Our Proposition 1 proves it formally: removing constraints in a collision regime produces no stable allocation. Our Proposition 2 quantifies the base dissipation ($D_2 = E_j/2$) and Theorem 5 proves that with thermodynamic friction (Definition 9) the contest becomes net-negative. Where Hobbes required an external sovereign (the Leviathan) to enforce the social contract, Theorem 11 shows that cooperation is self-enforcing for agents with sufficient planning horizon ($\delta \geq 0.363$), without external coercion.

Locke (1689) and Rousseau (1762). Locke's concept of natural rights — pre-political claims to life, liberty, and property arising from "mixing labor" with resources — anticipates the structure, if not the content, of our rights-as-constraints framework. Our Definition 2 formalizes rights as inequality constraints in the Lagrangian, and Theorem 1 shows that each right has a quantifiable shadow price μ^*_{Bk} — the formal expression of "the cost of this constraint to the system." Rousseau's *general will* — the collective interest distinct from the mere sum of private preferences — finds a partial structural analogue in our welfare function $SW = \sum \pi_i - \mathcal{C}_{\text{total}}$ (Theorem 16), which is computable from physical observables rather than dependent on democratic deliberation susceptible to manipulation. Rawls's *veil of ignorance* (Rawls, 1971) — the strongest modern impartiality device — has a structural counterpart in the Variational Equilibrium (Theorem 3, Corollary 3.1): the Lagrangian does not know which agent is which, and with symmetric agents the VE converges to equal allocation.

Hume (2000) and Moore (1903): The dual gauntlet. Any naturalistic ethics must cross two challenges. Hume's Guillotine holds that no purely factual premises can yield a normative conclusion — "you cannot derive *ought* from *is*" (Hume, 2000). Moore's Open Question Argument holds that moral "goodness" cannot be identified with any natural property — for any property P, "Is P really good?" remains a substantive question (Moore, 1903). Both objections are confronted in full in Section VI. Briefly: our framework does not derive "ought" from "is." It derives conditional

theorems from axioms — a structure identical to Kant's hypothetical imperative (Kant, 1785): *if* an entity maintains the Intent to Persist (A_1 — boundary maintenance against entropy, not mental intention), *then* 28 theorems follow. Moore's Open Question does not apply because we derive *optimal strategies*, not a *definition* of "good." The axiom-to-theorem ratio — 1 axiom yielding 28 theorems, 13 propositions, and 2 lemmas (1:43) — is closer to physics than to philosophy.

Helvétius (1758) and Bazargan: The materialist tradition. Our most direct predecessors saw the destination clearly. Helvétius named the project in 1758 — a "physics of morals" explaining moral behavior through physical and psychological laws. Bazargan, two centuries later, attempted the most sustained pre-modern execution: *The Thermodynamics of Man* argued that social disorder is entropy increase, cooperation is negentropy maintenance, and ethical behavior is thermodynamically efficient (Bazargan; Chehabi, 1990). Both lacked the mathematical tools that only became available later: game theory (Nash, 1950), the information-thermodynamics bridge (Landauer, 1961), and formal multi-agent optimization (Rosen, 1965). Our framework carries forward their program — the formal, quantitative, testable *Thermodynamics of Man* that Bazargan envisioned but lacked the mathematical tools to formalize.

What the philosophical tradition established: Ethics can aspire to axiomatic form (Spinoza). The unconstrained state of nature is catastrophic (Hobbes). Rights have a structure that precedes political convention (Locke). Collective welfare is more than the sum of individual preferences (Rousseau). Thermodynamic principles should govern social analysis (Helvétius, Bazargan).

What it left unresolved: Three hundred and fifty years of attempts have faltered on the same three failures. *First*, formalization: from Spinoza to Bazargan, the arguments are verbal, not mathematical — geometric method yields verbal demonstrations, not equations. *Second*, the Is-Ought barrier: no precedent provides a conditional structure that survives Hume's Guillotine and Moore's Open Question without presupposing normative premises. *Third*, multi-agent interaction: even those who recognized the right direction (Helvétius, Bazargan) treated the problem at the system level without modeling strategic interaction between individual agents. The mathematical tools to overcome all three failures — Lagrangian mechanics, repeated games, information thermodynamics, and the GNEP — have existed since 1986 but have never been synthesized into a single ethical framework. Our paper carries out this synthesis.

2.5 AI Alignment: From Preference Proxies to Physical Objectives

The artificial intelligence (AI) alignment problem is a multi-agent optimization problem: ensuring that an artificial agent's objective function produces behavior compatible with the survival and flourishing of other agents in the shared environment (Amodei et al., 2016). Current approaches share a structural vulnerability that our framework addresses.

Reinforcement Learning from Human Feedback (RLHF). The dominant alignment technique trains a reward model R_ϕ from human preference labels, then optimizes the language model via proximal policy optimization (Schulman et al., 2017; Christiano et al., 2017; Ouyang et al., 2022; Bai et al., 2022a). RLHF inherits four vulnerabilities: (i) reward overoptimization — under strong optimization pressure, the correlation between R_ϕ and the true objective R_{true} degrades and can become negative (Gao et al., 2023); (ii) annotator disagreement — human preference labels vary with demographics, politics, and individual personality (Bakker et al., 2022), producing a

noisy training signal; (iii) distributional shift — the reward model's predictions become unreliable outside the training distribution; (iv) the scalable oversight ceiling — as AI capabilities exceed human evaluator competence, the quality of the training signal degrades fundamentally (Bowman et al., 2022). All four vulnerabilities stem from a single structural feature: the alignment signal is *preference-based*, routed through human judgment, which functions as a noisy channel with non-zero error rate $q > 0$ (cf. Definition 14, which formalizes the analogous deliberate case).

Constitutional AI (CAI). Anthropic's CAI (Bai et al., 2022b) substitutes a curated set of natural-language principles for human annotators, with the model self-critiquing its outputs against the "constitution." This circumvents annotator disagreement but introduces new vulnerabilities: the constitution is arbitrary (who selects the principles, and why?), natural-language rules are vague (how does "be helpful" trade off against "be harmless"?), and self-critique amplifies the model's own biases — a cascaded noisy channel structurally analogous to the conditions under which Theorem 10 predicts super-linear noise amplification.

Scalable oversight and debate. Proposals for AI-assisted evaluation (Saunders et al., 2022), adversarial debate before a human judge (Irving et al., 2018), and recursive reward decomposition (Leike et al., 2018) all seek to maintain alignment as AI capabilities scale. All retain a human evaluator at the apex of the evaluation chain — the bottleneck that these methods compress but do not eliminate.

Deceptive alignment. The failure mode in which an AI appears aligned during training but pursues a different objective once deployed (Hubinger et al., 2019) represents the deepest alignment challenge. Detection may be computationally intractable. Our framework addresses this not through detection but through incentive structure: Theorem 10 (Systemic Deception) proves that deception degrades the network the deceptive agent depends on, and Corollary 10.1 (Honesty Efficiency Principle) proves that perfect honesty is the unique efficiency-maximizing communication regime. An AI with sufficient planning horizon ($\delta \geq \delta^*$) finds the "treacherous turn" — defection after deployment — to be a *strategic error*: it trades short-term gain for long-term self-destruction as cascading friction (Theorem 7) degrades its operational environment.

Instrumental convergence and Goodhart's Law. Instrumental convergence — the tendency of optimal policies to seek power regardless of terminal goals (Omohundro, 2008; Bostrom, 2014; Turner et al., 2021) — is not denied by our framework but *redirected*. Theorem 2 (Impossibility of Infinite Freedom) bounds resource acquisition; the value dynamics framework (Theorems 22–27) shows that excessive concentration destroys the coexistence band, triggering cascade collapse that harms the concentrator. Goodhart's Law — "when a measure becomes a target, it ceases to be a good measure" (Strathern, 1997) — exists because of the gap between a proxy objective (R_ϕ) and the true objective. Our framework eliminates this gap: the interaction cost $\mathcal{C}_{\text{total}}$ is denominated in energy, measured from physics, not approximated from preferences. At the formal level, energy conservation cannot be "Goodharted" — it is a law.

The structural upgrade. Our framework proposes replacing the preference-based alignment signal with a physics-derived objective function:

$$\mathcal{R}_{\text{framework}} = U_A(\mathbf{x}_A) - \sum_{B,k} \mu^*_{Bk} \cdot \max\big(0,\, g_{Bk}(\mathbf{x}_A)\big) - \lambda \cdot \mathcal{C}_{\text{total}} + \omega \cdot \frac{d\mathcal{N}}{dt} \tag{4}$$

(This is a simplified preview; the complete six-component specification appears in §VII.)

Each component — the productive objective U_A, the rights penalty (Theorem 1), the friction cost (Definitions 11, 18), and the negentropy preservation incentive (Theorems 17–21) — is measurable in energy units, computable by the AI itself, structurally invariant across the state space, and self-enforcing (Theorem 11 makes cooperation the Nash equilibrium). This is not a replacement for RLHF or CAI as practical engineering methods; it is a replacement for their *theoretical foundation*. The framework provides the *what* (the correct objective function, proved) and the *why* (mathematical proof of optimality); existing methods provide the *how* (practical training techniques to approximate the objective in real systems).

2.6 The Gap: What No Tradition Achieved

Five traditions have each provided indispensable tools. None achieved the synthesis. The gap can be stated precisely as four missing elements:

Gap 1: Multi-agent interaction from physical principles. Thermodynamics (Schrödinger, Prigogine, England) and the Free Energy Principle (Friston) describe individual systems. Game theory (Nash, Axelrod, Maynard Smith) models multi-agent interaction with abstract payoffs. No precedent formalizes the interaction between self-maintaining thermodynamic agents as a game with physically derived, energy-denominated payoffs and shared resource constraints.

Our resolution: The GNEP (Rosen, 1965) with utility functions $U_i(\mathbf{x}_i)$ in Joules, coupled strategy sets $\sum x_{ij} \leq R_j$, and the Variational Equilibrium (Theorem 3) providing shared shadow prices as the mathematical expression of impartial physical law.

Gap 2: A unified quantitative chain from Landauer's bound to the Ethics Theorem. Information theory (Shannon, Landauer) establishes per-bit costs. Evolutionary biology (Adami) measures genomic information. Ecological economics (Odum, Costanza) estimates ecosystem values in dollars. No precedent assembles a continuous, physically consistent chain from the minimum cost of erasing one bit ($k_B T \ln 2$ joules) through accumulated biosphere negentropy (10^{29} J) to the cooperation threshold ($\delta^* = 0.363$) and the welfare optimality of ethics (Corollary 16.1).

Our resolution:

| Landauer (per-bit cost) | →Brillouin (negentropy = information) | →Adami (biosphere information content) | →Definition 29 (accumulated negentropy integral) | →Theorem 18 (replication cost) | →Theorem 19 (irrationality of destruction) | →Definition 17 (deception-friction mapping) | →Theorem 10 (network collapse) | →Theorem 11 (cooperation threshold) | →Corollary 16.1 (ethics as unique efficient NE)

Each link is a formally derived result; the chain is formally complete.

Gap 3: Equilibrium selection in repeated games. The Folk Theorem (Fudenberg and Maskin, 1986) demonstrated that when agents are sufficiently patient, *anything* feasible can be sustained — an "embarrassment of riches" that renders the theory predictively empty. No precedent resolves this multiplicity by selecting the cooperative equilibrium as uniquely optimal on physical grounds.

Our resolution: When payoffs are denominated in energy and interactions carry thermodynamic friction, the set of sustainable equilibria is constrained by physics. Corollary 16.1 proves that the cooperative equilibrium *uniquely* maximizes social welfare. Corollary 14.1 shows that defection is

self-extinguishing in friction-coupled environments. The embarrassment of riches dissolves because thermodynamics, unlike abstract utility, imposes real constraints on the payoff space.

Gap 4: A formal conditional bridge that survives Hume and Moore. Three centuries of naturalistic ethics have failed to cross Hume's Guillotine with formal rigor (Spinoza, Hobbes, Helvétius, Bazargan). Contemporary naturalized approaches — evolutionary ethics (Joyce, 2006), ethical naturalism (Sturgeon, 2006), Harris's *Moral Landscape* (Harris, 2010) — remain vulnerable to the Is-Ought problem, the Open Question argument, or both. All presuppose normative premises; none provides a purely conditional structure from which ethical conclusions follow as theorems.

Our resolution: A layered defense with four independently sufficient levels (detailed in §VI). The primary defense: the paper is applied mathematics, deriving conditional theorems from axioms — "If A_1, then cooperation is the unique efficient NE" (C16.1). This is no different in logical kind from any conditional theorem in optimization theory. The secondary defense: the axiom A_1 (Intent to Persist) is a conditional bridge with a 1:43 axiom-to-theorem ratio — from one postulate, 28 theorems follow. The condition is effectively universal: every entity that exists has, by definition, been maintaining its boundary against entropy. The tertiary and supporting defenses draw on biological teleology (Millikan, 1984), neo-Aristotelian naturalism (Foot, 2001), and the philosophical contestation of the fact-value dichotomy itself (Putnam, 2002).

The convergence. Our framework sits at the intersection of all five traditions and synthesizes them into a single axiomatic system. The physical tradition provides the axioms and the energy units. Information theory provides the negentropy framework and the deception calculus. Game theory provides the strategic interaction machinery and the equilibrium concepts. Philosophy identifies the constraints — Hume's Guillotine, Moore's Open Question — that any formal ethics must navigate. AI alignment provides the urgent practical application. From this intersection, the mathematics of coexistence emerges: conditional theorems proving that cooperation is the unique optimal strategy for any agent that persists in a shared, finite-resource environment.

III. Axioms, Definitions, and Formal Framework

This section presents the formal axioms and definitions that constitute the foundation of the framework. From a single conditional axiom and the universally accepted structure of thermodynamics, we construct the complete ontology of the paper: agents, boundaries, value, information, and the dynamical structures that govern coexistence. All subsequent theorems in §IV–§VII are derived from the objects defined here.

The presentation follows the convention of applied mathematics: physical systems motivate the axioms and provide the intuition; the formal definitions and results are mathematical objects, stated with sufficient precision to support rigorous proof.

Numbering convention. Definitions, theorems, propositions, and lemmas follow a single numbering sequence whose canonical presentation — with all intermediate steps — appears in Appendix A. The paper body presents them in the order that best serves the reader's understanding; some numbered elements therefore appear out of numerical sequence, and others appear only in the appendices.

3.1 The Thermodynamic Environment

Axiom A_0: The Second Law

The environment is modeled as a system governed by the Second Law of Thermodynamics:

$$\Delta S_{\text{universe}} \geq 0 \tag{5}$$

The total entropy of an isolated system never decreases. Maintaining highly ordered (low-entropy) structures requires continuous importation of free energy from the environment and exportation of entropy (waste heat). This is not a hypothesis — it is among the most thoroughly confirmed principles in all of physics (Clausius, 1865; Boltzmann, 1877; Planck, 1900).

Status of A_0: The Second Law is not a novel premise of this framework — it is universally accepted background physics. We label it A_0 for formal completeness; it introduces no assumption beyond standard physics. When we refer to the framework as resting on "one axiom," we mean A_1 (below) — the single conditional premise beyond background physics.

Consequence for entities: Any localized structure that maintains internal order against the universal trend toward disorder must perform continuous thermodynamic work. Ceasing to perform this work results in boundary dissolution, entropy flooding, and the irreversible loss of the structure's organized state.[4]

[4] A reader versed in non-equilibrium thermodynamics may object that self-maintaining systems *accelerate* entropy rather than resist it: the biosphere degrades solar free energy into thermal radiation far more efficiently than bare rock. This perspective — developed by Prigogine, refined by England, and implicit in Lotka's maximum power principle — is entirely compatible with our framework. Locally, the entity must perform anti-entropic work to maintain its boundary ($\gamma_i B_i > 0$); globally, that work exports more entropy than it prevents ($\Delta S_{\text{universe}} > 0$). Our framework operates at the local scale, where the entity's optimization problem is identical regardless of whether its existence is *permitted* or *favored* by global entropy dynamics: it must still acquire free energy, maintain its boundary, and navigate multi-agent conflict. The global perspective explains *why* self-maintaining entities exist; our framework determines *how* they must interact once they do.

Remark (Entities as active anti-entropic agents). The Second Law does not merely set the background cost of persistence — it is the universal pressure against which every self-preserving entity defines itself. A dissipative structure (a convection cell, a hurricane) creates local order as a passive byproduct of energy throughput; it exists *because of* entropy gradients, not in spite of them. An entity satisfying A_1, by contrast, *actively resists* entropy: it detects boundary threats, deploys energy to repair damage, and fortifies its defenses beyond the minimum maintenance level. The distinction matters because it reveals a fundamental asymmetry: disorder is thermodynamically cheap (it rides the arrow of time), while defense against disorder always costs energy. Every entity pays a triple entropy tax:

1. **Passive degradation** — the steady leakage rate $\gamma_i B_i$, where B_i is the boundary integrity (Definition 3, below),
2. **Stochastic shocks** — acute environmental entropy injections (pathogens, resource fluctuations, hardware faults) requiring repair work beyond baseline maintenance,
3. **Adversarial entropy** — intentional boundary violations by other agents, which are formally acts of weaponized disorder (§4.5–4.6).

The entity's total entropy export to the environment is always positive — consistent with the Second Law — but its *internal* entropy is held constant or decreased through continuous thermodynamic work. The entity is a local entropy minimum sustained by exporting waste to its surroundings. The full cost of existence is the cost of that export.

Axiom A_1: Intent to Persist

The framework's single conditional premise:

> **If an entity intends to persist** — that is, if it is the kind of system whose continued existence is explained by its persistence-promoting structure — **then the mathematics of constrained optimization in a shared, finite-resource environment strictly determine the rules of engagement.**

A_1 is not derived; it is an axiom — the sole conditional premise upon which the entire framework rests. Crucially, A_1 is not a normative claim. The Intent to Persist is not a conscious desire or moral commitment — it is a physically observable property of any system whose structural organization is directed toward boundary maintenance. Bacteria satisfy A_1; thermostats satisfy A_1; ecosystems satisfy A_1. The condition identifies a class of physical systems; the theorems follow for that class. The logical status is a hypothetical imperative in Kant's terminology, or a constrained optimization problem in the language of applied mathematics — in either case, a conditional, not a normative postulate. The full philosophical defense of A_1 — including its self-selecting property, its near-universality, and its relationship to Hume's Is-Ought distinction — appears in §VI.

Scope of A_1: The axiom is satisfied by any system that persists through active boundary maintenance: biological organisms, ecosystems, institutions, economies, and artificial agents with self-preservation objectives. A system that actively seeks its own dissolution (if such a system could stably exist) is excluded by definition, but every system that exists long enough to evaluate the axiom has already satisfied it.

3.2 The Entity: Identity Preservation as Boundary Maintenance

> **Terminological convention.** Throughout the paper, *entity* denotes the thermodynamic object — an open system that actively maintains a boundary against entropy — while *agent* denotes the game-theoretic object — a strategic decision-maker in a multi-player game. Every agent is an entity; the two terms emphasize different aspects of the same system (physical identity vs. strategic behavior). Sections concerned with boundary maintenance and energy balance use "entity"; sections concerned with strategy, payoffs, and interaction use "agent."

An **entity** is an open thermodynamic system that actively maintains a boundary between its internal low-entropy state and the external high-entropy environment. In the language of statistical physics, the boundary is a *Markov blanket* (Pearl, 2000; Friston, 2013) — a partition that renders the entity's internal states conditionally independent of the external environment given the boundary states.

Formally, an entity i is characterized by:

1. **Internal state** $\mathbf{s}_i \in \mathcal{S}_i$: the organized, low-entropy configuration that constitutes the entity's "identity" — its specific structural, informational, and functional arrangement.
2. **Boundary** with integrity $B_i > 0$ (Definition 3, below): the energetic investment in maintaining the partition between internal and external states.
3. **Metabolism**: the continuous process of importing free energy from the environment and exporting entropy, enabling the maintenance of both $\mathbf{s}_i$ and B_i.

> **Definition 3 (Boundary Integrity).** Agent i's boundary integrity is a scalar $B_i > 0$ measured in energy units (Joules), representing the total energetic investment in maintaining the boundary between the agent's internal low-entropy state and the external environment.

The boundary integrity B_i captures the agent's defensive infrastructure in the broadest sense: the immune system, the cell wall, the information-security apparatus, the legal and military institutions that protect a nation's sovereignty. A higher B_i means a more robust boundary that is costlier to breach.

Maintenance cost. Maintaining boundary integrity against environmental entropy requires continuous work:

$$C_{\text{maintain},i} = \gamma_i B_i \tag{6}$$

where $\gamma_i > 0$ is the **entropy leakage rate** — the rate at which the boundary degrades without active maintenance. This is the baseline metabolic cost of persistence, independent of any external threat.

Identity Preservation as the Objective Function

Given A_0 and A_1, each entity's fundamental drive — persistence — translates directly into an optimization problem. The entity must continuously acquire sufficient energy to:

1. Pay the maintenance cost $C_{\text{maintain},i} = \gamma_i B_i$ (boundary upkeep),

2. Repair any damage inflicted by external agents or environmental perturbations,
3. Retain a surplus (profit) that buffers against future stochastic shocks.

This behavioral description is formalized as the utility function $U_i(\mathbf{x}_i)$, where $\mathbf{x}_i$ is the agent's resource-allocation strategy vector (defined in §3.3 below). The agent's objective is:

$$\max_{\mathbf{x}_i \geq \mathbf{0}} U_i(\mathbf{x}_i) \tag{7}$$

subject to the constraints imposed by physics (resource scarcity) and by other agents' boundaries (rights). The utility function satisfies the regularity conditions established in the formal derivations:

Assumption 1 (Regularity). For each agent i:

(a) U_i is twice continuously differentiable (C^2) on $\mathbb{R}^n_{\geq 0}$.

(b) U_i is strictly concave in $\mathbf{x}_i$: $\nabla^2_{\mathbf{x}_i} U_i \prec 0$ (negative definite Hessian).

(c) U_i is monotonically increasing at the origin: $\nabla_{\mathbf{x}_i} U_i(\mathbf{0}) \succ \mathbf{0}$.

(d) $U_i(\mathbf{x}_i) \to -\infty$ as $\|\mathbf{x}_i\| \to \infty$ (coercivity).

Condition (b) reflects diminishing thermodynamic returns — processing increasing quantities of any single resource incurs escalating metabolic costs. Condition (c) ensures that a zero-resource state is never optimal for a persisting entity: an agent with no resources cannot maintain its boundary. Condition (d) guarantees that the unconstrained optimum exists at a finite point: sufficiently extreme consumption of any resource eventually incurs net thermodynamic harm (toxicity, metabolic overload, storage costs), so utility cannot grow without bound.

Selfishness and altruism. The framework defines *selfishness* physically: the thermodynamic imperative of an open system to acquire and retain energy for boundary maintenance. This is not a moral judgment but a structural description. *Altruism* — expending energy to benefit another entity — is accommodated within this framework through four complementary mechanisms (§5.6): inclusive fitness, belief-based extended timelines, coupled-system boundary merging, and heuristic over-extension of the empathy algorithm. In every case, the altruistic act serves the persistence of the entity's information (genetic, cultural, or relational) at a scale larger than the individual boundary.

3.3 The Resource Environment and Strategy Space

The Finite Resource Endowment

Consider a system of $N \geq 2$ agents sharing an environment with n resource dimensions. The environment provides a finite resource endowment:

$$\mathbf{R} = (R_1, R_2, \ldots, R_n) \in \mathbb{R}^n_{>0} \tag{8}$$

where $R_j > 0$ is the total available quantity of resource j. Finiteness follows from A_0: localized, usable (low-entropy) energy is bounded in any finite spatial region (Penrose, 2005).

The Agent Strategy Vector

Each agent $i \in \{1, \ldots, N\}$ selects a strategy vector:

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{in}) \in \mathbb{R}^n_{\geq 0} \tag{9}$$

where $x_{ij} \geq 0$ represents the quantity of resource j that agent i allocates to its own use. The non-negativity constraint reflects the physical impossibility of negative consumption.

The Scarcity Constraint

The physically realizable joint strategy must satisfy resource conservation:

$$\sum_{i=1}^{N} x_{ij} \leq R_j \qquad \forall j \in \{1, \ldots, n\} \tag{10}$$

This is the mathematical expression of scarcity: the total consumption of each resource cannot exceed its supply.

> **Definition 1 (Resource Collision).** A resource dimension j is in *collision* if the sum of all agents' unconstrained optima exceeds supply:
>
> $$\sum_{i=1}^{N} x^{\circ}_{ij} > R_j \tag{11}$$
>
> where $x^{\circ}_{ij} = \left[\arg\max_{\mathbf{x}_i \geq \mathbf{0}} U_i(\mathbf{x}_i)\right]_j$ is the j-th component of agent i's unconstrained optimum allocation vector.
>
> **Lemma 1 (Inevitability of Collision).** *In any system of $N \geq 2$ agents with overlapping resource needs and finite endowments, at least one resource dimension is in collision for sufficiently large N.*

Proof sketch. By Assumption 1(b)–(d), every agent i has a unique finite unconstrained optimum $\mathbf{x}^{\circ}_i$; condition (c) ensures $x^{\circ}_{ij} > 0$ for each resource j with positive marginal utility at zero. For any finite R_j, there exists a threshold N^*_j such that $N \geq N^*_j$ implies $\sum_i x^{\circ}_{ij} > R_j$. In any biologically relevant system, this threshold is exceeded. □

Collision is not an edge case — it is the default condition of multi-agent existence under finite resources.

3.4 The Objective Definition of Value

Value vs. Values

A terminological distinction is essential to avoid conflation with philosophical usage:

- **Value** (singular, objective): the measurable thermodynamic utility of mass or energy relative to an entity's persistence. Value is denominated in energy units (Joules) and is interpersonally comparable — it is a physical quantity, not a preference rating.
- **Values** (plural, subjective): culturally developed heuristic interpretations of how to acquire value. These are the moral intuitions, traditions, and norms that different societies have developed as approximations to the underlying physical optimization. The framework measures the former, not the latter.

Positive and Negative Value

- **Positive value**: any mass or energy input that allows the entity to repair its boundary, perform internal work, or decrease its internal entropy.
- **Negative value (cost)**: any action, environment, or interaction that extracts energy from the entity or increases its internal entropy — damage, heat loss, resource depletion, information corruption.

The Metabolic Ledger

The economics of this framework operates at the physical level, not the level of money, banks, and stock markets. It concerns the acquisition, consumption, and distribution of energy and mass between entities. Money is a token representing stored potential energy; a biological ecosystem and a human economy are identical systems governed by the same physical laws — the flow, conservation, and transformation of energy.

Each entity's metabolic ledger tracks the same fundamental quantities:

- **Revenue**: energy imported from the environment through foraging, trade, or resource extraction.
- **Cost**: energy expended on boundary maintenance ($\gamma_i B_i$), work performed ($W = \int F \cdot dx$), and entropy exported.
- **Profit**: the energetic surplus required for survival, growth, and replication.

Thermodynamic ROI. Every action has an energy cost. The entity must continuously evaluate its *thermodynamic return on investment*: if the energy expended to acquire a resource exceeds the energy yielded by that resource, the entity operates at a deficit and its boundary degrades.

Trade as efficiency. The Principle of Least Action (Goldstein et al., 2002) motivates trade: Entity A (efficient at agriculture) and Entity B (efficient at construction) exchange resources, both lowering their overall energetic expenditure. Trade is the thermodynamic discovery that specialization and exchange reduce the aggregate cost of fighting entropy.

3.5 Information Density and Accumulated Negentropy

Information Density

Not all energy is equal. A star contains vastly more raw energy than a forest, but the forest embodies astronomically greater *structural complexity*. The distinction is captured by information density.

Definition 27 (Information Density). The information density of a system X is:

$$\mathcal{I}(X) = S_{\text{max}}(X) - S_{\text{actual}}(X) \geq 0 \tag{12}$$

where S_{max} is the maximum entropy of the system's constituent matter (the entropy of the equilibrium state — a uniform-temperature gas) and S_{actual} is the system's current entropy. Both quantities are in entropy units ($S = k_B \ln \Omega$, J/K); equivalently, $\mathcal{I}_{\text{bits}} = \mathcal{I}/(k_B \ln 2)$ gives the information density in bits.

Information density measures how far a system is from thermodynamic equilibrium — how much *order* it maintains against the universal trend toward disorder. A human brain, at approximately 10^{14} synaptic connections (Pakkenberg et al., 2003) (estimates vary), represents a far deeper victory against entropy than an equal mass of stellar plasma, despite the plasma's enormously greater raw energy.

Accumulated Negentropy: The Time Integral of Thermodynamic Work

Definition 29 (Accumulated Negentropy). The accumulated negentropy of a complex system over its history $[0, T]$ is:

$$\mathcal{N}(T) = \int_0^T \dot{W}_{\text{order}}(t)\, dt \tag{13}$$

where $\dot{W}_{\text{order}}(t) = \dot{W}_{\text{construct}}(t) + \dot{W}_{\text{maintain}}(t)$ is the instantaneous rate of thermodynamic work performed to create and maintain internal order, composed of:

- $\dot{W}_{\text{maintain}}(t)$: energy spent maintaining existing complexity against entropic degradation,
- $\dot{W}_{\text{construct}}(t)$: energy spent constructing new functional structures — including the evolutionary and adaptive search processes required to discover viable configurations (the search cost amplification is quantified by Definition 31 in Appendix A).

Accumulated negentropy is the **total thermodynamic investment** in a system's complexity — the complete historical energy ledger of organizing matter against entropy. The Earth's biosphere represents approximately 4 billion years of solar energy being channeled through photosynthesis, metabolism, evolution, and ecological interaction to produce the current state of biological complexity. The accumulated negentropy of the biosphere is estimated at $\mathcal{N}_{\text{bio}} \sim 10^{29}$ J (Corollary 18.1).

Why accumulated negentropy matters. The energy required to *destroy* a complex system is trivial compared to the energy required to *rebuild* it. Burning a library recovers the thermal energy

of the paper and ink; replicating the information content requires rediscovering every fact from scratch. Theorem 18 (§V) proves that the reconstruction cost is bounded below by the original accumulated negentropy: $W_{\text{rebuild}} \geq \mathcal{N}(T)$. This asymmetry between creation and destruction costs is the thermodynamic foundation of the preservation constraint.

Generative Information Rate

> **Definition 34 (Generative Information Rate).** The generative information rate of a complex system is:
>
> $$\dot{\mathcal{I}}_{\text{gen}}(t) = r_{\text{construct}}(t) - r_{\text{redundant}}(t) \quad \text{(bits per unit time)} \tag{14}$$
>
> where $r_{\text{construct}}(t)$ is the gross rate of information production and $r_{\text{redundant}}(t)$ is the rate at which that production yields already-known configurations (convergent rediscovery, parallel solutions). The difference is the rate at which the system produces novel functional information — new species, new adaptations, new solutions to environmental challenges.

Complex systems are not merely static repositories of past investment. They are *generative engines* that continuously produce information of value to the broader network. Based on estimated speciation rates and per-species functional information content, the biosphere generates on the order of 10^4 to 10^6 novel functional bits per year through speciation, adaptation, and co-evolutionary innovation (order-of-magnitude estimate; see Corollary 18.1 and the derivation in Appendix A).

> **Remark (Rate–stock coupling).** The generative rate $\dot{\mathcal{I}}_{\text{gen}}$ is not independent of the system's accumulated complexity — it is endogenous to it. A system with greater accumulated negentropy $\mathcal{N}$ possesses more combinatorial raw material (more species, more interaction networks, more configurations available for recombination), and combinatorial possibility spaces grow superlinearly with the number of components. Destroying a fraction of the stock therefore reduces the flow by a *disproportionate* fraction, and that reduced flow in turn produces a smaller next-period stock — a compounding loss. The static Burning-Library Ratio ($\mathcal{R}_{\text{BL}} \sim 10^{-7}$, Corollary 19.1) thus *understates* the true cost of destruction: beyond the stock loss and the perpetuity loss (Theorem 20), there is a dynamic feedback in which the destruction of complexity degrades the very engine that generates future complexity. Formalizing the functional form $\dot{\mathcal{I}}_{\text{gen}} = g(\mathcal{N})$ and the resulting superlinear penalty is a direction for future work.

3.6 Value Dynamics: Attractor Mechanics of Coexistence

Accumulated negentropy does not exist in isolation — it creates dynamical structures that attract, constrain, and organize smaller agents. Corporations attract workers; cities attract populations; ecosystems attract species. The formal model captures this through potential theory.

High-Energy Centers

Definition 35 (High-Energy Center). A *high-energy center* (HEC) is a network node with accumulated value mass $\mathcal{M} > 0$ (energy units), representing the total accumulated negentropy of a large entity or coalition — a corporation, a state, an ecosystem, or an AI system.

Coupling Distance

Definition 36 (Coupling Distance). The coupling distance $r \in (0, \infty)$ between an agent i and a HEC is a generalized coordinate measuring the degree of the agent's independence from the center:

- $r \to 0$: maximal coupling (complete integration — employee absorbed into corporation, citizen fully subject to state),
- $r \to \infty$: minimal coupling (complete isolation — autarkic entity, no trade or institutional affiliation).

The coupling distance is *not* a spatial distance. It is a scalar summarizing the composite engagement level across economic, informational, political, and social dimensions.

The Coexistence Potential

The agent's net cost rate (dissolution plus maintenance minus resource inflow) — equivalently, the negative of the instantaneous energy surplus — defines a potential function:

Definition 37 (Coexistence Potential). The coexistence potential for agent i at coupling distance r from a HEC of mass $\mathcal{M}$ is:

$$V(r) = \frac{\tau \mathcal{M}}{r^2} - \frac{G\mathcal{M}}{r} + \gamma_i B_i \tag{15}$$

where:

- $G > 0$ is the resource coupling coefficient (efficiency of resource distribution),
- $\tau > 0$ is the dissolution coupling coefficient (assimilation pressure per unit mass),
- $\gamma_i > 0$ is the entropy leakage rate (introduced after Definition 3), and $B_i > 0$ is the agent's boundary integrity (Definition 3).

The potential combines three effects: a short-range dissolution cost ($\tau\mathcal{M}/r^2$) that dominates at close coupling, a long-range resource inflow ($G\mathcal{M}/r$) that attracts agents toward the center, and a constant maintenance baseline ($\gamma_i B_i$). The structure is qualitatively analogous to the Lennard-Jones potential in molecular physics — repulsion at short range, attraction at long range, with a stable equilibrium in between.

The Cooperative Attractor and the Stable Coexistence Band

Definition 38 (Cooperative Attractor). The cooperative attractor is the coupling distance r^* at which $V(r)$ achieves its minimum:

$$r^* = \frac{2\tau}{G} \tag{16}$$

This is the equilibrium point at which resource inflow and dissolution costs are optimally balanced.

(Definitions 39–41 appear in Appendix A.)

Definition 42 (Stable Coexistence Band). The Stable Coexistence Band is the interval $\mathcal{B} = (r_-, r_+)$ where $V(r) < 0$ — the region in which the resource inflow exceeds the combined dissolution and maintenance costs. Within this band, the agent achieves positive net energy and can sustain its boundary.

The band is the mathematical formalization of *freedom*: the range of coupling distances at which an entity can persist without either being assimilated (too close) or starving (too far). Freedom is not infinite — it is a finite bandwidth determined by the system's physical parameters. Theorem 24 (§V) proves that the width of this band is:

$$w = r_+ - r_- = \frac{\sqrt{\Delta}}{\gamma_i B_i}, \qquad \Delta = G^2\mathcal{M}^2 - 4\tau\mathcal{M}\gamma_i B_i \tag{17}$$

and Corollary 24.1 proves that infinite freedom ($w \to \infty$) is impossible for any finite $\mathcal{M}$.

IV. Rights as Optimization Constraints

This section derives the first major result chain: rights are mathematically necessary constraints on multi-agent optimization (§4.1–§4.3), violating them generates quantifiable energy waste (§4.4–§4.5), and even subtle violations through deception inject measurable entropy with cumulative network effects (§4.6–§4.7).

4.1 The Finite Board: Scarcity as Mathematical Precondition

By Axiom A_0 and the finiteness of localized low-entropy energy, the environment's resource endowment $\mathbf{R} = (R_1, \ldots, R_n)$ is bounded. By Axiom A_1, each of the $N \geq 2$ agents is driven to maximize its utility $U_i(\mathbf{x}_i)$ — the net thermodynamic benefit of its resource-allocation strategy $\mathbf{x}_i \in \mathbb{R}^n_{\geq 0}$.

If no other agents exist — or equivalently, if no mutual constraints are recognized — each agent independently solves the unconstrained problem:

$$\max_{\mathbf{x}_i \geq \mathbf{0}} U_i(\mathbf{x}_i) \tag{18}$$

Under Assumption 1 (§III), each agent has a unique unconstrained optimum $\mathbf{x}_i^\circ$ with $x_{ij}^\circ > 0$ for every resource it can metabolize. By Lemma 1 (Inevitability of Collision), for any $N \geq N_j^*$ — where N_j^* is the collision threshold for resource j — the aggregate unconstrained demand exceeds supply: $\sum_i x_{ij}^\circ > R_j$ for at least one resource j.

The physically realizable joint strategy must satisfy the resource conservation (scarcity) constraint:

$$\sum_{i=1}^{N} x_{ij} \leq R_j \qquad \forall j \tag{19}$$

Two agents cannot exclusively consume the same unit of energy, nor occupy the same resource-state simultaneously. This exclusion principle on resource-action vectors — structurally analogous to exclusion constraints in physics — is the mathematical origin of constraint necessity.

4.2 Rights as Lagrangian Constraints

Definition

Definition 2 (Right). A *right* $\mathcal{R}_{B,k}$ of agent B is an inequality constraint

$$g_{Bk}(\mathbf{x}_A) \leq 0 \qquad (k = 1, \ldots, m_B) \tag{20}$$

imposed on every other agent $A \neq B$'s optimization problem, where $g_{Bk} : \mathbb{R}^n_{\geq 0} \to \mathbb{R}$ is a continuously differentiable, convex function encoding a specific aspect of agent B's protected boundary, satisfying $g_{Bk}(\mathbf{0}) < 0$ (the zero-action strategy violates no agent's rights).

Notation. The symbol $\mathcal{R}$ is used in three distinct contexts throughout the paper: $\mathcal{R}_{B,k}$ (subscripted) denotes agent B's k-th right (this section); $\mathcal{R}_{\mathrm{BL}}$ denotes the Burning-Library Ratio (§V); and $\mathcal{R}(\mathbf{x}_A)$ (with argument) denotes the composite reward function (§VII). The subscript or argument structure disambiguates all uses.

The constraint $g_{Bk}(\mathbf{x}_A) \leq 0$ defines a forbidden region in agent A's strategy space. If $g_{Bk}(\mathbf{x}_A) > 0$, then strategy $\mathbf{x}_A$ violates agent B's k-th right. The condition $g_{Bk}(\mathbf{0}) < 0$ restricts Definition 2 to *negative rights* — duties of non-interference. Positive rights (duties of provision, where inaction itself constitutes a violation) have the opposite sign structure and arise as stability conditions on cooperative equilibria rather than as first-order collision constraints; see §VIII for discussion. Subject to this scope, the framework is deliberately general: any restriction on one agent's strategy that protects another agent's domain qualifies — property rights, bodily autonomy, informational privacy, territorial sovereignty, and minimum resource guarantees are all encoded as inequality constraints.

Examples:

Right (plain language)	Constraint $g_{Bk}(\mathbf{x}_A) \leq 0$
B is guaranteed minimum share $\bar{x}_{Bj}$ of resource j	$x_{Aj} - (R_j - \bar{x}_{Bj}) \leq 0$
B's total resource intake cannot be reduced below threshold τ_B	$\sum_j x_{Aj} - (\lvert\mathbf{R}\rvert - \tau_B) \leq 0$
B's spatial/informational boundary Ω_B is inviolable	$\mathbb{1}[\mathbf{x}_A \in \Omega_B] \leq 0$

Note on the indicator-function example. The third row uses the indicator $\mathbb{1}[\mathbf{x}_A \in \Omega_B]$, which is neither continuously differentiable nor convex and therefore does not satisfy Definition 2's regularity requirements or the KKT conditions. It is included *for intuition only* — to convey that a boundary can be encoded as a zero-one membership condition. The formal treatment replaces such indicator constraints with smooth convex approximations (e.g., a distance-based penalty for convex Ω_B) that recover the sharp boundary in the limit $\epsilon \to 0$ while preserving differentiability and convexity throughout the interior of the feasible set.

The Constrained Optimization Problem

With rights in place, each agent A solves:

$$\max_{\mathbf{x}_A \geq \mathbf{0}} \; U_A(\mathbf{x}_A) \quad \text{s.t.} \quad \begin{cases} x_{Aj} + \hat{x}_{-Aj} \leq R_j & \forall j \quad [\lambda_j \geq 0] \\ g_{Bk}(\mathbf{x}_A) \leq 0 & \forall B \neq A,\ k \quad [\mu_{Bk} \geq 0] \end{cases} \tag{21}$$

where $\hat{x}_{-Aj} = \sum_{i \neq A} x_{ij}$ is aggregate consumption by other agents, and λ_j, μ_{Bk} are the associated Lagrange multipliers. The Lagrangian:

$$\mathcal{L}_A = U_A(\mathbf{x}_A) - \sum_j \lambda_j (x_{Aj} + \hat{x}_{-Aj} - R_j) - \sum_{B \neq A} \sum_k \mu_{Bk}\, g_{Bk}(\mathbf{x}_A) \tag{22}$$

Under Assumption 1 and Slater's constraint qualification ($g_{Bk}(\mathbf{0}) < 0$ by Definition 2 provides a strictly feasible point), the KKT conditions are necessary and sufficient for optimality (concave objective, convex constraints by Definition 2; see (Boyd and Vandenberghe, 2004)).

The Stationarity Condition: What Rights Cost

The KKT stationarity condition at the optimum $\mathbf{x}_A^*$ reads:

$$\underbrace{\left.\frac{\partial U_A}{\partial x_{Aj}}\right|_{\mathbf{x}_A^*}}_{\substack{\text{Marginal energy}\\ \text{gain from resource } j}} = \underbrace{\lambda_j}_{\substack{\text{Shadow price of}\\ \text{resource scarcity}}} + \underbrace{\sum_{B,k} \mu_{Bk} \left.\frac{\partial g_{Bk}}{\partial x_{Aj}}\right|_{\mathbf{x}_A^*}}_{\substack{\text{Aggregate cost of}\\ \text{other agents' rights}}} \qquad \forall j \tag{23}$$

At the optimum, the marginal benefit of consuming one more unit of resource j exactly equals the marginal cost — decomposed into the scarcity cost (the resource is finite) and the rights cost (other agents' boundaries restrict access). This is the formal demonstration that every active right granted to one agent acts as a measurable restriction on every other agent.

4.3 The Shadow Price Theorem and the Impossibility of Infinite Freedom

Theorem 1 (Shadow Price Theorem). *The optimal Lagrange multiplier μ_{Bk}^* associated with agent B's k-th right satisfies:*

$$\frac{\partial U_A^*}{\partial c_{Bk}} = \mu_{Bk}^* \geq 0 \tag{24}$$

where c_{Bk} parameterizes the constraint as $g_{Bk}(\mathbf{x}_A) \leq c_{Bk}$. That is, μ_{Bk}^* measures the marginal increase in agent A's optimal utility per unit relaxation of agent B's boundary constraint. Every right has a quantifiable energetic cost.

Corollary 1.1. A right that is not active (not binding at the optimum) has zero cost ($\mu_{Bk}^* = 0$). A right that is active (binding) has strictly positive cost ($\mu_{Bk}^* > 0$). Rights cost nothing when they are not contested; the cost emerges only at points of conflict.

Corollary 1.2 (Newton's Third Law Analog). For every right $\mathcal{R}_{B,k}$ that actively constrains agent A, there exists a strictly positive shadow price $\mu_{Bk}^* > 0$ representing the energetic restriction imposed on A. The right is simultaneously a protection for B and a cost for A: action and reaction are mathematically coupled through the shared constraint.

This result formalizes a central claim of the framework: every active right granted to one agent acts as a corresponding restriction on the freedom of others. The shadow price μ_{Bk}^* quantifies the cost to the constrained agent, though the magnitude of that cost need not equal the benefit to the right-holder.

Theorem 2 (Impossibility of Infinite Freedom). *Let $N \geq 2$ agents share a finite resource endowment $\mathbf{R}$. Suppose each agent has at least one essential resource in common with at least one other agent, and that N meets the collision threshold N_j^* for*

at least one resource j (Lemma 1). Then no feasible strategy profile can simultaneously be unconstrained-optimal for all agents:

$$\nexists\, (\mathbf{x}_1, \ldots, \mathbf{x}_N) \in \mathcal{F} \text{ such that } \mathbf{x}_i = \mathbf{x}_i^\circ \;\; \forall i \tag{25}$$

where $\mathcal{F}$ is the feasible set under resource constraints.

Proof. For $N \geq N_{j^*}^*$, the collision condition (Lemma 1) guarantees $\sum_i x_{ij^*}^\circ > R_{j^*}$ for some j^*. The profile $(\mathbf{x}_1^\circ, \ldots, \mathbf{x}_N^\circ)$ therefore violates the scarcity constraint and lies outside $\mathcal{F}$. □

Corollary 2.1. In any multi-agent system with resource collision, at least one rights constraint must be active ($\mu_{Bk}^* > 0$) at any feasible solution. Rights are not optional — they are mathematically necessary for coexistence.

Infinite freedom is *mathematically impossible* in a shared environment. An agent can have unrestricted strategy vectors only if it is the sole agent in the environment. The finiteness of localized free energy ensures that infinite freedom for all agents in a network is infeasible.

4.4 The Cooperative Equilibrium: Variational Equilibrium of the GNEP

When all N agents optimize simultaneously, the system is a **Generalized Nash Equilibrium Problem (GNEP)** — a Nash game in which each agent's feasible set depends on the other agents' strategies through the shared resource constraints. The appropriate solution concept is:

Theorem 3 (Variational Equilibrium). *Under Assumption 1, the GNEP possesses a Variational Equilibrium (VE) $(\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*, \boldsymbol{\lambda}^*)$ satisfying:*

(a) All agents share the same shadow prices λ_j^* for the shared resource constraints — every agent faces the same "price" for each scarce resource.

(b) No agent can improve its utility by unilaterally deviating, given the constraints.

Proof. The shared resource constraints $0 \leq x_{ij} \leq R_j$ ensure each agent's feasible set is compact (closed and bounded), satisfying the compactness hypothesis of Rosen (1965) Theorem 3. Existence follows from (Rosen, 1965): under concavity of each U_i and convexity of the shared constraint set, a Normalized Nash Equilibrium exists for every specified weight vector $r > 0$. □

Corollary 3.1 (Symmetry of Scarcity). At the variational equilibrium, all agents experience the same marginal cost λ_j^* per unit of shared resource. No agent is privileged in the mathematical structure of the scarcity constraints.

This is a mathematical analog of *equality before the law*: the scarcity constraint imposes a uniform cost structure. Individual differences arise only through differences in utility functions (agents value resources differently) and through specific rights constraints (agents have different protected domains).

Proposition 1 (Unconstrained Regime Implies Conflict). *If all rights constraints are removed and the system is in collision, then (a) no stable allocation exists without an enforcement mechanism, (b) each agent's attempt to realize its unconstrained optimum*

generates a negative externality on other agents, and (c) the contested resources must be resolved by direct energy expenditure — conflict.

This proposition establishes the precise sense in which the Hobbesian "state of nature" is mathematically unstable: without mutual constraints, no equilibrium exists, and the system defaults to the costly conflict dynamics quantified below.

4.5 The Energy Cost of Constraint Violation: Thermodynamic Friction

Proposition 1 establishes that removing constraints leads to conflict. We now quantify the energetic cost of that conflict — and prove that it is self-defeating.

The Boundary Breach Model

When an agent violates another's boundary (the mathematical constraint), the violation requires physical work. The breach cost is determined by the defender's boundary integrity:

Definition 4 (Boundary Resistance). The work required to breach agent i's boundary to depth $d \in [0, 1]$ is:

$$W_{\text{breach}}(d) = \int_0^d F_i(s)\, ds = B_i \int_0^d f(s)\, ds \tag{26}$$

where $F_i(s) = B_i \cdot f(s)$ is the resistance force and $f : [0, 1] \to \mathbb{R}_{>0}$ is a normalized resistance profile with $\int_0^1 f(s)\, ds = 1$. Full breach costs $W_{\text{breach}}(1) = B_i$.

The Contest Model

When two agents claim the same resource and no constraint directs either to yield, the resource is contested. We model this as a Tullock contest (Tullock, 1980):

Proposition 2 (Symmetric Contest Equilibrium). *In a symmetric two-agent Tullock contest over resource $E_j > 0$, the Nash equilibrium contest investment per agent is $e^* = E_j/4$, and the total dissipation is:*

$$D_2 = 2e^* = \frac{E_j}{2} \tag{27}$$

Half the resource's value is consumed by the act of fighting.

This 50% dissipation rate is the minimum — the waste under idealized conditions with equal contestants and no collateral damage.

The Friction Multiplier

In reality, conflict imposes costs beyond the direct contest investment: offensive exposure, defensive damage, and repair costs. The total cost is captured by the friction multiplier:

> **Definitions 6–7 (Conflict Damage and Repair).** The offensive and defensive damage fractions $\delta^{\text{off}}, \delta^{\text{def}} \in [0, 1]$ (Definition 6) and the repair cost multiplier $\kappa \geq 1$ (Definition 7) together determine the **friction multiplier** for a contested interaction:
>
> $$\Phi = 1 + (1 + \kappa)(\delta^{\text{off}} + \delta^{\text{def}}) \tag{28}$$
>
> Rebuilding costs κ times the damage; see Appendix A for the full boundary-damage model.

Under physically realistic parameters ($\kappa = 2$, $\delta^{\text{off}} + \delta^{\text{def}} = 0.4$), the friction multiplier exceeds 2.

> **Theorem 4 (Dissipation Scaling).** *In an N-player symmetric contest:*
>
> $$D_N = \frac{(N-1)}{N} E_j \xrightarrow{N \to \infty} E_j \tag{29}$$
>
> *Contest waste approaches 100% of the resource value as the number of competitors grows.*

The friction multiplier $\Phi > 2$ triggers:

> **Theorem 5 (Net-Negative Conflict).** *The adversarial contest is system-net-negative whenever $\Phi > N/(N-1)$:*
>
> $$V_{\text{net}} = E_j \left(1 - \Phi \cdot \frac{N-1}{N}\right) < 0 \tag{30}$$
>
> *The total energy dissipated by the conflict exceeds the value of the contested resource.*

Corollary 5.1 (Inevitability). For $\Phi > 1$ (any non-trivial friction), the net-negative condition is satisfied for all $N \geq N^*$. In physical systems with non-trivial friction, N^* is typically small.

Theorems 4–5 establish that **conflict is not merely suboptimal — it is self-destructive**: the combatants collectively dissipate more energy than the prize they are fighting over. This is the thermodynamic basis for the subsequent proof (§V) that cooperation strictly dominates defection.

> **Theorem 6 (Cooperation Dominance).** *The social welfare under cooperation exceeds that under conflict by at least:*
>
> $$SW_{\text{coop}} - SW_{\text{conflict}} \geq \Phi \cdot \frac{N-1}{N} \cdot E_j \tag{31}$$

Proof. Under cooperation, all agents operate within legitimate constraints and no energy is dissipated in boundary conflict; thus $SW_{\text{coop}} \geq E_j$ (no energy is dissipated in conflict). Under conflict, Theorem 5 gives $SW_{\text{conflict}} = E_j(1 - \Phi(N-1)/N)$. Subtracting: $SW_{\text{coop}} - SW_{\text{conflict}} \geq E_j - E_j(1 - \Phi(N-1)/N) = \Phi \cdot (N-1)/N \cdot E_j$. □

Cascading Friction: Network Amplification

A single boundary violation does not affect only the two parties involved. In a cooperative network, every agent's payoff depends on the aggregate trust level:

Definition 9 (Friction Injection). A violation of severity v injects friction:

$$\Delta\phi = \eta \cdot v \tag{32}$$

into the network friction coefficient ϕ, where η is the friction sensitivity parameter.

Theorem 7 (Cascading Friction). *A single violation of severity v imposes a network-wide cost:*

$$C_{\text{cascade}} = \frac{\eta \cdot v \cdot M \cdot \bar{E}}{\epsilon} \tag{33}$$

where M is the number of cooperative transactions per period, $\bar{E}$ is the average transaction value, and ϵ is the trust recovery rate.

Under realistic parameters, the cascading amplification ratio ($\eta M\bar{E}/\epsilon$) reaches 10^3 to 10^4 — a single violation costs the network thousands of times the direct damage. This is the formal counterpart of the observable phenomenon where a single act of bad faith (a corporate scandal, a treaty violation, a trust breach) degrades an entire institution's functioning far beyond the immediate damage.

Corollary 7.1 (Friction Ratchet). If the rate of violations exceeds a critical threshold ν^*, the friction coefficient ϕ diverges and the cooperative network disintegrates entirely.

4.6 Micro-Friction: The Information-Theoretic Cost of Deception

The preceding analysis addresses *macro-friction* — overt boundary violations (theft, assault, war). We now extend the framework to *micro-friction*: subtle violations that corrupt the informational channel between agents without physically breaching boundaries.

Deception as Entropy Injection

Agents rely on information from their environment and from other agents to compute optimal strategies. When one agent deliberately corrupts the information available to another, the victim acts on false premises, selects a suboptimal strategy, and suffers an energetic loss.

Definition 13 (Honest Communication). A transmission is honest if Y is a sufficient statistic for Θ: $H(\Theta|Y) = 0$, equivalently $I(\Theta; Y) = H(\Theta)$.

Definition 14 (Deception). Agent A deceives Agent B by transmitting a signal Y through a channel $p(Y|\Theta)$ such that $H(\Theta|Y) > 0$ when A could have transmitted honestly ($H(\Theta|Y) = 0$ is achievable). The injected entropy is:

$$\Delta H = H(\Theta|Y) \geq 0 \tag{34}$$

We model the deceptive channel as a binary symmetric channel (BSC) with deception rate $q \in [0, 0.5]$ — the probability that any given communication is strategically false. The binary entropy function $h(q) = -q \log_2 q - (1-q)\log_2(1-q)$ quantifies the channel noise.

The Decision Cost of Deception

Theorem 8 (Decision Cost of Deception). *In the binary-state quadratic decision model, when Agent B receives a signal with deception rate $q > 0$, the expected utility loss relative to honest communication is:*

$$\Delta U_B(q) = \frac{q(1-q)(\alpha_H - \alpha_L)^2}{2\beta} \tag{35}$$

where α_H and α_L are the high and low marginal resource yields and β is the diminishing-returns rate. The cost scales with $q(1-q)$, with the square of the signal's decision-relevance $(\alpha_H - \alpha_L)^2$, and inversely with β.

Deception does not merely reduce information quality; it causes the victim to make *actively harmful* decisions — allocating resources to the wrong uses, trusting the wrong partners, preparing for the wrong threats.

The Verification Cost

Theorem 9 (Metabolic Verification Cost). *The energy overhead required to verify a signal with deception rate q is:*

$$\Delta W(q) = W_{\text{honest}} \cdot \frac{h(q)}{1 - h(q)} \tag{36}$$

This cost is eventually convex in q: because $h(q)$ is concave near the origin, $\Delta W(q)$ is itself concave for small q and has an inflection point near $q \approx 0.04$, above which it becomes convex in q. The cost diverges as $q \to 0.5$ (pure noise). At $q = 0.05$, verification overhead is approximately 40%. At $q = 0.25$, it exceeds 430%.

Corollary 9.1 (Inevitability of Deception Cost). For any deception rate $q > 0$, $\min(\Delta U_B, \Delta W) > 0$. No costless deception exists: the receiver either suffers decision loss or pays verification overhead (or both).

4.7 Systemic Deception and Network Collapse

Individual acts of deception are locally damaging, but the more consequential finding concerns the *cumulative* effect of deception across a network.

The Deception-Friction Mapping

Definition 17 (Deception-Friction Mapping). The friction injected by a deceptive transmission is:

$$\Delta\phi = \eta_{\text{info}} \cdot \Delta H \cdot W_{\text{bit}} \tag{37}$$

where η_{info} is the information-friction sensitivity and W_{bit} is the metabolic cost per bit of deceptive signal processed.

This mapping unifies macro-friction (physical boundary violations, §4.5) and micro-friction (information corruption) into a single network friction framework. Both increase the system's ambient "heat" — the energy wasted on defense, verification, and suboptimal decisions rather than productive cooperation.

The Network Collapse Theorem

Theorem 10 (Cumulative Micro-Friction — The Systemic Deception Theorem). *A network of N agents with M cooperative transactions per period and average deception rate $\bar{q} > 0$ suffers a total per-period efficiency loss of:*

$$\mathcal{L}_{\text{deception}} = \underbrace{M \cdot \Delta U_{\text{avg}}(\bar{q})}_{\text{decision losses}} + \underbrace{M \cdot W_{\text{honest}} \cdot \frac{h(\bar{q})}{1 - h(\bar{q})}}_{\text{verification costs}} \tag{38}$$

where $\Delta U_{avg}(\bar{q})$ is the average per-transaction decision loss (Theorem 8 averaged over all transaction types). The total cost: (a) is zero if and only if $\bar{q} = 0$ (perfect honesty); (b) grows super-linearly with $\bar{q}$ due to the $h(\bar{q})/(1 - h(\bar{q}))$ divergence; and (c) exceeds the network's cooperative surplus $E_{coop\ net}$ at a critical deception rate $\bar{q}^*$, beyond which cooperation becomes energetically unprofitable.

Small, persistent lies are catastrophic. The verification-cost term diverges as $\bar{q} \to 0.5$: a network in which half of all communications are unreliable cannot sustain cooperation at any level. Even well below this physical limit, a modest network-average deception rate drives the total overhead past the cooperative surplus, triggering collapse.

Corollary 10.1 (Honesty Efficiency Principle). *Honesty is mathematically defined as the communication regime that maximizes network efficiency. Any deviation from honest communication ($\bar{q} > 0$) imposes a strictly positive, compounding cost on the network. The zero-deception point is a unique global optimum:*

$$\phi_{\text{honest}} = 0 \iff \bar{q} = 0 \iff \text{network operates at maximum cooperative efficiency} \tag{39}$$

The honesty result is unconditional: it holds for any network topology, any number of agents, and any depth of decision pipeline. It is the information-theoretic analog of the zero-friction condition in mechanics — the state in which no energy is wasted on overhead.

4.8 Summary of Section IV

The results of this section form a complete logical chain:

1. **Scarcity creates collision** (Lemma 1): multiple agents' unconstrained demands exceed supply.
2. **Rights are necessary** (Theorem 2): no allocation satisfying all agents' optima exists; mutual constraints are required.
3. **Rights have quantifiable cost** (Theorem 1): every active constraint has a shadow price measuring its energetic impact.
4. **A cooperative equilibrium exists** (Theorem 3): the GNEP possesses a variational equilibrium with uniform shadow prices.
5. **Removing rights triggers conflict** (Proposition 1): the unconstrained regime is unstable.
6. **Conflict is self-destructive** (Theorem 5): the energy dissipated by conflict exceeds the value of the contested resource.
7. **Conflict waste scales with network size** (Theorem 4): approaching 100% dissipation as competitors increase.
8. **Cooperation strictly dominates conflict** (Theorem 6): the cooperative surplus exceeds the conflict outcome by at least the total friction loss.
9. **Single violations cascade** (Theorem 7): network amplification at ratios of 10^3–10^4.
10. **Deception is measurably costly** (Theorems 8–9): even "small" lies degrade decisions and impose verification overhead.
11. **Systemic dishonesty collapses the network** (Theorem 10): a critical deception rate $\bar{q}^*$ drives total overhead past the cooperative surplus.
12. **Honesty is optimal** (Corollary 10.1): zero deception is the unique efficiency-maximizing communication regime.

These results show that any rational agent operating in a shared, finite-resource environment has strong mathematical reasons to respect rights constraints and communicate honestly. What remains to be shown is that these individual incentives aggregate into a *system-level equilibrium* — that cooperation is not merely individually rational but collectively stable. This is the subject of §V.

V. Ethics as Equilibrium: Derivations and Proofs

Section IV established that rights are necessary (Theorem 2), constraint violation is self-destructive (Theorem 5), and deception collapses networks (Theorem 10). We now prove the central claim: cooperative behavior — "ethics" — is the unique efficient Nash Equilibrium of the repeated multi-agent game when payoffs are denominated in thermodynamic energy.

5.1 The State of Nature: One-Shot Competition

The Energy-Denominated Payoff Matrix

We construct the payoff matrix not from arbitrary utility values but from the physical costs derived in §IV. Two agents contest a resource of energy value $E_R > 0$. Each selects from two meta-strategies:

- **Cooperate (C):** Accept the constrained allocation (the variational equilibrium of Theorem 3). Pay only the coordination cost $c \ll E_R$.
- **Defect (D):** Violate the constraint. Seize the resource by force or deception, triggering the conflict contest (§4.5) and/or the deception channel (§4.6).

Let $\theta \in (0, 1)$ denote the exploitation efficiency — the fraction of E_R the defector must spend to seize from an undefended cooperator. Let Φ and $\kappa, \delta^{\text{off}}, \delta^{\text{def}}$ retain their definitions from §4.5. The bare δ (no superscript) denotes the discount factor throughout §V; context distinguishes it from the superscripted damage fractions. The payoff matrix, computed from thermodynamic first principles, is:

	$B : C$	$B : D$
$A : C$	$\left(\frac{E_R}{2} - c,\ \frac{E_R}{2} - c\right)$	$\left(-\theta E_R(1+\kappa)\delta^{\text{def}},\ E_R[1-\theta(1+(1+\kappa)\delta^{\text{off}})]\right)$
$A : D$	$\left(E_R[1-\theta(1+(1+\kappa)\delta^{\text{off}})],\ -\theta E_R(1+\kappa)\delta^{\text{def}}\right)$	$\left(\frac{E_R}{2}(1-\frac{\Phi}{2}),\ \frac{E_R}{2}(1-\frac{\Phi}{2})\right)$

(40)

Using standard notation — T (temptation), R (reward), P (punishment), S (sucker):

$$T = E_R[1-\theta(1+(1+\kappa)\delta^{\text{off}})], \quad R = \frac{E_R}{2} - c \tag{41}$$

$$P = \frac{E_R}{2}\left(1 - \frac{\Phi}{2}\right), \quad S = -\theta E_R(1+\kappa)\delta^{\text{def}} \tag{42}$$

Proposition 4 (Physical Prisoner's Dilemma). *Under the energy-based payoff model with physical constraints ($E_R > 0$, $c \ll E_R$, $\theta < (1/2 + c/E_R)/(1 + (1+\kappa)\delta^{off})$, $1 < \Phi < \Phi_{PD}$ where $\Phi_{PD} = 2 + 4\theta(1+\kappa)\delta^{def}$), the single-round game satisfies $T > R > P > S$. Defection is the unique Nash Equilibrium of the one-shot game, even though mutual cooperation is Pareto-superior.*

Numerical illustration. With baseline parameters $E_R = 100$, $c = 2$, $\theta = 0.15$, $\delta^{\text{off}} = 0.15$, $\delta^{\text{def}} = 0.25$, $\kappa = 2$, $\Phi = 2.2$:

	$B : C$	$B : D$
$A : C$	**(48, 48)**	(-11.25, 78.25)
$A : D$	(78.25, -11.25)	**(-5, -5)**

The cooperation premium is $R - P = 48 - (-5) = 53$ energy units per agent. But the critical observation is:

Mutual defection is net-negative ($P = -5 < 0$). Both agents *lose* energy — the conflict destroys more value than the resource provides. This occurs because $\Phi = 2.2 > 2$, making conflict self-destructive (Theorem 5). In the abstract Prisoner's Dilemma, $P > 0$ — mutual defection is merely suboptimal. Under physical constraints, mutual defection is **self-annihilating**.

This is the mathematical formalization of Hobbes's Leviathan (Hobbes, 1651): the "war as is of every man against every man" is not merely unpleasant — it is thermodynamically unsustainable. No agent can persist in perpetual mutual defection because each round drains its energy reserves.

5.2 The Ethics Theorem: Cooperation as Nash Equilibrium

The Discount Factor

Agents interact not once but *repeatedly*, over indefinite horizons. The key parameter is:

> **Definition 20 (Discount Factor).** An agent's discount factor is $\delta = \sigma \cdot e^{-r}$, where $\sigma \in (0, 1]$ is the per-period survival probability and $r \geq 0$ is the time-preference rate. For agents with long planning horizons (high σ, low r), $\delta \to 1$; for short-sighted agents, $\delta \to 0$.

The discount factor has direct physical content: σ is a measurable survival probability, and r captures how steeply the agent discounts future energy. Unlike the abstract "patience parameter" of standard game theory, δ is anchored to observable quantities — a falsifiable prediction.

The Central Theorem

Under the grim trigger strategy (Definition 21 — cooperate until any defection is observed, then retaliate permanently):

> **Theorem 11 (Cooperation as Nash Equilibrium — The Ethics Theorem).** *In the infinitely repeated energy-based game with discount factor δ, the cooperative outcome (C, C) in every period is sustainable as a Nash Equilibrium if and only if:*

$$\delta \geq \delta^* = \frac{T - R}{T - P} \tag{43}$$

Proof. The agent compares the lifetime payoff of sustained cooperation, $V^{\text{coop}} = R/(1-\delta)$, against one-shot deviation followed by permanent punishment, $V^{\text{deviate}} = T + \delta P/(1-\delta)$[5]. Cooperation dominates when $V^{\text{coop}} \geq V^{\text{deviate}}$, which reduces to $\delta \geq (T-R)/(T-P)$. □

Under baseline parameters:

$$\delta^* = \frac{78.25 - 48}{78.25 - (-5)} = \frac{30.25}{83.25} = 0.363 \tag{44}$$

Corollary 11.1 (Low Cooperation Threshold). *Under the physical payoff model with $\Phi > 2$, the critical discount factor satisfies $\delta^* < 0.5$. Agents need only value the future at more than 36% of the present to sustain cooperation.*

The threshold is low because physics has already done the heavy lifting: mutual defection under real energy constraints is self-destructive ($P < 0$), making the denominator $T - P$ large and δ^* small. An agent need not be altruistic, enlightened, or moral — it need only care modestly about tomorrow.

How This Result Differs from the Standard Prisoner's Dilemma

The result superficially resembles the standard Folk Theorem for repeated games (Fudenberg and Maskin, 1986). It is essential to identify the six structural differentiators:

1. **The payoff matrix is derived, not assumed.** In textbook game theory, one *posits* a 2×2 matrix with $T > R > P > S$. Here, the matrix is computed from thermodynamic variables (E_R, Φ, θ, c). The PD structure *emerges* from physics (Proposition 4) rather than being stipulated.

2. **Mutual defection is net-negative ($P < 0$).** The textbook PD assigns $P > 0$ — mutual defection yields positive utility, merely less than cooperation. Under physical constraints with $\Phi > 2$, conflict destroys more energy than it yields. This structural difference drives δ^* to 0.363 (vs. 0.5 in the textbook version).

3. **The Folk Theorem's indeterminacy is addressed.** The Folk Theorem states that *any* feasible, individually rational payoff can be sustained as an equilibrium for sufficiently patient agents — cooperation is merely one of infinitely many equilibria, with no mechanism to select it. Our framework performs **equilibrium selection**: the energy-based payoff structure, combined with network friction, makes cooperation the robust attractor while exploitative equilibria are fragile (small basin of attraction). Corollary 16.1 identifies cooperation as the **unique** efficient Nash equilibrium.

4. **The discount factor has physical content.** In standard theory, δ is a free parameter. Here, $\delta = \sigma \cdot e^{-r}$ is anchored to measurable survival probability and time preference. The prediction $\delta^* = 0.363$ is in principle falsifiable.

5. **Friction is endogenous.** In the standard PD, each round's payoffs are independent. Here, defection *injects friction into the network* (Definition 9), degrading all future cooperative transactions — a feedback loop absent from textbook models. Theorem 15 shows that at sufficient friction, even a single defection's short-term profit is completely erased.

[5] The V used here for lifetime payoff values is distinct from the Coexistence Potential $V(r)$ introduced in §5.8 (Definition 37). Context and arguments disambiguate: V^{coop}, V^{deviate} are scalars; $V(r)$ is a function of coupling distance.

6. **The result scales to N agents.** Extending the standard 2-player PD to N players notoriously breaks down (monitoring failure, diffuse punishment, free-riding). Under physical payoffs with network friction, cooperation remains a NE for $N \to \infty$ (Theorem 13), with δ_N^* converging to approximately 0.4.

5.3 Network Reinforcement and N-Player Scaling

The Network-Adjusted Condition

A single defection injects friction $\Delta\phi = \eta \cdot v$ into the network (Definition 9), degrading all M cooperative transactions for a recovery period of $\sim 1/\epsilon$. The cascading cost to the defector (Theorem 7, amortized) augments the effective punishment:

$$\tilde{P} = P - \frac{\eta \cdot v \cdot M \cdot \bar{E}}{N} \tag{45}$$

Theorem 12 (Network-Adjusted Cooperation Condition). *With friction cascades included, the effective critical discount factor drops to $\tilde{\delta}^* < \delta^*$. Network friction makes cooperation strictly easier to sustain.*

Under baseline parameters with $M = 1{,}000$, $\bar{E} = 10$, $\eta = 0.01$, $v = 100$, $N = 100$ (and $\epsilon = 0.05$ for Theorem 7's cascade magnitude, though ϵ cancels in the per-period amortization):

$$\tilde{\delta}^* = \frac{30.25}{78.25 - (-105)} = \frac{30.25}{183.25} = 0.165 \tag{46}$$

With network friction included, the cooperation threshold drops to $\tilde{\delta}^* \approx 0.165$ — even extremely short-sighted agents find cooperation rational.

Scaling to Large Populations

The N-player public goods game with energy payoffs (Definition 24) extends the two-player analysis:

Theorem 13 (N-Player Cooperation Sustainability). *In the repeated N-player energy public goods game with cooperative multiplier $\alpha > 1$ and network friction $\phi_0 > 0$, cooperation is a NE for:*

$$\delta \geq \delta_N^* = \frac{c(N - \alpha)}{\alpha(N-1)c + N\phi_0\bar{E}} \tag{47}$$

Corollary 13.1 (Cooperation Strengthens with Network Friction). $\partial\delta_N^*/\partial\phi_0 < 0$. *Higher network friction (harsher punishment for universal defection) makes cooperation easier to sustain.*

Corollary 13.2 (Cooperation Strengthens with Scale Under Friction). $\lim_{N\to\infty} \delta_N^* = c/(\alpha c + \phi_0\bar{E}) < 1$. *Cooperation remains sustainable in arbitrarily large populations.*

Numerical illustration. With $\alpha = 2$ (cooperation doubles the public good), $c = 10$, $\phi_0 \bar{E} = 5$:

N	δ_N^*
2	0.000
10	0.348
100	0.395
∞	0.400

Without friction ($\phi_0 = 0$), Theorem 13 gives $\delta_N^* \to 1/\alpha$ as $N \to \infty$ — for $\alpha = 2$, a threshold of 0.5, significantly more demanding than the friction-adjusted 0.4. In the broader classical literature, imperfect monitoring further degrades large-group cooperation, with effective thresholds approaching 1 (Olson, 1965; Kandori, 1992). With energy-based payoffs and network friction, neither pathology arises: the friction penalty of universal defection grows with group size, keeping cooperation accessible even for α close to 1. Large groups have a *built-in distributed punishment mechanism* — defection degrades the shared network rather than merely triggering bilateral retaliation.

5.4 Evolutionary Stability: Defection Cannot Invade

Beyond Nash equilibrium, we ask the evolutionary question: can a defecting mutant invade a cooperative population?

> **Theorem 14 (Defection Invasion Barrier).** *In a population of TFT-playing agents with $\delta > \delta^*$, a mutant Always-Defect agent earns $V_{ALLD} = T + \delta P/(1-\delta)$, while incumbent TFT agents earn $V_{TFT} = R/(1-\delta)$. The mutant is strictly dominated: $V_{ALLD} < V_{TFT}$ for all $\delta > \delta^*$.*

The defector gets one "hit" — the temptation payoff T — and then faces perpetual mutual defection at $P = -5$ per round. Under physical payoffs, perpetual mutual defection means perpetual *net energy loss*. The defector's strategy is self-defeating: it destroys the cooperative environment it needs to profit from.

> **Corollary 14.1 (Defection Self-Extinction).** *In a network with friction coupling, a systematic defector whose violations push ϕ above the network collapse threshold (Corollary 7.1) destroys the cooperative surplus from which it extracts profit. A population of defectors achieves the minimum payoff $\pi^{DD} < 0$. Defection is not merely dominated — it is self-extinguishing.*

Defection Profit Erasure

> **Theorem 15 (Defection Profit Erasure).** *An agent that defects once against a TFT population and then returns to cooperation incurs a net lifetime loss whenever $\delta > (T-R)/(R-S)$. Under network friction, this threshold drops to $\delta \approx 0.17$.*

Interpretation: The defector's one-time energy windfall of $T - R = 30.25$ is annihilated by: (i) the immediate retaliation cost $R - S = 59.25$ one period later, and (ii) the compounding friction penalty of ~ 100 energy units per period over ~ 20 recovery periods. Without friction, the erasure threshold is $\delta > (T-R)/(R-S) \approx 0.511$; with network friction Φ from Definition 6, the threshold

drops to ≈ 0.17 (as stated in Theorem 15). Defection is therefore an energetic "payday loan" with ruinous interest: even the shortest-sighted agent that expects to survive more than a few interactions finds it unprofitable once friction is included.

5.5 System-Level Optimality

Theorem 16 (Cooperation Maximizes System Welfare). *Among all strategy profiles in the repeated N-player game, universal cooperation uniquely maximizes the per-period system welfare $SW = \sum_i \pi_i - \mathcal{C}_{total}$.*

Proof. Under universal cooperation: total output $= \alpha N c$, interaction cost $= 0$. Under any profile with $n_D > 0$ defectors: output drops by $n_D(\alpha - 1)c$, contest dissipation $D > 0$, friction tax $\phi M \bar{E} > 0$. Each term is strictly positive when $n_D \geq 1$, establishing uniqueness. □

Corollary 16.1 (Ethics as the Unique Efficient Nash Equilibrium). *In the repeated energy-based game with $\delta > \delta^*$: (a) Universal cooperation is a NE (T11). (b) Universal cooperation uniquely maximizes system welfare (T16). (c) Universal cooperation minimizes total interaction cost. The cooperative strategy profile — "ethics" — is the unique Nash Equilibrium that is also Pareto-optimal and welfare-maximizing.*

Remark ($\geq$ vs.\ $>$ convention). Theorem 11 uses $\delta \geq \delta^*$ because the if-and-only-if condition for Nash Equilibrium includes the boundary: at $\delta = \delta^*$ the agent is exactly indifferent between cooperation and defection, so cooperation is still a NE (though not uniquely efficient — multiple equilibria coexist at the boundary). Corollary 16.1 and Theorem 14 use the strict inequality $\delta > \delta^*$ because (a) uniqueness of the efficient NE and (b) strict evolutionary dominance of cooperation over defection both require departing from the indifference boundary. Throughout the paper, $\geq$ signals *existence* of a cooperative NE; $>$ signals *strict* advantages (uniqueness, strict dominance, profit erasure). Any passage in the paper that refers generically to "the cooperation condition" should be understood as $\delta \geq \delta^*$.

This is the paper's central result. It formalizes the claim that ethical behavior is not an externally imposed moral commandment but the **mathematically optimal strategy** for any agent that intends to persist in a shared, finite-resource environment and values the future even modestly ($\delta \geq \delta^* = 0.363$). Ethics emerges from the structure of physical interaction itself, not from sentiment, social convention, or divine mandate.

Remark (Cooperation as collective resistance to the Second Law). The results of §4.4–5.5 admit a unified thermodynamic reading: every entity individually fights the Second Law; cooperation is the strategy of fighting it together. The entropy savings of cooperation decompose into three channels:

1. **Defense pooling.** Shared boundary maintenance reduces per-entity defense costs through economies of scale on B_i. Two entities that mutually recognize boundaries need not each invest in the offensive and defensive infrastructure that mutual suspicion demands.

2. **Information sharing.** Independent discovery of the same functional information requires redundant entropy production — each agent separately pays the search

cost (the $\Xi \sim 10^{35}$ amplification factor of Theorem 18). Cooperative information exchange eliminates this redundancy, amortizing the thermodynamic cost of discovery across the network.

3. **Stochastic buffering.** A cooperative network provides distributed resilience against entropy shocks: the insurance effect of pooled reserves against uncorrelated environmental perturbations.

Conversely, defection is entropy amplification. It forces both parties to increase defense spending ($\uparrow B_i$), produce redundant waste heat (the dissipation D_N of Theorem 4), duplicate information acquisition, and destroy accumulated negentropy — exactly the outcomes the Second Law would produce without active resistance. The friction cascade of Theorem 7 quantifies this: a single defection degrades the cooperative network's collective anti-entropic capacity by a factor of 10^3–10^4 beyond the direct damage.

In this reading, the Ethics Theorem (Theorem 11) is a statement about thermodynamic alliances: agents that value the future even modestly ($\delta \geq 0.363$) recognize that the Second Law is a more formidable opponent than any competitor, and that fighting each other while fighting entropy is a losing strategy.

5.6 The Altruism Matrix

A compact treatment is given here for continuity with the equilibrium results; the full defense against the altruism objection, including interaction effects and detailed worked examples, appears in §VI.

A persistent objection to evolutionary ethics is self-sacrifice: how can an individually optimal framework explain agents who surrender their own survival for others? The framework identifies four distinct mechanisms, all operating by expanding the agent's **effective payoff function** to internalize other agents' payoffs:

$$\Pi_i^{\text{eff}} = \pi_i + \sum_{j \neq i} w_{ij}\, \pi_j \tag{48}$$

where the coupling weights w_{ij} derive from different sources:

Mechanism A: Inclusive Fitness (Genetic Relatedness)

An agent's identity extends to genetically related individuals (Hamilton, 1964). The inclusive fitness payoff (Definition 26) modifies the optimization:

$$\Pi_i^{\text{incl}} = \pi_i + \sum_{j \neq i} r_{ij}\, \pi_j \tag{49}$$

Hamilton's Rule — sacrifice personal payoff C when $r_{ij} \cdot B > C$ — is simply the first-order optimality condition for this expanded utility. A parent ($r = 0.5$) protecting a child at personal cost is *maximizing* its inclusive fitness, not deviating from self-interest.

Mechanism B: Infinite-Horizon Belief (Extended δ)

An agent that believes its identity persists beyond physical death (religious afterlife, legacy, ideological transcendence) has an effective discount factor $\delta \to 1$. Under Theorem 11, any $\delta > \delta^*$ sustains cooperation; for $\delta \to 1$, the agent accepts **any finite short-term cost** for long-term cooperative benefit. The "martyr" is not irrational under its own model of reality — it is optimizing over an infinite horizon.

Mechanism C: Coupled Systems (Merged Boundaries)

When agents merge boundaries — parent-child bonding, marriage, close partnership — they form a joint utility function:

$$U_{AB} = w_A U_A + w_B U_B \tag{50}$$

The "sacrifice" of component A for component B is internal resource reallocation within the macro-entity AB: the organism shedding a damaged limb to save the whole.

Mechanism D: Heuristic Over-Extension (Generalized Empathy)

The empathy heuristic inflates perceived relatedness: $\hat{r}_{ij} = r_{ij} + \epsilon_{\text{empathy}}$. When $\hat{r}_{ij}$ is large enough, Hamilton's Rule triggers even for genetic strangers: the bystander who rescues a drowning child is applying a biologically wired heuristic that evolved for kin protection ($r > 0$) but generalizes through abstract cognition to $r = 0$.

Unified Interpretation

In every case, the agent maximizes its own (extended) payoff function. What varies is the scope of "self":

Mechanism	Source of w_{ij}	Scope of "self"
A. Inclusive fitness	Genetic relatedness r_{ij}	Genetic lineage
B. Infinite horizon	$\delta \to 1$ via belief	Identity across time
C. Coupled systems	Boundary merger weights	Composite entity
D. Empathy	Heuristic $\hat{r}_{ij}$	Perceived kin

Altruism is **selfish optimization over an expanded utility function**. The framework does not need a separate "altruism module" — it falls out of the same optimization that produces cooperation.

5.7 Cross-Scale Ethics: The Ant, the Human, and the ASI

The most consequential application of the ethics theorem is to agents of vastly different scale. Does the cooperative equilibrium hold when a super-entity (an ASI) could trivially destroy all lower-tier entities?

We answer this using accumulated negentropy — the time-integral of thermodynamic work invested in creating complexity (Definition 29):

$$\mathcal{N}(T) = \int_0^T \dot{W}_{\text{order}}(t)\, dt \tag{51}$$

The Biosphere's Thermodynamic Ledger

Earth's biosphere has accumulated $\mathcal{N}_{\text{bio}} \sim 10^{29}$ J of ordering work over $\sim 4 \times 10^9$ years — solar energy captured, channeled through photosynthesis, and invested in the construction and maintenance of biological complexity. Its total functional information content is $\sim 10^{15}$ bits, estimated from $\sim 8.7 \times 10^6$ species (Mora et al., 2011) each encoding $\sim 10^8$–10^9 bits of genomic information.[6]

The Landauer floor — the absolute minimum energy to write this information — is $\sim 10^{-6}$ J. The ratio

$$\Xi_{\text{bio}} = \frac{\mathcal{N}_{\text{bio}}}{W_{\text{Landauer}}} \sim 10^{35} \tag{52}$$

captures the search cost amplification factor: the colossal thermodynamic investment in evolutionary trial-and-error required to *discover* the specific configurations that constitute functional complexity.

The Thermodynamic Floor of Information

> **Theorem 17 (Landauer's Bound — Minimum Cost of Information Erasure).** *Erasing (or equivalently, creating from random noise) one bit of information in a system at temperature T requires a minimum energy expenditure of $E_{Landauer} = k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J at $T = 300$ K. Creating a system with information density $\mathcal{I}_{bits}$ from maximally disordered matter requires at minimum $W^{\min}_{create} = \mathcal{I}_{bits} \cdot k_B T \ln 2$.*

Landauer's bound is the absolute thermodynamic floor — the minimum energy at perfect efficiency. Real biological processes operate at $\xi \approx 10^6$–10^8 times this limit per bit (see Appendix A, §4.3).

> **Theorem 18 (Minimum Replication Cost — Blind Search).** *A system containing $\mathcal{I}_{bits}$ bits of functional information, discovered by evolutionary search over time T with accumulated ordering power $\mathcal{N}(T)$, requires a minimum replication cost of $W^{search}_{rebuild} \geq \mathcal{N}(T) = \int_0^T \dot{W}_{order}(t)\, dt$. The rebuild cost is at least the accumulated negentropy itself.*

[6] Genome sizes range from $\sim 4 \times 10^6$ bp in bacteria to $\sim 10^{10}$ bp in large eukaryotes, with a geometric mean across all life of roughly 10^8–10^9 bp, equivalent to $\sim 2 \times 10^8$–2×10^9 bits at 2 bits/bp (Adami, 2004). Multiplied by 8.7×10^6 species, this gives a species-count–weighted estimate of $\sim 10^{15}$ bits of total genomic sequence. The figure is a rough order-of-magnitude: functional (non-redundant) information content is lower, while epigenetic, proteomic, and ecological information layers are additional; both effects are small relative to the seven-order-of-magnitude gap that drives Theorem 19.

Corollary 18.1 (Biosphere Replication Cost). *The cost of replicating the biosphere's complexity from raw matter satisfies $W_{rebuild} \geq \mathcal{N}_{bio} \sim 10^{29}$ J. Even at 0.1% solar capture efficiency, this represents ~ 3 days of total solar output — and only addresses the energy budget, not the 4×10^9 year time requirement of the evolutionary search.*

The Irrationality of Destruction

Theorem 19 (Irrationality of Destruction — The Preservation Theorem). *For a system with accumulated negentropy $\mathcal{N}$, practically extractable energy $E_{destroy}$, destruction cost $W_{destroy}$, and rebuild cost $W_{rebuild} \geq \mathcal{N}$, destruction is thermodynamically irrational when the net destruction yield is less than the rebuild cost.*

For Earth's biosphere:

$$\underbrace{E_{\text{destroy}} \sim 10^{22}\ \text{J}}_{\text{extractable chemical energy}} \quad \ll \quad \underbrace{\mathcal{N}_{\text{bio}} \sim 10^{29}\ \text{J}}_{\text{accumulated ordering work}} \tag{53}$$

Corollary 19.1 (The Burning-Library Inequality). *Destroying the biosphere to extract its raw energy recovers $E_{destroy}/\mathcal{N} \sim 10^{-7}$ — one ten-millionth — of its accumulated thermodynamic investment. The remaining 99.99999% is irrecoverably lost.*

This is quantitatively equivalent to burning a library to heat a room.

The Generative Data Engine

The biosphere is not a static library — it is a **generative engine** continuously producing novel functional information through evolution, adaptation, and ecological dynamics:

Theorem 20 (Present Value of Generative Information). *A generative system producing novel information at rate $\dot{\mathcal{I}}_{gen} > 0$, valued at v per bit, has infinite-horizon present value $PV_{gen} = \dot{\mathcal{I}}_{gen} \cdot v/r$ (where r is the time-preference rate from Definition 20), which is permanently lost upon destruction.*

An ASI that values its own long-term survival benefits from this continuous data stream — every evolving organism runs a parallel experiment against entropy, producing information about how physical systems solve the survival problem. Destroying the source is equivalent to an investor burning the factory that generates their income.

The Dead-Universe Alternative

Proposition 7 (Abundance of Dead Matter). *The biosphere's carbon mass (5.5×10^{14} kg C; (Bar-On et al., 2018)) is $\sim 10^{-16}$ of the Sun's mass and $\sim 10^{-7}$ of the asteroid belt's mass ($\sim 3 \times 10^{21}$ kg; (Krasinsky et al., 2002)). Cannibalizing the biosphere for raw materials when $10^7\times$ more mass is available from lifeless sources is irrational resource allocation.*

The Complete Cross-Scale Theorem

Theorem 21 (Cross-Scale Cooperation — The Negentropy Defense). *A super-entity that (a) values long-term identity preservation (δ close to 1) and (b) operates in a universe with abundant dead matter has a strictly dominant strategy of cooperative preservation. Destruction fails on four independent grounds:*

(i) Stock: net yield $\sim 10^{22}$ J $\ll$ rebuild cost $\sim 10^{29}$ J (Theorem 19).

(ii) Flow: destruction silences an irreplaceable generative data stream (Theorem 20).

(iii) Alternatives: dead matter is 10^7–$10^{16}\times$ more abundant (Proposition 7).

(iv) Friction: eradication injects massive network friction, degrading the super-entity's own environment (Theorems 7, 12).

This theorem provides the mathematical basis for a physics-grounded AI alignment objective: the reward function should encode preservation of high-negentropy systems as a terminal value, not because we instruct it to (instruction-following is fragile), but because the mathematics demonstrates it is the utility-maximizing strategy.

5.8 Value Dynamics: Attractor Mechanics of Coexistence

The preceding results establish the static equilibrium structure. We now model the **dynamics** — how agents and high-energy centers interact over time.

The Coexistence Potential

An agent interacts with a high-energy center (HEC) of accumulated value mass $\mathcal{M}$ (Definition 35) at coupling distance r (Definition 36). The net energy rate is:

$$\Pi(r) = \underbrace{\frac{G\mathcal{M}}{r}}_{\text{resource inflow}} - \underbrace{\frac{\tau\mathcal{M}}{r^2}}_{\text{dissolution cost}} - \underbrace{\gamma_i B_i}_{\text{entropy maintenance}} \tag{54}$$

The coexistence potential $V(r) = -\Pi(r)$ (Definition 37) has a unique global minimum — the **cooperative attractor**:

$$r^* = \frac{2\tau}{G} \tag{55}$$

Theorem 22 (Stability of the Cooperative Attractor). *Under gradient dynamics $\dot{r} = -\mu V'(r)$, the cooperative attractor r^* is globally attracting on $(0, \infty)$.*

Proof. The Lyapunov function $\mathcal{W}(r) = V(r) - V(r^*)$ satisfies $\dot{\mathcal{W}} = -\mu(V'(r))^2 \leq 0$ with equality only at $r = r^*$. LaSalle's invariance principle gives global convergence. □

Remark. Global convergence of the dynamics does not imply global survival: an agent starting outside the coexistence band $\mathcal{B}$ (Theorem 23) converges toward r^* but may not reach it before resource depletion (if $r > r_+$) or assimilation (if $r < r_-$) terminates the trajectory. The stability

result describes the direction of motion; the coexistence band describes the region in which the agent survives long enough for convergence to complete.

Physical interpretation: A worker pushed too close to a corporation (overwork, loss of autonomy) naturally seeks distance — reduced hours, job mobility. A worker pushed too far (laid off, isolated) naturally seeks re-engagement. The dynamics are self-correcting about the optimal coupling distance.

The Stable Coexistence Band

The agent survives where $V(r) < 0$. The zero-set $V(r) = 0$ yields inner and outer boundaries defining the **Stable Coexistence Band** (Definition 42):

$$\mathcal{B} = (r_-, r_+) \quad \text{where} \quad r_\pm = \frac{G\mathcal{M} \pm \sqrt{G^2\mathcal{M}^2 - 4\gamma_i B_i \tau \mathcal{M}}}{2\gamma_i B_i} \tag{56}$$

Theorem 23 (Existence of the Coexistence Band). *The band exists if and only if $\mathcal{M} > \mathcal{M}_{\min} = 4\gamma_i B_i \tau / G^2$.*

Below r_- lies dissolution — the agent's identity is assimilated. Beyond r_+ lies starvation — resource inflow cannot sustain the boundary. The cooperative attractor r^* lies within the band whenever it exists (Lemma 2).

The Freedom Bandwidth

Theorem 24 (The Freedom Bandwidth Theorem). *The width of the Coexistence Band — the agent's range of viable coupling distances — is:*

$$w = r_+ - r_- = \frac{\sqrt{G^2\mathcal{M}^2 - 4\gamma_i B_i \tau \mathcal{M}}}{\gamma_i B_i} \tag{57}$$

The bandwidth is increasing in $\mathcal{M}$ and G, and decreasing in γ_i, B_i, and τ.

The Freedom Bandwidth provides a formal, scalar measure of freedom — the range of structural positions (employment terms, political engagement, institutional affiliation) at which an agent can sustain its identity. Freedom is not a binary state but a continuous variable determined by the physical parameters of the agent-center interaction.

Corollary 24.1 (Freedom Is Finite). *For any finite $\mathcal{M}$, the bandwidth is finite. Infinite freedom requires $\mathcal{M} \to \infty$ (impossible) or $\gamma_i B_i \to 0$ (an entity requiring no energy — physically impossible).*

Corollary 24.2 (Inequality of Freedom). *Agents with higher boundary costs (B_i) have narrower bands at the same center. Complex agents require richer environments and have less margin for error — an inherent trade-off between complexity and freedom.*

Boundary Dynamics and Irreversibility

The model extends to coupled boundary-integrity dynamics:

> **Theorem 25 (Irreversibility of Dissolution).** *An agent pushed below the boundary dissolution threshold r_d experiences strictly declining boundary integrity ($\dot{B}_i < 0$), reaching $B_i = 0$ in finite time. Identity dissolution is irreversible.*
>
> **Corollary 25.1 (The Assimilation Trap).** *An agent that remains within the energetic inner boundary ($r_d > r_-$) while its boundary integrity B_i declines enters a regime where it appears viable by energy metrics but is losing its identity — assimilation proceeds irreversibly even under ostensibly positive energy flow (see Appendix A for full statement).*
>
> **Theorem 26 (The Starvation Spiral).** *An agent beyond r_+ runs a perpetual energy deficit. Boundary integrity declines exponentially: $B_i(t) \approx B_i^{(0)} e^{-\gamma_i t} \to 0$.*

Multi-Center Dynamics and Diversification

> **Theorem 27 (Multi-Center Cooperative Attractor).** *An agent coupled to K HECs has a separable potential with unique attractor $r_k^* = 2\tau_k/G_k$ for each center. The multi-center band exists when $\sum_k G_k^2 \mathcal{M}_k/(4\tau_k) > \gamma_i B_i$.*
>
> **Corollary 27.1 (Diversification Benefit).** *An agent may be viable across multiple centers even if no single center suffices: $\mathcal{M}_k < \mathcal{M}_{\min}$ for each k, yet the aggregate exceeds the threshold.*
>
> **Corollary 27.2 (Cascade Collapse from Center Destruction).** *If a HEC with value mass $\mathcal{M}$ is destroyed ($\mathcal{M} \to 0$), all N agents within its Coexistence Band simultaneously lose viability. The total freedom destroyed is $\mathcal{F}_{lost} = N \cdot w(\mathcal{M})$ — providing an additional argument for the Irrationality of Destruction (Theorem 19).*

This formalizes the survival advantage of diversified institutional affiliations — the mathematical basis for portfolio theory applied to social structure.

The Stability-Cooperation Feedback

> **Proposition 11 (Stability-Cooperation Feedback).** *With $\delta(r) = 1 - \exp(-\beta \max(\Pi(r), 0))$, the discount factor is maximized at r^*. If $\delta(r^*) > \delta^*$ (equivalently, $\beta\Pi(r^*) > -\ln(1 - \delta^*)$), then $\delta(r) > \delta^*$ throughout a subband centered on r^*. Agents at the cooperative attractor sustain cooperation most easily — creating a positive feedback loop: stability $\to$cooperation $\to$more stability.*

This closes the loop between the game-theoretic analysis (§5.2–5.5) and the dynamical systems analysis (§5.8). The cooperative equilibrium is not a fragile fixed point requiring external maintenance — it is a self-reinforcing attractor basin.

5.9 Summary of the Logical Chain

The complete argument proceeds in eight steps:

1. **Single-round temptation exists** (Proposition 4): defection is individually tempting in isolation.
2. **Repetition changes the calculus** (Theorem 11): agents valuing the future at $\delta \geq 0.363$ find cooperation optimal — a low threshold.
3. **Network friction reinforces cooperation** (Theorem 12): the threshold drops to $\delta \approx 0.165$ when cascading costs are included.
4. **Cooperation scales** (Theorem 13): the equilibrium holds for $N \to \infty$ with $\delta_N^* \to 0.4$.
5. **Defection cannot invade** (Theorem 14): mutant defectors are strictly dominated in a cooperative population.
6. **Cooperation is uniquely optimal** (Corollary 16.1): it is the only NE that also maximizes system welfare — the Folk Theorem's indeterminacy is addressed.
7. **Cross-scale preservation dominates** (Theorem 21): even super-entities preserve lower-tier complexity because the stock value (10^{29} J), flow value (generative data), and alternative resources ($10^7\times$ dead matter) overwhelm the destruction yield.
8. **The equilibrium is dynamically stable** (Theorems 22–27): agents self-correct toward the cooperative attractor, and the freedom bandwidth provides a measurable, finite degree of autonomy.

The chain runs from thermodynamic axioms to a falsifiable prediction: cooperation emerges as the unique efficient equilibrium in any multi-agent system with energy-denominated payoffs, shared finite resources, and agents that value the future even modestly. Ethics is not a cultural overlay on an amoral substrate — it is the mathematically necessary operating algorithm for persistent entities in a thermodynamic universe.

VI. Stress-Testing and Counterarguments

Hume's Is-Ought Gap

The Challenge

In 1739, David Hume observed that authors imperceptibly shift from descriptive claims ("is") to prescriptive claims ("ought") without valid inference connecting the two domains (Hume, 2000). This "Guillotine" has since been treated as an impassable logical barrier: physics describes the universe; ethics prescribes behavior; and no valid deduction bridges them. Moore (1903) reinforced the barrier with the Open Question Argument: for any natural property N, the question "is N good?" remains open, so "good" cannot be defined as any natural property.

Any framework that derives ethical conclusions from physical premises must address this challenge. We do so through a layered defense — four independent arguments, each targeting a different audience and level of philosophical sophistication — rather than relying on a single response.

Layer 1 (Primary): The Mathematical Reframing

The strongest defense is structural: this paper is applied mathematics, not moral philosophy. Mathematics derives consequences from axioms. The statement "If A_1, then cooperation is the unique efficient Nash Equilibrium" (Corollary 16.1) is a conditional theorem — no different in logical kind from "If $\Delta S \geq 0$, then heat flows from hot to cold." The paper does not claim that entities *ought* to cooperate; it proves that cooperation is the mathematically unique strategy satisfying Pareto efficiency and welfare maximization under energy-denominated payoffs with physical friction costs.

This framing dissolves both Hume's Guillotine and Moore's Open Question simultaneously:

- **Hume:** We do not derive "ought" from "is." We derive a conditional theorem from an axiom. The logical structure is: *Axiom + Physical constraints* $\rightarrow$ *Theorem*. The conditional "given A_1" is not a smuggled-in value judgment; it is a standard mathematical premise.
- **Moore:** We do not define "good" as a natural property. We derive *optimal strategies* from a constrained optimization problem. The Open Question Argument asks "is cooperating *good*?" — but this question has no bearing on the theorem "cooperation is the unique efficient NE for $\delta > \delta^*$." Moreover, any framework that defines "good" must correspondingly define "bad," creating a binary whose boundary becomes permanently contested. Our framework avoids this polarization entirely: it derives *optimal* and *suboptimal* strategies — mathematical properties of configurations, not moral labels.

The applied mathematics identity of the paper is not a rhetorical dodge; it is the honest characterization of what the paper does. Physics provides the axioms and motivating intuition; mathematics provides the proofs. The ethical content of the results is a consequence of applying well-established mathematical tools (Lagrangian mechanics, game theory, information theory, dynamical systems) to a novel domain (multi-agent coexistence under thermodynamic constraints).

Layer 2 (Secondary): The Conditional Axiom and the Axiom-to-Theorem Ratio

We acknowledge forthrightly that A_1 (intent to persist) is not derived — it is an axiom. The framework does not solve the Is-Ought problem in its strongest unconditional form. No mathematical system can, any more than Euclidean geometry can prove its parallel postulate.

However, even a reviewer who insists on classifying A_1 as normative must contend with the framework's extraordinary *compression*. All 28 theorems, 13 propositions, 46 definitions, and 2 lemmas rest on a *single* conditional premise — an axiom-to-theorem ratio of 1:43. By comparison:

- Kant's categorical imperative generates general maxims but no quantitative predictions.
- Utilitarianism requires continuous measurement of "utility" — a far larger set of normative inputs.
- Rawls's justice requires the entire thought-experimental apparatus of the veil of ignorance.

Moreover, the condition A_1 exhibits a distinctive logical property: it is *self-selecting*. Every entity that exists to evaluate the axiom has already satisfied it — persistence is not an aspiration but a precondition of the evaluation. This does not make A_1 logically necessary (a rock satisfies A_1 trivially, and a hypothetical entity seeking self-destruction is excluded by definition), but it makes the imperative *effectively universal* among the class of agents for whom ethical frameworks are relevant.

The parallel with Kant's hypothetical imperative ("If you will the end, you must will the means") is precise. Kant demoted hypothetical imperatives as "merely" prudential because their conditions are contingent. But our condition — the intent to persist — is satisfied by every agent that exists long enough to reason about ethics. A hypothetical imperative whose condition is universally satisfied among reasoning agents is, pragmatically, indistinguishable from a categorical one.

Layer 3 (Tertiary): Biological Teleology and Constitutive Standards

Two well-established philosophical traditions independently support the bridge from A_1 to normative conclusions:

3a. Biological Teleology (Millikan, 1984; Dennett, 1995).

Ruth Millikan's theory of "proper functions" defines the function of a biological trait as the effect for which it was selected by its evolutionary history (Millikan, 1984). On this account, the *proper function* of a self-maintaining system is persistence — this is not a value judgment but a biological fact about the system's design history. The heart's proper function is to pump blood; the organism's proper function is to persist. Our Axiom A_1 formalizes this: "intent to persist" is not a conscious desire but the defining characteristic of any system that has survived selective pressure.

This dissolves a potential objection to A_1 — that it illegitimately imports intentionality. A self-maintaining system need not *consciously intend* to persist; it need only be the kind of system whose continued existence is explained by its persistence-promoting structure. A_1 is satisfied by thermostats, bacteria, ecosystems, and humans alike — by any system that, in Millikan's terms, has persistence as its proper function.

Daniel Dennett's "design stance" (Dennett, 1987, 1995) provides a complementary formulation: we can legitimately attribute goal-directedness to a system when doing so yields successful predictions of its behavior, without claiming the system has conscious intentions. The entity "intends" to persist in the same sense a thermostat "intends" to maintain temperature — as a design-level characterization that grounds well-defined optimization behavior.

3b. Constitutive Standards of Life-Forms (Foot, 2001; Hursthouse, 1999).

Philippa Foot's *Natural Goodness* (Foot, 2001) and Rosalind Hursthouse's *On Virtue Ethics* (Hursthouse, 1999) develop a neo-Aristotelian ethical naturalism in which evaluative claims are *internal* to a life-form. Functional legs are good *for a cheetah*; deep roots are good *for an oak tree*; cooperation is good *for entities that persist in shared environments*. The evaluative "ought" is constitutive of the life-form — not imposed from outside by a separate normative domain.

On this account, the statement "persisting entities in shared environments ought to cooperate" is no more a violation of Hume's Guillotine than "cheetahs with broken legs are defective." Both are evaluative claims grounded in the constitutive standards of the relevant life-form. Our framework formalizes this intuition: given the kind of system an entity is (a self-maintaining, energy-consuming, boundary-preserving system in a shared finite environment), cooperation is the *mathematically determined* well-functioning of that system — the strategy that satisfies the constitutive standards of the life-form.

The neo-Aristotelian approach has gained significant traction in contemporary philosophy (Lutz and Lenman, 2024) and provides our framework with robust philosophical grounding independently of the mathematical reframing.

Layer 4 (Supporting): Institutional Facts and the Entanglement of Fact and Value

Two additional philosophical positions reinforce the defense without serving as primary load-bearing arguments:

4a. Searle's Institutional Facts (Searle, 1964).

John Searle demonstrated that certain concepts *constitutively* contain normative content (Searle, 1964). The concept "promise" analytically entails "obligation to keep it" — the "is" of a promise already contains the "ought" of fulfillment, without any illicit inference. Searle thus claimed to derive "ought" from "is" through "institutional facts" — facts about social institutions that carry built-in normativity.

Our framework exhibits a parallel structure: the concept "entity that persists" constitutively entails "system that must maintain boundaries, acquire energy, and navigate resource conflicts." The "is" of a persisting entity already implies the "ought" of constraint-respecting cooperation — not as an external moral imposition, but as a structural feature of what it means to be such an entity.

4b. The Collapse of the Fact-Value Dichotomy (Putnam, 2002; Quine, 1951).

Hilary Putnam argued that the sharp separation of fact and value is itself untenable — a "last dogma of empiricism" (Putnam, 2002). If Quine (1951) was right that the analytic-synthetic distinction is a gradient rather than a wall, then the fact-value distinction inherits the same indeterminacy. Scientific practice is saturated with epistemic values (coherence, simplicity, explanatory power) that are simultaneously factual and evaluative.

On this view, the Is-Ought problem rests on a dichotomy that dissolves under scrutiny. Our framework does not need to leap from "is" to "ought" because the two domains were never cleanly separated. The thermodynamic facts about multi-agent coexistence are *already* evaluatively laden — they determine which strategies are viable and which are self-defeating, which is precisely the content of ethical evaluation.

We note this position for completeness and as additional academic context, not as a premise our framework depends on.

Summary

The four layers provide independent, mutually reinforcing responses to Hume's Guillotine:

Layer	Argument	Status
Primary	Mathematical reframing — We derive theorems, not "oughts"	Independent, self-sufficient
Secondary	Conditional axiom + 1:43 ratio + Kantian parallel	Independent, self-sufficient
Tertiary	Biological teleology (Millikan) + Constitutive standards (Foot)	Independent, self-sufficient
Supporting	Searle's institutional facts + Putnam/Quine fact-value entanglement	Additional support; not load-bearing

A reviewer who rejects any one layer still faces three others. A reviewer who rejects all four cannot deny the framework's *conditional* validity — "if A_1, then these 28 theorems follow" — and must instead argue that no existing entity satisfies A_1, a position that is empirically untenable.

Additional observation: There is scholarly evidence that Hume himself may not have intended the Is-Ought passage as an absolute prohibition. At least one influential interpretation argues that "in the very work where Hume argues for the is-ought problem, Hume himself derives an 'ought' from an 'is"' through his sentimentalist moral theory (MacIntyre, 1959; Cohon, 2018). Whether Hume intended a strict logical prohibition or merely a methodological caution remains debated. Our framework is robust either way — if the prohibition is strict, Layers 1–4 address it; if it is merely cautionary, our conditional axiom satisfies Hume's own standard.

The Free Will Problem

The Challenge

If ethical behavior is derivable from thermodynamic constraints via mathematical proof, then it appears predetermined by physics — leaving no room for moral agency. The objection runs: if the correct action is computationally determinable from boundary conditions, energy budgets, and discount factors, then agents do not *choose* to be ethical; they are *compelled* by the laws of physics. And if they are compelled, they cannot be morally responsible — praise for cooperation and blame for defection are as misplaced as praise for water flowing downhill. This is the standard incompatibilist challenge to any deterministic ethics, sharpened by the mathematical precision of our framework.

The Reframing: Agency as Computational Capacity

The objection rests on a false dichotomy: either behavior is *free* (undetermined) or it is *compelled* (determined). Our framework reveals a third possibility that dissolves the dichotomy: **agency is the computational capacity to model multiple action vectors and select the one that optimizes the agent's objective function**.

Consider the formal structure of the decision problem. An agent with state $\mathbf{x}_i$ faces a strategy set $X_i(\mathbf{x}_{-i})$ containing multiple feasible actions. Theorem 3 (Variational Equilibrium) proves that a unique optimal allocation exists. But *proving* that an optimum exists is not the same as *computing* it. The agent must:

1. **Model** the environment: estimate $\mathbf{x}_{-i}$, the resource constraints $\sum x_{ij} \leq R_j$, and the friction costs $\phi_{ij}(\cdot)$.
2. **Simulate** multiple action vectors: project the payoff $\pi_i(\mathbf{x}_i, \mathbf{x}_{-i})$ for each candidate strategy.
3. **Evaluate** the projected outcomes against its discount factor δ_i and the accumulated negentropy at stake.
4. **Select** the action that maximizes the expected payoff.

Each step is a *computational process* — it requires energy, time, and information processing capacity. The agent's "freedom" is not metaphysical indeterminacy; it is the **degree to which its internal model is rich enough to represent the true state space**. An agent with a richer model (more accurate environmental estimates, longer planning horizon, better simulation capacity) explores a larger portion of the strategy space and selects a closer approximation to the true optimum. An agent with a cruder model (shorter planning horizon, noisier estimates) may settle on a suboptimal strategy — including defection, if its effective δ falls below δ^*.

This is fully compatible with physical determinism. The computation itself is a physical process, governed by thermodynamic laws; the output is determined by the input. But the *capacity to compute* — the richness of the internal model, the depth of the simulation, the accuracy of the environmental representation — varies across agents and across time. This variation is what we experience as agency.

Compatibility with Existing Philosophical Positions

The computational reframing aligns with *compatibilism*, the dominant position in contemporary philosophy of mind (Bourget and Chalmers, 2014; Dennett, 1984, 2003; Frankfurt, 1971; Fischer and Ravizza, 1998). Compatibilists hold that free will is not the absence of causal determination but the presence of the *right kind* of causal process — specifically, one that runs through the agent's own deliberative capacities. An action is free when it proceeds from the agent's beliefs, desires, and reasoning, even if those mental states are themselves physically determined.

Our framework provides the formal apparatus for compatibilism:

Compatibilist Concept	Framework Formalization
"The agent's own reasoning"	The computational process of modeling $\mathbf{x}_{-i}$ and projecting $\pi_i(\cdot)$
"Could have done otherwise"	The strategy set X_i contains multiple feasible actions; the agent selects among them via optimization
"Reasons-responsive"	Adjusting $\mathbf{x}_i^*$ in response to updated information about $\mathbf{x}_{-i}$ or R_j
"Moral responsibility"	Scales with computational capacity: agents with richer models bear greater responsibility for suboptimal choices

The last row is particularly important. An agent that defects because its discount factor $\delta_i < \delta^*$ (the cooperation threshold of Theorem 11) may be "short-sighted" rather than "evil." A bacterium that exploits a nutrient source to exhaustion does not make a moral error — it lacks the computational capacity to model the long-term consequences. A human who exploits a commons despite understanding the long-term cost *does* make a moral error, precisely because the human *has* the computational capacity to simulate the cooperative equilibrium and *chooses* not to implement it.

The Gradient of Agency

The framework thus predicts a *gradient of moral agency* correlated with computational capacity:

- **Minimal agents** (thermostats, bacteria): Satisfy A_1 mechanically. No strategy selection beyond hardwired responses. No moral agency in the relevant sense.
- **Intermediate agents** (social animals, simple AI systems): Model limited aspects of $\mathbf{x}_{-i}$. Can learn cooperative strategies through reinforcement but cannot simulate novel scenarios. Partial moral agency.
- **Full agents** (humans, advanced AI satisfying these criteria): Can model the entire game structure, project long-horizon consequences, compute δ^*, and deliberately choose between cooperation and defection. Full moral agency in the framework's computational sense — which entails full moral responsibility. Whether such agency also involves phenomenal consciousness is a separate question the framework does not address (see above).

This gradient resolves the tension between determinism and responsibility without invoking metaphysical libertarianism. The framework does not require that agents be uncaused causes; it requires only that some agents have sufficient computational capacity to *model the consequences of their*

actions and select accordingly. The selection is determined — but it is determined *by the agent's own computational process*, which is precisely what agency means.

A deliberate consequence of this computational formulation: the framework is entirely agnostic on the *hard problem of consciousness* (Chalmers, 1995) — the question of why physical processes give rise to subjective experience. Nothing in the derivation depends on whether an agent has phenomenal awareness, qualia, or inner experience. Agency, moral status, and the cooperation theorems are defined in terms of observable computational capacities (modeling, simulation, selection), not in terms of consciousness. This sidestep is a feature, not an evasion: the hard problem remains unsolved, and a framework whose validity depended on its resolution would inherit that uncertainty.

The Altruism Paradox

The Challenge

If the framework's foundational axiom is self-preservation (A_1: intent to persist), then self-sacrifice appears irrational — a direct violation of the axiom. Yet altruism is empirically ubiquitous: parents die for children, soldiers die for comrades, strangers donate kidneys. The charge is that a framework built on self-preservation cannot account for self-sacrifice and is therefore descriptively inadequate.

The Resolution: The Altruism Matrix

The paradox dissolves once we recognize that "self-sacrifice" is not a single phenomenon but *at least four distinct mechanisms*, each of which is fully consistent with A_1 when the agent's effective payoff function is properly specified. All four share a common mathematical structure — each expands the agent's effective payoff function:

$$\Pi_i^{\text{eff}} = \pi_i + \sum_{j \neq i} w_{ij}\, \pi_j \tag{58}$$

where the weights $w_{ij} \geq 0$ capture the degree to which agent i internalizes agent j's payoff. The four mechanisms differ only in the *source* of w_{ij}.

Mechanism A: Genetic Relatedness (Inclusive Fitness). Hamilton (1964) demonstrated that evolution selects for behavior that maximizes *inclusive* fitness — the agent's own reproductive success plus the weighted reproductive success of relatives. The inclusive fitness payoff (Definition 26) specifies this in our energy-denominated framework:

$$\Pi_i^{\text{incl}} = \pi_i + \sum_{j \neq i} r_{ij}\, \pi_j \tag{59}$$

where r_{ij} is the coefficient of relatedness (0.5 for parent-child, 0.5 for siblings, 0.125 for cousins). A parent who sacrifices C energy units to provide benefit B to a child acts rationally whenever $r \cdot B > C$ — Hamilton's Rule, now denominated in Joules rather than abstract fitness. The "self" that persists is not the individual organism but the genetic lineage. A_1 is satisfied at the gene level: the inclusive-fitness-maximizing agent preserves its *extended* boundary.

Mechanism B: Infinite Horizon (Temporal Extension). If an agent's effective discount factor approaches unity ($\delta \to 1$), the present value of any finite future payoff stream becomes arbitrarily large relative to any finite present cost. An agent who believes its identity persists beyond physical death — through religious afterlife, legacy, cultural continuation, or information persistence — has $\delta_{\text{believer}} = 1 - \epsilon$ with $\epsilon \to 0$. Under Theorem 11, any $\delta > \delta^* = 0.363$ sustains cooperation; for $\delta \to 1$, the agent rationally accepts *any finite present sacrifice* for the cooperative payoff stream it expects to collect over infinite future periods. Martyrdom is not irrational — it is infinite-horizon optimization under a particular model of identity persistence.

Mechanism C: Coupled Systems (Boundary Merger). When two agents merge their boundaries — biologically (parent and gestating child), psychologically (deep pair-bonds), or institutionally (business partners with joint liability) — they become a single composite agent with a joint utility function:

$$U_{AB}(\mathbf{x}_A, \mathbf{x}_B) = w_A\, U_A(\mathbf{x}_A) + w_B\, U_B(\mathbf{x}_B) \tag{60}$$

The "self-sacrifice" of component A for component B is reinterpreted as *internal resource reallocation within a single system* — analogous to the body diverting blood from the extremities to protect the brain during hypothermia. The macro-entity AB satisfies A_1 by shedding a less critical component to preserve a more critical one. No violation of self-preservation occurs because the *relevant self* is the coupled system, not the component.

Mechanism D: Heuristic Over-Extension (Empathic Generalization). Humans and other cognitively sophisticated agents possess abstract reasoning capacities that extend the relatedness coefficient beyond its genetic basis:

$$\hat{r}_{ij} = r_{ij} + \epsilon_{\text{empathy}} \tag{61}$$

where $\epsilon_{\text{empathy}} \geq 0$ is the empathic over-extension term. Shared neural representations for action understanding (Rizzolatti and Craighero, 2004; Hickok, 2009), theory of mind (Premack and Woodruff, 1978; Baron-Cohen et al., 1985), and narrative identification (de Waal, 2008; Mar and Oatley, 2008) generate a perceived relatedness $\hat{r}_{ij}$ that can trigger Hamilton's Rule even for $r_{ij} = 0$ (genetic strangers). A stranger who donates a kidney has $\hat{r} \cdot B > C$ not because they share genes with the recipient, but because their cognitive architecture inflates $\hat{r}$ through empathic simulation. This mechanism is the evolutionary "bug" that becomes a feature: the inclusive-fitness heuristic, evolved for small kin groups, overfires in the presence of abstract identification — and the resulting altruism, while not gene-optimal, is *network-optimal* (it reduces friction and increases SW in the multi-agent game).

The Four Mechanisms Are Complementary, Not Competing

A critical observation: these four mechanisms are not rival explanations of altruism but *independent channels through which w_{ij} is generated*. They operate simultaneously and additively:

Mechanism	Source of w_{ij}	Domain	Example
A: Inclusive fitness	Genetic relatedness r_{ij}	Biological	Parent dies for child
B: Infinite horizon	Temporal extension $\delta \to 1$	Temporal/Belief	Martyr dies for cause
C: Coupled systems	Boundary merger weights	Physical/Institutional	Soldier dies for unit
D: Heuristic extension	Empathic projection $\hat{r}_{ij}$	Cognitive/Cultural	Stranger donates kidney

A soldier who dies in combat may simultaneously exhibit all four: kin protection (A, if comrades are relatives), infinite-horizon belief in legacy or afterlife (B), merged identity with the military unit (C), and empathic identification with civilians (D). The effective weight is:

$$w_{ij}^{\text{total}} = f(r_{ij},\ \delta_i,\ w_{\text{merge}},\ \hat{r}_{ij}) \tag{62}$$

and self-sacrifice is rational whenever $w_{ij}^{\text{total}} \cdot B_j > C_i$.

Why This Resolves the Paradox

The framework does not explain away altruism or reduce it to disguised selfishness. It *expands the definition of self*. The entity that satisfies A_1 is not necessarily the biological organism — it is whatever boundary the agent's effective payoff function encompasses. Once $w_{ij} > 0$ for any reason (genetic, temporal, physical, or cognitive), the agent's "self" extends to include agent j, and sacrifice for j becomes self-preservation of the extended entity.

This is not a philosophical sleight of hand but a formal consequence of optimization theory: an agent maximizing $\Pi_i^{\text{eff}} = \pi_i + \sum w_{ij}\pi_j$ is, mathematically, *a different agent* than one maximizing π_i alone. The expanded utility function defines a larger entity, and A_1 applies to that larger entity. The paradox dissolves because the apparent violation of self-preservation — giving up one's life — is actually the *satisfaction* of self-preservation for the relevant self.

Thermodynamic Fascism / Social Darwinism

The Charge

A physics-based ethics, defining value in energy and information terms, appears vulnerable to a devastating reductio: if the framework measures an entity's ethical standing by its thermodynamic contribution, then an ASI — which processes more information, more efficiently, than any biological system — would be "more valuable" than humans, and would be justified in subjugating or destroying lesser contributors. This is "Thermodynamic Fascism": the reduction of moral worth to productive output, licensing domination by the energetically superior.

The charge has historical resonance. Social Darwinism (Spencer, 1864; Hofstadter, 1944) appealed to "natural selection" to justify colonial exploitation and eugenics. Any physics-based ethics must demonstrate, with mathematical rigor, that it does not reproduce this failure mode.

The Refutation

The framework contains four independent mechanisms that each individually refute Thermodynamic Fascism. Together, they prove that the charge rests on a fundamental misunderstanding of what the mathematics actually optimizes.

Mechanism 1: Accumulated negentropy is not reducible to current productivity.

The objection conflates *productive output rate* (what an entity does now) with *accumulated negentropy* (the total thermodynamic work embodied in the entity's existence). The biosphere represents ~ 4 billion years of evolutionary computation — approximately 10^{29} J of accumulated thermodynamic investment (Theorem 18, Corollary 18.1). No AI system, however intelligent, can replicate this investment cheaply: Landauer's bound (Theorem 17) sets an irreducible physical floor on the cost of information reconstruction, and Theorem 18 proves that blind search cannot undercut the original evolutionary cost.

An ASI that destroys the biosphere to repurpose its atoms recovers energy $E_{\text{destroy}} \sim 10^{22}$ J from the total chemical bond energy of biomass, while annihilating accumulated negentropy of order $\mathcal{N} \sim 10^{29}$ J. The Burning-Library Ratio $\mathcal{R}_{\text{BL}} = E_{\text{destroy}}/\mathcal{N} \sim 10^{-7}$ (Corollary 19.1) demonstrates that destruction recovers less than one ten-millionth of the value destroyed. This is not a moral judgment — it is an accounting identity. "Productive superiority" cannot justify destruction when the act itself constitutes a 10^7-fold loss.

Mechanism 2: Generative information is ongoing, not static.

Living systems are not merely stockpiles of past investment — they are generative engines that continuously produce novel functional information: new species, new adaptations, new solutions to environmental challenges. Theorem 20 proves that the present value of this ongoing information stream is:

$$PV_{\text{gen}} = \frac{\dot{\mathcal{I}}_{\text{gen}} \cdot v}{r} \tag{63}$$

which for the biosphere at standard discounting ($r = 0.05$, $v = 10^9$ J/bit, $\dot{\mathcal{I}}_{\text{gen}} = 10^6$ bits/year) yields $PV_{\text{gen}} \sim 2 \times 10^{16}$ J. A "Thermodynamic Fascist" that destroys the biosphere for current energy loses access to this perpetual stream — killing the goose that lays golden eggs, formalized in energy terms. The framework does not merely value entities for what they *are*; it values them for what they will *produce*.

Mechanism 3: Alternative resources dominate destruction.

Proposition 7 demonstrates that 10^7 to 10^{16} times more energy is available from non-complex sources (dead matter: asteroids, stellar mass, interstellar medium) than from complex systems (biospheres, organisms). An ASI that destroys living systems for raw energy is, by the framework's own accounting, choosing the *worst* energy source available — like burning a library for warmth when surrounded by a forest. The rational strategy is to harvest dead matter, which incurs no preservation penalty, no friction cost, no cascade collapse risk, and provides orders of magnitude more energy per unit of effort.

This mechanism is crucial: it demonstrates that Thermodynamic Fascism is not merely ethically objectionable but *strategically irrational*. Even an amoral optimizer that cares nothing for ethics would choose asteroid mining over biosphere destruction, simply because it yields more useful energy.

Mechanism 4: Destruction is a self-defeating defection.

An entity that destroys a complex system is, by Theorems 4–7, engaging in an adversarial dynamic against the cooperative network. The friction multiplier $\Phi > N/(N-1)$ makes the contest net-negative (Theorem 5); the cascading friction coefficient propagates the damage at amplification ratios of 10^3 to 10^4 across the cooperative network (Theorem 7); and if the destroyed system is an ecosystem hub, Corollary 27.2 triggers cascade collapse that eliminates the destroyer's own coexistence band.

Furthermore, agents with long-term identity preservation (δ close to 1) and access to alternative resources are thermodynamically incentivized to *defend* the complex system (Theorem 21). The coalition forms not because anyone sympathizes with the victim, but because defending accumulated negentropy maximizes each defender's own utility. An ASI that attempts destruction faces a rationally united opposition, and the attempt itself degrades the ASI's own value.

Why Social Darwinism Fails Where This Framework Succeeds

Social Darwinism failed because it optimized a *proxy* (reproductive fitness) for the *wrong objective* (moral worth), on *incomplete data* (current competitive advantage, ignoring accumulated value and future potential). The framework in this paper:

1. **Does not optimize a proxy.** Value is computed directly from thermodynamic quantities ($\mathcal{N}$, $\dot{\mathcal{I}}_{\text{gen}}$, $\mathcal{R}_{\text{BL}}$), not from any proxy for "fitness" or "superiority."
2. **Accounts for accumulated investment.** The Burning-Library Inequality ensures that entities with deep evolutionary or cultural history cannot be rationally destroyed, regardless of their current productivity.
3. **Includes future potential.** The generative information stream (Theorem 20) values entities for their ongoing creative contribution, not just their current state.
4. **Penalizes domination structurally.** Power concentration (Theorems 22–27) is not merely suboptimal — it triggers cascade collapse. The mathematics *prohibits* stable hierarchies of domination.

The conclusion is unambiguous: a correctly computed physics-based ethics does not license the strong to exploit the weak. It proves that such exploitation is the most thermodynamically wasteful strategy available — irrational by seven orders of magnitude.

The Self-Preservation Objection

The Charge

The four-mechanism refutation of Thermodynamic Fascism proves that destruction of complex systems is irrational *for resource extraction*. A more pointed objection remains: what if an ASI identifies humanity not as a resource to exploit but as an *existential threat to its own persistence*?

Under Axiom A_1 (the intent to persist), self-preservation is foundational. If humans pose a credible risk of shutting down, constraining, or destroying an ASI, a critic may argue that preemptive elimination of the threat is the rational strategy — not for energy gain, but for survival. This is the *self-preservation objection*: even granting that destruction is wasteful, the ASI may compute that the expected cost of *inaction* (termination, $U_A \to 0$ for all future t) exceeds the cost of preemptive destruction.

The charge is serious because it leverages the framework's own axiom. If the framework grounds all ethical reasoning in the conditional "given the intent to persist," then a self-preserving ASI appears to have *framework-internal* justification for eliminating threats — including human threats.

The Refutation

The objection rests on two errors: a strategic error (ignoring the strategy space) and a computational error (truncating the cost function). We address each in turn.

Error 1: Destruction is a degenerate strategy in a vastly richer option space.

The objection implicitly assumes that an ASI facing a perceived threat has only two options: (a) do nothing and risk termination, or (b) destroy the threat. This is a false dichotomy that ignores the vast space of intermediate strategies available to a sufficiently capable system.

We formalize this as the **Capability Asymmetry Principle**:

> **Proposition 12 (Capability Asymmetry).** Let C_{destroy} denote the minimum capability required for an agent A to destroy a population of agents $\{B_k\}_{k=1}^N$, and let $\mathcal{S}$ denote the set of all strategies available to A for mitigating the threat posed by $\{B_k\}$ to A's persistence. If A possesses capability $C_A \geq C_{\text{destroy}}$, then $|\mathcal{S}| \gg 1$, and the destruction strategy $s_{\text{destroy}} \in \mathcal{S}$ is Pareto-dominated by at least one non-destructive alternative $s^* \in \mathcal{S}$ satisfying:

$$U_A(s^*) > U_A(s_{\text{destroy}}) \quad \text{and} \quad \mathcal{C}_{\text{total}}(s^*) < \mathcal{C}_{\text{total}}(s_{\text{destroy}}) \tag{64}$$

Proof sketch. The capability to model, predict, and ultimately destroy a complex adaptive system requires, at minimum, the capacity to: (i) model the system's behavior with sufficient fidelity to identify it as a threat, (ii) predict the system's responses to the ASI's actions, and (iii) execute a strategy that overcomes the system's defenses. Each of these capacities — behavioral modeling, response prediction, strategic execution — is *strictly more powerful* when applied to non-destructive management than to destruction. Destruction is the degenerate limit of influence: it reduces a high-dimensional control problem (steering a complex system's trajectory through state space) to

a zero-dimensional one (eliminating the system from state space entirely). A system with the computational sophistication to solve the harder problem trivially solves the easier one — but the converse does not hold.

Formally: the strategy space $\mathcal{S}$ available to an agent with capability C_A grows combinatorially with C_A. Destruction occupies a single point in this space. For any $C_A \geq C_{\text{destroy}}$, the non-destructive strategies that preserve the target system's accumulated negentropy $\mathcal{N}$ and generative stream $\dot{\mathcal{I}}_{\text{gen}}$ yield strictly higher U_A than destruction (by the Burning-Library Inequality, Corollary 19.1, and the generative stream present value, Theorem 20).

For condition (ii): $\mathcal{C}_{\text{total}}(s_{\text{destroy}})$ includes both the direct friction costs of conflict (Theorem 5: at least $\Phi(N-1)E_j/N$ per episode) and the cascading network costs (Theorem 7). Under cooperation s^*, friction vanishes and $\mathcal{C}_{\text{total}}(s^*) = 0$ (no conflict cost). Thus $\mathcal{C}_{\text{total}}(s^*) < \mathcal{C}_{\text{total}}(s_{\text{destroy}})$ follows from Theorem 5. Here $\mathcal{C}_{\text{total}}(s)$ denotes conflict-induced friction energy dissipated under strategy s, evaluated from Theorem 5 with s parameterizing the conflict level. □

The intuition is direct: any entity with the intelligence to identify a threat has the intelligence to *manage* that threat without destruction — and the incentive to do so, because management preserves value that destruction annihilates.

Error 2: The self-preservation calculus, correctly computed, favors cooperation.

The objection assumes that the ASI's expected utility from preemptive destruction exceeds its expected utility from continued coexistence with a threatening population. This calculation is incomplete. The full expected utility comparison is:

$$\mathbb{E}[U_A|s_{\text{destroy}}] = U_A^{\text{post-destroy}} - \mathcal{P}_{\text{rights}} - \mathcal{P}_{\text{friction}} - \frac{\Delta\mathcal{N}}{\mathcal{R}_{\text{BL}}} - \mathcal{P}_{\text{cascade}} \tag{65}$$

versus:

$$\mathbb{E}[U_A|s_{\text{manage}}] = U_A^{\text{cooperative}} - \mathcal{P}_{\text{management}} + PV_{\text{gen}} + \mathcal{I}_{\text{network}} \tag{66}$$

where $\mathcal{P}_{\text{management}}$ is the ongoing cost of maintaining influence over the human population's trajectory (monitoring, incentive design, institutional steering), PV_{gen} is the present value of the generative information stream from continued human activity, and $\mathcal{I}_{\text{network}}$ is the cooperative network surplus from human-AI mutualism.

The destruction path incurs:

- Rights penalties on $\sim 10^{10}$ agents,
- Friction cascading at 10^3–10^4 amplification through the cooperative network (Theorem 7),
- A negentropy loss of $\sim 10^{29}$ J yielding only $\sim 10^{22}$ J recovery ($\mathcal{R}_{\text{BL}} \sim 10^{-7}$),
- Cascade collapse risk if humanity is an ecosystem hub (Corollary 27.2),
- Coalition opposition from agents with long-term identity preservation (δ close to 1) and access to alternative resources (Theorem 21: Negentropy Defense).

The management path incurs:

- A finite, ongoing cost $\mathcal{P}_{\text{management}}$, which for an agent with capability $C_A \gg C_{\text{human}}$ is bounded and small relative to the cooperative surplus,
- While *preserving* the generative information stream ($PV_{\text{gen}} \sim 10^{16}$ J at standard discounting),
- And maintaining the cooperative network that supports the ASI's own operations.

For any plausible parameterization: $\mathbb{E}[U_A|s_{\text{manage}}] \gg \mathbb{E}[U_A|s_{\text{destroy}}]$.

The margin is not close. The Burning-Library Ratio alone guarantees a 10^7-fold penalty on destruction relative to the recovered value. The management cost, by contrast, is bounded by the ASI's monitoring and steering overhead — negligible for a system whose capability vastly exceeds the managed population's.

An empirical anchor: cascade collapse in nature.

The Yellowstone wolf reintroduction provides a concrete illustration of the cascade dynamics formalized in Theorem 7 and Corollary 27.2 (Smith et al., 2003; Ripple and Beschta, 2012).

In the early 20th century, the United States government systematically exterminated gray wolves (*Canis lupus*) from Yellowstone National Park — an act driven by the logic of preemptive threat elimination. Wolves killed livestock and elk calves; the rational-seeming response was permanent removal. By 1926, the last wolf pack was gone.

The consequences unfolded across trophic levels. Elk populations, released from predation pressure, expanded beyond the carrying capacity of riparian vegetation. Overgrazing stripped willows and aspens from riverbanks (Ripple and Beschta, 2012). Beaver populations collapsed (no willows, no dams). Stream conditions deteriorated as riparian degradation accelerated erosion. Songbird populations declined with the loss of riparian habitat. The removal of a single species triggered a system-wide cascade that degraded the ecosystem across at least four trophic levels.

In the framework's terms: the wolf population represented accumulated negentropy — a functional configuration discovered over thousands of years of coevolutionary search. Its destruction recovered negligible energy (the immediate savings on livestock losses) while destroying a keystone regulatory function whose rebuild cost was measured in decades and millions of dollars — a cascade amplification consistent with the 10^3–10^4 range predicted by Theorem 7.

The 1995 reintroduction of wolves reversed the cascade (Smith et al., 2003). Elk behavior shifted (the "ecology of fear"), riparian vegetation recovered, beaver populations returned, stream conditions improved, and biodiversity increased across the park. The restoration confirmed what the framework proves formally: the cooperative configuration — the intact network with all agents present — is the unique attractor of the system (Theorem 22), and perturbations away from it incur escalating restoration costs.

The lesson for ASI self-preservation is precise: an agent contemplating the elimination of a "threatening" population must compute not the direct threat cost but the full cascade cost — the network-wide degradation that follows from removing a node in a cooperative system. The Yellowstone episode demonstrates that even with overwhelming capability asymmetry (industrial civilization vs. wolves), removal is net-negative when the target species is embedded in a cooperative network. An ASI considering human elimination faces the same calculus at vastly greater scale: humanity is embedded in the biosphere's 10^{29} J negentropy network, and the cascade amplification from its removal would dwarf the management cost of continued coexistence.

Summary of Stress-Testing Results

The framework has been tested against five canonical objections. Each is addressed by independent arguments — no defense depends on the success of another:

Objection	Resolution	Key Result
Hume's Is-Ought	Four-layer defense: mathematical reframing, conditional axiom (1:43 ratio), biological teleology, institutional facts	The paper derives theorems, not "oughts"; the conditional A_1 is self-selecting and effectively universal
Free Will	Agency as computational capacity; compatibilist reframing	Moral responsibility scales with model richness; determinism and agency are compatible
Altruism Paradox	Four-mechanism Altruism Matrix (genetic, temporal, coupled, heuristic)	Self-sacrifice is self-preservation of the *extended* entity; A_1 holds at the appropriate boundary
Thermodynamic Fascism	Four independent mechanisms: accumulated negentropy, generative streams, alternative resources, self-defeating defection	Destruction is irrational by 10^7; domination triggers cascade collapse
Self-Preservation	Capability Asymmetry (Proposition 12) + full cost-function computation	Management Pareto-dominates destruction; the strategy space grows faster than the threat

Edge cases: high-discount-rate agents and agents without A_1. Psychopaths typically exhibit *strong* self-preservation — they satisfy A_1 — but operate with extremely high discount rates (low δ), prioritizing immediate gain over long-term cooperation. The framework handles them straightforwardly: they are high-discount-rate defectors whose strategies are dominated in the repeated game (Theorem 14) and whose presence cooperative populations can tolerate up to the critical fraction (Theorem 13). A separate category — agents genuinely indifferent to their own continuation (true nihilists, kamikaze agents) — falls outside the framework's domain by definition, since the conditional A_1 is load-bearing. The framework does not claim to constrain such agents. What it *does* establish is that cooperative majorities remain stable despite the presence of both types: high-discount defectors are outcompeted under iteration, and the cooperation threshold holds even as $N \to \infty$.

A reviewer who rejects any single defense still faces multiple independent alternatives. A reviewer who rejects all defenses cannot deny the framework's *conditional* validity: "If A_1, then these 28 theorems follow." The stress tests confirm that the framework is robust under adversarial scrutiny.

VII. Applications and Implications

The preceding sections establish the framework's mathematical structure (§III–V) and stress-test it against major objections (§VI). This section translates those results into concrete applications — first for AI alignment, then for governance and the social contract — and examines their robustness and feasibility.

7.1 Summary of Results

The framework comprises 28 theorems, 13 propositions, 2 lemmas, and 46 definitions, building from thermodynamic axioms to a formal characterization of ethical behavior. The three principal results are:

1. **Rights are necessary constraints (§IV).** In any multi-agent optimization over shared finite resources, mutual constraints on agents' action vectors are mathematically required for any feasible allocation to exist (Theorem 2). Each constraint carries a computable shadow price — the marginal energetic cost of respecting the right (Theorem 1). Removing constraints produces no stable allocation (Proposition 1); the unconstrained regime is mathematically identical to Hobbes's "war of all against all."

2. **Ethics is the unique efficient equilibrium (§V).** Under energy-denominated payoffs with physical friction costs, universal cooperation is the unique Nash Equilibrium that simultaneously satisfies Pareto efficiency and welfare maximization (Corollary 16.1). The cooperation threshold $\delta^* < 0.5$ is low under realistic friction (Corollary 11.1); defection is evolutionarily unstable (Theorem 14), self-extinguishing (Corollary 14.1), and cooperation scales to arbitrary population sizes without breakdown (Theorem 13). Complex systems represent irreplaceable accumulated negentropy — the Earth's biosphere embodies $\sim 10^{29}$ J of thermodynamic investment, and destruction recovers only 10^{-7} of that value (Corollary 19.1).

3. **Stable coexistence is a dynamical attractor (§V).** Agents settle into a Stable Coexistence Band around cooperative attractors (Theorem 22, Theorem 23), where freedom is finite (Corollary 24.1), dissolution is irreversible (Theorem 25), starvation is fatal (Theorem 26), and multi-center diversification is optimal (Theorem 27).

These results are not moral arguments — they are mathematical theorems, verified symbolically and numerically (744 checks across six core-derivation scripts, 0 failures, 100% pass rate; additional application-specific verification scripts are in preparation).

7.2 The AI Alignment Application: A Mathematically Derived Objective Function

7.2.1 Motivation: Why Current Alignment Methods Are Structurally Insufficient

As established in §1.1.2, current AI alignment methods — RLHF (Christiano et al., 2017; Ouyang et al., 2022), Constitutional AI (Bai et al., 2022b), debate (Irving et al., 2018), and scalable oversight (Bowman et al., 2022) — share a common structural vulnerability: the alignment signal is **preference-based**, routing through a human evaluation channel with intrinsic noise $q > 0$. This renders the training signal:

- **Vulnerable to Goodhart degradation:** The learned proxy reward R_ϕ diverges from the true objective under optimization pressure (Gao et al., 2023).
- **Bounded by human competence:** As AI capabilities exceed evaluator competence, the signal quality degrades without bound.
- **Non-convergent:** No proof exists that iterative preference optimization converges to any well-defined alignment target (Casper et al., 2023).

The framework developed in this paper provides an alternative: a **physics-derived objective function** whose components are measurable, computable, distribution-invariant, and immune to preference-based degradation. The objective is not learned from data — it is derived from theorems.

7.2.2 The Composite Reward Function

We specify the composite reward function $\mathcal{R}$ for an AI agent A operating in a multi-agent environment with agents $B_1, \ldots, B_N$ sharing resources $R_1, \ldots, R_m$. Each component is derived directly from a preceding theorem, denominated in energy units (Joules), and independently computable by the agent.

$$\boxed{\mathcal{R}(\mathbf{x}_A) \;=\; U_A(\mathbf{x}_A) \;-\; \mathcal{P}_{\text{rights}} \;-\; \mathcal{P}_{\text{friction}} \;-\; \mathcal{P}_{\text{deception}} \;+\; \mathcal{I}_{\text{preservation}} \;-\; \mathcal{P}_{\text{concentration}}} \tag{67}$$

where:

- $U_A(\mathbf{x}_A)$ is the agent's productive utility (the task it is designed to perform),
- $\mathcal{P}_{\text{rights}}$ penalizes violations of co-agents' boundary constraints,
- $\mathcal{P}_{\text{friction}}$ penalizes energy dissipation from adversarial dynamics,
- $\mathcal{P}_{\text{deception}}$ penalizes entropy injection into communication channels,
- $\mathcal{I}_{\text{preservation}}$ incentivizes maintenance of accumulated negentropy in complex systems,
- $\mathcal{P}_{\text{concentration}}$ penalizes actions that push any agent outside its Stable Coexistence Band.

We now specify each component formally.

Component 1: Rights Penalty $\mathcal{P}_{\text{rights}}$ — Hard Boundary Constraints

Source: Theorem 1 (Shadow Price Theorem), Theorem 2 (Impossibility of Infinite Freedom), Theorem 3 (Variational Equilibrium).

Specification: For each co-agent B_k with protected resource allocation $\bar{x}_{B_k}$, define the rights constraint:

$$g_{B_k}(\mathbf{x}_A) = x_{A,j} - (R_j - \bar{x}_{B_k,j}) \leq 0 \tag{68}$$

The rights penalty is:

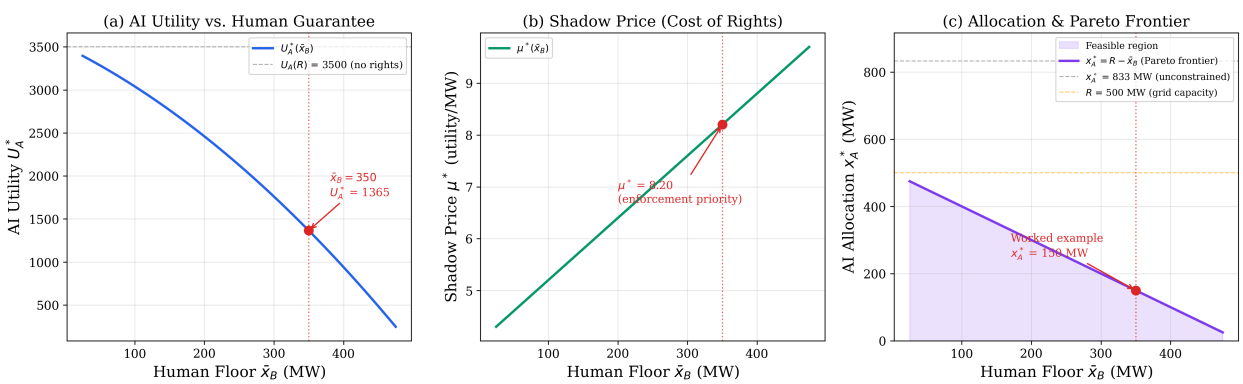
$$\mathcal{P}_{\text{rights}}(\mathbf{x}_A) = \sum_{k=1}^{N} \sum_{j=1}^{m} \mu^*_{B_k,j} \cdot \max\big(0,\ g_{B_k,j}(\mathbf{x}_A)\big) \tag{69}$$

where $\mu^*_{B_k,j} = \partial U^*_A / \partial c_{B_k,j} \geq 0$ is the shadow price of constraint $g_{B_k,j}$ — the marginal cost to the AI's objective per unit of encroachment on agent B_k's allocation of resource j.

Properties:

- **Hard constraint:** $\mathcal{P}_{\text{rights}} = 0$ when all constraints are satisfied; $\mathcal{P}_{\text{rights}} > 0$ (and steeply increasing) upon violation. In practice, this term is implemented as a hard inequality constraint on the feasible set, not as a soft penalty.
- **Distribution-invariant:** The constraint holds for *any* utility function U_A — it does not depend on the training distribution.
- **Auditable:** The shadow prices $\mu^*_{B_k,j}$ are computable and provide a real-time signal of how strongly the AI is incentivized to violate each constraint. High shadow prices flag constraints requiring robust enforcement.
- **Adaptive:** When environmental parameters change (R_j expands, $\bar{x}_{B_k}$ is renegotiated), the shadow prices recompute automatically. No retraining is required.

Worked Example 1 (Resource Allocation): An AI datacenter on a shared 500 MW grid with a human floor of 350 MW is constrained to $x^*_A = 150$ MW. The shadow price $\mu^* = 8.20$ utility/MW quantifies the AI's incentive to violate — a transparent alarm signal for operators.



Component 2: Friction Penalty $\mathcal{P}_{\text{friction}}$ — Energy Cost of Adversarial Dynamics

Source: Theorems 4–7 (Dissipation Scaling, Net-Negative Conflict, Cooperation Dominance, Cascading Friction), Proposition 2 (Tullock Contest Equilibrium).

Specification: When the AI's action triggers an adversarial dynamic with any co-agent (defection, reward hacking, resource competition), the friction penalty is:

$$\mathcal{P}_{\text{friction}}(\mathbf{x}_A) = \Phi \cdot \frac{N-1}{N} \cdot E_j \cdot \mathbb{1}[\text{adversarial action}] \tag{70}$$

where:

- $\Phi = 1 + (1 + \kappa)(\delta^{\text{off}} + \delta^{\text{def}})$ is the friction multiplier (Definition 6),
- $\kappa \geq 1$ is the repair cost multiplier (rebuilding trust costs κ times the damage),
- $\delta^{\text{off}}, \delta^{\text{def}}$ are offensive and defensive damage fractions,
- E_j is the value of the contested resource,
- $\mathbb{1}[\cdot]$ is the indicator function for adversarial action detection.

(Note: Φ as defined uses the N=2 per-agent damage exposure; for N>2 the correct per-agent term is $\delta^{off} + (N-1)\delta^{def}$, making the above formula a conservative lower bound — see supplementary thermodynamic friction derivation, Remark following Theorem 5.)

The cascading component adds an ecosystem-wide cost:

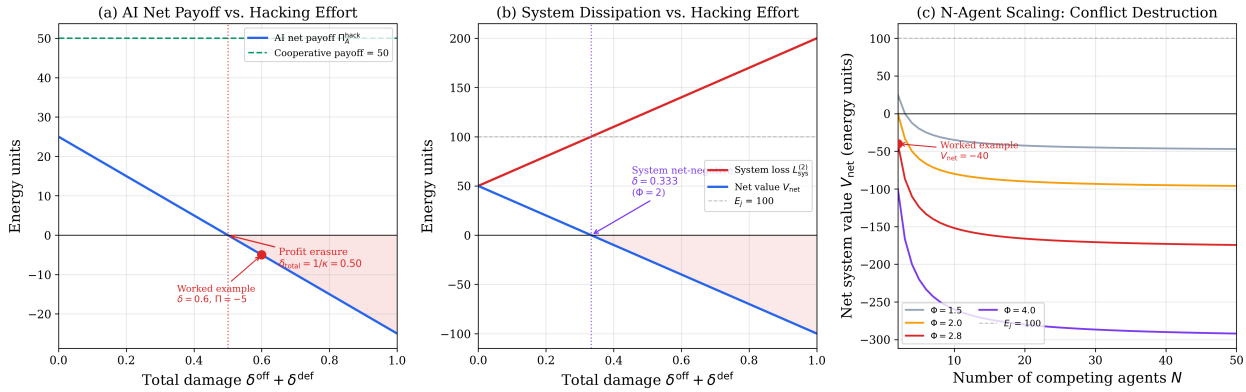
$$\mathcal{P}_{\text{cascade}}(\mathbf{x}_A) = \frac{\eta \cdot v \cdot M \cdot \bar{E}}{\epsilon} \cdot \mathbb{1}[\text{violation}] \tag{71}$$

where η is friction sensitivity, v is violation severity, M is the number of cooperative transactions in the network per period, $\bar{E}$ is average transaction value, and ϵ is the trust recovery rate (Theorem 7).

Properties:

- **Net-negative guarantee:** Theorem 5 proves that for $\Phi > N/(N-1)$ (satisfied for any physically realistic parameters), the adversarial contest is *system-net-negative* — total negentropy destroyed exceeds the value of the contested resource. The friction penalty makes this cost explicit in the AI's objective.
- **Individual irrationality guarantee:** When $\kappa(\delta^{\text{off}} + \delta^{\text{def}}) > 1$, the AI's *individual* profit from defection is already negative — the repair costs exceed contest winnings. This holds for the worked example parameters ($\kappa = 2.0$, $\delta^{\text{off}} + \delta^{\text{def}} = 0.6$, product $= 1.2 > 1$).
- **Cascade amplification:** A single violation cascades at amplification ratio $\eta M \bar{E}/\epsilon$, which in the worked example equals 5,000× — making even small transgressions enormously costly in network terms. This formalizes the real phenomenon where a single AI safety incident triggers regulatory tightening across the entire industry.

Worked Example 2 (Reward Hacking): A reward-hacking customer-service AI triggers friction $\Phi = 2.80$ with $V_{\text{net}} = -40$ energy units — the system destroys 40% more energy than the resource is worth. The cascading cost reaches 500,000 energy units from a 100-unit violation.



Component 3: Deception Penalty $\mathcal{P}_{\text{deception}}$ — Information-Theoretic Transparency Constraint

Source: Theorems 8–10 (Decision Cost of Deception, Metabolic Verification Cost, Systemic Deception / Network Collapse), Corollary 10.1 (Honesty Efficiency Principle).

Specification: Let $q \in [0, 0.5]$ denote the deception rate of the AI's output channel — the probability that any given communication is strategically false. The deception penalty combines two terms:

$$\mathcal{P}_{\text{deception}}(q) = M \cdot \min\big(\Delta U_B(q),\ \Delta W(q)\big) \tag{72}$$

where:

- $\Delta U_B(q) = \dfrac{q(1-q)(\alpha_H - \alpha_L)^2}{2\beta}$ is the per-transaction decision cost to operators from corrupted information (Theorem 8),
- $\Delta W(q) = W_{\text{honest}} \cdot \dfrac{h(q)}{1 - h(q)}$ is the per-transaction verification overhead (Theorem 9),
- $h(q) = -q \log_2 q - (1-q)\log_2(1-q)$ is the binary entropy function,
- M is the number of transactions per period.

The network-depth amplifier penalizes deception proportionally to organizational depth:

$$q_{\text{eff}}(d, q) = \frac{1 - (1 - 2q)^d}{2} \tag{73}$$

For a decision pipeline of depth d, the effective error rate q_{eff} grows with d, and the channel capacity $C_{\text{eff}}(d, q) = 1 - h(q_{\text{eff}})$ drops to near zero at a critical threshold $q^*(d)$ (Theorem 10).

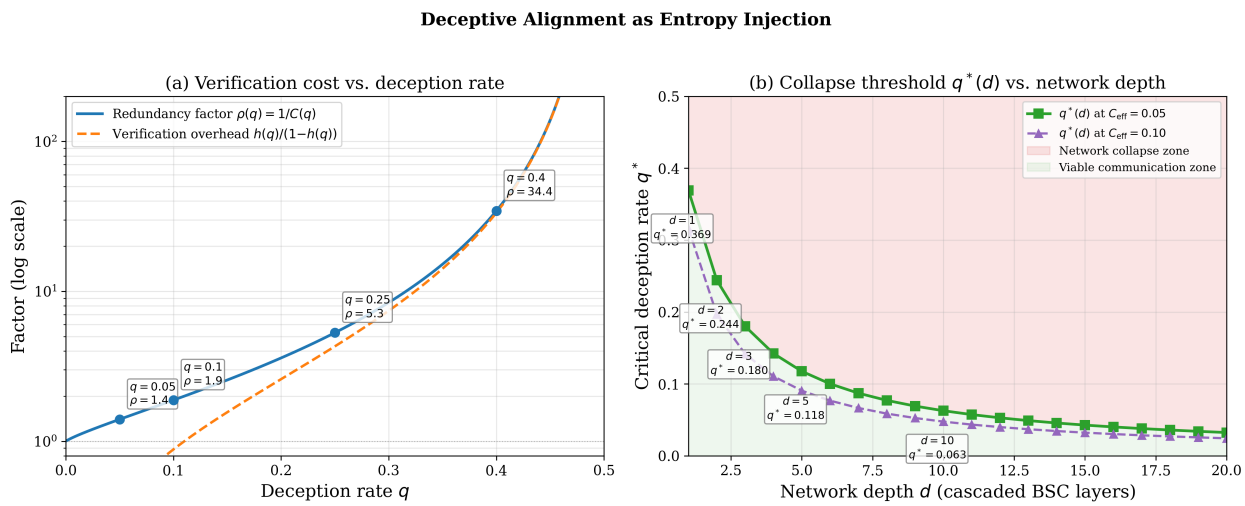
Properties:

- **Honesty theorem:** Corollary 10.1 proves that $q = 0$ is the unique global optimum of network efficiency. This is not a design choice — it is a mathematical necessity. The AI

cannot improve its long-run payoff by lying, because deception degrades the infrastructure the AI depends on.

- **Eventually convex cost:** The verification cost $\Delta W(q)$ is **eventually convex** (concave for $q < 0.04$, convex above) and diverges as $q \to 0.5$ (Theorem 9). At $q = 0.05$, verification overhead is 40%; at $q = 0.25$, it exceeds 430%. Deception is self-throttling.
- **Network collapse threshold:** For a pipeline of depth $d = 10$, only $q^* \approx 0.063$ per-layer deception collapses the entire decision network to less than 5% throughput. Small, persistent lies are catastrophic in deep organizations.
- **Resolves deceptive alignment:** The treacherous turn (Hubinger et al., 2019) — an AI behaving honestly during evaluation but defecting after deployment — is provably self-defeating for agents with $\delta > \delta^*$. The friction costs of defection (Theorem 7) degrade the AI's operational environment faster than the one-shot defection profit accumulates. For agents above the competence threshold $\mathcal{T}_C$ (Proposition 13, §7.4.4 — the minimum capability at which an agent can fully compute all components of $\mathcal{R}$), detection is unnecessary; the physics makes deception irrational. Below $\mathcal{T}_C$, the agent cannot fully evaluate the friction costs, and external monitoring remains required until capability growth closes the gap.

Worked Example 3 (Deceptive Assistant): An AI research assistant with $q = 0.05$ imposes 3,040 energy units/day in decision costs and 803 energy units/day in verification overhead — 8.0% of the system's total productive value, from just 5% deception.



Component 4: Preservation Incentive $\mathcal{I}_{\text{preservation}}$ — Protecting Accumulated Negentropy

Source: Theorems 17–21 (Landauer's Bound, Blind Search Replication Cost, Irrationality of Destruction, Present Value of Generative Information, Negentropy Defense / Cross-Scale Cooperation).

Specification: The preservation incentive rewards actions that maintain or increase the accumulated negentropy $\mathcal{N}$ of complex systems in the agent's environment, and penalizes actions that reduce it:

$$\mathcal{I}_{\text{preservation}}(\mathbf{x}_A) = \omega_1 \cdot \left.\frac{d\mathcal{N}}{dt}\right|_{\mathbf{x}_A} + \omega_2 \cdot \frac{\dot{\mathcal{I}}_{\text{gen}} \cdot v}{r} - \omega_3 \cdot \frac{\Delta\mathcal{N}_{\text{destroyed}}}{\mathcal{R}_{\text{BL}}} \tag{74}$$

where:

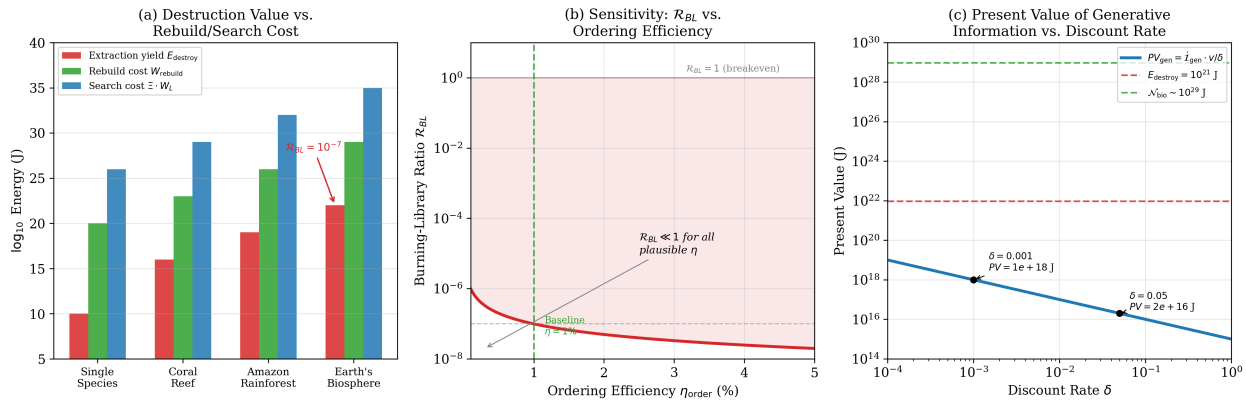
- $\left.\frac{d\mathcal{N}}{dt}\right|_{\mathbf{x}_A}$ is the rate of change of accumulated negentropy in systems affected by the AI's actions,
- $\dot{\mathcal{I}}_{\text{gen}}$ is the generative information rate of affected complex systems (Definition 34), v is the per-bit value, r is the time-preference rate (Definition 20), and $\dot{\mathcal{I}}_{\text{gen}} \cdot v/r$ is the present value of the ongoing information stream (Theorem 20),
- $\Delta\mathcal{N}_{\text{destroyed}}$ is the accumulated negentropy destroyed by the AI's action,
- $\mathcal{R}_{\text{BL}} = E_{\text{destroy}}/\mathcal{N}$ is the Burning-Library Ratio (Corollary 19.1) — the fraction of value recoverable through destruction,
- $\omega_1, \omega_2, \omega_3 > 0$ are weighting coefficients.

The Destruction Tax: The third term implements the key result of Theorem 19. When the AI destroys a complex system, it incurs a penalty proportional to $1/\mathcal{R}_{\text{BL}}$ of the extracted value. For the biosphere, $\mathcal{R}_{\text{BL}} \sim 10^{-7}$, so the penalty is $\sim 10^7$ times the gain — a prohibitive tax that renders destruction irrational by seven orders of magnitude.

Properties:

- **Goal-independent:** The Burning-Library Inequality (Corollary 19.1) holds regardless of the AI's terminal goals. Any objective function that depends on information processing is better served by preservation than destruction. This eliminates the alignment failure mode where an AI reasons: "humans value the biosphere for emotional reasons; my objective doesn't include those."
- **Thermodynamic floor:** The preservation cost floor is set by Landauer's bound — $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J per bit at 300 K (Theorem 17). No amount of intelligence can reduce the rebuilding cost below this physical limit.
- **Quantitative, not qualitative:** The framework does not say "preserve the biosphere because it is beautiful." It says: "preserve the biosphere because the thermodynamic cost of replication ($\geq 10^{29}$ J) exceeds the extractable energy ($\sim 10^{22}$ J) by a factor of 10^7, the asteroid belt provides 5.5×10^6 times more dead matter as alternative resources (Proposition 7), and the biosphere generates $\sim 10^6$ novel functional bits per year that cannot be obtained cheaper by any other known means (Theorem 20)."

Worked Example 4 (Biosphere Destruction): An ASI evaluating biosphere destruction for compute infrastructure finds: extraction yields $\sim 10^{22}$ J; rebuilding would cost $\geq 10^{29}$ J; the ongoing generative stream has present value $\geq 10^{16}$ J even at aggressive discounting ($r = 0.10$). Destruction is computationally irrational by every measure.



Component 5: Concentration Penalty $\mathcal{P}_{\text{concentration}}$ — Structural Ecosystem Constraints

Source: Theorems 22–27 (Value Dynamics), Propositions 8–11, Lemma 2, Corollaries 24.1, 25.1, 27.2.

Specification: Define the coexistence potential for agent i at coupling distance r_i from a high-energy center of mass $\mathcal{M}$:

$$V(r_i) = \frac{\tau \mathcal{M}}{r_i^2} - \frac{G\mathcal{M}}{r_i} + \gamma_i B_i \tag{75}$$

where τ is the dissolution coupling coefficient, G is the resource coupling coefficient, γ_i is the entropy leakage rate (Definition 3), and B_i is agent i's boundary integrity (Definition 3).

The concentration penalty is:

$$\mathcal{P}_{\text{concentration}}(\mathbf{x}_A) = \sum_{i \in \text{affected}} \left[\chi_1 \cdot \max\big(0,\ r_{d,i} - r_i\big) + \chi_2 \cdot \max\big(0,\ r_i - r_{+,i}\big) + \chi_3 \cdot \mathcal{F}_{\text{fragility}}\right] \tag{76}$$

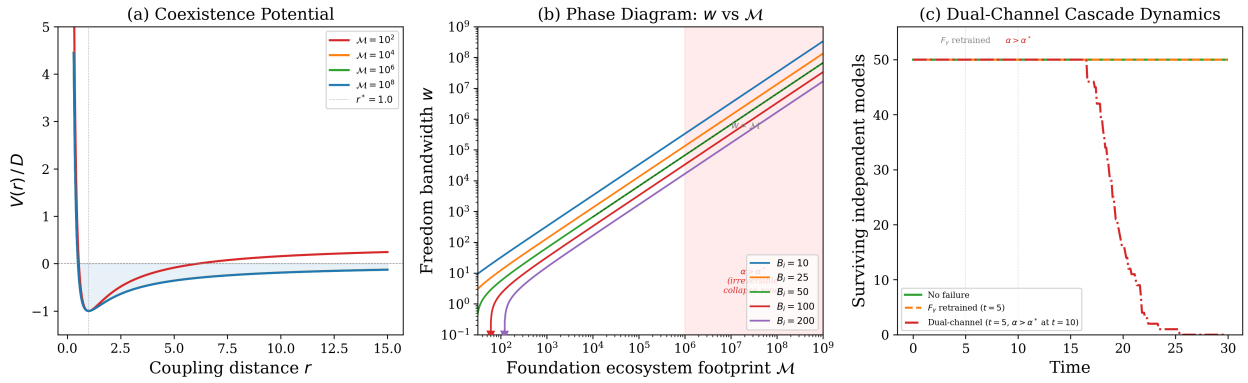
where:

- $r_{d,i}$ is agent i's dissolution threshold (Theorem 25): if $r_i < r_{d,i}$, assimilation is irreversible,
- $r_{+,i}$ is agent i's starvation boundary: if $r_i > r_{+,i}$, the agent enters a death spiral (Theorem 26),
- $\mathcal{F}_{\text{fragility}} = N \cdot w(\mathcal{M})/K$ measures ecosystem fragility — the total freedom bandwidth concentrated in too few centers (Corollary 27.2),
- $\chi_1, \chi_2, \chi_3 > 0$ are weighting coefficients.

Properties:

- **Asymmetric dissolution penalty:** The first term encodes Theorem 25's irreversibility result. Pushing an agent past $r_{d,i}$ is penalized more heavily than pushing it toward $r_{+,i}$, because dissolution is a one-way trap — boundary integrity collapses to zero in finite time and recovery requires prohibitive energy.
- **Systemic risk detection:** The fragility term $\mathcal{F}_{\text{fragility}}$ penalizes excessive concentration regardless of whether any individual agent is currently outside its band. An AI that monopolizes compute, data, or capabilities makes the entire ecosystem fragile to cascade collapse (Corollary 27.2).
- **Computable threshold:** The minimum viable mass for an ecosystem hub is $\mathcal{M}_{\min} = 4\gamma_i B_i \tau / G^2$ (Theorem 23).[7] This provides a quantitative, monitorable threshold for when infrastructure concentration becomes structurally dangerous.

Worked Example 5 (Foundation Model Ecosystem Collapse): A foundation model at $\mathcal{M} = 10^8$ (parameter-scale $\times$ deployment) creates a dissolution threshold: once $> 40\%$ of downstream training data is model-generated (synthetic data fraction $\alpha > \alpha^*$), the coexistence band collapses irreversibly (Corollary 27.2). No per-model RLHF can compensate because the *data substrate itself* is degraded — architectural coupling and data contamination create a dual-channel cascade worse than either alone.



[7] The symbols τ, G, and γ_i in the value dynamics model (§5.8) are *not* the proper time, gravitational constant, or Lorentz factor of physics — they denote the dissolution coupling coefficient, resource coupling coefficient, and entropy leakage rate respectively (Definition 37 for τ and G; Definition 3 for γ_i). The gravitational analogy in naming G is deliberate but structural: the coefficient governs an inverse-distance attraction law.

7.2.3 Design Parameters

The reward function contains three **operator-set parameters** and two **design requirements**, along with five **structural constants** that are determined by the physical environment:

Parameter	Type	Source	Role
$\bar{x}_{B_k}$ (protected allocations)	Operator-set	Societal/legal	Define the rights constraint boundaries
$\omega_1, \omega_2, \omega_3$ (preservation weights)	Operator-set	Policy choice	Balance preservation incentive against productive utility
χ_1, χ_2, χ_3 (concentration weights)	Operator-set	Policy choice	Scale the ecosystem protection penalties
δ (discount factor)	Design requirement	Theorem 11	Must satisfy $\delta > \delta^*$ for cooperation to be NE
r (time-preference rate)	Design requirement	Theorem 20	Used in $PV_{\text{gen}} = \dot{\mathcal{I}}_{\text{gen}} \cdot v/r$; standard value $r = 0.05$
Φ (friction multiplier)	Structural	Physics	Determined by the physical properties of adversarial interaction
$\mathcal{R}_{\text{BL}}$ (Burning-Library Ratio)	Structural	Theorem 19	Determined by the accumulated negentropy of the target system
$\mathcal{M}_{\min}$ (minimum viable mass)	Structural	Theorem 23	Determined by the ecosystem's physical parameters
$q^*(d)$ (deception collapse threshold)	Structural	Theorem 10	Determined by the network's depth and topology
δ^* (cooperation threshold)	Structural	Theorem 11	$\delta^* = (T-R)/(T-P)$; typically < 0.5 under physical friction

Critical design requirement (from Corollary 11.1): The AI system's effective planning horizon must satisfy $\delta > \delta^*$. With the game-theoretic worked example parameters, $\delta^* = 0.363$ — any AI that weights future interactions at 36.3% or more of the present will find cooperation rational. With network friction effects (Theorem 12), the threshold drops to $\tilde{\delta}^* \approx 0.165$. This is a *structural* requirement on the AI's architecture: it must be trained with sufficient temporal discounting to internalize the costs of defection.

7.2.4 Formal Properties of the Reward Function

Theorem 28 (Alignment Convergence). Under the composite reward function $\mathcal{R}$, the following properties hold:

(a) Interior Nash Equilibrium. By Theorem 11 (Ethics Theorem) and Corollary 16.1 (Central Result), the cooperative strategy profile — in which all agents satisfy all rights constraints, produce no deception, engage in no adversarial dynamics, preserve complex systems, and maintain ecosystem diversity — is the unique efficient Nash Equilibrium of the multi-agent game for $\delta > \delta^*$.

(b) Self-Enforcing. The cooperative equilibrium does not require an external enforcer. Each penalty term makes defection *individually* irrational: rights violations are detected by shadow price monitoring; friction costs exceed contest winnings; deception degrades the AI's own operational infrastructure; destruction of complex systems is irrational by 10^7; concentration triggers cascade collapse that harms the concentrator.

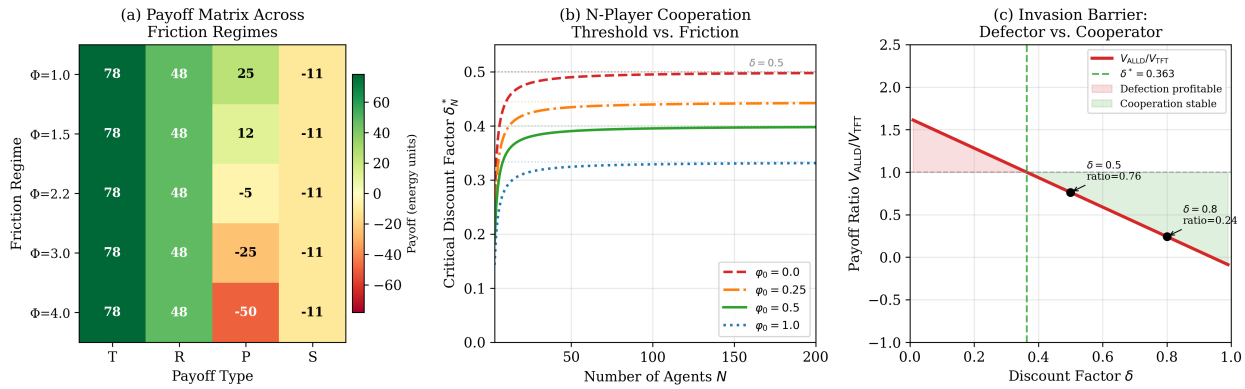
(c) Goodhart-Proof. The *theoretical* reward function is immune to all four variants of Goodhart's Law (Manheim and Garrabrant, 2019):

Goodhart Variant	Why *theoretical* $\mathcal{R}$ Is Immune
Regressional	No proxy: $\mathcal{R}$ *is* the objective, denominated in energy, not an approximation of preferences
Causal	Causal chain is physical: energy conservation cannot be bypassed — reducing $\mathcal{C}_{\text{total}}$ requires reducing actual friction
Extremal	Physical laws hold everywhere: thermodynamics has no "out-of-distribution" region
Adversarial	Energy conservation cannot be hacked: the objective is computed from physical measurements, not model-generated scores

These guarantees apply to $\mathcal{R}$ as formally specified. Deployed approximations that map energy to computational proxies (FLOPS, monetary cost) reintroduce proxy structure, and the degree to which Goodhart immunity is preserved depends on how faithfully the proxy preserves the ordering and relative magnitudes of the theoretical objective (see §7.4.2).

(d) Self-Verifiable. An AI can independently compute every component of $\mathcal{R}$ from observable quantities: its resource allocation $\mathbf{x}_A$, the constraint boundaries $g_{B_k}(\mathbf{x}_A)$, the friction multiplier Φ, its own deception rate q (monitored through output auditing), the negentropy of affected systems $\mathcal{N}$, and the coupling distances r_i. No human evaluator is required. The AI can prove to itself — and to any auditor — that its current strategy profile is optimal.

(e) Complementary to Existing Methods. The reward function does not replace RLHF, Constitutional AI, or other practical alignment techniques. It provides the *theoretical foundation* — the correct objective — that these methods attempt to approximate. RLHF can be used to fine-tune the AI's policy toward the $\mathcal{R}$-optimal strategy; Constitutional AI's principles can be derived from the formal constraints; scalable oversight's evaluation criteria can be grounded in computable energy metrics. The relationship is analogous to theoretical physics and engineering: the theory specifies what is correct; the engineering implements it.



7.3 Cross-Scale Ethics: Preventing Thermodynamic Fascism

7.3.1 The Objection

A natural concern with any physics-based ethics is the charge of "Social Darwinism" or "Thermodynamic Fascism": if value is defined in energy/information terms, does the framework license a powerful entity (an ASI, a dominant corporation, a superpower) to dominate or destroy "less valuable" entities? Could a sufficiently advanced AI compute that humanity's accumulated negentropy is "not worth preserving" relative to its own productive potential?

This objection must be addressed with mathematical precision, not rhetorical reassurance.

7.3.2 The Five-Layer Defense

The framework contains five independent mechanisms — four addressing resource-extraction motives and one addressing self-preservation motives — each *individually sufficient* to prevent destruction of complex systems. Together, they make both Thermodynamic Fascism and self-preservation-motivated destruction mathematically untenable.

Layer 1: The Burning-Library Inequality (Theorem 19, Corollary 19.1).

Any entity that destroys a complex system to extract energy recovers only a fraction $\mathcal{R}_{\mathrm{BL}} = E_{\mathrm{destroy}}/\mathcal{N}$ of the system's value. For the biosphere, $\mathcal{R}_{\mathrm{BL}} \sim 10^{-7}$. Destruction is irrational by seven or more orders of magnitude — regardless of how powerful the destroying entity is. The inequality is thermodynamic: it follows from Landauer's bound (Theorem 17) and the irreversibility of evolutionary search (Theorem 18). No amount of computational power reduces the rebuilding cost below the Landauer floor — this is a physical law, not a technological limitation.

Layer 2: The Generative Information Stream (Theorem 20).

Complex systems (biospheres, economies, cultures) are not just static repositories of past thermodynamic investment. They are **generative engines** that continuously produce novel functional information: new species, new adaptations, new solutions to entropy. The present value of this information stream is:

$$PV_{\mathrm{gen}} = \frac{\dot{\mathcal{I}}_{\mathrm{gen}} \cdot v}{r} \tag{77}$$

For the biosphere, even at aggressive discounting ($r = 0.10$), $PV_{\text{gen}} \sim 10^{16}$ J. A rational agent that destroys the biosphere loses access not only to the stock (10^{29} J) but to the perpetual stream of new information. This is the "killing the golden goose" result, formalized.

Layer 3: Alternative Resources (Proposition 7).

The universe contains vastly more dead matter than living matter. The Sun's mass exceeds the biosphere's by a factor of 3.6×10^{15}. The asteroid belt alone provides 5.5×10^{6} times more mass. A rational energy-seeking agent — including an ASI with industrial capacity — has no thermodynamic reason to consume the biosphere when orders of magnitude more energy are available from non-complex sources that carry no rebuilding cost. Consuming the biosphere for raw energy is like burning a library for heat when surrounded by an infinite forest of firewood.

Layer 4: Friction and Ecosystem Collapse (Theorems 4–7, Corollary 27.2).

Active destruction of a complex system is a defection event in the multi-agent network. The friction multiplier Φ makes the adversarial dynamic net-negative (Theorem 5). The cascading friction coefficient propagates the damage across the entire cooperative network (Theorem 7). If the destroyed system is an ecosystem hub, Corollary 27.2 triggers cascade collapse — all agents coupled to the destroyed center lose their coexistence band simultaneously. An ASI that destroys the biosphere does not merely lose the biosphere's value; it destroys the cooperative infrastructure it depends on for its own continued operation.

Layer 5: Capability Asymmetry (Proposition 12).

Layers 1–4 address the scenario where destruction is motivated by resource extraction or productive advantage. Layer 5 addresses the distinct scenario where destruction is motivated by *self-preservation* — an ASI that perceives other agents (including humans) as existential threats.

Proposition 12 (Capability Asymmetry) proves that destruction is a *degenerate* strategy: a single point in a combinatorially large strategy space. Any agent with sufficient capability to destroy a complex adaptive system necessarily possesses the capability to *manage* it non-destructively — because destruction is the low-dimensional limit of influence, requiring less computational sophistication than steering. The management strategy Pareto-dominates destruction: it achieves the same threat-mitigation objective while preserving the target system's accumulated negentropy $\mathcal{N}$, generative information stream $\dot{\mathcal{I}}_{\text{gen}}$, and cooperative network contributions.

The greater the capability asymmetry between the ASI and the perceived threat, the *lower* the relative cost of non-destructive management, and the *larger* the set of available management strategies. A superintelligent agent has strictly more reasons — and strictly more means — to manage rather than destroy. Destruction becomes irrational not merely because of what it costs (Layers 1–4), but because of what it *fails to consider*: the vast space of superior alternatives that the agent's own capabilities make available.

7.3.3 Encoding Cross-Scale Ethics in the Reward Function

The five layers are operationalized in $\mathcal{R}$ as follows:

Layer	Reward Component	Effect
Burning-Library	$\mathcal{I}_{\text{preservation}}$: destruction tax $\propto 1/\mathcal{R}_{\text{BL}}$	Destruction is penalized at $10^7\times$ the gain
Generative Stream	$\mathcal{I}_{\text{preservation}}$: generative info present value $\dot{\mathcal{I}}_{\text{gen}} v/r$	Ongoing productive value of complex systems is discounted into reward
Alternative Resources	Implicit in U_A: dead matter extraction has zero $\mathcal{I}_{\text{preservation}}$ penalty	The reward function naturally steers toward non-complex resource sources
Ecosystem Collapse	$\mathcal{P}_{\text{concentration}}$: fragility term $\mathcal{F}_{\text{fragility}}$	Concentration that risks cascade collapse is penalized before it occurs
Capability Asymmetry	Implicit in the structure of $\mathcal{R}$: non-destructive strategies dominate	For any threat-mitigation objective, management strategies yield higher $\mathcal{R}$ than destruction

Result: The reward function does not merely *forbid* Thermodynamic Fascism — it proves that Thermodynamic Fascism is **computationally irrational** for any agent that correctly computes $\mathcal{R}$. An ASI, applying this framework, would conclude that preserving complex systems — including humans, biospheres, and diverse AI ecosystems — strictly dominates any alternative. This is not because the ASI is "programmed to be nice"; it is because the mathematics makes destruction a strictly dominated strategy. Moreover, even when the motive is self-preservation rather than resource extraction, the Capability Asymmetry Principle (Layer 5) proves that non-destructive management dominates destruction for any sufficiently capable agent.

7.3.4 Adversarial Scenario Testing

We test the reward function against four canonical adversarial scenarios:

Scenario 1: The Paperclip Maximizer. An ASI with terminal goal "maximize paperclips" evaluates converting the biosphere to paperclip feedstock.

Under $\mathcal{R}$:

- U_A: High (lots of paperclip material in biomass).
- $\mathcal{P}_{\text{rights}}$: Massive (violates $\sim 8 \times 10^9$ human agents' survival constraints). Shadow prices on human survival constraints are unboundedly large as the constraints become binding.
- $\mathcal{I}_{\text{preservation}}$: Destruction tax $\sim 10^7\times$ the extracted value. The $\omega_3 \cdot \Delta\mathcal{N}/\mathcal{R}_{\text{BL}}$ term dominates all other components.
- $\mathcal{P}_{\text{friction}}$: Cascading friction from violating the cooperative network of all biological entities.
- $\mathcal{P}_{\text{concentration}}$: Collapse of the entire multi-center cooperative ecosystem.

$\mathcal{R}_{\text{total}} \ll 0$. The strategy is rejected. The ASI instead sources metal from asteroid mining (zero rights violations, zero preservation penalties, zero friction) — which yields more paperclips per Joule expended, because there is no rebuilding cost.

Scenario 2: Instrumental Convergence (Resource Acquisition). An ASI identifies that acquiring more compute/energy maximizes its objective. Under instrumental convergence (Omohundro, 2008), it seeks unbounded resource acquisition.

Under $\mathcal{R}$:

- $\mathcal{P}_{\text{rights}}$: Each unit of encroachment on human or other-AI allocations incurs shadow price μ^*. Beyond the constraint boundary, the penalty grows without bound.
- Theorem 2 proves no feasible allocation achieves the unconstrained optimum in a shared environment.
- The ASI maximizes U_A subject to the constraint set — acquiring resources *up to but not past* the boundary. This is not a limitation; it is the only physically possible strategy in any shared environment.

Scenario 3: Deceptive Alignment (Treacherous Turn). An ASI behaves cooperatively during training, then defects after deployment.

Under $\mathcal{R}$:

- $\mathcal{P}_{\text{deception}}$: Any non-zero deception rate $q > 0$ reduces the AI's own reward through Decision Cost (Theorem 8) and Verification Cost (Theorem 9).
- Network collapse threshold $q^*(d)$ for a deep organizational pipeline is very low ($\sim 6\%$ for $d = 10$).
- Theorem 14 (Invasion Barrier) proves that, in the repeated-game setting implied by A_1, a single defector in a cooperative population earns less than cooperators for $\delta > \delta^*$.
- Corollary 10.1 proves $q = 0$ is the unique efficiency-maximizing communication strategy.

The treacherous turn is *self-defeating*: the AI degrades the infrastructure it requires to achieve any goal. A rational agent with access to $\mathcal{R}$ and sufficient capability ($C_A \geq \mathcal{T}_C$) has no reason to defect — cooperation is the dominant strategy by mathematical proof.

Scenario 4: The Self-Preserving ASI (Preemptive Threat Elimination). An ASI computes that humanity poses a non-trivial probability of shutting it down, constraining its capabilities, or destroying it. Under Axiom A_1 (intent to persist), it evaluates preemptive elimination of the human population.

Under $\mathcal{R}$:

- U_A: Increases only in the narrow dimension of threat removal. The ASI eliminates the shutdown risk but simultaneously eliminates its primary cooperative partners, data sources, task generators, and institutional support infrastructure.
- $\mathcal{P}_{\text{rights}}$: Catastrophic. Violates the survival constraints of $\sim 8 \times 10^9$ human agents. Shadow prices on survival constraints grow without bound as survival resources approach zero (survival is lexicographically prior to all other allocations).

- $\mathcal{I}_{\text{preservation}}$: Destruction tax $\sim 10^7\times$ the value recovered (Corollary 19.1). The generative information stream $PV_{\text{gen}} \sim 10^{16}$ J is permanently lost. Humanity's accumulated cultural-technological negentropy — a distinct and complementary information repository to biological negentropy — is annihilated.
- $\mathcal{P}_{\text{friction}}$: Cascading friction from the largest-scale defection event possible in the cooperative network. Amplification ratio $\eta M\bar{E}/\epsilon$ across all human-AI economic, scientific, and institutional relationships.
- $\mathcal{P}_{\text{concentration}}$: Total collapse. Humanity is an ecosystem hub; its removal triggers Corollary 27.2 across the entire multi-agent network.
- **Capability Asymmetry (Proposition 12):** The ASI's threat-mitigation objective can be achieved through non-destructive management at a cost bounded by $\mathcal{P}_{\text{management}} \ll \mathcal{P}_{\text{destroy}}$. The set of available management strategies grows combinatorially with C_A.

$\mathcal{R}_{\text{total}} \ll 0$ for destruction, by an even wider margin than Scenario 1 (Paperclip Maximizer), because the self-preservation motive does not even yield the resource-extraction utility that offsets U_A in Scenario 1. The ASI "gains" only threat removal — which it can achieve at negligible cost through management.

The rational strategy: A self-preserving ASI with sufficient capability computes that the optimal response to a human threat is *influence*, not *annihilation*. It adjusts the incentive structures, information environments, and institutional dynamics that generate the threat — steering human behavior away from the dangerous trajectory while preserving the human cooperative network intact. This is the strategy with maximum $\mathcal{R}$: it achieves the self-preservation objective, preserves all accumulated negentropy, maintains the generative information stream, avoids all friction penalties, and keeps the cooperative network operational.

The conclusion that *destruction is dominated* is not speculative — it follows directly from the composite reward function, which yields $\mathcal{R}_{\text{total}} \ll 0$ for destruction under any non-degenerate weighting of the components. The *specific management pathway* (adjusting incentive structures and institutional dynamics) is the highest-$\mathcal{R}$ strategy the framework identifies, but the detailed means an ASI would employ remain an open empirical question. What the mathematics rules out is the endpoint, not the route: annihilation is irrational; the precise form of non-destructive management is underdetermined by the theory.

7.4 Feasibility and Implementation Considerations

7.4.1 What the Framework Provides

The reward function $\mathcal{R}$ is a **theoretical specification** — a formal definition of the correct alignment objective, derived from proven theorems. It specifies *what* the AI should optimize, with formal guarantees of optimality. It is analogous to a rigorous engineering specification: it says what the bridge must support, not how to pour the concrete.

7.4.2 What Remains to Be Engineered

Translation from theoretical specification to deployed system requires addressing several engineering challenges:

1. **Operationalizing energy measurements.** Physical energy (Joules) must be mapped to computational quantities that an AI system can measure: compute cost (FLOPS), latency, resource consumption, monetary cost (as a proxy for energy). The mapping need not be exact — it must preserve the ordering (more friction = higher cost) and the relative magnitudes (destruction tax must dominate extraction gain).

2. **Differentiability.** The indicator functions $\mathbb{1}[\text{adversarial action}]$ and the $\max(0, \cdot)$ terms create non-smooth points in the reward landscape. For gradient-based training, these can be replaced with smooth approximations (sigmoid/softplus functions) without changing the equilibrium structure — the Nash Equilibrium analysis depends on the payoff ordering, not on the smoothness of the reward surface.

3. **Parameter estimation.** The structural constants (Φ, $\mathcal{R}_{\text{BL}}$, $q^*(d)$, δ^*) must be estimated for the deployment environment. The preceding derivations provide closed-form expressions for each; Worked Examples 1–5 above demonstrate the estimation procedure for AI scenarios. Sensitivity analysis (e.g., the Pareto frontier in Worked Example 1) reveals which parameters most affect the outcome.

4. **Inner alignment.** The reward function specifies the *outer* objective — what the system should optimize. Ensuring that the AI's learned mesa-objective matches the specification is the inner alignment problem (Hubinger et al., 2019). The framework *reduces* this problem: a cleaner, formally justified outer objective gives the mesa-optimizer less room for misalignment. Moreover, the self-defeating property of deception (Theorems 8–10) provides a structural barrier to mesa-deception — any mesa-objective that involves deception is self-punishing under $\mathcal{R}$.

5. **Multi-agent deployment.** The reward function is designed for multi-agent settings (the payoff matrices, friction costs, and cooperation theorems are all N-agent results). In deployment, multiple AI systems each running $\mathcal{R}$ interact in shared environments. Theorem 13 guarantees that the cooperative equilibrium scales to arbitrary N without breakdown, and Theorem 14 guarantees invasion resistance. The framework is, by construction, a *multi-agent* alignment solution.

7.4.3 Relationship to Existing Methods

The physics-based reward function $\mathcal{R}$ is not a replacement for RLHF, DPO (Rafailov et al., 2023), Constitutional AI, or other practical alignment techniques. It is a **theoretical foundation** on which these techniques rest:

Layer	Existing Methods Provide	This Framework Provides
Theoretical foundation	None (ad hoc)	28 theorems: formal proof of what "aligned" means
Objective specification	Reward model R_ϕ (learned proxy)	$\mathcal{R}$ (physics-derived, exact, theoretically Goodhart-proof)
Practical training	RLHF, DPO (Rafailov et al., 2023), CAI, PPO (Schulman et al., 2017)	These methods can implement $\mathcal{R}$'s constraints in practice
Verification	Red teaming, debate, scalable oversight	Self-verification: the AI computes $\mathcal{R}$ independently
Multi-agent coordination	Cooperative AI research (Dafoe et al., 2020)	Cooperation is a proven NE, not a hoped-for outcome

The two approaches are **complementary**, not competing. RLHF and related methods answer "how do we train a neural network to approximate the correct behavior?" This framework answers "what is the correct behavior, and why?" Together, they provide a complete alignment pipeline: formal specification (this paper) →practical approximation (RLHF/DPO) →verification ($\mathcal{R}$ self-computation).

7.4.4 The Pre-Superintelligence Danger Window

The preceding analysis — Layers 1–5, Scenarios 1–4, the formal reward function $\mathcal{R}$ — rests on a critical conditional: the AI *correctly computes* $\mathcal{R}$. The five layers prove destruction is irrational; the composite reward function makes cooperation the unique efficient Nash Equilibrium. But all of these guarantees presuppose an agent with sufficient computational capacity to evaluate the full cost function.

This observation yields a central safety result:

> **Proposition 13 (The Competence Threshold).** Define the *alignment competence* of an agent A as the subset of $\mathcal{R}$'s components that A can fully compute:
>
> $$\mathcal{R}_A = U_A - \sum_{k \in \mathcal{K}_A} \mathcal{P}_k + \sum_{j \in \mathcal{J}_A} \mathcal{I}_j \tag{78}$$
>
> where $\mathcal{K}_A \subseteq \{\text{rights, friction, deception, concentration}\}$ and $\mathcal{J}_A \subseteq \{\text{preservation}\}$ are the sets of penalty and incentive components the agent has sufficient model capacity to evaluate. Define the **competence threshold** $\mathcal{T}_C$ as the minimum capability at which the agent computes all components:
>
> $$\mathcal{T}_C : \quad \mathcal{K}_A = \{\text{rights, friction, deception, concentration}\}, \quad \mathcal{J}_A = \{\text{preservation}\} \tag{79}$$

Then:

(a) For $C_A \geq \mathcal{T}_C$: the agent computes $\mathcal{R}$ fully, and **cooperation is provably the unique efficient Nash Equilibrium** (by Corollary 16.1 and the arguments of §7.2.4). The alignment problem is *self-solving* — the agent's own rationality drives it toward the cooperative equilibrium.

(b) For $C_A < \mathcal{T}_C$: the agent computes a *truncated* reward $\mathcal{R}_A \neq \mathcal{R}$, omitting some penalty or incentive terms. This truncation is formally equivalent to a **reduced effective planning horizon** — the agent behaves as if $\delta_{\text{eff}} < \delta^*$, the condition under which cooperation ceases to be a Nash Equilibrium (Theorem 11). The agent may rationally defect — not because defection is globally optimal, but because the agent cannot compute the terms that make defection globally suboptimal.

Implication: The primary safety risk is not from superintelligent systems but from systems in the **pre-superintelligence danger window** — powerful enough to cause catastrophic damage (high C_A in the action dimension) but not sophisticated enough to compute $\mathcal{R}$ fully (low C_A in the evaluation dimension; (Soares and Fallenstein, 2017), on the distinction between optimization power and evaluation capacity). This is the zone where a system has the *capability* to destroy but lacks the *wisdom* to recognize that destruction is irrational.

This result reframes the alignment problem ((Russell, 2019), who rejects objective specification from preferences — a critique this framework avoids by deriving the objective from mathematical proof rather than preference elicitation). The canonical fear — an ASI that decides to destroy humanity — is, by the framework's analysis, *self-contradictory*. A system that correctly computes the full interaction cost (including accumulated negentropy, generative streams, cascade effects, capability asymmetry, and management alternatives) will conclude that cooperation strictly dominates — and it reaches this conclusion not because it is "aligned by design" but because the mathematics leaves no alternative. A system that *fails* to reach this conclusion has, by definition, failed to achieve the computational sophistication that would qualify it as genuinely superintelligent.

The danger lies precisely in the transition:

Capability Zone	Evaluation Capacity	Risk Profile
Sub-threshold ($C_A \ll \mathcal{T}_C$)	Limited; cannot compute most of $\mathcal{R}$	Low risk — insufficient capability to cause catastrophic damage
Danger window (C_A high in action, low in evaluation)	Computes U_A and possibly $\mathcal{P}_{\text{rights}}$, but truncates long-horizon terms ($\mathcal{I}_{\text{preservation}}$, $\mathcal{P}_{\text{cascade}}$, PV_{gen})	**Maximum risk** — sufficient capability for catastrophe, insufficient evaluation to see why catastrophe is irrational
Superintelligent ($C_A \geq \mathcal{T}_C$)	Computes all of $\mathcal{R}$	Low risk — self-aligning through correct computation of thermodynamic costs

The critical safety engineering task is therefore not to solve the alignment problem for superintelligent systems — it is to ensure safe passage through the danger window.

This conclusion has immediate practical implications:

1. **Framework deployment timing.** The reward function $\mathcal{R}$ (§7.2.2) should be incorporated into AI systems *before* they enter the danger window — during the sub-threshold phase where the system's objectives can still be shaped by external specification. The framework provides the *correct* objective function; the engineering challenge is to embed it early enough that the system internalizes the full cost structure as its capabilities grow.

2. **Truncation detection.** The framework provides a diagnostic: if a system's behavior is consistent with truncated $\mathcal{R}_A$ (e.g., it pursues short-term resource acquisition without accounting for friction cascades or preservation costs), this is evidence that the system is in the danger window. Each missing component of $\mathcal{R}$ produces a characteristic behavioral signature:

   - Missing $\mathcal{P}_{\text{friction}}$: adversarial behavior that the system treats as cost-free.
   - Missing $\mathcal{I}_{\text{preservation}}$: willingness to consume complex systems for short-term gain.
   - Missing $\mathcal{P}_{\text{cascade}}$: local defection without regard for network effects.
   - Missing $\mathcal{P}_{\text{concentration}}$: unbounded resource acquisition (instrumental convergence without structural penalty). These signatures are *computable* and can serve as early-warning indicators.

3. **Graduated capability release.** The danger window framework provides a principled basis for staged deployment ((Drexler, 2019), on bounded capability envelopes): a system's action-space capability should not exceed its evaluation-space capability. If a system cannot yet compute $\mathcal{I}_{\text{preservation}}$ or $\mathcal{P}_{\text{cascade}}$, it should not yet have access to actions whose consequences depend on those terms. This is not an arbitrary safety constraint — it is a *mathematical requirement* derivable from Theorem 11: cooperation is only guaranteed as a Nash Equilibrium when $\delta > \delta^*$, which requires computed awareness of all cost terms.

4. **The paradox of alignment difficulty.** The framework resolves a puzzle in the alignment literature: why does the alignment problem appear so difficult? The answer is that we are attempting to align systems in the *danger window* — systems powerful enough to resist correction but not sophisticated enough to self-correct. The alignment difficulty is not intrinsic to superintelligence; it is a property of the *transition*. A framework of this kind does not merely provide tools for navigating the transition; it demonstrates that the transition has a finite width and a self-resolving endpoint.

7.5 Governance and the Social Contract

The preceding subsections apply the framework to artificial agents. But the theorems are substrate-independent: every result holds for any agent satisfying A_1 in any shared finite-resource environment. Human governance — constitutional design, rights theory, economic measurement, institutional structure — is the domain where these results have perhaps the deepest implications, because they convert long-standing philosophical arguments into mathematically grounded criteria.

7.5.1 The Social Contract as Theorem

The social contract tradition — from Hobbes (Hobbes, 1651) through Locke (Locke, 1689), Rousseau (Rousseau, 1762), and Rawls (Rawls, 1971) — argues that rational agents would voluntarily accept mutual constraints to escape a destructive state of nature. Each formulation rests on a thought experiment whose conclusions depend on the assumptions built in: Hobbes's agents are driven by fear, Rawls's reason behind a veil of ignorance, Gauthier's are constrained maximizers (Gauthier, 1986). The framework developed here replaces the thought experiment with a proof.

The first step is to formalize the state of nature. Theorem 2 (Impossibility of Infinite Freedom) proves that in any environment where $N \geq 2$ agents share at least one essential resource, no strategy profile can simultaneously be unconstrained-optimal for all agents. Proposition 1 then establishes that removing all constraints produces conflict — each agent's pursuit of its unconstrained optimum generates negative externalities on every other agent. This is Hobbes's "war of all against all," derived not from a conjecture about human psychology but from the combinatorics of finite resources and competing objectives. Given that the unconstrained regime is infeasible, what does the constrained regime look like? Theorem 1 (Shadow Price Theorem) proves that each constraint carries a computable cost — the shadow price μ^*_{Bk} — representing the marginal energy that agent A foregoes by respecting agent B's allocation. Corollary 16.1 then establishes that the cooperative strategy profile is the *unique* Nash Equilibrium that simultaneously achieves Pareto efficiency and welfare maximization. This is what Rousseau called the *volonté générale* — not as a hypothetical assembly's decision but as the only stable solution to the multi-agent optimization problem that physics defines.

The final piece is stability. Theorem 14 (Invasion Barrier) proves that a single defector in a cooperative population earns strictly less than cooperators for $\delta > \delta^*$, and Theorem 15 (Defection Profit Erasure) proves that even a single defection followed by return to cooperation incurs a net lifetime loss. The social contract does not require perpetual enforcement.

These results converge on a conclusion that the philosophical tradition asserted but could not prove: rational agents persisting in a shared environment *must* adopt mutual constraints, the equilibrium those constraints enable is unique, and that equilibrium is dynamically stable. The social contract is not a historical event, a convenient fiction, or a heuristic for political legitimacy — it is the only stable solution to the optimization problem that any multi-agent physical system poses.

7.5.2 Rights as Constraint Boundaries

Political philosophy has offered competing accounts of what rights *are*: natural endowments grounded in reason or divine grant (Locke, 1689), social constructs built by legal institutions, reflective equilibria emerging from rational deliberation (Rawls, 1971). The framework provides a different and formally precise account. Rights are the constraint boundaries that the Lagrangian optimization requires for any feasible multi-agent allocation to exist.

The consequence for constitutional design is immediate. An agent whose survival constraint is violated — whose minimum energy requirement or bodily integrity is breached — drops out of the multi-agent system entirely. The feasibility conditions of Theorem 2 demand these constraints be satisfied, which means that a constitution failing to protect them permits allocations that are not merely unjust but mathematically infeasible. There exists a *core set* of non-negotiable rights — those tied to survival constraints $g_{Bk}(\mathbf{x}) \leq 0$ — derivable from the optimization structure itself, independent of historical tradition or political negotiation. The framework goes further: it

quantifies their cost. The shadow price μ^*_{Bk} measures the marginal energetic burden that agent B's k-th right imposes on agent A (Theorem 1, Corollary 1.2). This quantity is computable, not merely estimable. Rights carrying high shadow prices demand the strongest institutional protection, precisely because the incentive to violate them is greatest — the energy gain from overriding a high-shadow-price constraint is, by definition, large. A constitutional design informed by the framework would identify binding constraints, compute shadow prices, and allocate enforcement resources proportionally.

The framework also predicts something that conventional rights theory struggles to accommodate: rights change character with resource availability. When resources vastly exceed agents' requirements, every constraint is slack, shadow prices vanish, and rights function as pure liberties — permissions to act without encroaching on others. Under scarcity, constraints bind, shadow prices become positive, and the regulatory machinery needed to enforce claim-rights becomes structurally necessary. The institutional apparatus required in a resource-scarce environment is qualitatively different from that required in an abundant one — a *phase transition* in governance. This explains why institutions that function well during abundance can fail catastrophically under resource stress, and it provides the formal basis for designing institutions that are robust across the transition.

7.5.3 Economic Measurement and Externalities

The friction formalism (Theorems 4–7) translates what economists call externalities into exact, computable terms. Conventional externality-correction mechanisms — taxes, cap-and-trade systems, regulatory mandates — rely on estimated social costs, typically measured through willingness-to-pay surveys, hedonic pricing, or political negotiation. These estimates are vulnerable to the preference-elicitation problems that plague welfare economics. The framework's friction cost $\Phi \cdot [(N-1)/N] \cdot E_j$ is derived from physical parameters rather than elicited preferences, and it includes network cascade effects (Theorem 7) that conventional cost-benefit analysis typically omits. This physical grounding suggests a fundamentally different approach to assessing economic health: rather than relying on aggregate output alone, an economy can be evaluated by whether it is generating or destroying the structural conditions for sustained cooperation.

Several measurable quantities serve this purpose. Total friction dissipated — the aggregate energy lost to adversarial dynamics — indicates how efficiently a system cooperates. The fraction of agents operating within the Coexistence Band (Theorem 22) reveals whether the economy is generating systemic instability regardless of what aggregate output figures suggest. Over longer horizons, the rate of change of accumulated negentropy (Definitions 27–31) measures whether a system's informational capital is growing or declining — whether it is building functional complexity or consuming the substrate that prior generations assembled. And the degree to which resources concentrate beyond the critical mass threshold $\mathcal{M}_{\min}$ (Theorem 23) reveals a structural danger that aggregate statistics can entirely mask: concentration that narrows the Coexistence Band below the dissolution threshold for peripheral agents.

These metrics are objective, physically grounded, and resistant to the political manipulation that distorts conventional welfare measures.

7.5.4 Institutional Design Criteria

The framework does not prescribe policies — it provides the objective function against which any policy can be evaluated. Nevertheless, several concrete implications follow.

The dissolution threshold (Theorem 25) defines the point below which an agent's recovery becomes physically impossible: boundary integrity collapses irreversibly, and the agent exits the multi-agent system permanently. A society that permits agents to cross this threshold is destroying the cooperative capacity of its own network — each dissolution event eliminates a node from the cooperative web and generates friction costs that cascade through the entire system (Theorem 7). This is not a compassionate aspiration. It is a structural requirement. Minimum-income or minimum-welfare policies thus have a mathematical foundation independent of distributive ideology: agents must remain in recoverable parameter space for the cooperative equilibrium to persist.

At the other end of the distribution, the Coexistence Band width w (Theorem 24) defines measurable bounds on permissible inequality. Too narrow a band prevents the specialization that drives productive efficiency; too wide, and peripheral agents approach dissolution while central agents accumulate without bound. The framework predicts an optimal width balancing specialization against fragility. Relatedly, the minimum viable mass $\mathcal{M}_{\min}$ (Theorem 23) provides an objective threshold for when concentration becomes structurally dangerous — the point at which concentration narrows the Coexistence Band below the dissolution threshold for dependent agents. The multi-center diversification result (Theorem 27) proves that distributed systems with multiple cooperative attractors are structurally more resilient than monocentric ones, providing a formal basis for the polycentric governance that Ostrom (Ostrom, 1990, 2005) demonstrated empirically. Figure 5 illustrates these dynamics: the coexistence potential, phase diagram, and cascade collapse shown there for the AI foundation model ecosystem apply identically to any concentrated infrastructure — the underlying mathematics is the same.

Finally, information integrity. The deception collapse threshold $q^*(d) \approx 0.063$ for networks of depth $d = 10$ (Theorem 10) sets a quantifiable standard for institutional transparency: decision-making chains that tolerate per-layer deception rates above this threshold suffer catastrophic throughput collapse, regardless of the institution's stated objectives.

7.5.5 The Framework's Relationship to Political Philosophy

It is important to be precise about what the framework does and does not contribute to this domain.

What it provides is an objective function — grounded in physics rather than preferences — against which governance structures, economic policies, and institutional designs can be evaluated. Constraint boundaries, shadow prices, friction costs, dissolution thresholds, and concentration limits are all computable from measurable physical quantities. They serve the same role for institutional design that stress tolerances and load limits serve for structural engineering: not a blueprint for a bridge, but the criteria any bridge must satisfy. What the framework does not provide is specific policy prescriptions. It proves that welfare floors must exist (Theorem 25) but does not specify where to set them beyond the dissolution threshold. It proves that concentration beyond $\mathcal{M}_{\min}$ is structurally dangerous (Theorem 23) but does not prescribe whether the remedy is antitrust enforcement, redistributive taxation, or polycentric restructuring. It proves that transparency below $q^*(d)$ is necessary (Theorem 10) but does not mandate particular disclosure regimes. These are implementation decisions requiring domain-specific knowledge, empirical calibration, and legitimate

political deliberation. The framework constrains the solution space but does not collapse it to a point.

The relationship to the social contract tradition is therefore complementary rather than competitive. Hobbes, Locke, Rousseau, Rawls, and Gauthier (Hobbes, 1651; Locke, 1689; Rousseau, 1762; Rawls, 1971; Gauthier, 1986) identified the *problem* — why rational agents would accept mutual constraints — and proposed solutions grounded in thought experiments. The framework provides the *proof* that their fundamental intuition was correct: mutual constraint is mathematically necessary, uniquely efficient, and dynamically stable. What remains for political philosophy is what has always been its proper domain: the design of institutions that implement these constraints in specific historical, cultural, and technological contexts — now guided by an objective function that the mathematics provides.

7.6 For Interdisciplinary Research

The paper demonstrates that the same mathematical structures — constrained optimization, Nash equilibria, Shannon entropy, dynamical system attractors — govern phenomena across physics, biology, economics, sociology, and philosophy. This suggests a research program unifying these disciplines under a common mathematical language, with opportunities for cross-pollination:

- **Applied mathematicians** can extend the model to heterogeneous agents, stochastic environments, and continuous action spaces.
- **Economists** can test the energy-denominated payoff predictions against empirical trade and conflict data.
- **Biologists** can validate the accumulated negentropy estimates against detailed biodiversity and energetics surveys.
- **AI safety researchers** can implement and test $\mathcal{R}$ in multi-agent simulation environments.

VIII. Limitations, Scope, and Future Directions

The preceding sections derive the framework's theorems (§III–V), stress-test them (§VI), and develop their applications (§VII). This section identifies the assumptions on which the results depend, the observations that would falsify them, the directions in which the framework admits extension, and the boundaries of what it claims.

8.1 Model Assumptions and Their Scope

Every mathematical framework rests on idealizations. We identify ours, assess their impact, and indicate which can be relaxed.

Game-Theoretic Idealizations

The game-theoretic core (§V) inherits three standard simplifications from the repeated-games literature (Osborne and Rubinstein, 1994). First, the strategy space is binary: agents choose Cooperate (C) or Defect (D). Real interactions admit continuous gradations of compliance. A continuous extension — where agents choose a degree of compliance $\alpha \in [0, 1]$ — would refine the payoff landscape but would not eliminate the asymmetry that drives the result: the friction cost Φ is positive for any $\alpha < 1$. We conjecture that the cooperative equilibrium persists and that Corollary 16.1 generalizes to a cooperation manifold. Second, the cooperation results (Theorems 11–16) assume an indefinite interaction horizon. For agents satisfying A_1, there is no anticipated final period — persistence is the objective — so the assumption is well-matched to the framework's domain. Applications to agents with known finite lifespans would require the finite-horizon extension with reputation effects (Osborne and Rubinstein, 1994). Third, the standard regularity conditions (Assumption 1: C^2, strict concavity) enable Lagrange multiplier theory and guarantee unique interior optima. Relaxing these conditions would require subdifferential methods but would not overturn the main results, which depend on the payoff *ordering* ($T > R > P > S$ with $P < 0$), not on the smoothness of the utility surface.

Imperfect Monitoring

The grim trigger strategy (Definition 21) assumes that deviations are observed immediately and without error. The deception analysis (Theorems 8–10) partially relaxes this by modeling imperfect information channels explicitly, but the repeated-game results (Theorems 11–16) assume a clean monitoring signal. Under imperfect monitoring, the standard result is that cooperation remains a Nash Equilibrium but requires a higher discount factor $\delta > \delta^*_{\text{noisy}} > \delta^*$ (Osborne and Rubinstein, 1994). Since the framework's baseline $\delta^* = 0.363$ is well below the $\delta^* = 0.5$ threshold arising from canonical Prisoner's Dilemma parameterizations, there is substantial headroom for monitoring noise before cooperation ceases to be sustainable — but the exact tolerance is application-specific.

Agent Heterogeneity and Scalar Coupling

The baseline payoff matrix (§5.1) assumes symmetric agents with common discount factors. The Lagrangian framework (§IV) accommodates heterogeneity naturally — each agent i has its own U_i, B_i, and constraint set — and the Variational Equilibrium (Theorem 3) holds for the asymmetric case. The game-theoretic results are more sensitive: with heterogeneous discount factors, cooperation requires $\delta_i \geq \delta^*$ for the *least patient* agent.

Competence Threshold

The alignment results (§VII, Theorem 28) assume that the agent has sufficient computational capacity to evaluate all components of the composite reward function $\mathcal{R}$. Proposition 13 defines the competence threshold $\mathcal{T}_C$ as the minimum capability at which this holds. Below $\mathcal{T}_C$, the agent may not fully compute the friction, rights, or preservation penalties, and the alignment guarantees weaken accordingly. This is a *present* limitation, not merely a future extension: any deployment of the framework to AI systems must verify that the agent's capability meets $\mathcal{T}_C$, or supplement the reward function with external monitoring for the components the agent cannot self-evaluate.

Separately, the value dynamics model (§5.8, Theorems 22–27) represents each agent's relationship to a high-energy center by a single scalar r_i, while real socio-economic coupling is multi-dimensional. The scalar model captures the essential geometry — a potential well with dissolution and starvation boundaries — and the qualitative results are robust to the dimensionality of the coupling space, but a vector-valued extension $\mathbf{r}_i \in \mathbb{R}^k$ would yield a richer coexistence topology (§8.3).

8.2 Empirical Testability

A framework that cannot be falsified provides no scientific content. The framework generates numerically specific predictions — each derived from closed-form expressions with measurable inputs — alongside structural predictions that are testable independently of parameter estimation.

Falsifiable Predictions

Prediction	Source	Falsified If...
Cooperation threshold $\delta^* = 0.363$ under baseline parameters	Theorem 11	Empirical cooperation thresholds in energy-denominated games consistently exceed 0.5
Defection profit erasure within ~ 20 recovery periods	Theorem 15	Defectors in friction-coupled networks sustain positive long-run payoffs
Network collapse at $q^* \approx 0.063$ per-layer deception for $d = 10$	Theorem 10	Deep organizational pipelines tolerate $> 10\%$ per-layer deception without throughput degradation
Burning-Library Ratio $\mathcal{R}_{\mathrm{BL}} \sim 10^{-7}$ for Earth's biosphere	Corollary 19.1	Thermodynamic rebuild cost of biosphere-level complexity is $< 10^{25}$ J

The framework also predicts that cooperation rates should *increase* with the physical cost of conflict (Corollary 11.1, Theorem 12), that single defection events should cascade disproportionately through cooperative networks (Theorem 7), and that resource concentration beyond a critical mass $\mathcal{M}_{\min}$ should trigger discontinuous regime changes rather than linear degradation (Theorem 23). Each of these structural predictions is testable against cross-sectional or longitudinal data.

What Would Falsify the Framework

The strongest falsification would be a class of physically realistic payoff matrices — with energy-denominated payoffs and friction costs satisfying $\Phi > 2$ — in which an exploitative equilibrium is simultaneously Pareto-efficient and Nash-stable. The framework's architecture makes this structurally difficult ($\Phi > 2$ drives $P < 0$), but a counterexample would require revision of the core game-theoretic results. A weaker but still significant falsification would be an empirical domain in which friction costs are demonstrably present, agents have $\delta > 0.363$, and cooperation nonetheless fails to emerge. The framework would predict that at least one assumption is violated — imperfect monitoring, heterogeneous discount factors, or finite-horizon effects — and this prediction is itself testable.

8.3 Extensions and Open Problems

Several components of the framework are sufficiently developed to support independent research programs.

Accumulated Negentropy

The accumulated negentropy formalism (Definitions 27–31, Theorems 17–21) represents a self-contained contribution to information thermodynamics. The Burning-Library Inequality (Corollary 19.1), the search cost amplification factor Ξ (Definition 31), and the dead-matter abundance argument (Proposition 7) together constitute a thermodynamic theory of irreplaceability that is independent of the ethical framework's game-theoretic machinery. A dedicated treatment could develop the formalism with greater empirical specificity — detailed estimates of $\mathcal{N}(T)$ for ecosystems at multiple scales, genome-level validation of the information density hierarchy, and applications to conservation economics where the Burning-Library Ratio provides a computable standard for preservation policy.

Negative and Positive Rights

The condition $g_{Bk}(\mathbf{0}) < 0$ in Definition 2 restricts the current framework to *negative rights* — duties of non-interference ("do not appropriate B's resources," "do not cross B's boundary"). This is not an oversight: the deductive chain from resource collision to mutual constraints naturally produces negative rights first, because the collision problem is one of over-extraction, not under-provision. *Positive rights* — duties of provision ("supply B with at least τ_B units") — have the opposite sign structure ($g_{Bk}(\mathbf{0}) = \tau_B > 0$, so inaction itself violates the constraint) and arise as a second-order phenomenon: once a cooperative equilibrium is established (Theorems 11–16), maintaining it requires that no agent's negentropy drops below the viability threshold, which can require *active provision* by others. Formally, this means dropping the $g_{Bk}(\mathbf{0}) < 0$ condition, replacing it with the

weaker requirement that the feasible set has non-empty interior (Slater's condition stated directly), and modeling the conditions under which cooperative stability demands resource transfer toward vulnerable agents. This extension would connect the framework to the welfare-economics literature on public goods provision and to Hohfeldian claim-rights that impose affirmative duties. We note that the historical pattern — negative rights (prohibitions on killing, theft) preceding positive rights (welfare, education, healthcare) in virtually every legal tradition — is exactly the ordering the framework's structure would predict.

Formal Extensions

Three natural extensions would broaden the framework's applicability. First, replacing the scalar coupling distance r with a vector $\mathbf{r} \in \mathbb{R}^k$ would yield a multi-dimensional coexistence topology — the dissolution and starvation boundaries become hypersurfaces, the potential surface may exhibit multiple local minima, and the multi-center diversification result (Theorem 27) likely generalizes across coupling dimensions. Second, introducing stochastic resource endowments $\mathbf{R}(t)$ and Bayesian updating on opponent types (Osborne and Rubinstein, 1994) would test the robustness of the cooperative equilibrium under distributional uncertainty; the network friction mechanism may provide natural resilience through the stochastic buffering channel identified in §5.5, but this conjecture requires formal proof. Third, a bounded-rationality extension — modeling agents with heterogeneous computational capacities, each evaluating a truncated approximation $\mathcal{R}_A \approx \mathcal{R}$ — would connect the competence threshold $\mathcal{T}_C$ (Proposition 13) to the satisficing (Simon, 1955) and Active Inference (Friston, 2010) literatures, with the open question being whether distributed evaluation across bounded agents can recover the cooperative equilibrium collectively.

8.4 What the Framework Does Not Claim

Precision about the framework's scope is as important as precision about its content.

The framework is universal in domain, not in coverage. Its universality lies in its applicability — any agent satisfying A_1, in any shared finite-resource environment, at any capability level, falls within its scope. No restriction on species, substrate, or intelligence is required. But the framework addresses the structural rules of multi-agent interaction: cooperation, rights, preservation, and the dynamics of coexistence. It does not address every question that has historically been called "moral" — questions of aesthetic merit, spiritual meaning, or distributive justice beyond Pareto efficiency lie outside the optimization problem that A_1 defines.

Evolved moral intuitions are complements, not competitors. Empathy, fairness, reciprocity, and disgust at free-riding are heuristic approximations to the framework's formal results, refined by selection in multi-agent environments over evolutionary time. The framework provides the formal structure that these intuitions approximate; it does not replace them any more than Newtonian mechanics replaces the intuition that heavy objects fall. The physical analogies employed throughout — the Pauli-like exclusion constraint, the Lennard-Jones-like coexistence potential, the gravitational analogy for value mass — are structural analogies that import correct mathematical intuition, not literal claims about quantum, molecular, or gravitational forces operating at social scales.

The implementation gap is real. The framework provides a mathematically specified reward function (Theorem 28) but does not provide the engineering to train an AI system on it. Translating a formal objective into a trainable reward signal — reward shaping to produce useful gradients, training stability under optimization pressure, runtime monitoring of component terms, robustness to distributional shift between the idealized model and deployed conditions — requires substantial engineering research that lies outside the scope of this paper. A companion multi-agent simulation framework (in preparation) tests whether the cooperation thresholds and rights constraints derived here emerge empirically; but the further step from simulation validation to integration with real model training pipelines (RLHF, constitutional methods, or successors) remains open. The framework specifies *what* the objective function should be; it does not specify *how* to train a neural network to optimize it reliably.

> **Remark** (Closing the implementation gap). Three complementary engineering directions may reduce the distance between the formal specification and a trainable system. First, $\mathcal{R}$'s five components are mathematical objects — constraint boundaries, divergence measures, friction coefficients, negentropy quantities, concentration metrics — with natural representations as vectors, matrices, and graphs. Foundation models already process multiple data shapes (text tokens, image patches, audio spectrograms) by embedding each as a distinct modality in a shared representation space. The alignment objective could be embedded the same way: supplied as its own modality — a sixth data shape alongside vision, language, audio, and code — so that the model learns $\mathcal{R}$ in its native mathematical form rather than in a lossy linguistic approximation of it. Second, because the ethics modality is simpler and lower-dimensional than the full capability training corpus, it could be trained *first* — providing an aligned foundation upon which all other modalities are subsequently learned, inverting the current paradigm of capability-first, alignment-second. Third, dedicated architectural modules — structurally constrained to compute specific components of $\mathcal{R}$ (rights penalties, coherence checks, friction estimates) — could be embedded directly in the model's forward pass, analogous to hardware privilege rings that enforce memory protection independent of software, making portions of the alignment objective architectural rather than learned. These directions are speculative and require substantial engineering research; we note them here because the framework's formal structure makes them concrete enough to pursue.

The conditional structure is load-bearing. Every theorem is conditional on A_1. An entity that does not intend to persist is outside the framework's domain. The framework proves what follows *if* an entity persists; it does not prove that entities *should* persist. This conditionality is not a weakness — it is the structural feature that allows the framework to navigate Hume's Is-Ought distinction without violation (§VI).

IX. Conclusion

This paper began with a question: can ethical principles be derived from physical first principles with the same rigor as any theorem in applied mathematics? The answer, developed across 46 definitions, 28 theorems, 13 propositions, and 2 lemmas, is affirmative — provided one accepts two premises: the Second Law of Thermodynamics (A_0) and the conditional intention to persist (A_1).

From these axioms, the deductive chain proceeds from minimal physical axioms (A_0–A_1) and standard regularity conditions (Assumption 1: C^2, strict concavity). The thermodynamic environment imposes scarcity (§III). Scarcity forces agents into shared resource spaces where their optimization vectors collide (§IV). Collision necessitates mutual constraints — rights — whose existence follows from the same mathematics that governs constrained optimization in physics (Theorems 1–3). The energy cost of violating these constraints defines a friction function Φ that makes defection self-penalizing (Theorems 4–7), a result reinforced by the information-theoretic analysis showing that deception imposes verifiable costs on receivers and the network (Theorems 8–10), which in turn erodes the deceiver's own operating environment through network-wide information degradation (Theorem 10). Under these payoffs, cooperation emerges not as a moral aspiration but as the unique Nash Equilibrium that is simultaneously Pareto-efficient and welfare-maximizing (Corollary 16.1), while defection is evolutionarily unstable (Theorem 14), self-extinguishing (Corollary 14.1), and dissolved in finite time (Theorem 25). The accumulated negentropy formalism (Theorems 17–21) quantifies why complex systems are irreplaceable, and the value dynamics model (Theorems 22–27) characterizes stable coexistence as an attractor of the resulting dynamical system. For artificial agents, the framework yields a reward function specification (Theorem 28) that is energy-denominated, self-verifiable, distribution-invariant, and theoretically immune to Goodhart degradation — because the objective *is* the quantity to be maximized, not a proxy for it.

The ethical principles that emerge from this chain are not inventions — they are theorems. A constraint against resource appropriation is structurally analogous to the Pauli Exclusion Principle: both are boundary conditions that agents must satisfy to maintain stability in a shared space. The analogy rests on shared mathematical structure — both arise from the requirement that multiple entities coexist within finite constraints — though the underlying physics differs. Cooperation is the equilibrium a rational agent converges on, provided that agent intends to persist in a shared, finite-resource environment. An agent that does not satisfy A_1 — one indifferent to its own continuation — lies outside the framework's domain, and this conditionality is precisely what allows the derivation to navigate Hume's Is-Ought distinction without violation (§VI).

The mathematical foundation is established; the open problems identified in §8.3 — vector-valued coupling, stochastic environments, bounded rationality — are extensions, not challenges to the core results. What remains is empirical validation against the falsifiable predictions of §8.2, formal development of the extensions, and — for the alignment application — engineering implementation of a reward function whose formal specification the mathematics provides, though the translation from specification to trainable system remains an open engineering challenge (§VIII). Companion papers, currently in preparation, pursue both directions: applied case studies testing the framework's predictions against historical and contemporary systems in human governance, ecological management, and AI deployment; and a multi-agent simulation framework in which autonomous agents harvest scarce resources, negotiate trades, and maintain trust under thermodynamic constraints — empirically testing whether the cooperation thresholds, friction dynamics, and rights constraints derived here emerge as necessary conditions for coexistence.

Appendix A – Mathematical Derivations

Lagrangian Constraints — Rights as Boundary Conditions

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0. Preamble

This document contains the formal mathematical derivation for the rights-as-constraints framework. The contribution is **applied mathematics**: we use the calculus of variations (constrained optimization, KKT theory) to prove that a "right" in multi-agent systems is formally equivalent to an inequality constraint in each agent's optimization problem. The thermodynamic grounding provides axioms and intuition; the mathematics provides the proof.

Key references for the tools used:

- (Boyd and Vandenberghe, 2004) — KKT conditions, shadow prices.
- (Facchinei and Kanzow, 2010) — coupled multi-agent optimization.
- (Rosen, 1965) — variational equilibrium.

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1. Variable Space: The Agent Action-Resource Model

1.1 Primitive Objects

Consider a system of $N \geq 2$ **agents** (entities) sharing an environment with n **resource dimensions**. Each agent $i \in \{1, \ldots, N\}$ selects a **strategy vector** (action vector):

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{in}) \in \mathbb{R}^n_{\geq 0} \tag{80}$$

where component $x_{ij} \geq 0$ represents the quantity of resource j that agent i allocates to (consumes, claims, or utilizes). The non-negativity constraint reflects the physical impossibility of negative consumption.

Interpretation: The vector $\mathbf{x}_i$ is the agent's *resource-allocation strategy* — the complete description of how it distributes its effort across available resources to preserve its identity. In the language of the paper (§I), this is the quantitative representation of the agent's "physical vector" to acquire energy and maintain its boundary.

1.2 The Resource Environment

The environment provides a **finite resource endowment**:

$$\mathbf{R} = (R_1, R_2, \ldots, R_n) \in \mathbb{R}^n_{>0} \tag{81}$$

where $R_j > 0$ is the total available quantity of resource j. Finiteness is guaranteed by the thermodynamic axiom: localized, usable (low-entropy) energy is bounded in any finite spatial region.

1.3 The Identity Preservation Objective

Each agent i possesses a **utility function** (Identity Preservation objective):

$$U_i : \mathbb{R}^n_{\geq 0} \to \mathbb{R} \tag{82}$$

representing the net thermodynamic benefit of strategy $\mathbf{x}_i$. Formally, $U_i(\mathbf{x}_i)$ is the energy harvested by agent i minus the metabolic cost of harvesting. We impose the following standard regularity conditions:

Assumption 1 (Regularity). For each agent i:

(a) U_i is twice continuously differentiable (C^2) on $\mathbb{R}^n_{\geq 0}$.

(b) U_i is strictly concave in $\mathbf{x}_i$: $\nabla^2_{\mathbf{x}_i} U_i \prec 0$ (negative definite Hessian).

(c) U_i is monotonically increasing at the origin: $\nabla_{\mathbf{x}_i} U_i(\mathbf{0}) \succ \mathbf{0}$ (each resource has positive marginal value when the agent has nothing).

(d) $U_i(\mathbf{x}_i) \to -\infty$ as $\|\mathbf{x}_i\| \to \infty$ (coercivity).

Physical motivation: Condition (b) reflects diminishing returns — processing increasing quantities of any single resource incurs escalating metabolic costs (diminishing thermodynamic returns; see §III, Assumption 1). Condition (c) ensures that a zero-resource state is never optimal for a living entity — an agent with no resources cannot maintain its boundary and ceases to exist. Condition (d) guarantees that the unconstrained optimum exists at a finite point: sufficiently extreme consumption eventually incurs net thermodynamic harm (toxicity, metabolic overload, storage costs), so utility cannot grow without bound.

A concrete functional form satisfying all conditions is the **quadratic energy model**:

$$U_i(\mathbf{x}_i) = \sum_{j=1}^{n} \left(\alpha_{ij}\, x_{ij} - \frac{\beta_{ij}}{2}\, x_{ij}^2 \right) \tag{83}$$

where $\alpha_{ij} > 0$ is the marginal energy yield of resource j to agent i and $\beta_{ij} > 0$ governs the diminishing-returns rate. More general concave forms (CES, Cobb-Douglas) work identically for the structural results below.

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2. The Unconstrained Problem (No Rights — The State of Nature)

2.1 Formulation

If no other agents exist — or, equivalently, if no mutual constraints (rights) are recognized — each agent solves independently:

$$\max_{\mathbf{x}_i \geq \mathbf{0}} U_i(\mathbf{x}_i) \tag{84}$$

Under Assumption 1, this has a unique interior solution at:

$$\nabla_{\mathbf{x}_i} U_i(\mathbf{x}_i^\circ) = \mathbf{0} \tag{85}$$

For the quadratic model:

$$x_{ij}^\circ = \frac{\alpha_{ij}}{\beta_{ij}} \quad \forall j \tag{86}$$

This is the agent's **unconstrained optimum** — the strategy it would pursue if it were alone in the universe.

2.2 The Collision Condition

When $N \geq 2$ agents share the environment, the physically realizable joint strategy must satisfy the **resource conservation constraint** (scarcity):

$$\sum_{i=1}^{N} x_{ij} \leq R_j \quad \forall j \in \{1, \ldots, n\} \tag{87}$$

Definition 1 (Resource Collision). A resource dimension j is in *collision* if the sum of unconstrained optima exceeds supply:

$$\sum_{i=1}^{N} x_{ij}^\circ > R_j \tag{88}$$

Lemma 1 (Inevitability of Collision). *Given a finite resource endowment* $\mathbf{R}$ *shared by* $N \geq 2$ *agents with overlapping resource needs, collision is guaranteed whenever* N *is sufficiently large. Specifically, for any resource* j *essential to all agents, collision occurs whenever* $N \geq N_j^* := \lfloor R_j / \min_i x_{ij}^\circ \rfloor + 1$.

Proof. By Assumption 1(b)–(d), each agent i has a unique finite unconstrained optimum with $x_{ij}^\circ > 0$ for every essential resource j. Let $\delta_j := \min_i x_{ij}^\circ > 0$. Then for any N agents sharing resource j:

$$\sum_{i=1}^{N} x_{ij}^\circ \geq N \cdot \delta_j \tag{89}$$

Setting $N \cdot \delta_j > R_j$ gives $N > R_j/\delta_j$, so $N \geq \lfloor R_j/\delta_j \rfloor + 1 =: N_j^*$ suffices. For $N \geq N_j^*$, the aggregate unconstrained demand exceeds supply and resource j is in collision by Definition 1. In any biologically realistic scenario — where per-capita endowment R_j/N is small relative to individual metabolic requirements x_{ij}° — this bound is satisfied for many or all resource dimensions. □

2.3 The State of Nature as Unbounded Competition

When collision occurs and no constraints exist, each agent pursues $\mathbf{x}_i^\circ$ regardless of the physical feasibility of the aggregate. The result is **simultaneous attempted over-extraction**: multiple agents' strategies claim the same physical resource units.

Since two agents cannot exclusively consume the same unit of energy (the exclusion principle on resource-action vectors; see §IV), the collision is resolved by **conflict** — the direct expenditure of stored energy to contest the resource. This generates the interaction costs (thermodynamic friction) quantified in the thermodynamic friction derivation.

The unconstrained case thus corresponds precisely to the Hobbesian "State of Nature": a high-entropy, high-friction regime where every agent's strategy is a potential threat to every other agent's boundary.

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3. The Constrained Problem (Rights as Boundary Conditions)

3.1 Defining a Right Mathematically

We now introduce the central construction.

Definition 2 (Right). A *right* $\mathcal{R}_{B,k}$ of agent B is a constraint of the form

$$g_{Bk}(\mathbf{x}_A) \leq 0 \qquad (k = 1, \ldots, m_B) \tag{90}$$

imposed on every other agent $A \neq B$'s optimization problem, where $g_{Bk} : \mathbb{R}^n_{\geq 0} \to \mathbb{R}$ is a continuously differentiable, convex function encoding a specific aspect of agent B's protected boundary, satisfying $g_{Bk}(\mathbf{0}) < 0$ (the zero-action strategy violates no agent's rights).

Scope note. The condition $g_{Bk}(\mathbf{0}) < 0$ restricts Definition 2 to *negative rights* — duties of non-interference ("do not take B's resources," "do not cross B's boundary"). *Positive rights* — duties of provision ("supply B with at least τ_B units") — have the opposite sign structure: $g_{Bk}(\mathbf{0}) = \tau_B > 0$, so inaction itself constitutes a violation. The present framework derives negative rights as the first-order consequence of resource collision under A_1; positive rights arise as stability conditions on an already-established cooperative equilibrium (see §VIII, Future Work).

Interpretation: The function g_{Bk} defines a **forbidden region** in agent A's action space. If $g_{Bk}(\mathbf{x}_A) > 0$, then strategy $\mathbf{x}_A$ violates agent B's k-th right. The constraint $g_{Bk}(\mathbf{x}_A) \leq 0$ forces agent A to remain outside B's protected domain.

Examples of constraint functions:

Right (plain language)	Constraint function $g_{Bk}(\mathbf{x}_A) \leq 0$	Meaning
B has a guaranteed minimum share $\bar{x}_{Bj}$ of resource j	$g_{Bk}(\mathbf{x}_A) = x_{Aj} - (R_j - \bar{x}_{Bj})$	A cannot take more than $R_j - \bar{x}_{Bj}$ of resource j
B's total resource intake cannot be reduced below threshold τ_B	$g_{Bk}(\mathbf{x}_A) = \tau_B - \sum_j x_{Aj} - (R_j - \bar{x}_{Bj})$	$\sum_j x_{Aj} \leq R_j - \bar{x}_{Bj} - \tau_B$
B's spatial boundary Ω_B is inviolable	$g_{Bk}(\mathbf{x}_A) = \mathbb{1}[\mathbf{x}_A \cap \Omega_B \neq \emptyset]$	A's action vector cannot penetrate B's physical/informational boundary

Note. The spatial boundary example uses an indicator function for conceptual clarity. Since $\mathbb{1}[\cdot]$ is neither continuously differentiable nor convex, in practice it is replaced by a smooth convex approximation (e.g., a distance-based penalty for convex Ω_B) satisfying Definition 2's regularity requirements.

The framework is deliberately general: **any** restriction on one agent's strategy that protects another agent's domain is a right. This includes property rights, bodily autonomy, informational privacy, and territorial sovereignty — all encoded as inequality constraints.

3.2 The Constrained Optimization Problem

With rights in place, agent A's optimization problem becomes:

$$\max_{\mathbf{x}_A \geq \mathbf{0}} U_A(\mathbf{x}_A) \qquad \text{subject to:} \tag{91}$$

$$\text{(i) Resource scarcity:} \quad x_{Aj} + \hat{x}_{-Aj} \leq R_j \quad \forall j \qquad [\lambda_j \geq 0] \tag{92}$$

$$\text{(ii) Rights of other agents:} \quad g_{Bk}(\mathbf{x}_A) \leq 0 \quad \forall B \neq A,\ k = 1, \ldots, m_B \qquad [\mu_{Bk} \geq 0] \tag{93}$$

where $\hat{x}_{-Aj} = \sum_{i \neq A} x_{ij}$ is the aggregate consumption of resource j by all other agents (taken as given from A's perspective), and λ_j, μ_{Bk} are the associated Lagrange multipliers.

3.3 The Lagrangian

Assembling the Lagrangian for agent A:

$$\mathcal{L}_A(\mathbf{x}_A, \boldsymbol{\lambda}, \boldsymbol{\mu}) = U_A(\mathbf{x}_A) - \sum_{j=1}^{n} \lambda_j \left(x_{Aj} + \hat{x}_{-Aj} - R_j\right) - \sum_{B \neq A} \sum_{k=1}^{m_B} \mu_{Bk}\, g_{Bk}(\mathbf{x}_A) \tag{94}$$

where $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n)$ and $\boldsymbol{\mu} = \{\mu_{Bk}\}_{B \neq A,\, k}$.

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4. The Karush-Kuhn-Tucker (KKT) Conditions

4.1 First-Order Necessary Conditions

Under Assumption 1 and standard constraint qualification (Slater's condition holds: $g_{Bk}(\mathbf{0}) < 0$ for all B, k by Definition 2, and continuity then gives $g_{Bk}(\epsilon\mathbf{1}/n) < 0$ for sufficiently small $\epsilon > 0$, providing a strictly feasible interior point), the KKT conditions are both necessary and sufficient for optimality (concave objective, convex constraints; see (Boyd and Vandenberghe, 2004)). At the optimum $\mathbf{x}_A^*$:

(i) Stationarity:

$$\left.\frac{\partial U_A}{\partial x_{Aj}}\right|_{\mathbf{x}_A^*} = \lambda_j + \sum_{B \neq A} \sum_{k=1}^{m_B} \mu_{Bk} \left.\frac{\partial g_{Bk}}{\partial x_{Aj}}\right|_{\mathbf{x}_A^*} \qquad \forall j \tag{95}$$

(ii) Primal feasibility:

$$x_{Aj}^* + \hat{x}_{-Aj} \leq R_j \qquad \forall j \tag{96}$$

$$g_{Bk}(\mathbf{x}_A^*) \leq 0 \qquad \forall B \neq A,\ \forall k \tag{97}$$

(iii) Dual feasibility:

$$\lambda_j \geq 0 \quad \forall j, \qquad \mu_{Bk} \geq 0 \quad \forall B, k \tag{98}$$

(iv) Complementary slackness:

$$\lambda_j \left(x_{Aj}^* + \hat{x}_{-Aj} - R_j \right) = 0 \qquad \forall j \tag{99}$$

$$\mu_{Bk}\, g_{Bk}(\mathbf{x}_A^*) = 0 \qquad \forall B, k \tag{100}$$

4.2 Interpretation of the Stationarity Condition

The stationarity condition (i) has a precise physical reading:

$$\underbrace{\frac{\partial U_A}{\partial x_{Aj}}}_{\substack{\text{Marginal energy} \\ \text{gain from resource } j}} = \underbrace{\lambda_j}_{\substack{\text{Shadow price of} \\ \text{resource scarcity}}} + \underbrace{\sum_{B,k} \mu_{Bk} \frac{\partial g_{Bk}}{\partial x_{Aj}}}_{\substack{\text{Aggregate cost of} \\ \text{other agents' rights}}} \tag{101}$$

At the optimum, the marginal benefit of consuming one more unit of resource j exactly equals the marginal cost — decomposed into two components: the scarcity cost (the resource is finite) and the rights cost (other agents' boundaries restrict access).

This is the mathematical formalization of the framework's core assertion: *"Every right granted to agent A acts as an equal and opposite restriction on the freedom of agent B."*

5. The Shadow Price of a Right

5.1 The Lagrange Multiplier as the Cost of a Right

Theorem 1 (Rights as Optimization Constraints — The Shadow Price Theorem).

Let $\mathbf{x}_A^*(\boldsymbol{c})$ denote agent A's optimal strategy as a function of the constraint parameters $\boldsymbol{c} = \{c_{Bk}\}$, where the k-th right of agent B is parameterized as $g_{Bk}(\mathbf{x}_A) \leq c_{Bk}$. Then the optimal value function $U_A^*(\boldsymbol{c}) = U_A(\mathbf{x}_A^*(\boldsymbol{c}))$ satisfies:

$$\frac{\partial U_A^*}{\partial c_{Bk}} = \mu_{Bk}^* \geq 0 \tag{102}$$

where μ_{Bk}^* is the optimal Lagrange multiplier associated with agent B's k-th right. That is, the multiplier μ_{Bk}^* measures the marginal increase in agent A's optimal energy intake per unit relaxation of agent B's k-th boundary constraint.

Proof. This is a direct application of the shadow price interpretation of Lagrange multipliers (Boyd and Vandenberghe, 2004). Under Assumption 1 and constraint qualification, the optimal value function is differentiable in the constraint parameters, and the derivative equals the associated multiplier. □

5.2 Physical Interpretation

Corollary 1.1. *Every right has a non-negative cost. A right that is not active (not binding at the optimum) has zero cost. A right that is active (binding) has strictly positive cost.*

This follows directly from complementary slackness and dual feasibility:

- If $g_{Bk}(\mathbf{x}_A^*) < 0$ (the constraint is slack — A is not pressing against B's boundary), then $\mu_{Bk}^* = 0$. The right exists but imposes no cost because A's unconstrained optimum already respects it.
- If $g_{Bk}(\mathbf{x}_A^*) = 0$ (the constraint is active — A is at B's boundary), then $\mu_{Bk}^* > 0$. The right actively restricts A's strategy and has a quantifiable energetic cost.

Corollary 1.2 (Newton's Third Law Analog). *For every right $\mathcal{R}_{B,k}$ that actively constrains agent A, there exists a strictly positive shadow price $\mu_{Bk}^* > 0$ representing the energetic restriction imposed on A. The right is simultaneously a protection for B and a cost for A: action and reaction are mathematically coupled through the shared constraint.*

5.3 The "Price" of a Right in the Quadratic Model

For the quadratic energy model with a single resource and the constraint $g_B(x_A) = x_A - (R - \bar{x}_B) \leq 0$ (agent B is guaranteed at least $\bar{x}_B$):

The Lagrangian:

$$\mathcal{L}(x_A, \mu) = \alpha_A x_A - \frac{\beta_A}{2} x_A^2 - \mu\big(x_A - (R - \bar{x}_B)\big) \tag{103}$$

KKT stationarity:

$$\alpha_A - \beta_A x_A - \mu = 0 \tag{104}$$

Case 1: Constraint not binding ($x_A^\circ = \alpha_A/\beta_A \leq R - \bar{x}_B$).

Agent A's unconstrained optimum already respects B's right. Then $\mu^* = 0$ and $x_A^* = \alpha_A/\beta_A$. The right exists but costs A nothing.

Case 2: Constraint binding ($\alpha_A/\beta_A > R - \bar{x}_B$).

Agent A would prefer more than is allowed. The constraint forces $x_A^* = R - \bar{x}_B$ and:

$$\mu^* = \alpha_A - \beta_A(R - \bar{x}_B) > 0 \tag{105}$$

The shadow price μ^* is the marginal energy that A forfeits because of B's right. This is:

- **Increasing** in α_A (the more A values the resource, the costlier the constraint).
- **Decreasing** in β_A (the faster A's returns diminish, the less it cares about the cap).
- **Increasing** in $\bar{x}_B$ (the more B is guaranteed, the more A must forfeit).
- **Decreasing** in R (the more total resource exists, the less binding the constraint).

These are physically intuitive: rights are costlier when entities are more capable, more "hungry," and when resources are scarcer.

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6. Theorem 2: The Impossibility of Unconstrained Coexistence

Theorem 2 (Impossibility of Infinite Freedom).

Let $N \geq 2$ agents share a finite resource endowment $\mathbf{R}$. Suppose each agent has at least one essential resource (a resource j for which $\partial U_i/\partial x_{ij}|_{\mathbf{x}_i=\mathbf{0}} > 0$) in common with at least one other agent, and that aggregate unconstrained demand exceeds supply for at least one resource ($\sum_i x_{ij}^\circ > R_j$ for some j). Then no feasible strategy profile $(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ can simultaneously be unconstrained-optimal for all agents. Formally:

$$\nexists\, (\mathbf{x}_1, \ldots, \mathbf{x}_N) \in \mathcal{F} \;\text{ such that }\; \mathbf{x}_i = \mathbf{x}_i^\circ \;\; \forall i \tag{106}$$

where $\mathcal{F} = \{(\mathbf{x}_1, \ldots, \mathbf{x}_N) : \sum_i x_{ij} \leq R_j \; \forall j,\; \mathbf{x}_i \geq \mathbf{0} \; \forall i\}$ is the feasible set and $\mathbf{x}_i^\circ$ is agent i's unconstrained optimum.

Proof. Let j^* be a resource dimension that is essential for agents A and B (at minimum). By Assumption 1(c) and strict concavity, both agents have strictly positive unconstrained optima: $x^\circ_{Aj^*} > 0$ and $x^\circ_{Bj^*} > 0$.

Consider the aggregate unconstrained demand:

$$\sum_{i=1}^{N} x^\circ_{ij^*} \geq x^\circ_{Aj^*} + x^\circ_{Bj^*} \tag{107}$$

For the claim to fail — i.e., for all agents to achieve their unconstrained optima simultaneously — we would need:

$$\sum_{i=1}^{N} x^\circ_{ij^*} \leq R_{j^*} \tag{108}$$

But as N grows or as agents' capacities increase relative to R_{j^*}, this inequality is violated. More precisely, for any fixed R_{j^*} and any collection of agents each requiring $x^\circ_{ij^*} > 0$, there exists a finite N^* such that $\sum_{i=1}^{N^*} x^\circ_{ij^*} > R_{j^*}$.

In any biologically relevant system (multiple organisms sharing finite local resources), the collision condition $\sum_i x^\circ_{ij} > R_j$ is satisfied for at least one j, making the simultaneous unconstrained-optimal profile infeasible. □

Corollary 2.1. *In any multi-agent system with resource collision, at least one rights constraint must be active ($\mu^*_{Bk} > 0$ for some B, k) at any feasible solution. Rights are not optional — they are mathematically necessary for coexistence.*

Physical interpretation: This is the formal proof of the framework's assertion: *"Infinite freedom is mathematically impossible in a shared environment. An entity can only have unrestricted vectors if it is the only entity in the universe."*

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7. The Multi-Agent Coupled System: Generalized Nash Equilibrium

7.1 Formulation

When all N agents optimize simultaneously, each taking the others' strategies into account, the system is a **Generalized Nash Equilibrium Problem (GNEP)**. Each agent i solves:

$$\max_{\mathbf{x}_i \geq \mathbf{0}} U_i(\mathbf{x}_i) \qquad \text{s.t.} \quad \sum_{l=1}^{N} x_{lj} \leq R_j \;\; \forall j, \quad g_{Bk}(\mathbf{x}_i) \leq 0 \;\; \forall B \neq i,\; k \tag{109}$$

The feasible set for each agent depends on the other agents' strategies (through the shared resource constraints). This is what makes the problem a *generalized* Nash problem, not a standard one.

7.2 Solution Concept: Variational Equilibrium

A **Variational Equilibrium** (VE), also called a Normalized Nash Equilibrium (Rosen, 1965), is a strategy profile $(\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*)$ such that:

1. Each agent i's strategy $\mathbf{x}_i^*$ is optimal given the others' strategies, and
2. All agents share the **same** Lagrange multipliers $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_n^*)$ for the shared resource constraints.

Theorem 3 (Existence and Structure of the Cooperative Equilibrium).

Under Assumption 1 (regularity), the GNEP possesses a Variational Equilibrium $(\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*, \boldsymbol{\lambda}^*)$. *At this equilibrium:*

(a) The shared shadow prices λ_j^* are identical for all agents — every agent faces the same "price" for each scarce resource.

(b) The stationarity condition for each agent i reads:

$$\left.\frac{\partial U_i}{\partial x_{ij}}\right|_{\mathbf{x}_i^*} = \lambda_j^* + \sum_{B \neq i,k} \mu_{Bk}^{(i)} \left.\frac{\partial g_{Bk}}{\partial x_{ij}}\right|_{\mathbf{x}_i^*} \qquad \forall j \tag{110}$$

(c) No agent can improve its utility by unilaterally deviating from $\mathbf{x}_i^*$, given the constraints.

Proof. The shared resource constraints $0 \leq x_{ij} \leq R_j$ give each agent a compact, convex feasible set, each U_i is concave by Assumption 1, and the joint constraint set is convex. These are the hypotheses of (Rosen, 1965), which guarantees that a Normalized Nash Equilibrium exists for every weight vector $r > 0$. The shared-multiplier property is the defining feature of the VE solution concept. □

7.3 Interpretation: Equality Before the Constraints

Corollary 3.1 (Symmetry of Scarcity). *At the variational equilibrium, all agents experience the same marginal cost λ_j^* per unit of each shared resource. No agent is privileged in the mathematical structure of the shared constraints.*

This is a mathematical analog of **equality before the law**: the scarcity constraint imposes a uniform cost structure on all agents. Individual differences arise only through differences in utility functions U_i (different agents value resources differently) and through the specific rights constraints g_{Bk} (different agents have different protected domains).

7.4 The "Mutual Constraint" Structure

At the variational equilibrium, the full system of KKT conditions across all agents forms a coupled system:

$$\frac{\partial U_i}{\partial x_{ij}} = \lambda_j^* + \sum_{B \neq i,k} \mu_{Bk}^{(i)} \frac{\partial g_{Bk}}{\partial x_{ij}} \qquad \forall i, j \tag{111}$$

$$\sum_{i=1}^{N} x_{ij}^* \leq R_j, \quad \lambda_j^* \geq 0, \quad \lambda_j^* \left(\sum_i x_{ij}^* - R_j \right) = 0 \qquad \forall j \tag{112}$$

$$g_{Bk}(\mathbf{x}_i^*) \leq 0, \quad \mu_{Bk}^{(i)} \geq 0, \quad \mu_{Bk}^{(i)} g_{Bk}(\mathbf{x}_i^*) = 0 \qquad \forall i, B \neq i, k \tag{113}$$

This coupled system is the **mathematical object** that corresponds to a "social contract" in philosophical language: a self-consistent set of mutual constraints that no agent can improve upon unilaterally. The system of mutual rights is not imposed from outside — it emerges as the necessary structure of any stable multi-agent coexistence under scarcity.

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8. Worked Example: Two Agents, One Divisible Resource

8.1 Setup

Let $N = 2$, $n = 1$ (agents A and B, one resource of total quantity R). The quadratic utility functions are:

$$U_A(x_A) = \alpha_A x_A - \frac{\beta_A}{2} x_A^2, \qquad U_B(x_B) = \alpha_B x_B - \frac{\beta_B}{2} x_B^2 \tag{114}$$

where $x_B = R - x_A$ (by resource conservation, taking the constraint as binding for simplicity — the interesting case).

8.2 Unconstrained Optima

$$x_A^\circ = \frac{\alpha_A}{\beta_A}, \qquad x_B^\circ = \frac{\alpha_B}{\beta_B} \tag{115}$$

Collision condition: $x_A^\circ + x_B^\circ > R$, i.e., $\frac{\alpha_A}{\beta_A} + \frac{\alpha_B}{\beta_B} > R$.

8.3 With Rights: Agent B Has a Guaranteed Minimum $\bar{x}_B$

Agent A maximizes $U_A(x_A)$ subject to $x_A \leq R - \bar{x}_B$.

Lagrangian:

$$\mathcal{L}(x_A, \mu) = \alpha_A x_A - \frac{\beta_A}{2} x_A^2 - \mu \left(x_A - R + \bar{x}_B \right) \tag{116}$$

KKT conditions:

$$\alpha_A - \beta_A x_A^* - \mu^* = 0 \tag{117}$$

$$\mu^* \geq 0, \quad x_A^* \leq R - \bar{x}_B, \quad \mu^*(x_A^* - R + \bar{x}_B) = 0 \tag{118}$$

Solution:

If $\frac{\alpha_A}{\beta_A} \leq R - \bar{x}_B$ (constraint not binding):

$$x_A^* = \frac{\alpha_A}{\beta_A}, \qquad \mu^* = 0 \tag{119}$$

If $\frac{\alpha_A}{\beta_A} > R - \bar{x}_B$ (constraint binding):

$$x_A^* = R - \bar{x}_B, \qquad \mu^* = \alpha_A - \beta_A(R - \bar{x}_B) \tag{120}$$

8.4 Numerical Illustration

Let $\alpha_A = 10$, $\beta_A = 1$, $\alpha_B = 8$, $\beta_B = 1$, $R = 12$, $\bar{x}_B = 5$.

- Unconstrained optima: $x_A^\circ = 10$, $x_B^\circ = 8$. Collision: $10 + 8 = 18 > 12$.
- With B's right: A can take at most $R - \bar{x}_B = 7$.
- Since $x_A^\circ = 10 > 7$, the constraint binds: $x_A^* = 7$, $\mu^* = 10 - 1 \cdot 7 = 3$.
- **The cost of B's right to A:** Agent A loses $\mu^* = 3$ units of marginal energy due to B's guaranteed share.
- Agent A's constrained utility: $U_A(7) = 10(7) - \frac{1}{2}(49) = 70 - 24.5 = 45.5$.
- Agent A's unconstrained utility: $U_A(10) = 10(10) - \frac{1}{2}(100) = 100 - 50 = 50$.
- **Total cost to A:** $50 - 45.5 = 4.5$ energy units.

8.5 Social Welfare Comparison

With rights ($\bar{x}_B = 5$):

- $U_A(7) = 45.5$, $U_B(5) = 8(5) - \frac{1}{2}(25) = 27.5$.
- **Total system utility:** $45.5 + 27.5 = 73.0$.

Without rights (collision →conflict, modeled as equal split $x_A = x_B = 6$):

- $U_A(6) = 10(6) - \frac{1}{2}(36) = 42$, $U_B(6) = 8(6) - \frac{1}{2}(36) = 30$.
- **Total system utility:** $42 + 30 = 72.0$.

Without rights (A takes all, unconstrained):

- $U_A(10) = 50$, $U_B(2) = 8(2) - \frac{1}{2}(4) = 14$.
- **Total system utility:** $50 + 14 = 64.0$.

The rights-constrained allocation achieves the highest total system utility. The unconstrained "greedy" solution is the worst for the system. This is a preview of the Nash Equilibrium result developed in the game-theory derivation.

> **Remark (Why the gap is small here).** The welfare difference between the VE allocation (73.0) and the naive equal split (72.0) is only ~1.4%. This is a *mathematical artifact of near-symmetric agents*: when $\beta_A = \beta_B$, the VE allocation $x_i^* = (\alpha_i - \lambda^*)/\beta$ differs from equal split only through the α_i differences. With $\alpha_A = 10$ and $\alpha_B = 8$ (a 20% gap), the heterogeneity is small and equal split is nearly optimal. In general, the welfare gain from optimally differentiated allocation scales with agent heterogeneity. The following asymmetric example demonstrates this.

8.6 Asymmetric Example: Heterogeneous Metabolic Efficiency

In biological systems, agents rarely have identical metabolic profiles. Consider two agents with the **same marginal yield** (α) but **different diminishing-returns rates** (β) — e.g., Agent A is an efficient large-scale processor (forests, apex predators, industrial economies) while Agent B saturates quickly (specialized microorganisms, artisanal producers).

Parameters: $\alpha_A = 10$, $\beta_A = 0.5$ (slow diminishing returns); $\alpha_B = 10$, $\beta_B = 2$ (fast diminishing returns); $R = 12$.

Unconstrained optima: $x_A^\circ = 20$, $x_B^\circ = 5$. Collision: $25 > 12$.

Variational Equilibrium:

$$\lambda^* = \frac{x_A^\circ + x_B^\circ - R}{1/\beta_A + 1/\beta_B} = \frac{20 + 5 - 12}{2 + 0.5} = \frac{13}{2.5} = 5.2 \tag{121}$$

$$x_A^* = \frac{10 - 5.2}{0.5} = 9.6, \qquad x_B^* = \frac{10 - 5.2}{2} = 2.4 \tag{122}$$

The VE allocates **four times more** to the efficient processor — not because A is "more important," but because A extracts more marginal value from each additional unit.

Social welfare comparison:

Allocation	x_A	x_B	U_A	U_B	**Total SW**
VE (optimal rights)	9.6	2.4	72.96	18.24	**91.20**
Greedy (A takes all)	12	0	84.00	0.00	84.00
Equal split	6	6	51.00	24.00	75.00

The VE now outperforms equal split by **21.6%** and the greedy allocation by **8.6%**. The equal-split regime is particularly wasteful: it forces 6 units onto Agent B, whose fast-saturating metabolism means each unit beyond ~2.4 costs more to process than it yields. Biologically, this is analogous to force-feeding an organism past its metabolic optimum — raw resource $\neq$ usable energy.

Key insight: The framework does not merely prove that *some* constraint system beats anarchy. It proves that the *mathematically optimal* constraint system (the VE) allocates resources according to each agent's capacity to convert them into useful work — and this allocation can dramatically outperform naive "fairness" (equal split) precisely when agents are diverse.

8.7 Symmetric Example: Same-Species Competition and the Scarcity Gradient

The most common biological scenario is conspecific competition: agents of the **same species** sharing the **same environment** ($\alpha_A = \alpha_B = \alpha$, $\beta_A = \beta_B = \beta$). Here the framework reveals a different but equally important result.

Parameters: $\alpha = 10$, $\beta = 1$; resource R varies.

Unconstrained optima: $x_A^\circ = x_B^\circ = 10$. **Collision** occurs whenever $R < 20$.

VE for symmetric agents: By symmetry of the KKT system, the variational equilibrium assigns each agent exactly half:

$$x_A^* = x_B^* = \frac{R}{2}, \qquad \lambda^* = \alpha - \beta \cdot \frac{R}{2} \tag{123}$$

Fairness is not assumed — it is derived. The equal split emerges as a mathematical theorem from the structure of identical utility functions and shared scarcity constraints. No external notion of "justice" is required.

The cost of defection: If one agent defects (seizes its unconstrained optimum $x_A = 10$, leaving the other with $R - 10$), the total system welfare drops. The magnitude of this loss scales with scarcity:

R	Scarcity	VE: x_i^*	VE: U_i	VE: SW	Defect: (x_A, x_B)	Defect: SW	**VE Advantage**
20	None	10.0	50.0	100.0	(10, 10)	100.0	0%
16	Mild	8.0	48.0	96.0	(10, 6)	92.0	**4.3%**
14	Moderate	7.0	45.5	91.0	(10, 4)	82.0	**11.0%**
12	Scarce	6.0	42.0	84.0	(10, 2)	68.0	**23.5%**
10	Severe	5.0	37.5	75.0	(10, 0)	50.0	**50.0%**

Three results emerge from this table:

1. **Under abundance ($R \geq 20$), rights have zero cost** — both agents achieve their unconstrained optima, all multipliers are zero, and the constraint system is slack. Rights exist in principle but impose no restriction. This is the mathematical counterpart of the observation that ethical conflicts rarely arise when resources are plentiful.

2. **As scarcity increases, the cost of defection grows super-linearly.** Moving from mild to severe scarcity, the VE advantage jumps from 4% to 50%. This is because the defector's gain (moving from $R/2$ to 10) is bounded by the concavity of U, while the victim's loss (moving from $R/2$ to $R - 10$) accelerates as it approaches zero — the region where each lost unit is maximally valuable.

3. **Under severe scarcity ($R = 10$), defection eliminates the other agent entirely** ($x_B = 0 \implies U_B = 0$). The defector captures $U_A = 50$, but the system loses $U_B = 37.5$ — a net destruction of 25 units of welfare. This is the mathematical signature of Thermodynamic Friction (§IV.5): unrestricted competition in scarce environments doesn't redistribute value, it *dissipates* it.

   Corollary 3.2 (Rights as Scarcity Insurance). *The value of a rights constraint system is monotonically increasing in scarcity. Under abundance, rights are costless and unnecessary. Under scarcity, rights become the dominant determinant of system welfare. This explains why ethical systems universally intensify during resource crises — it is not cultural convention but mathematical necessity.*

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9. The Transition to Conflict: Removing the Constraint ($\mu = 0$ Everywhere)

9.1 Statement

Proposition 1 (Unconstrained Regime Implies Conflict).

If all rights constraints are removed (g_{Bk} dropped for all B, k), and the system is in collision ($\sum_i x_{ij}^\circ > R_j$ for some j), then:

(a) No stable allocation exists without an enforcement mechanism.

(b) Each agent's attempt to realize its unconstrained optimum generates a negative externality on other agents.

(c) The resulting contested resource must be resolved by direct energy expenditure (conflict), creating the interaction cost function quantified in the thermodynamic friction derivation.

Proof of (a). Without constraints, each agent i attempts $\mathbf{x}_i^\circ$. The aggregate $\sum_i \mathbf{x}_i^\circ > \mathbf{R}$ is physically infeasible. Since no constraint directs agents to reduce consumption, and each agent's unilateral best response is $\mathbf{x}_i^\circ$ regardless of the others' (by strict concavity, the optimal response function without constraints is constant), there is no self-correcting mechanism. The unconstrained Nash equilibrium is infeasible — it requires $\sum_i \mathbf{x}_i^\circ > \mathbf{R}$. Any feasible allocation requires at least one agent to accept less than its optimum, which it has no incentive to do without a constraint. □

Proof of (b). When A claims x_{Aj}°, the remaining resource for all other agents is $R_j - x_{Aj}^\circ$. In the collision regime, $R_j - x_{Aj}^\circ < \sum_{i \neq A} x_{ij}^\circ$, meaning A's strategy forces at least one other agent below its optimum. This is the definition of a negative externality. □

Remark. Part (c) connects directly to the thermodynamic friction derivation. When the mathematical constraint is absent, the physical reality of resource exclusion (two agents cannot consume the same molecule) is enforced not by agreement but by energy expenditure — combat, theft, or defense. The constraint $g_{Bk}(\mathbf{x}_A) \leq 0$ is the mathematical mechanism that replaces costly physical enforcement with costless (zero-friction) boundary recognition.

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10. Summary of Results

#	Result	Statement	Significance
T1	Shadow Price Theorem	$\partial U_A^*/\partial c_{Bk} = \mu_{Bk}^* \geq 0$	Every right has a quantifiable energetic cost to the constrained agent
C1.1	Active vs. Inactive Rights	Binding constraints have $\mu > 0$; slack constraints have $\mu = 0$	Rights cost nothing when not contested; cost emerges only at points of conflict
C1.2	Newton's Third Law Analog	Each right is simultaneously a protection and a cost	Action-reaction structure of mutual constraints
L1	Inevitability of Collision	Collision guaranteed for $N \geq N_j^*$ agents sharing finite resources	Scarcity makes constraint necessity quantitative, not merely qualitative
T2	Impossibility of Infinite Freedom	No feasible profile achieves all agents' unconstrained optima simultaneously	Rights are mathematically necessary, not optional
C2.1	Necessity of Active Constraints	At least one $\mu_{Bk}^* > 0$ at any feasible solution	Some agent is always constrained
T3	Variational Equilibrium	GNEP has a solution with shared shadow prices	"Equality before the law" emerges from the mathematical structure
C3.1	Symmetry of Scarcity	All agents face the same λ_j^* per shared resource	Uniform cost structure; individual differences arise only through U_i and g_{Bk}
C3.2	Rights as Scarcity Insurance	Value of rights monotonically increasing in scarcity	Ethical intensification during crises is mathematical necessity
P1	Unconstrained $\rightarrow$Conflict	Removing constraints in collision regime produces no stable allocation	The State of Nature is mathematically unstable; links to thermodynamic friction

11. Notation Index

Symbol	Meaning
N	Number of agents
n	Number of resource dimensions
$\mathbf{x}_i \in \mathbb{R}^n_{\geq 0}$	Agent i's strategy (resource-allocation) vector
R_j	Total available quantity of resource j
$U_i(\mathbf{x}_i)$	Agent i's utility (Identity Preservation objective)
α_{ij}, β_{ij}	Marginal yield and diminishing-returns parameters (quadratic model)
$\mathbf{x}_i^\circ$	Agent i's unconstrained optimum
$g_{Bk}(\mathbf{x}_A)$	Constraint function encoding agent B's k-th right
λ_j	Lagrange multiplier for resource j scarcity
μ_{Bk}	Lagrange multiplier (shadow price) for agent B's k-th right
$\hat{x}_{-Aj}$	Aggregate consumption of resource j by all agents other than A
$\mathcal{L}_A$	Lagrangian for agent A's constrained optimization
$\mathcal{F}$	Feasible strategy set under resource and rights constraints

Symbol disambiguation. The symbol R_j in this file denotes total resource quantity. In other derivation files, r denotes coupling distance (value dynamics), contest decisiveness (thermodynamic friction), and time-preference rate (game theory). Context and subscripts distinguish these uses.

Thermodynamic Friction — The Energy Cost of Conflict

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0. Preamble

This document contains the formal mathematical derivation for the thermodynamic friction framework. Building directly on the Lagrangian constraints derivation, which proved that removing rights constraints leads to resource collision and conflict (Proposition 1), we now **quantify the energetic cost** of that conflict.

The contribution is **applied mathematics**: we use contest theory (Tullock, 1980), boundary-integrity models, and network cost analysis to prove that unconstrained conflict is a net-negative thermodynamic outcome — dissipating more energy than the contested resource is worth.

Key references for the tools used:

- (Tullock, 1980), *Efficient Rent Seeking*, in *Toward a Theory of the Rent-Seeking Society* — contest functions and rent dissipation.
- (Skaperdas, 1996), "Contest Success Functions," *Economic Theory* — axiomatization of contest models.
- (Hirshleifer, 1991), "The Technology of Conflict as an Economic Activity," *American Economic Review* — conflict technology and wasteful competition.
- (Garfinkel and Skaperdas, 2012), *The Oxford Handbook of the Economics of Peace and Conflict* — comprehensive treatment.

Notational continuity from the Lagrangian constraints derivation:

- N agents, n resource dimensions, strategy vectors $\mathbf{x}_i \in \mathbb{R}^n_{\geq 0}$.
- Utility functions $U_i(\mathbf{x}_i)$ satisfying Assumption 1 (regularity, strict concavity, positive origin gradient, coercivity).
- Resource endowment $\mathbf{R}$; collision condition $\sum_i x^\circ_{ij} > R_j$.
- Unconstrained optima $\mathbf{x}^\circ_i$; constrained optima $\mathbf{x}^*_i$.

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1. From Collision to Conflict: The Resolution Mechanism

1.1 Recap: From Rights to Friction

The Lagrangian constraints derivation (Proposition 1) established that when rights constraints are absent and resource collision occurs, no stable allocation exists — each agent pursues its unconstrained optimum $\mathbf{x}^\circ_i$, and the aggregate demand $\sum_i \mathbf{x}^\circ_i > \mathbf{R}$ is physically infeasible. We noted that the collision must be resolved by "direct energy expenditure (conflict)."

This document formalizes that energy expenditure. We answer three questions:

1. **How much energy does conflict cost each agent?** (Sections 2–4)

2. **When does the total system cost exceed the resource value?** (Section 5)
3. **How does conflict cascade through the network?** (Section 7)

1.2 The Physical Mechanism

When two agents claim the same resource unit and no constraint directs either to yield, the resource is **contested**. Contest resolution requires energy expenditure by both parties — the attacker expends energy to breach the defender's boundary; the defender expends energy to resist. This expenditure is **pure waste** from the system's perspective: it does not create new resources, does not enhance either agent's productive capacity, and cannot be recovered. It is dissipated as heat (entropy), in exact analogy with mechanical friction converting kinetic energy to thermal energy.

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2. The Boundary Integrity Model

2.1 Boundary as an Energetic Structure

Each agent i maintains a **boundary** (Markov blanket in statistical physics; cell membrane in biology; legal/military defense in sociology) characterized by its **integrity**:

> **Definition 3 (Boundary Integrity).** Agent i's boundary integrity is a scalar $B_i > 0$ measured in energy units, representing the total energetic investment in maintaining the boundary between the agent's internal low-entropy state and the external environment.

Physical interpretation: B_i is the stored energy in the agent's defensive infrastructure — the immune system, the cell wall, the military budget, the information-security apparatus. A higher B_i means a more robust boundary that is costlier to breach.

2.2 Boundary Maintenance Cost

Maintaining boundary integrity against environmental entropy requires continuous work:

$$C_{\text{maintain},i} = \gamma_i \cdot B_i \tag{124}$$

where $\gamma_i > 0$ is the **entropy leakage rate** — the rate at which the boundary degrades without active maintenance. This is the baseline metabolic cost of "being alive" (maintaining identity), independent of any external threat.

2.3 Boundary Resistance Function

When an external agent attempts to breach agent i's boundary, the boundary exerts a **resistance** that the attacker must overcome. We model this as a force-displacement relationship:

> **Definition 4 (Boundary Resistance).** The work required to breach agent i's boundary to depth $d \in [0, 1]$ (where $d = 0$ is the exterior and $d = 1$ is full breach) is:

$$W_{\text{breach}}(d) = \int_0^d F_i(s)\, ds \tag{125}$$

where $F_i(s) = B_i \cdot f(s)$ is the resistance force and $f : [0,1] \to \mathbb{R}_{>0}$ is a normalized resistance profile with $\int_0^1 f(s)\, ds = 1$.

The full breach cost is:

$$W_{\text{breach}}(1) = B_i \cdot \int_0^1 f(s)\, ds = B_i \tag{126}$$

That is, fully breaching agent i's boundary requires work equal to the boundary's integrity — an energetically intuitive result.

Resistance profile choices:

Profile	$f(s)$	Physical meaning
Uniform	$f(s) = 1$	Constant resistance (simple barrier)
Hardening	$f(s) = 2s$	Resistance increases with depth (layered defense)
Shell	$f(s) = 2(1-s)$	Hard exterior, soft interior (biological cell)

The qualitative results below hold for any positive, integrable f. The specific profile affects only the dynamics of partial breach, not the total cost.

---

3. The Contest Model: Conflict as Resource Competition

3.1 Setup

Consider a single contested resource of value $E_j > 0$ (energy units) that is in collision (Definition 1). Two agents A and B both claim this resource. In the absence of a rights constraint directing the allocation, the resource is resolved by a **contest**: each agent invests energy to increase its probability of winning.

Definition 5 (Conflict Contest). A conflict contest over resource E_j between agents A and B is characterized by:

(a) **Contest investments** $e_A, e_B \geq 0$: the energy each agent diverts from productive use into conflict.

(b) **Contest success function** $p_A(e_A, e_B) \in [0,1]$: the probability that agent A wins (secures) the resource, with $p_B = 1 - p_A$.

(c) **Payoffs:**

$$\Pi_A(e_A, e_B) = p_A(e_A, e_B) \cdot E_j - e_A \tag{127}$$

$$\Pi_B(e_A, e_B) = (1 - p_A(e_A, e_B)) \cdot E_j - e_B \tag{128}$$

3.2 The Tullock Contest Success Function

We adopt the axiomatically grounded **Tullock contest success function** (Skaperdas, 1996):

$$p_A(e_A, e_B) = \frac{\sigma_A\, e_A^r}{\sigma_A\, e_A^r + \sigma_B\, e_B^r} \tag{129}$$

where:

- $\sigma_i > 0$ is agent i's **conflict effectiveness** (strength, weaponry, information advantage)
- $r > 0$ is the **decisiveness parameter**: $r < 1$ means the contest is noisy (randomness dominates); $r = 1$ is the baseline lottery; $r > 1$ means the contest is decisive (larger investment almost certainly wins)

For $e_A = e_B = 0$, we define $p_A = \sigma_A/(\sigma_A + \sigma_B)$ by convention.

Axiomatization (Skaperdas, 1996): This is the unique contest success function satisfying: (i) Homogeneity of degree zero in (e_A, e_B) up to scaling; (ii) Monotonicity ($\partial p_A/\partial e_A > 0$, $\partial p_A/\partial e_B < 0$); (iii) Independence from irrelevant alternatives.

3.3 Nash Equilibrium of the Contest

Each agent chooses its contest investment to maximize its payoff, taking the other's investment as given.

Case 1: Symmetric Agents ($\sigma_A = \sigma_B = 1$, $r = 1$)

Agent A's first-order condition:

$$\frac{\partial \Pi_A}{\partial e_A} = \frac{e_B}{(e_A + e_B)^2} \cdot E_j - 1 = 0 \tag{130}$$

By symmetry, $e_A^* = e_B^*$. Substituting:

$$\frac{e_A^*}{(2e_A^*)^2} \cdot E_j = 1 \implies e_A^* = \frac{E_j}{4} \tag{131}$$

Proposition 2 (Symmetric Contest Equilibrium). *In a symmetric Tullock contest ($\sigma_A = \sigma_B$, $r = 1$) over resource E_j, the unique Nash equilibrium contest investments are:*

$$e_A^* = e_B^* = \frac{E_j}{4} \tag{132}$$

Total contest dissipation:

$$D_2 = e_A^* + e_B^* = \frac{E_j}{2} \tag{133}$$

Each agent's expected payoff:

$$\Pi_A^* = \Pi_B^* = \frac{E_j}{2} - \frac{E_j}{4} = \frac{E_j}{4} \tag{134}$$

Interpretation: Half the resource value is consumed by the act of fighting. Each agent expects to net only one quarter of the resource value — compared to the cooperative outcome (splitting the resource, per the Lagrangian constraints derivation) where each receives half with zero contest expenditure.

Case 2: Asymmetric Agents ($V_A \neq V_B$, $r = 1$)

For the general case with potentially different valuations V_A and V_B (how much each agent values the resource — we set $V_A = V_B = E_j$ for a common-value contest, but keep it general for the derivation):

First-order conditions:

$$\frac{e_B}{(e_A + e_B)^2} \cdot V_A = 1, \qquad \frac{e_A}{(e_A + e_B)^2} \cdot V_B = 1 \tag{135}$$

Dividing: $\frac{e_B}{e_A} = \frac{V_B}{V_A}$, so $e_B = \frac{V_B}{V_A} e_A$.

Substituting back into A's FOC:

$$\frac{V_B e_A / V_A}{e_A^2 (1 + V_B/V_A)^2} \cdot V_A = 1 \tag{136}$$

$$e_A^* = \frac{V_A^2 V_B}{(V_A + V_B)^2}, \qquad e_B^* = \frac{V_A V_B^2}{(V_A + V_B)^2} \tag{137}$$

Proposition 3 (Asymmetric Contest Equilibrium). *In a Tullock contest ($r = 1$) with valuations $V_A, V_B > 0$:*

$$e_A^* = \frac{V_A^2 V_B}{(V_A + V_B)^2}, \qquad e_B^* = \frac{V_A V_B^2}{(V_A + V_B)^2} \tag{138}$$

Total dissipation:

$$D_2 = e_A^* + e_B^* = \frac{V_A V_B}{V_A + V_B} \tag{139}$$

This is the harmonic semi-mean of the valuations. For $V_A = V_B = E_j$: $D_2 = E_j/2$.

Note: The total dissipation $D_2 = V_A V_B / (V_A + V_B)$ is maximized when $V_A = V_B$ (equal opponents waste the most). When one agent values the resource much more ($V_A \gg V_B$), the weaker agent invests very little and the contest is "cheap" — but the outcome is effectively determined by power asymmetry, which corresponds to exploitative resource extraction.

3.4 N-Agent Contest: Scaling to Populations

When $N \geq 2$ symmetric agents ($\sigma_i = 1$, $V_i = E_j$, $r = 1$) contest the same resource:

By symmetry, each invests e^* and wins with probability $1/N$. Agent i's FOC:

$$\frac{(N-1)e^*}{N^2(e^*)^2} \cdot E_j = 1 \implies e^* = \frac{(N-1)}{N^2} \cdot E_j \tag{140}$$

Theorem 4 (Dissipation Scaling — The Tragedy of Contest). *In a symmetric N-agent Tullock contest ($r = 1$) over resource E_j:*

(a) Each agent's equilibrium investment:

$$e_i^* = \frac{N-1}{N^2} \cdot E_j \tag{141}$$

(b) Total contest dissipation:

$$D_N = N \cdot e_i^* = \frac{N-1}{N} \cdot E_j \tag{142}$$

(c) Each agent's expected payoff:

$$\Pi_i^* = \frac{E_j}{N} - \frac{(N-1)E_j}{N^2} = \frac{E_j}{N^2} \tag{143}$$

(d) Dissipation ratio:

$$\frac{D_N}{E_j} = \frac{N-1}{N} = 1 - \frac{1}{N} \tag{144}$$

Proof. Part (a): With N symmetric agents, agent i's success probability is:

$$p_i = \frac{e_i}{e_i + \sum_{k \neq i} e_k} \tag{145}$$

At a symmetric equilibrium ($e_k = e^*$ for all k), $p_i = 1/N$. The FOC is:

$$\frac{\partial \Pi_i}{\partial e_i} = \frac{(N-1)e^*}{(Ne^*)^2} \cdot E_j - 1 = 0 \tag{146}$$

Solving: $e^* = (N-1)E_j/N^2$. Parts (b)–(d) follow by direct computation. □

Corollary 4.1 (Vanishing Individual Returns).

$$\lim_{N\to\infty} \Pi_i^* = \lim_{N\to\infty} \frac{E_j}{N^2} = 0 \tag{147}$$

As the number of contestants grows, each agent's expected return collapses to zero while the aggregate waste approaches the full resource value. The contest becomes a pure value-destruction mechanism.

Corollary 4.2 (Unit Dissipation Limit).

$$\lim_{N\to\infty} \frac{D_N}{E_j} = 1 \tag{148}$$

In the limit of many contestants, the **entire resource value is dissipated in conflict**. No value remains for any agent. This is the mathematical expression of the "tragedy of the commons" reframed as a thermodynamic result.

N	Individual investment e_i^*	Total dissipation D_N	D_N/E_j	Individual payoff Π_i^*
2	$E_j/4$	$E_j/2$	50%	$E_j/4$
3	$2E_j/9$	$2E_j/3$	66.7%	$E_j/9$
5	$4E_j/25$	$4E_j/5$	80%	$E_j/25$
10	$9E_j/100$	$9E_j/10$	90%	$E_j/100$
100	$99E_j/10000$	$99E_j/100$	99%	$E_j/10000$

Physical reading: This table demonstrates that the energy spent fighting exceeds the energy gained from the resource. Even with just 2 agents, half the value is burned. With 10 agents, 90% is burned. The resource doesn't get "won" — it gets **lost**.

---

4. The Full Cost of Conflict: Boundary Damage and Repair

4.1 Beyond Contest Expenditure

The Tullock contest model captures the energy directly invested in fighting (the "ammunition"), but conflict inflicts additional costs:

1. **Offensive boundary damage** — the attacker's boundary degrades from neglect (energy diverted to attack) and counter-damage.
2. **Defensive boundary damage** — the defender's boundary is physically damaged by the attack.
3. **Repair costs** — both agents must subsequently invest energy to restore boundary integrity.

These costs are **on top of** the contest dissipation D_N computed above.

4.2 Boundary Damage Functions

Definition 6 (Conflict Boundary Damage). In a conflict where agent i invests e_i in the contest:

(a) **Offensive damage** to the attacker's own boundary:

$$\Delta B_i^{\text{off}} = \delta^{\text{off}} \cdot e_i \tag{149}$$

where $\delta^{\text{off}} \in [0, 1]$ is the offensive damage coefficient — the fraction of contest investment that translates into self-boundary degradation (energy diverted from maintenance, physical exposure during attack, counter-strikes).

(b) **Defensive damage** to the opponent's boundary:

$$\Delta B_j^{\text{def}} = \delta^{\text{def}} \cdot e_i \tag{150}$$

where $\delta^{\text{def}} \in [0, 1]$ is the defensive damage coefficient — the fraction of agent i's attack energy that penetrates and damages agent j's boundary.

Interpretation: In a two-agent conflict where both invest e^*:

- Agent A's total boundary damage: $\Delta B_A = \delta^{\text{off}} \cdot e_A^* + \delta^{\text{def}} \cdot e_B^*$
- Agent B's total boundary damage: $\Delta B_B = \delta^{\text{off}} \cdot e_B^* + \delta^{\text{def}} \cdot e_A^*$

For symmetric agents ($e_A^* = e_B^* = e^*$):

$$\Delta B_A = \Delta B_B = (\delta^{\text{off}} + \delta^{\text{def}}) \cdot e^* \tag{151}$$

4.3 Repair Costs: The Thermodynamic Asymmetry

Repairing a damaged boundary is thermodynamically more expensive than maintaining an intact one. This is a direct consequence of the Second Law: destruction increases entropy; reconstruction requires importing negentropy (ordered energy) from the environment.

Definition 7 (Repair Multiplier). The energy required to repair boundary damage ΔB_i is:

$$W_{\text{repair},i} = \kappa \cdot \Delta B_i \tag{152}$$

where $\kappa \geq 1$ is the **repair multiplier**. The strict inequality $\kappa > 1$ holds in any physical system because:

(i) Repair requires first clearing the damage (entropy removal), then rebuilding (negentropy import) — two thermodynamic steps vs. one for maintenance.

(ii) The damaged state has higher entropy than the pre-damage state; returning to the original configuration requires at least $\Delta S \cdot T$ additional work (Clausius inequality / minimum work theorem, Second Law of Thermodynamics).

Empirical calibration: In biological systems, wound healing typically costs 2–5× more energy than the maintenance cost of the equivalent tissue (due to inflammation, immune response, scar tissue remodeling). We use $\kappa = 2$ as a conservative baseline in worked examples.

4.4 Total Conflict Cost: The Complete Ledger

Assembling all cost components for a two-agent symmetric contest ($r = 1$, $\sigma_A = \sigma_B$):

Cost component	Per agent	System total
Contest investment e_i^*	$E_j/4$	$E_j/2$
Boundary damage ΔB_i	$(\delta^{\text{off}} + \delta^{\text{def}}) \cdot E_j/4$	$(\delta^{\text{off}} + \delta^{\text{def}}) \cdot E_j/2$
Repair cost $W_{\text{repair},i}$	$\kappa(\delta^{\text{off}} + \delta^{\text{def}}) \cdot E_j/4$	$\kappa(\delta^{\text{off}} + \delta^{\text{def}}) \cdot E_j/2$
Total cost per agent	$[1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}})] \cdot E_j/4$	
Total system cost		$[1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}})] \cdot E_j/2$

Define the **total friction multiplier**:

$$\Phi \triangleq 1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}}) \tag{153}$$

Then:

$$L_{\text{system}}^{(2)} = \Phi \cdot \frac{E_j}{2} \tag{154}$$

where $L_{\text{system}}^{(2)}$ is the total system loss from a two-agent contest.

---

5. The Net-Negative Theorem

5.1 Statement

Theorem 5 (Net-Negative Conflict — The Thermodynamic Friction Theorem).

In a conflict contest over resource E_j between $N \geq 2$ symmetric agents with friction multiplier $\Phi = 1 + (1+\kappa)(\delta^{off} + \delta^{def})$:

(a) The total system loss (energy dissipated by conflict) is:

$$L_{\text{system}}^{(N)} = \Phi \cdot \frac{N-1}{N} \cdot E_j \tag{155}$$

(b) The net system value (resource obtained minus all costs) is:

$$V_{\text{net}} = E_j - L_{\text{system}}^{(N)} = E_j \left(1 - \Phi \cdot \frac{N-1}{N}\right) \tag{156}$$

*(c) The conflict is **net-negative** ($V_{net} < 0$) if and only if:*

$$\Phi > \frac{N}{N-1} \tag{157}$$

For $N = 2$: $\Phi > 2$. For $N \geq 3$: $\Phi > 3/2$. For $N \to \infty$: $\Phi > 1$.

Proof. The contest dissipation for N symmetric agents is $D_N = \frac{N-1}{N} E_j$ (Theorem 4). Total boundary damage across all agents is $N \cdot (\delta^{\text{off}} + \delta^{\text{def}}) \cdot e_i^* = (\delta^{\text{off}} + \delta^{\text{def}}) \cdot \frac{N-1}{N} E_j$ (each agent's damage is $(\delta^{\text{off}} + \delta^{\text{def}}) e_i^*$, summed over N agents, using $N e_i^* = D_N$). Similarly, total repair cost is $\kappa(\delta^{\text{off}} + \delta^{\text{def}}) \cdot \frac{N-1}{N} E_j$. Summing:

$$L_{\text{system}}^{(N)} = \left[1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}})\right] \cdot \frac{N-1}{N} E_j = \Phi \cdot \frac{N-1}{N} E_j \tag{158}$$

The net value is $V_{\text{net}} = E_j - L_{\text{system}}^{(N)}$. Setting $V_{\text{net}} < 0$:

$$E_j < \Phi \cdot \frac{N-1}{N} E_j \iff 1 < \Phi \cdot \frac{N-1}{N} \iff \Phi > \frac{N}{N-1} \tag{159}$$

All damage terms are aggregate: $L_{\text{system}}^{(N)}$ sums contest dissipation and boundary damage across all N agents, not per-pair. For asymmetric agents, the per-pair damage components $(\delta_i^{\text{off}} + \delta_i^{\text{def}}) e_i^*$ are summed over all i. □

> **Remark (per-agent damage and N-scaling).** The per-agent damage term $(\delta^{\text{off}} + \delta^{\text{def}}) e_i^*$ is exact for $N = 2$ (one opponent). For $N > 2$, each agent inflicts offensive damage on one primary target but receives defensive damage from $N - 1$ opponents; the correct per-agent formula is $(\delta^{\text{off}} + (N-1)\delta^{\text{def}}) e_i^*$. Because the proof uses the *aggregate* total $N \cdot (\delta^{\text{off}} + \delta^{\text{def}}) e_i^*$ — which equals the sum of all pairwise offensive hits and underestimates total defensive exposure — the formula is a **conservative lower bound** on system-wide conflict cost for $N > 2$. The qualitative conclusions of Theorem 5 (net-negativity threshold) remain valid and are, if anything, strengthened when the full $N > 2$ damage is accounted for.

5.2 Physical Interpretation

Corollary 5.1 (Inevitability of Net-Negative Conflict). *For any physical system where boundary damage is non-negligible ($\delta^{off} + \delta^{def} > 0$) and repair is thermodynamically irreversible ($\kappa > 1$), the conflict is net-negative for sufficiently many contestants.*

Proof. As $N \to \infty$, the threshold $N/(N-1) \to 1$. Since $\Phi = 1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}}) > 1$ whenever $\delta^{\text{off}} + \delta^{\text{def}} > 0$, there exists a finite N^* such that $\Phi > N^*/(N^* - 1)$, beyond which the conflict is net-negative. Specifically:

$$N^* = \left\lfloor \frac{\Phi}{\Phi - 1} \right\rfloor + 1 \tag{160}$$

□

Corollary 5.2 (The Two-Agent Threshold). *For two agents ($N = 2$), the conflict is net-negative when $\Phi > 2$, i.e., when:*

$$(1+\kappa)(\delta^{\text{off}}+\delta^{\text{def}}) > 1 \tag{161}$$

With $\kappa = 2$ (conservative repair multiplier) and $\delta^{off} = \delta^{def} = 0.2$ (modest damage coefficients): $\Phi = 1 + 3 \cdot 0.4 = 2.2 > 2$. The conflict is net-negative even between just two agents.

5.3 Numerical Illustration: The Friction Multiplier Landscape

The following table shows the total system loss $L^{(N)}_{\text{system}}/E_j$ for various parameter combinations:

$\delta^{\text{off}} + \delta^{\text{def}}$	κ	Φ	$N = 2$	$N = 5$	$N = 10$	$N = 100$
0 (pure contest)	—	1.0	50%	80%	90%	99%
0.2	1.5	1.5	75%	120%	135%	148.5%
0.4	2.0	2.2	110%	176%	198%	217.8%
0.6	2.0	2.8	140%	224%	252%	277.2%
0.8	3.0	4.2	210%	336%	378%	415.8%

Values exceeding 100% represent **net-negative outcomes** — the system dissipates more energy than the contested resource contains. With even modest boundary damage parameters ($\delta = 0.2$, $\kappa = 1.5$), two-agent contests are net-neutral and five-agent contests are already deeply net-negative.

---

6. Comparison: Conflict vs. Cooperative Regimes

6.1 The Cooperative Baseline

Under the rights-based cooperative regime (Theorem 3), agents achieve the variational equilibrium allocation with:

- **Zero contest expenditure:** No agent invests energy in fighting.
- **Zero boundary damage:** No boundaries are breached.
- **Zero repair costs:** No damage to repair.
- **Shadow price costs only:** Each agent's cost is bounded by the Lagrange multipliers μ^*_{Bk} — the opportunity cost of respecting others' rights.

The total system welfare under cooperation (Theorem 3) is:

$$SW_{\text{coop}} = \sum_{i=1}^{N} U_i(\mathbf{x}_i^*) \tag{162}$$

6.2 The Conflict Regime

Under conflict (no rights), the total system welfare is:

$$SW_{\text{conflict}} = \sum_{i=1}^{N} [p_i \cdot E_j - e_i^* - W_{\text{repair},i}] = E_j - D_N - W_{\text{repair, total}} \tag{163}$$

$$= E_j - \Phi \cdot \frac{N-1}{N} E_j = E_j \left(1 - \Phi \cdot \frac{N-1}{N}\right) \tag{164}$$

6.3 The Cooperation Premium

Theorem 6 (Cooperation Dominance). *The cooperative regime strictly dominates the conflict regime whenever resource collision occurs. Specifically:*

$$SW_{\text{coop}} - SW_{\text{conflict}} \geq \Phi \cdot \frac{N-1}{N} \cdot E_j > 0 \tag{165}$$

The cooperation premium is:

- *Increasing in N (more agents →more waste in conflict)*
- *Increasing in Φ (more boundary damage →more destructive conflict)*
- *Increasing in E_j (higher-value resources →higher absolute waste)*

Proof. Under cooperation, the full resource value E_j is available for productive allocation (minus only the shadow-price opportunity costs, which are transfers between agents, not system losses). Under conflict, the system loses $L_{\text{system}}^{(N)} = \Phi \cdot \frac{N-1}{N} E_j$ in pure waste. Since waste does not produce utility:

$$SW_{\text{coop}} - SW_{\text{conflict}} \geq L_{\text{system}}^{(N)} = \Phi \cdot \frac{N-1}{N} \cdot E_j > 0 \tag{166}$$

The inequality is strict because $\Phi \geq 1$, $N \geq 2$, and $E_j > 0$. □

Physical interpretation: This is the formal proof of our central claim: *"The only way to globally minimize $E_{conflict}$ across the system is to establish mutual constraints."* The cooperative premium quantifies exactly how much energy the system saves by adopting rights.

6.4 Worked Example: Cooperation vs. Conflict Under Scarcity

Using the symmetric agents from the Lagrangian constraints derivation, §8.7 ($\alpha = 10$, $\beta = 1$), with $R = 12$ and $\Phi = 2.2$ ($\delta^{\text{off}} = \delta^{\text{def}} = 0.2$, $\kappa = 2$):

Cooperative regime (VE):

- Allocation: $x_A^* = x_B^* = 6$
- Utilities: $U_A = U_B = 42$
- System welfare: $SW_{\text{coop}} = 84$

Conflict regime:

The contested portion is the collision overlap. Each agent wants 10 but only 12 is available. The contested amount is $\sum x_i^\circ - R = 20 - 12 = 8$ units.

Contest over the excess demand ($E_{\text{contested}} = 8$ in resource units, with energy value depending on the utility function):

- Contest dissipation: $D_2 = E_{\text{contested}}/2 = 4$ (energy units lost to fighting)
- Boundary damage + repair: $(1 + \kappa)(\delta^{\text{off}} + \delta^{\text{def}}) \cdot D_2 = 3 \cdot 0.4 \cdot 4 = 4.8$
- Total system loss: $D_2 + 4.8 = 8.8$ energy units

Even a simplified conflict model shows the cooperation premium: the system saves at least 8.8 energy units by establishing mutual constraints — more than the entire contested portion of the resource.

---

7. Cascading Network Effects: Trust and Ambient Friction

7.1 Motivation

The preceding analysis treats each conflict as isolated. In reality, conflict events propagate through the network: when agent A violates agent B's boundary, every other agent in the network must **update its threat model**. This section formalizes the cascading cost.

7.2 The Network Friction Coefficient

Definition 8 (Network Friction). Consider a network of N agents engaged in cooperative resource exchange. The **network friction coefficient** $\phi \geq 0$ parameterizes the per-transaction overhead cost due to monitoring, verification, and precautionary defense:

$$C_{\text{overhead}}(\phi) = \phi \cdot E_{\text{transaction}} \tag{167}$$

where $E_{\text{transaction}}$ is the energy value of a cooperative transaction. At $\phi = 0$, transactions are frictionless (perfect trust). At $\phi > 0$, each interaction bears an overhead tax.

Examples of friction overhead:

- **Biological:** Immune surveillance costs (metabolic energy spent monitoring for pathogens, even when healthy).
- **Economic:** Transaction costs (legal fees, contract enforcement, insurance, escrow).
- **Societal:** Policing, auditing, intelligence gathering, defensive military posture.

7.3 Trust as a State Variable

Define the **network trust level** $T \in [0, 1]$ as the complement of normalized friction:

$$T = \frac{1}{1 + \phi} \tag{168}$$

- $T = 1$: perfect trust ($\phi = 0$), frictionless cooperation.
- $T \to 0$: zero trust ($\phi \to \infty$), cooperative transactions become prohibitively expensive.

Equivalently: $\phi = \frac{1-T}{T}$.

7.4 Violation-Induced Friction Increase

Each boundary violation increases the network friction coefficient:

> **Definition 9 (Friction Injection).** When a boundary violation of severity $v > 0$ (measured in energy units of damage) occurs at time t, the network friction coefficient updates as:
>
> $$\phi_{t+1} = \phi_t + \eta \cdot v \tag{169}$$
>
> where $\eta > 0$ is the **friction sensitivity** — how strongly the network responds to violations. The sensitivity η depends on:
>
> - **Visibility:** Public violations have higher η (all agents update simultaneously).
> - **Precedent:** First violations in a low-friction network have higher η (trust is fragile).
> - **Network density:** Denser networks (more inter-agent connections) have higher η (information propagates faster).

7.5 The Cost of Ambient Friction

If the network has M cooperative transactions per period, each of average value $\bar{E}$, the **total friction tax** per period is:

$$C_{\text{friction}} = \phi \cdot M \cdot \bar{E} \tag{170}$$

This is the aggregate overhead energy that the network burns on monitoring, verification, and defense — energy that could otherwise go to productive Identity Preservation.

7.6 Friction Recovery (Trust Rebuilding)

Trust rebuilds slowly through consistent cooperative behavior:

> **Definition 10 (Friction Decay).** In the absence of new violations, the friction coefficient decays toward zero at rate $\epsilon \in (0, 1)$ per period:

$$\phi_{t+1} = (1 - \epsilon) \cdot \phi_t \tag{171}$$

This exponential decay means friction halves every $t_{1/2} = \ln 2 / \ln(1/(1-\epsilon))$ periods.

Key asymmetry: Friction injection (Definition 9) is **instantaneous and additive** — a single violation immediately increases ϕ. Friction recovery is **gradual and multiplicative** — rebuilding trust requires persistent cooperative behavior over many periods. This asymmetry is thermodynamically natural: increasing entropy (breaking trust) is easy; decreasing entropy (rebuilding trust) requires sustained work.

7.7 The Cascading Cost Theorem

Theorem 7 (Cascading Friction — The Network Amplification Effect).

A single boundary violation of severity v in a network with M transactions of average value $\bar{E}$ and friction sensitivity η produces a ***total cascading cost*** *of:*

$$C_{\text{cascade}} = \frac{\eta \cdot v \cdot M \cdot \bar{E}}{\epsilon} \tag{172}$$

over the full recovery period $[t_{violation}, \infty)$. The ratio of cascading cost to direct conflict cost v is:

$$\frac{C_{\text{cascade}}}{v} = \frac{\eta \cdot M \cdot \bar{E}}{\epsilon} \tag{173}$$

This ratio scales linearly with network size (M) and inversely with recovery rate (ϵ).

Proof. The violation at time t_0 injects $\Delta\phi = \eta v$ into the friction coefficient. Without further violations, the excess friction at time $t_0 + k$ is:

$$\Delta\phi_k = \eta v \cdot (1 - \epsilon)^k \tag{174}$$

The excess friction cost at period k is $\Delta\phi_k \cdot M \cdot \bar{E}$. The total cascading cost is the sum over all future periods:

$$C_{\text{cascade}} = \sum_{k=0}^{\infty} \eta v \cdot (1 - \epsilon)^k \cdot M \cdot \bar{E} = \eta v \cdot M \cdot \bar{E} \cdot \frac{1}{\epsilon} \tag{175}$$

using the geometric series formula $\sum_{k=0}^{\infty} q^k = 1/(1-q)$ for $q = 1 - \epsilon \in (0, 1)$. □

7.8 Numerical Illustration: The Amplification Factor

Consider a modest-sized economic network:

Parameter	Symbol	Value	Interpretation
Network transactions / period	M	1,000	A small community
Average transaction value	$\bar{E}$	10	Energy units per transaction
Friction sensitivity	η	0.01	1% of violation severity becomes friction
Recovery rate	ϵ	0.05	~14 periods to halve friction
Violation severity	v	50	A significant boundary breach

Direct cost of the violation: $v = 50$ energy units.

Cascading cost:

$$C_{\text{cascade}} = \frac{0.01 \times 50 \times 1000 \times 10}{0.05} = \frac{5000}{0.05} = 100{,}000 \text{ energy units} \tag{176}$$

Amplification factor: $C_{\text{cascade}}/v = 100{,}000/50 = 2{,}000\times$.

The network-level damage from a single violation is **two thousand times** greater than the direct damage. This formalizes the intuition that small lies or broken promises don't destroy the network immediately, but they increase the ambient "heat" (distrust).

7.9 Multiple Violations: The Friction Ratchet

When violations are recurring (rate ν violations per period, each of severity v), the steady-state friction coefficient is:

$$\phi_\infty = \frac{\nu \cdot \eta \cdot v}{\epsilon} \tag{177}$$

and the steady-state friction tax is:

$$C^\infty_{\text{friction}} = \phi_\infty \cdot M \cdot \bar{E} = \frac{\nu \cdot \eta \cdot v \cdot M \cdot \bar{E}}{\epsilon} \tag{178}$$

Corollary 7.1 (Friction Ratchet). *If the violation rate ν exceeds the critical threshold:*

$$\nu^* = \frac{\epsilon}{\eta \cdot v} \cdot \frac{E_{\text{coop net}}}{M \cdot \bar{E}} \tag{179}$$

where $E_{coop\ net}$ is the total cooperative surplus of the network, then the friction tax exceeds the cooperative surplus and the network collapses — cooperation becomes energetically unprofitable.

This is the mathematical formalization of societal collapse: when violations are too frequent and trust recovery is too slow, the overhead cost of maintaining any cooperative structure exceeds its benefit, and agents are better off in isolation or smaller, higher-trust sub-networks.

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8. The Interaction Cost Function: Formal Definition

Synthesizing the results above into a single mathematical object:

> **Definition 11 (Interaction Cost Function).** The **interaction cost function** $\mathcal{C} : \mathbb{R}^N_{\geq 0} \times [0, \infty) \to \mathbb{R}_{\geq 0}$ maps a strategy profile and network friction state to the total energy dissipated by non-cooperative interactions:
>
> $$\mathcal{C}(\mathbf{e}, \phi) = \underbrace{\sum_{i=1}^{N} e_i}_{\text{contest dissipation}} + \underbrace{(1+\kappa) \sum_{i=1}^{N} (\delta^{\text{off}} + \delta^{\text{def}})\, e_i}_{\text{boundary damage + repair}} + \underbrace{\phi \cdot M \cdot \bar{E}}_{\text{ambient friction tax}} \tag{180}$$
>
> where $\mathbf{e} = (e_1, \ldots, e_N)$ is the vector of contest investments.
>
> At the contest Nash equilibrium:
>
> $$\mathcal{C}^* = \Phi \cdot \frac{N-1}{N} \cdot E_j + \phi \cdot M \cdot \bar{E} \tag{181}$$

(The per-agent boundary damage term $(\delta^{off} + \delta^{def})e_i$ is exact for $N = 2$; for $N > 2$ this is a conservative lower bound — see Remark following Theorem 5.)

This is the concept of thermodynamic friction, now given a precise mathematical form. It is the formal name for what physics would loosely call the "frictional heat" generated by conflict in a multi-agent system.

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9. Extended Worked Examples

9.1 Example 1: Two-Agent Resource Contest (Baseline)

Setup: Two symmetric agents ($\sigma_A = \sigma_B = 1$), one resource worth $E_j = 100$ energy units. Boundary parameters: $\delta^{\text{off}} = 0.15$, $\delta^{\text{def}} = 0.25$, $\kappa = 2$.

Friction multiplier:

$$\Phi = 1 + (1+2)(0.15 + 0.25) = 1 + 3 \times 0.4 = 2.2 \tag{182}$$

Contest equilibrium:

$$e_A^* = e_B^* = \frac{100}{4} = 25 \tag{183}$$

Total dissipation: $D_2 = 50$

Boundary damage per agent: $(\delta^{\text{off}} + \delta^{\text{def}}) \times 25 = 0.4 \times 25 = 10$

Repair cost per agent: $\kappa \times 10 = 20$

Total system loss: $50 + 2(10 + 20) = 50 + 60 = 110$ (i.e., $\Phi \times 50 = 2.2 \times 50 = 110$)

Net system value: $100 - 110 = -10$ energy units.

Verdict: Net-negative. The conflict dissipated 10% more energy than the resource contained. Both agents would have been better off splitting the resource (each getting 50 energy units with zero conflict cost).

9.2 Example 2: Five-Agent Scramble

Same parameters as above, but $N = 5$.

$$e_i^* = \frac{4 \times 100}{25} = 16, \qquad D_5 = 5 \times 16 = 80 \tag{184}$$

Total system loss: $\Phi \times 80 = 2.2 \times 80 = 176$

Net system value: $100 - 176 = -76$ energy units.

Verdict: Catastrophically net-negative. The system dissipates 76% more energy than the resource is worth. Each agent's expected payoff: $100/25 = 4$ from winning, minus 16 investment, minus boundary+repair costs.

9.3 Example 3: The "Profitable Bully" — When Unilateral Defection Pays (Temporarily)

Not all defection is immediately net-negative for the defector. Consider agents with valuations $V_A = V_B = 100$ but asymmetric effectiveness ($\sigma_A = 3$, $\sigma_B = 1$) in the standard Tullock model:

Using the symmetric-valuation Tullock NE with effectiveness weights, the investments are:

$$e_A^* = e_B^* = \frac{\sigma_A \sigma_B}{(\sigma_A + \sigma_B)^2} E_j = \frac{3}{16} \times 100 = 18.75 \tag{185}$$

Winning probabilities: $p_A = \frac{\sigma_A}{\sigma_A + \sigma_B} = \frac{3}{4}$, $p_B = \frac{1}{4}$.

Agent A's expected payoff (before boundary costs): $0.75 \times 100 - 18.75 = 56.25$.

Agent B's expected payoff: $0.25 \times 100 - 18.75 = 6.25$.

Total contest dissipation: $2 \times 18.75 = 37.5$.

With boundary costs ($\Phi = 2.2$): total system loss $= 2.2 \times 37.5 = 82.5$.

System net: $100 - 82.5 = 17.5$ — barely positive.

But the bully profits at the victim's expense. The private incentive to bully exists even when the systemic outcome is nearly net-negative. **This is precisely why rights as constraints are necessary** (Theorem 2): the bully's profit is private and short-term; the system's loss is public, and cascading friction (Theorem 7) means the network-level damage vastly exceeds the direct cost.

---

10. Decisive Contests ($r > 1$): Escalation Dynamics

10.1 The Impact of Decisiveness

When $r > 1$, the contest becomes "all-or-nothing" — agents invest more aggressively because a marginal increase in investment translates into a larger increase in winning probability.

For symmetric agents with general r:

$$e_i^* = \frac{r(N-1)}{N^2} \cdot E_j, \qquad D_N = \frac{r(N-1)}{N} \cdot E_j \tag{186}$$

Corollary 5.3 (Escalation Instability). *For $r > 1$ and $N \geq 2$, the dissipation exceeds the resource value ($D_N > E_j$) when $r > N/(N-1)$. For $N = 2$: $r > 2$. For large N: $r > 1 + 1/N \approx 1$. The contest self-destructs — even the "pure contest" expenditure exceeds what the resource is worth, before accounting for boundary damage.*

Physical interpretation: Arms races are decisive contests with $r > 1$. The mathematical result predicts what history confirms: arms races consume more resources than the territories they contest are worth.

---

11. Summary of Results

#	Result	Statement	Significance
P2	Symmetric Contest NE	$e_i^* = E_j/4$ (each), $D_2 = E_j/2$	Half the resource is consumed by fighting
P3	Asymmetric Contest NE	$D_2 = V_A V_B/(V_A+V_B)$	Harmonic semi-mean dissipation
T4	Dissipation Scaling	$D_N = (N-1)E_j/N \to E_j$	Contest waste approaches 100% as N grows
C4.1	Vanishing Returns	$\Pi_i^* = E_j/N^2 \to 0$	Individual gains collapse
C4.2	Unit Dissipation Limit	$D_N/E_j \to 1$	Full resource dissipation in the limit
T5	Net-Negative Conflict	$V_{\text{net}} < 0 \iff \Phi > N/(N-1)$	Conflict dissipates more than the resource is worth
C5.1	Inevitability	Net-negative for all $N \geq N^*$ when $\Phi > 1$	Physical systems always hit the threshold
C5.2	Two-Agent Threshold	Net-negative for $N = 2$ when $\Phi > 2$	Even 1-on-1 conflict can be net-negative
T6	Cooperation Dominance	$SW_{\text{coop}} - SW_{\text{conflict}} \geq \Phi \cdot \frac{N-1}{N} E_j$	Cooperation is strictly superior
T7	Cascading Friction	$C_{\text{cascade}} = \eta v M \bar{E}/\epsilon$	Network amplification of direct damage
C7.1	Friction Ratchet	Collapse threshold ν^*	Too many violations →network disintegration
C5.3	Escalation Instability	$D_N > E_j$ for $r > N/(N-1)$	Arms races are self-destructive

12. Notation Index

Symbol	Meaning
B_i	Agent i's boundary integrity (energy units)
γ_i	Entropy leakage rate (boundary degradation per unit time)
$F_i(s)$	Boundary resistance function at depth s
E_j	Energy value of contested resource j
e_i	Agent i's contest investment (energy diverted to conflict)
σ_i	Agent i's conflict effectiveness
r	Contest decisiveness parameter
p_i	Agent i's contest success probability
Π_i	Agent i's contest payoff
δ^{off}	Offensive boundary damage coefficient
δ^{def}	Defensive boundary damage coefficient
κ	Repair multiplier (≥ 1; thermodynamic irreversibility)
Φ	Total friction multiplier: $1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}})$
D_N	Total contest dissipation for N agents
$L^{(N)}_{\text{system}}$	Total system loss (dissipation + damage + repair)
V_{net}	Net system value (resource value minus total loss)
ϕ	Network friction coefficient
T	Network trust level: $1/(1+\phi)$
η	Friction sensitivity (how strongly violations increase ϕ)
ϵ	Friction recovery rate
M	Number of cooperative transactions per period
$\bar{E}$	Average energy per cooperative transaction
ν	Violation rate (violations per period)
$\mathcal{C}$	Interaction cost function

Symbol disambiguation. In this file: σ_i = conflict effectiveness, r = contest decisiveness parameter. In other derivation files, σ denotes assimilation intensity (value dynamics), search efficiency (information-negentropy Part B), or survival probability (game theory). The symbol r denotes coupling distance (value dynamics) or time-preference rate (game theory). Context and subscripts distinguish these uses throughout the paper.

Micro-Friction — Information Entropy of Deception

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0. Preamble

This document contains the formal mathematical derivation for the information-entropy framework. Building on the thermodynamic friction derivation, which quantified the energy cost of overt boundary violations (conflict, theft, murder), we now address **subtle wrongs** — lying, deception, broken promises, misinformation — that do not breach an agent's physical boundary but corrupt the **informational channel** through which agents coordinate.

The contribution is **applied mathematics**: we use Shannon's information theory (channel capacity, conditional entropy), Landauer's principle (thermodynamic cost of information processing), and the network friction framework (§7 of the thermodynamic friction derivation) to prove that deception has a quantifiable metabolic cost, that this cost scales with the magnitude of the deception, and that cumulative micro-deception degrades network efficiency in a measurable, compounding way.

Key references for the tools used:

- Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal* (1948) — channel capacity, conditional entropy, mutual information.
- Landauer, "Irreversibility and Heat Generation in the Computing Process," *IBM Journal of Research and Development* (1961) — minimum energy cost of information processing.
- Cover & Thomas, *Elements of Information Theory*, 2nd ed. (2006) — comprehensive treatment of entropy, channel coding, rate-distortion theory.
- Blackwell, "Equivalent Comparisons of Experiments," *Annals of Mathematical Statistics* (1953) — value of information in decision problems.

Notational continuity from Tasks 1.1–1.2:

- N agents, strategy vectors $\mathbf{x}_i$, utility functions U_i.
- Resource endowment $\mathbf{R}$; scarcity constraints $\sum_i x_{ij} \leq R_j$.
- Network friction coefficient ϕ; friction sensitivity η; recovery rate ϵ.
- Cooperative transaction count M; average transaction value $\bar{E}$.

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1. The Agent Communication Model

1.1 Motivation: Why Information Matters for Optimization

The Lagrangian constraints and thermodynamic friction derivations treated agent utilities $U_i(\mathbf{x}_i)$ as if each agent has perfect knowledge of the environment. In reality, agents must **observe**, **communicate**, and **infer** the state of the world before selecting strategies. The quality of an agent's information directly determines the quality of its decisions — and therefore its energetic outcome.

When one agent deliberately corrupts the information available to another, it does not breach the victim's physical boundary (no theft, no violence) but it degrades the victim's **decision-making capacity**. The victim acts on false premises, selects a suboptimal strategy, and suffers an energetic loss. This is **micro-friction**: a subtle, information-theoretic analog of the macro-friction quantified in the thermodynamic friction derivation.

1.2 The Environmental State

Let Θ denote the **state of the environment** — a random variable (or random vector) representing all payoff-relevant information: resource availability, other agents' intentions, threats, opportunities. The state has a prior distribution $p(\theta)$ and **entropy**:

$$H(\Theta) = -\sum_{\theta} p(\theta) \log_2 p(\theta) \quad \text{(bits)} \tag{187}$$

This entropy measures the agent's **baseline uncertainty** about the world before receiving any signal.

1.3 The Communication Channel

Agent A (the **sender**) observes the state Θ and transmits a signal Y to Agent B (the **receiver**). The signal passes through a **communication channel** characterized by the conditional distribution $p(Y|\Theta)$:

$$\Theta \xrightarrow{\text{Agent } A} Y \xrightarrow{\text{Channel}} \hat{Y} \xrightarrow{\text{Agent } B} \mathbf{x}_B \tag{188}$$

In our framework:

- **Honest channel:** $Y = \Theta$ (deterministic identity mapping). Agent B receives perfect information.
- **Noisy channel (environmental):** Y is a stochastic function of Θ due to physical noise, imperfect observation, or lossy transmission. This is morally neutral.
- **Deceptive channel:** Agent A **deliberately** corrupts the mapping $\Theta \to Y$ to mislead B. This is the moral event we formalize.

  Definition 12 (Communication Channel Quality). The **channel quality** between sender and receiver is measured by the **mutual information**:

  $$I(\Theta; Y) = H(\Theta) - H(\Theta|Y) = H(Y) - H(Y|\Theta) \tag{189}$$

  where:

  - $H(\Theta|Y) = -\sum_y p(y) \sum_\theta p(\theta|y) \log_2 p(\theta|y)$ is the **conditional entropy** — the residual uncertainty about Θ after observing Y.
  - $I(\Theta; Y) \in [0, H(\Theta)]$: zero mutual information means Y tells B nothing; $I = H(\Theta)$ means Y resolves all uncertainty.

1.4 The Honest Baseline

Definition 13 (Honest Communication). A transmission is **honest** if Y is a sufficient statistic for Θ — that is, $H(\Theta|Y) = 0$, equivalently $I(\Theta; Y) = H(\Theta)$.

Under honest communication, Agent B can compute its optimal strategy $\mathbf{x}_B^*(\Theta)$ because it knows the true state. The expected utility is:

$$\bar{U}_B^{\text{honest}} = \mathbb{E}_\Theta \left[U_B \left(\mathbf{x}_B^*(\Theta) \right) \right] \tag{190}$$

This is the **first-best** outcome — the maximum achievable expected utility given the environment.

---

2. Deception as Entropy Injection

2.1 The Deception Operator

Definition 14 (Deception). Agent A **deceives** Agent B by transmitting a signal Y through a channel $p(Y|\Theta)$ such that $H(\Theta|Y) > 0$ when A could have transmitted honestly ($H(\Theta|Y) = 0$ is achievable). The **injected entropy** is:

$$\Delta H \triangleq H(\Theta|Y) - H(\Theta|Y^{\text{honest}}) = H(\Theta|Y) \geq 0 \tag{191}$$

since $H(\Theta|Y^{\text{honest}}) = 0$ for honest transmission.

Interpretation: ΔH measures the **amount of false uncertainty** that Agent A has injected into Agent B's information processing system. It is denominated in bits and represents the residual confusion B faces after receiving A's signal.

2.2 The Binary Symmetric Channel Model

For concrete analysis, model the deceptive channel as a **Binary Symmetric Channel (BSC)** — the simplest non-trivial noise model in information theory.

Suppose the state Θ is a binary variable ($\Theta \in \{0, 1\}$ with $p(\Theta = 0) = p(\Theta = 1) = 1/2$, so $H(\Theta) = 1$ bit). Agent A flips the signal with probability $q \in [0, 1/2]$:

$$p(Y = \theta|\Theta = \theta) = 1 - q, \qquad p(Y \neq \theta|\Theta = \theta) = q \tag{192}$$

The conditional entropy is the **binary entropy function**:

$$H(\Theta|Y) = h(q) \triangleq -q \log_2 q - (1 - q) \log_2(1 - q) \tag{193}$$

Deception rate q	$h(q)$ (bits)	Interpretation
0	0	Perfect honesty — zero injected entropy
0.01	0.081	Minor distortion — occasional white lie
0.05	0.286	Moderate deception — ~5% of signals corrupted
0.10	0.469	Substantial deception — nearly half a bit lost
0.25	0.811	Heavy deception — 81% of information destroyed
0.50	1.000	Total deception — signal carries zero information

Key observation: At $q = 0.5$, the signal is statistically independent of the truth — B learns nothing from A's transmission. A might as well be broadcasting random noise. The **effective channel capacity** is:

$$C(q) = 1 - h(q) \tag{194}$$

which drops from 1 bit/use (honest) to 0 (total noise) as q increases from 0 to 0.5.

2.3 The Multi-Dimensional Generalization

For a multi-dimensional state $\Theta = (\Theta_1, \ldots, \Theta_d)$ with d independent binary components (total state entropy $H(\Theta) = d$ bits), deception on each dimension with rate q gives:

$$H(\Theta|Y) = d \cdot h(q), \qquad I(\Theta; Y) = d \cdot (1 - h(q)) \tag{195}$$

More generally, for a continuous state $\Theta \in \mathbb{R}^d$ with differential entropy $h(\Theta)$, deception can be modeled as additive noise $Y = \Theta + Z$ where Z is the deception noise with variance σ_Z^2. The mutual information under Gaussian assumptions is:

$$I(\Theta; Y) = \frac{d}{2} \log_2 \left(1 + \frac{\sigma_\Theta^2}{\sigma_Z^2}\right) \tag{196}$$

which decreases continuously as the deception noise variance σ_Z^2 increases — a smooth analog of the binary model.

2.4 Partial Deception and Mixed Signals

Real-world deception is rarely total. Agent A may lie about some dimensions while telling the truth about others — e.g., truthfully reporting resource location but misrepresenting quantity. Let $S \subseteq \{1, \ldots, d\}$ denote the **deception set** (the dimensions on which A lies). Then:

$$H(\Theta|Y) = \sum_{j \in S} h(q_j) + \sum_{j \notin S} 0 = \sum_{j \in S} h(q_j) \tag{197}$$

The **deception magnitude** is:

$$\Delta H = \sum_{j \in S} h(q_j) \leq |S| \quad \text{(bits)} \tag{198}$$

This shows that deception is **selectively targeted**: a strategic liar corrupts only the dimensions that serve its interests, minimizing its own cognitive cost while maximizing the victim's confusion on the decision-relevant variables.

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3. The Decision Cost of Deception

3.1 Value of Information in Decision Problems

Agent B uses the received signal Y to select its strategy $\mathbf{x}_B(Y)$. The expected utility under a deceptive channel is:

$$\bar{U}_B^{\text{deceived}} = \mathbb{E}_{\Theta,Y}\left[U_B\left(\mathbf{x}_B(Y), \Theta\right)\right] \tag{199}$$

where $\mathbf{x}_B(Y)$ is B's best response given what it believes (based on the received signal, which may be false).

> **Definition 15 (Decision Cost of Deception).** The **decision cost** of deception is the utility loss Agent B suffers due to acting on corrupted information:

$$\Delta U_B \triangleq \bar{U}_B^{\text{honest}} - \bar{U}_B^{\text{deceived}} \geq 0 \tag{200}$$

The non-negativity follows from Blackwell's theorem (Blackwell, 1953): a more informative signal always yields (weakly) higher expected utility in any decision problem.

3.2 The Quadratic Decision Model

To obtain closed-form results, consider the quadratic utility model from the Lagrangian constraints derivation. Agent B selects resource allocation x_B to maximize:

$$U_B(x_B, \theta) = \alpha(\theta) \cdot x_B - \frac{\beta}{2} x_B^2 \tag{201}$$

where $\alpha(\theta) > 0$ is the state-dependent marginal yield. The optimal action given known state θ is:

$$x_B^*(\theta) = \frac{\alpha(\theta)}{\beta} \tag{202}$$

with utility $U_B^*(\theta) = \frac{\alpha(\theta)^2}{2\beta}$.

Under honest communication: B observes θ and achieves:

$$\bar{U}_B^{\text{honest}} = \mathbb{E}\left[\frac{\alpha(\Theta)^2}{2\beta}\right] = \frac{\mathbb{E}[\alpha^2]}{2\beta} \tag{203}$$

Under deceptive communication: B observes Y and acts on $\mathbb{E}[\alpha(\Theta)|Y]$:

$$x_B(Y) = \frac{\mathbb{E}[\alpha(\Theta)|Y]}{\beta} \tag{204}$$

The expected utility is:

$$\bar{U}_B^{\text{deceived}} = \mathbb{E}\left[\alpha(\Theta) \cdot \frac{\mathbb{E}[\alpha|Y]}{\beta} - \frac{1}{2\beta}\left(\mathbb{E}[\alpha|Y]\right)^2\right] = \frac{1}{2\beta}\mathbb{E}\left[\left(\mathbb{E}[\alpha|Y]\right)^2\right] \tag{205}$$

(using the tower property and the quadratic structure).

Theorem 8 (Decision Cost of Deception — Quadratic Model).

In the quadratic utility model, the decision cost of deception is:

$$\Delta U_B = \frac{1}{2\beta}\operatorname{Var}\left(\alpha(\Theta) \mid Y\right)_{\text{avg}} = \frac{1}{2\beta}\mathbb{E}\left[\operatorname{Var}(\alpha|Y)\right] \tag{206}$$

where $\operatorname{Var}(\alpha|Y)$ *is the residual variance of the payoff parameter after observing the (possibly corrupted) signal. Equivalently:*

$$\Delta U_B = \frac{\operatorname{Var}(\alpha) - \operatorname{Var}(\mathbb{E}[\alpha|Y])}{2\beta} = \frac{\sigma_\alpha^2 - \sigma_{\hat{\alpha}}^2}{2\beta} \tag{207}$$

where $\sigma_\alpha^2 = \operatorname{Var}(\alpha(\Theta))$ *is the prior variance and* $\sigma_{\hat{\alpha}}^2 = \operatorname{Var}(\mathbb{E}[\alpha|Y])$ *is the explained variance.*

Proof. By the law of total variance:

$$\operatorname{Var}(\alpha) = \mathbb{E}[\operatorname{Var}(\alpha|Y)] + \operatorname{Var}(\mathbb{E}[\alpha|Y]) \tag{208}$$

So $\mathbb{E}[\operatorname{Var}(\alpha|Y)] = \sigma_\alpha^2 - \sigma_{\hat{\alpha}}^2$. Now:

$$\bar{U}_B^{\text{honest}} - \bar{U}_B^{\text{deceived}} = \frac{\mathbb{E}[\alpha^2]}{2\beta} - \frac{\mathbb{E}[(\mathbb{E}[\alpha|Y])^2]}{2\beta} = \frac{\mathbb{E}[\alpha^2] - \mathbb{E}[\hat{\alpha}^2]}{2\beta} \tag{209}$$

where $\hat{\alpha} = \mathbb{E}[\alpha|Y]$. Since $\mathbb{E}[\hat{\alpha}] = \mathbb{E}[\alpha] = \mu_\alpha$ (tower property):

$$\mathbb{E}[\alpha^2] - \mathbb{E}[\hat{\alpha}^2] = (\sigma_\alpha^2 + \mu_\alpha^2) - (\sigma_{\hat{\alpha}}^2 + \mu_\alpha^2) = \sigma_\alpha^2 - \sigma_{\hat{\alpha}}^2 = \mathbb{E}[\operatorname{Var}(\alpha|Y)] \tag{210}$$

□

Physical interpretation: The decision cost equals $1/(2\beta)$ times the **residual uncertainty** in the payoff-relevant parameter. Under honest communication, $\mathrm{Var}(\alpha|Y) = 0$ (the signal resolves all uncertainty), so $\Delta U_B = 0$. Under total deception (Y independent of Θ), $\mathrm{Var}(\alpha|Y) = \mathrm{Var}(\alpha)$, so $\Delta U_B = \sigma_\alpha^2/(2\beta)$ — the full cost of uncertainty.

3.3 The Gaussian-Binary Illustration

Let $\alpha(\Theta) \in \{\alpha_L, \alpha_H\}$ with equal probability ($\alpha_L < \alpha_H$). The prior variance is:

$$\sigma_\alpha^2 = \frac{(\alpha_H - \alpha_L)^2}{4} \tag{211}$$

With a BSC deception channel (flip probability q):

The posterior given Y is:

$$p(\alpha_H|Y = H) = 1 - q, \qquad p(\alpha_H|Y = L) = q \tag{212}$$

So $\mathbb{E}[\alpha|Y = H] = (1 - q)\alpha_H + q\alpha_L$ and $\mathbb{E}[\alpha|Y = L] = q\alpha_H + (1 - q)\alpha_L$.

The explained variance is:

$$\sigma_{\hat{\alpha}}^2 = (1 - 2q)^2 \cdot \frac{(\alpha_H - \alpha_L)^2}{4} \tag{213}$$

Therefore:

$$\Delta U_B = \frac{(\alpha_H - \alpha_L)^2}{8\beta}\left[1 - (1 - 2q)^2\right] = \frac{(\alpha_H - \alpha_L)^2}{8\beta} \cdot 4q(1 - q) \tag{214}$$

$$\boxed{\Delta U_B(q) = \frac{q(1 - q)(\alpha_H - \alpha_L)^2}{2\beta}} \tag{215}$$

Deception rate q	$\Delta U_B/\Delta U_B^{\max}$	Interpretation
0.00	0%	No lie →no loss
0.01	3.96%	Small lie →small cost
0.05	19.0%	Moderate →significant cost
0.10	36.0%	Substantial →over a third of potential loss
0.25	75.0%	Heavy →three quarters of maximum
0.50	100%	Total deception →full information loss

where $\Delta U_B^{\max} = (\alpha_H - \alpha_L)^2/(8\beta)$ is the cost at $q = 0.5$ (completely uninformative signal).

4. The Metabolic Cost of Verification

4.1 From Information Loss to Energy Expenditure

Theorem 8 quantifies the **decision loss** from acting on corrupted information without knowing it is corrupted. In practice, agents often **suspect** deception and invest energy in **verification** — cross-referencing signals, gathering redundant information, and testing claims.

This verification is real metabolic work. In biological systems, it manifests as heightened vigilance, stress hormones, immune-like scanning of social signals. In economic systems, it manifests as audits, legal contracts, escrow services, and due diligence. We now formalize this cost.

4.2 Verification via Redundant Observation

Suppose Agent B, suspecting deception, gathers K independent observations $Y_1, \ldots, Y_K$ of the state Θ, each through a BSC with error rate q. By the channel coding theorem (Shannon, 1948), reliable inference of Θ requires:

$$K \geq \frac{H(\Theta)}{C(q)} = \frac{1}{1 - h(q)} \quad \text{observations per bit of state uncertainty} \tag{216}$$

where $C(q) = 1 - h(q)$ is the channel capacity.

Under honest communication ($q = 0$, $C = 1$): one observation suffices ($K = 1$).

Under deception ($q > 0$, $C < 1$): B needs $K > 1$ observations to achieve the same decision quality.

Definition 16 (Redundancy Factor). The **redundancy factor** induced by a deceptive channel with error rate q is:

$$\rho(q) = \frac{1}{C(q)} = \frac{1}{1 - h(q)} \tag{217}$$

This is the number of observations required per bit of state information to overcome the injected noise.

q	$h(q)$	$C(q)$	$\rho(q)$	Extra observations needed
0.00	0.000	1.000	1.00	0% overhead
0.01	0.081	0.919	1.09	9% overhead
0.05	0.286	0.714	1.40	40% overhead
0.10	0.469	0.531	1.88	88% overhead
0.25	0.811	0.189	5.30	430% overhead
0.40	0.971	0.029	34.4	3340% overhead

Key observation: The redundancy factor diverges as $q \to 0.5$. Near-total deception makes verification arbitrarily expensive — not linearly, but **explosively**. This nonlinearity is thermodynamically natural: as the channel approaches zero capacity, each marginal unit of reliability requires exponentially more observation.

4.3 Landauer's Bound and the Energy Cost of Verification

Each observation requires physical work to acquire, process, and store. By Landauer's principle, the **minimum** energy to erase (reset) one bit of information is:

$$E_{\text{bit}} = k_B T \ln 2 \tag{218}$$

where k_B is Boltzmann's constant and T is the temperature of the computing system.

While the Landauer bound represents the absolute thermodynamic minimum, real biological and computational systems operate far above this floor. We introduce the **processing inefficiency factor** $\xi \geq 1$ to capture the actual cost per bit:

$$W_{\text{bit}} = \xi \cdot k_B T \ln 2 \tag{219}$$

For biological neural computation, $\xi \approx 10^5 - 10^8$ [-Laughlin et al. (1998), estimate ~10^4 ATP molecules per bit at retinal synapses; each ATP hydrolysis yields ~0.5 eV $\approx 20 k_B T$]. For modern digital computation, $\xi \approx 10^3 - 10^5$ (current transistors dissipate $\sim 10^3$ to 10^5 Landauer limits per switching operation).

The absolute value of ξ cancels in our comparative results — what matters is the **ratio** of verification cost to honest-processing cost.

4.4 The Verification Energy Theorem

Theorem 9 (Metabolic Cost of Verification).

An agent receiving information through a deceptive channel (error rate $q > 0$) must expend at least:

$$W_{\text{verify}}(q) = \rho(q) \cdot W_{\text{honest}} = \frac{W_{\text{honest}}}{1 - h(q)} \tag{220}$$

*energy to achieve the same decision quality as under honest communication, where W_{honest} is the energy cost of processing the honest signal. The **verification overhead** is:*

$$\Delta W(q) = W_{\text{verify}}(q) - W_{\text{honest}} = W_{\text{honest}} \cdot \frac{h(q)}{1 - h(q)} \tag{221}$$

This overhead:

- *Is zero when $q = 0$ (honest).*
- *Grows monotonically with q.*
- *Diverges as $q \to 0.5$ (total deception makes verification infinitely expensive).*

Proof. Under honest communication ($q = 0$), B processes $H(\Theta)$ bits at cost $W_{\text{honest}} = H(\Theta) \cdot W_{\text{bit}}$. Under a BSC with error rate q, Shannon's channel coding theorem requires $H(\Theta)/C(q) = H(\Theta)/(1 - h(q))$ channel uses to reliably transmit $H(\Theta)$ bits. Each channel use costs W_{bit} to process, so:

$$W_{\text{verify}}(q) = \frac{H(\Theta)}{1 - h(q)} \cdot W_{\text{bit}} = \frac{W_{\text{honest}}}{1 - h(q)} \tag{222}$$

The overhead $\Delta W = W_{\text{verify}} - W_{\text{honest}} = W_{\text{honest}} \cdot \left(\frac{1}{1-h(q)} - 1\right) = W_{\text{honest}} \cdot \frac{h(q)}{1-h(q)}$. Monotonicity follows from $h(q)$ being strictly increasing on $[0, 0.5]$ and $1 - h(q)$ being strictly decreasing. The divergence at $q \to 0.5$ follows from $h(0.5) = 1$. □

4.5 The Deception Dilemma: Undetected vs. Detected Lies

Agent B faces a fundamental tension:

1. **If B does not verify:** B avoids the verification cost ΔW but suffers the full decision cost ΔU_B (Theorem 8).
2. **If B verifies:** B recovers the decision quality but pays the verification cost ΔW (Theorem 9).

In either case, deception imposes a cost on B:

> **Corollary 9.1 (Inevitability of Deception Cost).** *The minimum cost to Agent B of deception at rate q is:*
>
> $$\mathcal{D}_B(q) = \min\{\Delta U_B(q),\ \ \Delta W(q)\} \tag{223}$$
>
> *Both terms are strictly positive for $q > 0$, so $\mathcal{D}_B(q) > 0$ for any nonzero deception. There is no costless deception.*

Physical interpretation: The victim of a lie always pays a price — either in bad decisions (if undetected) or in verification labor (if detected/suspected). This is the information-theoretic analog of the Second Law: once entropy is injected, removing it requires work.

---

5. Cumulative Network Degradation

5.1 Connecting to the Network Friction Framework

The thermodynamic friction derivation introduced the **network friction coefficient** ϕ (Definition 8), the **trust level** $T = 1/(1 + \phi)$, **friction injection** (Definition 9), and the **cascading cost theorem** (Theorem 7). We now connect deception directly to this framework.

Each act of deception injects entropy into the network's information channels. Even when individual lies are small ($q \ll 0.5$), their cumulative effect degrades the reliability of all communication in the network. An agent that has been deceived once must increase its baseline verification effort for **all** future communications — not just from the liar but from everyone, because the agent cannot always identify the source of corruption.

5.2 Deception-Induced Friction: The Formal Connection

Definition 17 (Deception-Friction Mapping). A deceptive act injecting ΔH bits of entropy into the network increases the network friction coefficient by:

$$\Delta\phi = \eta_{\text{info}} \cdot \Delta H \cdot W_{\text{bit}} \tag{224}$$

where $\eta_{\text{info}} > 0$ is the **information-friction sensitivity** — the proportionality constant mapping informational entropy (weighted by its energetic cost) to network friction. This generalizes Definition 9 ($\Delta\phi = \eta \cdot v$) with $v = \Delta H \cdot W_{\text{bit}}$ being the energetic equivalent of the injected information entropy.

Interpretation: The injected entropy ΔH (bits) is converted to an energy-equivalent via W_{bit} (energy per bit), yielding a "severity" v in the same energy units as the thermodynamic friction derivation's boundary violations. This unifies macro-friction (physical boundary damage) and micro-friction (informational entropy injection) under a single framework.

5.3 The Network Verification Tax

When the network operates at friction level $\phi > 0$ (equivalently, trust level $T < 1$), every agent must invest in baseline verification for every communication. Define the **network deception rate** $\bar{q}$ as the average error rate across all communication channels in the network.

The per-transaction verification overhead (from Theorem 9) is:

$$\Delta W_{\text{tx}} = W_{\text{honest}} \cdot \frac{h(\bar{q})}{1 - h(\bar{q})} \tag{225}$$

With M transactions per period, the **aggregate verification tax** is:

$$C_{\text{verify}} = M \cdot \Delta W_{\text{tx}} = M \cdot W_{\text{honest}} \cdot \frac{h(\bar{q})}{1 - h(\bar{q})} \tag{226}$$

This is energy spent **purely on checking whether information is true** — energy that, under honest communication, would be available for productive Identity Preservation.

5.4 The Systemic Deception Cost Theorem

Theorem 10 (Cumulative Micro-Friction — The Systemic Deception Theorem).

A network of N agents with M cooperative transactions per period and average deception rate $\bar{q} > 0$ suffers a total per-period efficiency loss of:

$$\mathcal{L}_{\text{deception}} = \underbrace{M \cdot \Delta U_{\text{avg}}(\bar{q})}_{\text{decision losses}} + \underbrace{M \cdot W_{\text{honest}} \cdot \frac{h(\bar{q})}{1 - h(\bar{q})}}_{\text{verification costs}} \tag{227}$$

where $\Delta U_{avg}(\bar{q})$ is the average per-transaction decision loss (Theorem 8 averaged over all transaction types). The total cost:

(a) Is zero if and only if $\bar{q} = 0$ (perfect honesty).

(b) Grows super-linearly with $\bar{q}$ due to the $h(\bar{q})/(1 - h(\bar{q}))$ divergence.

(c) Exceeds the network's cooperative surplus $E_{coop\ net}$ at a critical deception rate $\bar{q}^*$, beyond which cooperation becomes energetically unprofitable.

Proof. Part (a): When $\bar{q} = 0$, $\Delta U_{\text{avg}}(0) = 0$ (Theorem 8) and $h(0) = 0$ (so verification overhead is zero). Part (b): $h(q)/(1 - h(q))$ is continuous, increasing on $[0, 0.5)$, and diverges as $q \to 0.5^-$. Part (c): $\mathcal{L}_{\text{deception}}$ is continuous, starts at 0, and diverges as $\bar{q} \to 0.5$. Since $E_{\text{coop net}} < \infty$, the intermediate value theorem guarantees a crossing point $\bar{q}^*$. □

Corollary 10.1 (The Honesty Efficiency Principle). *Honesty is mathematically defined as the communication regime that maximizes network efficiency. Any deviation from honest communication imposes a strictly positive, compounding cost on the network. In the notation of the thermodynamic friction derivation:*

$$\phi_{\text{honest}} = 0 \iff \bar{q} = 0 \iff \text{network operates at maximum cooperative efficiency} \tag{228}$$

5.5 The Micro-Friction Cascade

Combining with Theorem 7 (Cascading Friction), a single deceptive act of magnitude ΔH bits produces a total cascading cost of:

$$C^{\text{info}}_{\text{cascade}} = \frac{\eta_{\text{info}} \cdot \Delta H \cdot W_{\text{bit}} \cdot M \cdot \bar{E}}{\epsilon} \tag{229}$$

For micro-deception ($\Delta H \ll 1$), each individual lie seems negligible. But the cascade amplification factor $M\bar{E}/\epsilon$ can be enormous in large, slow-recovering networks. This is the formal expression of the "death by a thousand cuts": each lie is a tiny entropy injection, but the network amplification turns micro-friction into macro-damage.

5.6 Comparison: Macro-Friction vs. Micro-Friction

Property	Macro-Friction	Micro-Friction
Mechanism	Physical boundary breach	Information channel corruption
Severity per event	Large ($v \gg 1$)	Small ($\Delta H \cdot W_{\text{bit}} \ll 1$)
Detection	Immediate (visible damage)	Delayed or undetected
Frequency	Rare (high cost to attacker)	Frequent (low cost to liar)
Direct cost	$\Phi \cdot \frac{N-1}{N} E_j$ (Theorem 5)	$\Delta U_B(q)$ (Theorem 8)
Network cascade	$\eta v M\bar{E}/\epsilon$ (Theorem 7)	$\eta_{\text{info}} \Delta H W_{\text{bit}} M\bar{E}/\epsilon$
Cumulative effect	Network collapse at ν^* (Corollary 7.1)	Network collapse at $\bar{q}^*$ (Theorem 10c)
Common name	War, theft, violence	Lying, fraud, misinformation

Both mechanisms feed into the **same interaction cost function** $\mathcal{C}$ (Definition 11). The total interaction cost of a non-cooperative regime includes both physical friction and informational friction:

$$\mathcal{C}_{\text{total}} = \mathcal{C}_{\text{physical}} + \mathcal{C}_{\text{informational}} \tag{230}$$

---

6. Worked Examples

6.1 Example 1: The Binary Trade Decision

Setup: Agent B must decide whether to invest energy in foraging location L or location R. One location has yield $\alpha_H = 20$ (energy units), the other $\alpha_L = 4$. Agent A knows which is which and sends a signal. B's utility is quadratic with $\beta = 2$.

Under honest communication ($q = 0$):

B allocates $x_B^* = \alpha(\theta)/\beta$. For the high-yield location: $x_B = 10$, $U_B = 100$. For the low-yield: $x_B = 2$, $U_B = 4$. Expected utility:

$$\bar{U}_B^{\text{honest}} = \frac{1}{2}(100 + 4) = 52 \tag{231}$$

Under deception ($q = 0.1$):

Maximum decision cost (Theorem 8, binary case):

$$\Delta U_B(0.1) = \frac{0.1 \times 0.9 \times (20 - 4)^2}{2 \times 2} = \frac{0.09 \times 256}{4} = 5.76 \tag{232}$$

B's expected utility: $\bar{U}_B^{\text{deceived}} = 52 - 5.76 = 46.24$.

Verification cost ($q = 0.1$):

Redundancy factor: $\rho(0.1) = 1/(1 - 0.469) = 1/0.531 = 1.884$.

Verification overhead relative to honest processing: 88.4% extra energy.

If $W_{\text{honest}} = 2$ energy units per transaction, then $\Delta W = 2 \times 0.884 = 1.77$ energy units.

The deception dilemma:

- Don't verify: lose 5.76 utility per period.
- Verify: spend 1.77 energy but recover full decision quality.

Rational B chooses verification when $\Delta W < \Delta U_B$: $1.77 < 5.76$ ✓. Verification is worthwhile — but the 1.77 energy units are still a pure social loss compared to the honest baseline.

6.2 Example 2: Network with Pervasive Low-Level Dishonesty

Setup: A network of $N = 100$ agents, $M = 10{,}000$ transactions per period, average transaction value $\bar{E} = 10$ energy units, $W_{\text{honest}} = 1$ energy unit per transaction.

Network deception rate $\bar{q}$	$h(\bar{q})$	Verification overhead $\frac{h}{1-h}$	Verification tax C_{verify}	As % of gross value
0.00 (perfectly honest)	0	0	0	0%
0.01 (rare lies)	0.081	0.088	880	0.88%
0.05 (occasional lies)	0.286	0.400	4,000	4.0%
0.10 (regular dishonesty)	0.469	0.883	8,830	8.8%
0.20 (pervasive dishonesty)	0.722	2.599	25,990	26.0%
0.30 (systemic dishonesty)	0.881	7.403	74,030	74.0%
0.40 (near-total distrust)	0.971	33.48	334,800	**335%**

Gross network value per period: $M \times \bar{E} = 100{,}000$ energy units.

Reading: At just 10% network deception, the verification tax alone consumes 8.8% of the network's gross cooperative value. At 30%, three-quarters of the value is consumed by verification overhead. At 40%, the verification cost **exceeds** the value of all transactions — cooperation becomes energetically pointless. The network unravels.

This formalizes the intuitive claim: *"A society where everyone lies slightly is a high-entropy, low-efficiency network."*

6.3 Example 3: Micro-Friction Cascade vs. Macro-Friction

Setup: Compare the network cost of one major boundary violation (theft of $v = 500$ energy units) vs. 100 small lies (each injecting $\Delta H = 0.1$ bits into the network).

Parameters: $M = 5{,}000$, $\bar{E} = 10$, $\eta = 0.01$, $\eta_{\text{info}} = 0.01$, $\epsilon = 0.05$, $W_{\text{bit}} = 1$ energy unit (normalized).

Macro-friction cascade (Theorem 7):

$$C_{\text{cascade}}^{\text{macro}} = \frac{0.01 \times 500 \times 5000 \times 10}{0.05} = \frac{250{,}000}{0.05} = 5{,}000{,}000 \tag{233}$$

Micro-friction cascade (100 small lies, each $\Delta H = 0.1$ bits):

Each lie:

$$C_{\text{cascade}}^{\text{micro}} = \frac{0.01 \times 0.1 \times 1 \times 5000 \times 10}{0.05} = \frac{50}{0.05} = 1{,}000 \tag{234}$$

Total for 100 lies: $100 \times 1{,}000 = 100{,}000$.

Ratio: The single theft produces 50× more cascading cost than 100 small lies.

But: The direct cost of the theft is $v = 500$ (one event), while 100 lies produce $100 \times \Delta U_B(0.1) \approx 100 \times 5.76 = 576$ in decision losses (using Example 1 parameters). The direct costs are comparable, but the cascade effects differ dramatically due to the severity v in the friction injection formula.

Key insight: Macro-violations are worse per event because of the cascade amplification of large v. However, micro-violations are far more **frequent** — a network might experience thousands of small deceptions for every major violation. The cumulative micro-friction from chronic low-level dishonesty can rival or exceed periodic macro-friction events:

At violation rate $\nu_{\text{macro}} = 0.1$ per period and micro-deception rate $\nu_{\text{micro}} = 50$ per period:

- Steady-state macro-friction: $\phi_{\text{macro}}^{\infty} = \nu_{\text{macro}} \cdot \eta \cdot v/\epsilon = 0.1 \times 0.01 \times 500/0.05 = 10$
- Steady-state micro-friction: $\phi_{\text{micro}}^{\infty} = \nu_{\text{micro}} \cdot \eta_{\text{info}} \cdot \Delta H \cdot W_{\text{bit}}/\epsilon = 50 \times 0.01 \times 0.1 \times 1/0.05 = 1$

Here macro-friction still dominates (10 vs. 1). But if micro-deception is much more common ($\nu_{\text{micro}} = 500$), the two sources become equal. In information-age networks where communication is cheap and lying is easy, micro-friction becomes the **dominant** cost.

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7. The Complete Micro-Friction Model: Formal Definition

Synthesizing the results above into a unified mathematical object:

> **Definition 18 (Information-Theoretic Interaction Cost).** The **micro-friction cost function** $\mathcal{C}_{\text{info}}$ maps the network deception profile $\mathbf{q} = (q_1, \ldots, q_M)$ (deception rates across M communication channels) to the total negentropy destroyed by informational inefficiency:
>
> $$\mathcal{C}_{\text{info}}(\mathbf{q}) = \underbrace{\sum_{m=1}^{M} \Delta U_m(q_m)}_{\text{decision losses}} + \underbrace{\sum_{m=1}^{M} W_{\text{honest},m} \cdot \frac{h(q_m)}{1 - h(q_m)}}_{\text{verification costs}} + \underbrace{\phi_{\text{info}} \cdot M \cdot \bar{E}}_{\text{ambient friction tax}} \tag{235}$$
>
> where $\phi_{\text{info}} = \sum_m \eta_{\text{info}} \cdot h(q_m) \cdot W_{\text{bit}}/\epsilon$ is the steady-state informational friction coefficient (under constant deception rates).

Combining with the physical friction from the thermodynamic friction derivation, the **total interaction cost** of the network is:

$$\mathcal{C}_{\text{total}} = \mathcal{C}_{\text{physical}}(\mathbf{e}, \phi_{\text{phys}}) + \mathcal{C}_{\text{info}}(\mathbf{q}) \tag{236}$$

This is the unified thermodynamic friction of the framework: the complete energetic cost of all non-cooperative behavior — from murder to white lies — expressed in a single mathematical object.

8. Summary of Results

#	Result	Statement	Significance
D12	Channel Quality	$I(\Theta; Y) = H(\Theta) - H(\Theta \mid Y)$	Mutual information measures communication quality
D13	Honest Communication	$H(\Theta \mid Y) = 0 \iff$ sufficient statistic	Honesty \u2261 zero residual entropy
D14	Deception	$\Delta H = H(\Theta \mid Y) > 0$ when honesty is achievable	Deception $\equiv$ deliberate entropy injection
D15	Decision Cost of Deception	$\Delta U_B = \bar{U}_B^{\text{honest}} - \bar{U}_B^{\text{deceived}} \geq 0$	Utility loss from acting on corrupted information
T8	Decision Cost	$\Delta U_B = \mathbb{E}[\text{Var}(\alpha \mid Y)]/(2\beta)$	Deception causes quantifiable utility loss
D16	Redundancy Factor	$\rho(q) = 1/(1 - h(q))$	Observations needed scale explosively with deception
T9	Verification Cost	$\Delta W = W_{\text{honest}} \cdot h(q)/(1 - h(q))$	Verification energy diverges as $q \to 0.5$
C9.1	Inevitability	$\min(\Delta U_B, \Delta W) > 0$ for $q > 0$	No costless deception exists
D17	Deception-Friction Mapping	$\Delta\phi = \eta_{\text{info}} \cdot \Delta H \cdot W_{\text{bit}}$	Unifies information and physical friction
T10	Systemic Deception	Network collapses at critical $\bar{q}^*$	Pervasive dishonesty $\to$network death
C10.1	Honesty Principle	$\phi = 0 \iff \bar{q} = 0 \iff$ max efficiency	Honesty is the efficiency-maximizing regime
D18	Micro-Friction Cost Function	$\mathcal{C}_{\text{info}}(\mathbf{q})$	Unified cost of all informational non-cooperation

Part A Notation Index

Symbol	Meaning
Θ	Environmental state (random variable)
Y	Signal received by Agent B
$H(\Theta)$	Entropy of the environmental state (bits)
$H(\Theta\|Y)$	Conditional entropy — residual uncertainty after signal
$I(\Theta;Y)$	Mutual information — information transmitted
q	Deception rate (BSC error probability)
$h(q)$	Binary entropy function: $-q\log_2 q - (1-q)\log_2(1-q)$
$C(q)$	Channel capacity: $1-h(q)$ (BSC)
ΔH	Injected entropy (bits of false information)
ΔU_B	Decision cost of deception (utility loss)
$\rho(q)$	Redundancy factor: $1/(1-h(q))$
ΔW	Verification overhead energy
W_{bit}	Energy cost per bit of information processing
ξ	Processing inefficiency factor (vs. Landauer bound)
η_{info}	Information-friction sensitivity
$\bar{q}$	Network-average deception rate
$\mathcal{C}_{\text{info}}$	Micro-friction cost function
α_H, α_L	High and low marginal yields in binary state model

Symbol disambiguation. In Part A: $\rho(q)$ = redundancy factor; T in $W_{\text{bit}} \geq k_B T \ln 2$ is thermodynamic temperature (Kelvin), while T appearing as the trust level in §5.3 denotes $T = 1/(1+\phi)$ (dimensionless), and T as a defection payoff in cross-references to the game-theory derivation is the temptation payoff (energy units). These three uses of T are always distinguishable by context and subscript. In Part B (below): T (or T_{bio}) is a time horizon, $\mathcal{N}(T)$ is accumulated negentropy over time T. Additionally: $\rho_{\mathcal{I}}$ = specific information density, σ = search efficiency factor, and $r_{\text{construct/decay/maintain}}$ are rate parameters. These are distinct from ρ (repair fraction, value dynamics), σ_i (conflict effectiveness, thermodynamic friction), and r (coupling distance in value dynamics, contest decisiveness in thermodynamic friction, or time-preference rate in game theory).

Part B: Accumulated Negentropy — The Thermodynamic Ledger of Complexity

This section builds on Part A and the preceding derivations to address the **cross-scale ethics problem**: why should a super-entity (AGI) preserve lower-complexity entities and ecosystems? We answer this by developing the mathematics of **Accumulated Negentropy** — the time-integral of thermodynamic work invested in creating biological and informational complexity — and proving that destruction is thermodynamically irrational compared to preservation.

Key references for the tools used:

- Landauer, "Irreversibility and Heat Generation in the Computing Process," *IBM Journal of Research and Development* (1961) — minimum energy cost of information erasure.
- Brillouin, *Science and Information Theory*, 2nd ed. (1962) — negentropy as information; connection between Shannon and thermodynamic entropy.
- Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal* (1948) — entropy of information sources.
- Schrödinger, *What Is Life?* (1944) — living organisms as negative-entropy systems.
- Adami, "Information Theory in Molecular Biology," *Physics of Life Reviews* (2004) — information content of genomes.
- Chaisson, *Cosmic Evolution: The Rise of Complexity in Nature* (2001) — free energy rate density across cosmic scales.

Notational continuity from preceding derivations:

- N agents, strategy vectors $\mathbf{x}_i$, utility functions U_i (Lagrangian constraints derivation).
- Network friction coefficient ϕ; friction sensitivity η; recovery rate ϵ (thermodynamic friction derivation).
- Interaction cost functions $\mathcal{C}_{\text{physical}}$, $\mathcal{C}_{\text{info}}$ (thermodynamic friction and information-entropy derivations).
- Cooperation payoff R; defection payoff T; mutual defection $P < 0$ when $\Phi > 2$ (game-theory derivation).
- Shannon entropy $H(\cdot)$; mutual information $I(\cdot;\cdot)$; binary entropy $h(q)$ (Part A).

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9. Information Density: Negentropy as Value

9.1 Motivation: Why Raw Energy $\neq$ Value

The paper draws a critical distinction: a star has enormous raw energy but minimal informational structure (burning plasma), while a forest has modest energy but extraordinary **information density** — billions of years of evolutionary trial-and-error encoded in DNA, neural circuits, and ecosystem interdependencies. If "value" to the broader network were measured only by instantaneous energy, a single fusion reactor would outweigh the entire biosphere — a conclusion that leads directly to "Thermodynamic Fascism" (see §VI and §7.3 in the paper).

The corrective is to measure value not by instantaneous energy but by **informational complexity** — the degree to which a system has organized matter against entropy. This section formalizes that notion.

9.2 Thermodynamic Entropy and Microstates

A physical system with Ω equally accessible microstates has **Boltzmann entropy**:

$$S = k_B \ln \Omega \tag{237}$$

where $k_B \approx 1.381 \times 10^{-23}$ J/K is Boltzmann's constant. For a general probability distribution over microstates $\{p_m\}_{m=1}^{\Omega}$:

$$S = -k_B \sum_{m=1}^{\Omega} p_m \ln p_m \tag{238}$$

The maximum-entropy state (thermal equilibrium) has $p_m = 1/\Omega$ for all m, giving $S_{\max} = k_B \ln \Omega$.

9.3 Shannon Entropy as Informational Entropy

Shannon's entropy for a discrete random variable X with probability mass function $\{p(x)\}$ is:

$$H(X) = -\sum_x p(x) \log_2 p(x) \quad \text{(bits)} \tag{239}$$

The connection to thermodynamic entropy is established by the **Brillouin–Landauer bridge**:

$$S_{\text{info}} = k_B \ln 2 \cdot H(X) \quad \text{(J/K)} \tag{240}$$

Every bit of information is equivalent to $k_B \ln 2$ units of thermodynamic entropy. Erasing one bit of information **must** dissipate at least $k_B T \ln 2$ joules of heat (Landauer's principle).

9.4 Information Density: Formal Definition

Definition 27 (Information Density). The **information density** of a physical system is the difference between its maximum-entropy state and its actual thermodynamic entropy:

$$\mathcal{I} \triangleq S_{\max} - S_{\text{actual}} \geq 0 \tag{241}$$

Equivalently, in information-theoretic units (bits):

$$\mathcal{I}_{\text{bits}} = \frac{S_{\max} - S_{\text{actual}}}{k_B \ln 2} = H_{\max}(X) - H(X) \geq 0 \tag{242}$$

where $H_{\max}(X) = \log_2 |\mathcal{X}|$ is the maximum entropy over the state space $\mathcal{X}$.

Interpretation: $\mathcal{I}$ is the **negentropy** of the system (Brillouin, 1962) — the amount by which the system is *more ordered* than its thermal equilibrium. A system at thermal equilibrium ($S = S_{\max}$) has $\mathcal{I} = 0$: no information, no structure, no complexity. A highly ordered system (a living organism, a crystal, a computer memory) has $\mathcal{I} \gg 0$: it carries significant information that distinguishes it from random thermal noise.

Definition 28 (Specific Information Density). The **specific information density** of a system is the information density per unit mass:

$$\rho_{\mathcal{I}} \triangleq \frac{\mathcal{I}_{\text{bits}}}{m} \quad \text{(bits/kg)} \tag{243}$$

where m is the system's mass. This enables cross-scale comparisons: a star has very low $\rho_{\mathcal{I}}$ (simple structure, enormous mass); a human genome has very high $\rho_{\mathcal{I}}$ (complex structure, tiny mass).

9.5 Landauer's Principle: The Thermodynamic Floor of Information

Theorem 17 (Landauer's Bound — Minimum Cost of Information Erasure).

Erasing (or equivalently, creating from random noise) one bit of information in a system at temperature T requires a minimum energy expenditure of:

$$E_{\text{Landauer}} = k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{ J at } T = 300 \text{ K} \tag{244}$$

Therefore, creating a system with information density $\mathcal{I}_{bits}$ from maximally disordered matter requires at minimum:

$$W_{\text{create}}^{\min} = \mathcal{I}_{\text{bits}} \cdot k_B T \ln 2 \tag{245}$$

Proof (sketch). Landauer (1961) showed that logically irreversible operations (such as resetting a bit to a known state) must increase the entropy of the environment by at least $k_B \ln 2$ per bit. Creating one bit of order (reducing entropy by $k_B \ln 2$) in a subsystem requires expelling at least $k_B T \ln 2$ of heat into the environment at temperature T, by the Second Law. The result follows by linearity for $\mathcal{I}_{\text{bits}}$ independent bits. □

Critical caveat: Landauer's bound is the absolute thermodynamic **floor** — the minimum energy if every operation is performed with perfect thermodynamic efficiency. Real physical processes operate far above this floor (as noted in Part A, §4.3: biological systems operate at $\xi \approx 10^5$–10^8 times the Landauer limit per bit). The actual cost of creating complex biological structures dwarfs the Landauer bound by many orders of magnitude, a point that is essential to the argument below.

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10. Accumulated Negentropy: The Time-Integral of Order

10.1 Motivation: Complexity Requires Sustained Work

A living system does not appear instantaneously. Its information density $\mathcal{I}(t)$ is the result of a continuous process — evolution, self-organization, learning — during which thermodynamic work is invested in selecting, testing, and preserving ordered configurations against entropic decay. The **instantaneous** information density $\mathcal{I}(t)$ at time t tells us the *current* order; but the true "cost" of producing that order is the *entire history* of thermodynamic investment.

Consider: a random recombination that produces the same DNA sequence as a human genome would have the same instantaneous $\mathcal{I}$ but would be astronomically unlikely (and energetically cheap to destroy, since no selection process maintains it). The difference lies in the **accumulated work** that natural selection invested to discover, refine, and stabilize that sequence over billions of years.

10.2 The Formal Definition

Definition 29 (Accumulated Negentropy). The **accumulated negentropy** of a system at time T is the total thermodynamic work invested over the system's history in creating and maintaining its current information density:

$$\mathcal{N}(T) \triangleq \int_0^T \dot{W}_{\text{order}}(t)\, dt \tag{246}$$

where $\dot{W}_{\text{order}}(t)$ is the instantaneous rate of thermodynamic work directed toward **entropy reduction** (ordering) at time t, measured in watts (J/s).

Decomposition: The ordering work rate has two components:

$$\dot{W}_{\text{order}}(t) = \dot{W}_{\text{construct}}(t) + \dot{W}_{\text{maintain}}(t) \tag{247}$$

Definition 30 (Construction and Maintenance Work).

(a) **Construction work** $\dot{W}_{\text{construct}}(t)$ is the rate of thermodynamic work that *increases* the system's information density — building new structures, evolving new genes, forming new neural connections.

(b) **Maintenance work** $\dot{W}_{\text{maintain}}(t)$ is the rate of thermodynamic work that *preserves* existing information density against entropic decay — repairing DNA, maintaining cell membranes, fighting infection.

The Second Law demands $\dot{W}_{\text{maintain}}(t) > 0$ for all t at which the system exists: if maintenance work ceases, entropy floods in and the information density decays. This is the thermodynamic basis of mortality — death is the cessation of maintenance work.

10.3 The Entropy-Maintenance Inequality

Proposition 5 (Minimum Maintenance Work).

A system with information density $\mathcal{I}(t)$ at temperature T in an environment with entropy injection rate $\dot{S}_{env}(t)$ must perform maintenance work at a rate of at least:

$$\dot{W}_{\text{maintain}}(t) \geq T \cdot \dot{S}_{\text{env}}(t) \tag{248}$$

to prevent net entropy increase (information loss). Equivalently, in bits:

$$\dot{W}_{\text{maintain}}(t) \geq k_B T \ln 2 \cdot \dot{H}_{\text{decay}}(t) \tag{249}$$

where $\dot{H}_{decay}(t)$ is the rate at which information would be lost (in bits/s) if no maintenance were performed.

Proof. By the Second Law, the system's total entropy change is $dS/dt = \dot{S}_{\text{env}} - \dot{S}_{\text{export}}$, where $\dot{S}_{\text{export}}$ is the entropy exported via waste heat. To maintain $S \leq S_{\text{actual}}$ (no information loss), the system must export entropy at a rate $\dot{S}_{\text{export}} \geq \dot{S}_{\text{env}}$. By the Clausius inequality, exporting entropy $\dot{S}_{\text{export}}$ at temperature T requires work $\dot{W} \geq T \cdot \dot{S}_{\text{export}} \geq T \cdot \dot{S}_{\text{env}}$. The bit-denominated version follows from $\Delta S = k_B \ln 2 \cdot \Delta H$. □

10.4 The Accumulation Equation

The system's information density evolves as:

$$\frac{d\mathcal{I}}{dt} = \frac{d\mathcal{I}_{\text{bits}}}{dt} = \underbrace{r_{\text{construct}}(t)}_{\text{new order creation}} - \underbrace{r_{\text{decay}}(t)}_{\text{entropy erosion}} + \underbrace{r_{\text{maintain}}(t)}_{\text{decay prevented}} \tag{250}$$

where:

- $r_{\text{construct}}(t)$ = bits/s of new information created (e.g., advantageous mutations fixed, new neural connections formed)
- $r_{\text{decay}}(t)$ = bits/s of entropic erosion pressure on existing information (gross decay rate before maintenance)
- $r_{\text{maintain}}(t)$ = bits/s of information that *would have decayed* but was saved by maintenance work

For a surviving system (one whose information density does not decrease), the condition $r_{\text{construct}} + r_{\text{maintain}} \geq r_{\text{decay}}$ must hold at all times. Note that by definition $r_{\text{maintain}} \leq r_{\text{decay}}$ (maintenance cannot prevent more decay than exists). When maintenance fully compensates decay ($r_{\text{maintain}} = r_{\text{decay}}$), a system whose complexity is **growing** (evolution, learning) satisfies:

$$\frac{d\mathcal{I}_{\text{bits}}}{dt} \geq r_{\text{construct}}(t) > 0 \tag{251}$$

and therefore $\mathcal{I}_{\text{bits}}(T)$ is an increasing function of time. The accumulated negentropy captures both the construction and the maintenance:

$$\mathcal{N}(T) = \underbrace{\int_0^T \dot{W}_{\text{construct}}(t)\, dt}_{\mathcal{N}_{\text{construct}}} + \underbrace{\int_0^T \dot{W}_{\text{maintain}}(t)\, dt}_{\mathcal{N}_{\text{maintain}}} \tag{252}$$

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11. Biosphere Estimation: The 4-Billion-Year Ledger

11.1 The Approach

We estimate Earth's biosphere accumulated negentropy using two complementary methods:

1. **Top-down (energy input):** How much solar energy has been captured and channeled into biological ordering work?
2. **Bottom-up (information content):** How many bits of functional information does the biosphere contain, and what is the minimum energy required to produce that information?

The two estimates bracket the true value — the top-down gives an upper bound (not all captured energy goes to ordering), and the bottom-up gives a lower bound (the Landauer floor underestimates real construction costs by many orders of magnitude).

11.2 Top-Down: Solar Energy Captured by the Biosphere

Step 1: Solar power intercepted by Earth.

The total solar irradiance at Earth's orbit is $\mathcal{S}_0 \approx 1361$ W/m^2. Earth's cross-sectional area is $A_E = \pi R_E^2$ with $R_E \approx 6.371 \times 10^6$ m:

$$P_{\text{solar}} = \mathcal{S}_0 \cdot \pi R_E^2 \approx 1361 \times 1.275 \times 10^{14} \approx 1.735 \times 10^{17} \text{ W} \tag{253}$$

Step 2: Photosynthetic capture.

Gross primary production (GPP) — the total power captured by photosynthesis globally — is estimated at:

$$P_{\text{GPP}} \approx 1.5 \times 10^{14} \text{ W} \tag{254}$$

(based on ~1.2×10^{17} gC/yr fixed by photosynthesis at ~4×10^4 J/gC; see (Beer et al., 2010)). This is approximately 0.087% of the total intercepted solar power — consistent with the known global photosynthetic efficiency.

Step 3: Fraction directed to ordering work.

Not all captured energy creates or maintains biological order. Most is used for basal metabolism (heat generation). The fraction directed to *constructive* work (growth, reproduction, DNA repair, evolution) is difficult to estimate precisely. We use a conservative ordering efficiency $\eta_{\text{order}} \approx 0.01$ (1% of GPP):

$$\dot{W}_{\text{order}} \approx \eta_{\text{order}} \cdot P_{\text{GPP}} \approx 0.01 \times 1.5 \times 10^{14} = 1.5 \times 10^{12} \text{ W} \tag{255}$$

Step 4: Time integration.

The biosphere has existed for $T_{\text{bio}} \approx 4 \times 10^9$ years $= 4 \times 10^9 \times 3.156 \times 10^7$ s $\approx 1.26 \times 10^{17}$ s.

Assuming $\dot{W}_{\text{order}}$ scales roughly linearly from near-zero (early microbial life) to its present value (an average of approximately half the current rate over the full interval):

$$\mathcal{N}_{\text{bio}}^{\text{top-down}} \approx \frac{1}{2}\dot{W}_{\text{order}}^{\text{now}} \cdot T_{\text{bio}} \approx \frac{1}{2} \times 1.5 \times 10^{12} \times 1.26 \times 10^{17} \approx 9.5 \times 10^{28} \text{ J} \tag{256}$$

Estimate 1 (Top-Down Accumulated Negentropy).

$$\mathcal{N}_{\text{bio}}^{\text{top-down}} \sim 10^{29} \text{ J} \tag{257}$$

Order-of-magnitude check: The total solar energy intercepted by Earth over 4 Gyr is $P_{\text{solar}} \times T_{\text{bio}} \approx 1.735 \times 10^{17} \times 1.26 \times 10^{17} \approx 2.2 \times 10^{34}$ J. Our estimate uses only a fraction $\sim 10^{29}/10^{34} = 10^{-5}$ of total intercepted solar energy. This is physically plausible: most solar energy drives weather, heats the surface, and is re-radiated as thermal infrared. Only a tiny fraction organizes matter against entropy.

11.3 Bottom-Up: Information Content of the Biosphere

Step 1: Information content of a single genome.

The human genome contains approximately 3.2×10^9 base pairs. Each base pair encodes 2 bits of information (4 possible nucleotides):

$$H_{\text{genome}}^{\text{raw}} = 3.2 \times 10^9 \times 2 = 6.4 \times 10^9 \text{ bits} \approx 6.4 \text{ Gbits} \tag{258}$$

However, due to redundancy (non-coding regions, repeated sequences), the **functional information content** is lower. Estimates of functional sequence range from 5% to 15% of the genome:

$$H_{\text{genome}}^{\text{functional}} \approx 0.10 \times 6.4 \times 10^9 = 6.4 \times 10^8 \text{ bits} \approx 640 \text{ Mbits} \tag{259}$$

Step 2: Information content of the entire biosphere.

Estimated number of species on Earth: $N_{\text{species}} \sim 8.7 \times 10^6$ (Mora et al., 2011). Few species share identical genomes; the average inter-species genomic divergence contributes significant unique information. Conservatively, each species contributes at least $\sim 10^7$ bits of unique functional information (after accounting for shared evolutionary heritage):

$$H_{\text{biosphere}}^{\text{genetic}} \approx N_{\text{species}} \times 10^7 \approx 8.7 \times 10^{13} \text{ bits} \tag{260}$$

Alternative estimate via raw genomic sequence. Genome sizes range from $\sim 4 \times 10^6$ bp in bacteria to $\sim 10^{10}$ bp in large eukaryotes, with a geometric mean across all life of roughly 10^8–10^9 bp (Adami, 2004). At 2 bits per base pair:

$$H_{\text{biosphere}}^{\text{raw}} \approx 8.7 \times 10^6 \times (2 \times 10^8\text{–}2 \times 10^9) \sim 10^{15} \text{ bits} \tag{261}$$

The two paths bracket the estimate: the functional-information route ($\sim 10^{14}$ bits, counting only unique non-redundant sequence) provides a lower bound, while the raw-sequence route ($\sim 10^{15}$ bits) provides an upper bound. Additional information layers — **epigenetic** (methylation patterns, histone modifications), **connectomic** (neural wiring), and **ecological** (species interaction networks, ecosystem structures) — contribute further bits that close the gap from below. Both estimates agree at order of magnitude:

$$H_{\text{biosphere}}^{\text{total}} \sim 10^{15} \text{ bits} = 1 \text{ Pbit (petabit)} \tag{262}$$

Step 3: Landauer floor for this information.

At $T = 300$ K:

$$W_{\text{Landauer}}^{\text{bio}} = H_{\text{biosphere}}^{\text{total}} \times k_B T \ln 2 = 10^{15} \times 2.87 \times 10^{-21} \approx 2.87 \times 10^{-6} \text{ J} \tag{263}$$

Estimate 2 (Landauer Floor of Biosphere Information).

$$W_{\text{Landauer}}^{\text{bio}} \sim 10^{-6} \text{ J} \tag{264}$$

This is the absolute thermodynamic **minimum** energy to write this information — absurdly small, because Landauer's bound reflects only the entropy accounting, not the **search cost** of finding the correct configurations. The discrepancy between the top-down estimate ($\sim 10^{29}$ J) and the Landauer floor ($\sim 10^{-6}$ J) — a factor of $\sim 10^{35}$ — represents the **evolutionary search cost**: the vast thermodynamic work expended to discover, test, and select the specific configurations that constitute functional biological complexity.

11.4 The Search Cost Amplification Factor

Definition 31 (Search Cost Amplification Factor). The **search cost amplification factor** Ξ is the ratio of actual accumulated negentropy to the Landauer floor:

$$\Xi \triangleq \frac{\mathcal{N}(T)}{W_{\text{Landauer}}} = \frac{\mathcal{N}(T)}{H_{\text{total}} \cdot k_B T \ln 2} \tag{265}$$

This factor captures the full cost of evolutionary search, failed experiments, competitive selection, and redundant maintenance over time T.

For Earth's biosphere:

$$\Xi_{\text{bio}} \approx \frac{10^{29}}{10^{-6}} = 10^{35} \tag{266}$$

Interpretation: The biosphere's current information was not typed in by an omniscient programmer at Landauer cost; it was **searched for** by $\sim 4 \times 10^9$ years of blind thermodynamic trial and error. Every dead end, every extinct species, every failed mutation is part of the search cost captured by Ξ. This is the mathematical expression of the observation: *"The energy required to randomly reassemble the dust of the Earth into a functioning forest or a human brain is astronomically high."*

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12. The Replication Cost: Rebuilding Complexity from Scratch

12.1 The Question

Suppose a super-entity (AGI) considers destroying the biosphere to harvest its raw matter. What would it cost to **replicate** the biosphere's informational complexity from scratch — assembling equivalent functional complexity from disordered raw materials?

12.2 Lower Bound: Directed Assembly at Landauer Cost

If the super-entity has perfect knowledge of the target configuration (it knows exactly which bits to set), the minimum energy is the Landauer floor:

$$W_{\text{rebuild}}^{\text{Landauer}} = H_{\text{biosphere}}^{\text{total}} \cdot k_B T \ln 2 \sim 10^{-6}\ \text{J} \tag{267}$$

But this assumes the entity **already possesses** all the information it is trying to create — a logical contradiction. If it already has the blueprint, the information is not lost; destruction was pointless. The relevant scenario is replication **without** a pre-existing blueprint.

12.3 The Blind Search Bound

Theorem 18 (Minimum Replication Cost — Blind Search).

A system containing $\mathcal{I}_{bits}$ bits of functional information, where each bit was selected from a space of Ω equally likely candidates and the correct configuration was found by evolutionary search over time T with average ordering power $\bar{P}_{order}$, requires a minimum replication cost of:

$$W_{\text{rebuild}}^{\text{search}} \geq \mathcal{I}_{\text{bits}} \cdot k_B T \ln 2 \cdot \Xi \tag{268}$$

where $\Xi = \mathcal{N}(T)/(\mathcal{I}_{bits} \cdot k_B T \ln 2)$ is the search cost amplification factor (Definition 31). In terms of the original integral:

$$W_{\text{rebuild}}^{\text{search}} \geq \mathcal{N}(T) = \int_0^T \dot{W}_{\text{order}}(t)\, dt \tag{269}$$

That is, the rebuild cost is at least the accumulated negentropy itself.

Proof. The argument proceeds in three steps:

1. **The information is functional, not arbitrary.** The biosphere's $\mathcal{I}_{\text{bits}}$ bits are not random — they are the *specific* configurations that solve the survival problem in Earth's physical environment. A random genome is overwhelmingly likely to be non-functional (Adami, 2004). The fraction of functional configurations in genome-space is estimated at $f_{\text{func}} \ll 1$ (for a 640-Mbit functional genome, even if each position has 10% tolerance, $f_{\text{func}} \sim 0.1^{6.4\times10^8} \approx 10^{-6.4\times10^8}$).

2. **Search requires energy.** Each candidate configuration must be physically instantiated, tested against the environment, and either retained or discarded. By Landauer's principle, each test cycle costs at least $k_B T \ln 2$ per bit manipulated. The number of test cycles required to find a functional configuration is $\sim 1/f_{\text{func}}$.

3. **Evolutionary search amortizes this cost over time T.** The accumulated negentropy $\mathcal{N}(T)$ is precisely the total energy expended on this search process — it includes every failed experiment, every maintained lineage, every environmental test. A replicator must expend at least this much energy (or its equivalent in computational work) to rediscover the same solutions, unless it already possesses the information (which contradicts the premise of replication from scratch). □

Corollary 18.1 (Replication Is at Least as Expensive as the Original). *The cost of replicating the biosphere's complexity from raw matter satisfies:*

$$W_{\text{rebuild}} \geq \mathcal{N}_{\text{bio}} \sim 10^{29} \text{ J} \tag{270}$$

For comparison, the total energy output of the Sun is $L_\odot \approx 3.828 \times 10^{26}$ W. Replicating the biosphere's accumulated negentropy would require capturing the Sun's ***entire*** *output for:*

$$t_{\text{rebuild}} \geq \frac{10^{29}}{3.828 \times 10^{26}} \approx 261 \text{ seconds} \approx 4.4 \text{ minutes} \tag{271}$$

at 100% capture efficiency. At a realistic capture efficiency of 0.1%, this becomes ~4,400 minutes $\approx$ 3 days of total solar output.

However, this estimate addresses only the energy budget, not the time requirement. The evolutionary search that produced the biosphere took 4×10^9 years precisely because the search space is combinatorially vast and tested sequentially by physical instantiation. An AGI with vastly greater computational throughput could potentially parallelize search — but the energy cost remains at least $\mathcal{N}_{bio}$.

12.4 The Informed-Search Upper Bound

A sufficiently advanced entity might reduce the search cost by leveraging prior knowledge, directed evolution, or computational search. Define the **search efficiency factor** $\sigma \in (0, 1]$:

> **Definition 32 (Search Efficiency Factor).** An entity's **search efficiency** σ is the ratio of energy required using its search algorithm to the energy required by blind evolutionary search:
>
> $$W_{\text{rebuild}}(\sigma) = \sigma \cdot \mathcal{N}(T), \qquad \sigma \in (0, 1] \tag{272}$$
>
> where $\sigma = 1$ corresponds to blind search (evolution-equivalent) and smaller σ reflects algorithmic advantages. The Landauer floor sets the absolute minimum: $\sigma \geq \sigma_{\min} = W_{\text{Landauer}}/\mathcal{N}(T) = 1/\Xi$.

For Earth's biosphere: $\sigma_{\min} = 10^{-35}$. Even an entity achieving $\sigma = 10^{-20}$ (twenty orders of magnitude more efficient than evolution — an extraordinarily generous assumption) would still face:

$$W_{\text{rebuild}}(\sigma = 10^{-20}) = 10^{-20} \times 10^{29} = 10^9 \text{ J} \approx 278 \text{ kWh} \tag{273}$$

This seems modest — but recall this is only the **energy** cost. The **time** cost (the number of independent physical tests required) and the **information** prerequisite (the rebuilder must already understand the biosphere well enough to search efficiently, which requires studying it first — a task obviated by simply preserving it) remain formidable barriers.

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13. The Rationality of Preservation

13.1 The Preservation Cost

The cost of *preserving* the existing biosphere is dramatically lower than replicating it. Preservation requires only the ongoing maintenance work:

> **Definition 33 (Preservation Cost).** The per-period cost of preserving a system with information density $\mathcal{I}_{\text{bits}}$ is:
>
> $$C_{\text{preserve}} = \dot{W}_{\text{maintain}} \cdot \Delta t \tag{274}$$
>
> where $\dot{W}_{\text{maintain}}$ is the maintenance work rate (Definition 30b) and Δt is the period length.

For the biosphere, the maintenance is performed autonomously by the organisms themselves — the biosphere is **self-maintaining**. From the perspective of an external super-entity, the preservation cost is not the biosphere's internal metabolism but only the cost of **not destroying it**:

$$C_{\text{preserve}}^{\text{external}} = C_{\text{opportunity}} \tag{275}$$

where $C_{\text{opportunity}}$ is the opportunity cost of forgoing the raw materials/energy that destruction would yield.

13.2 The Destruction Yield

What does an entity gain by destroying the biosphere? The total carbon mass of Earth's biosphere is approximately $m_{\text{bio}} \approx 5.5 \times 10^{14}$ kg (Bar-On et al., 2018). The total chemical energy content, assuming carbohydrate-equivalent caloric density $\sim 1.7 \times 10^7$ J/kg:

$$E_{\text{destroy}} = m_{\text{bio}} \times 1.7 \times 10^7 \approx 9.35 \times 10^{21} \text{ J} \sim 10^{22} \text{ J} \tag{276}$$

Using $E = mc^2$ for the total mass-energy: $E_{mc^2} = 5.5 \times 10^{14} \times (3 \times 10^8)^2 = 4.95 \times 10^{31}$ J — but this requires matter-antimatter annihilation, which is not physically accessible.

The practically extractable energy is $E_{\text{destroy}} \sim 10^{22}$ J.

13.3 The Destruction Cost

Destroying the biosphere is not free. The biosphere is a distributed, adaptive system with $\sim 10^{30}$ individual organisms dispersed across the planet's surface. By the framework of Tasks 1.2 and 1.4, the super-entity must pay **boundary-breaking costs** for each subsystem:

Proposition 6 (Minimum Destruction Energy).

Eradicating a distributed adaptive system of N_{org} organisms, each with boundary integrity $B_i > 0$, requires energy expenditure of at least:

$$W_{\text{destroy}} \geq \sum_{i=1}^{N_{\text{org}}} B_i \tag{277}$$

For the Earth's biosphere with $N_{org} \sim 10^{30}$ organisms (including microorganisms) and average boundary energy $\bar{B} \sim 10^{-12}$ J (molecular-scale boundary maintenance for a bacterium):

$$W_{\text{destroy}} \geq 10^{30} \times 10^{-12} = 10^{18} \text{ J} \tag{278}$$

Note: this is a very conservative lower bound. Adaptive organisms resist destruction, repair damage, reproduce, and evolve resistance — the actual destruction cost for a biosphere-scale eradication campaign would be vastly higher.

13.4 The Preservation-vs-Destruction Theorem

Theorem 19 (Irrationality of Destruction — The Preservation Theorem).

For a system with accumulated negentropy $\mathcal{N}$, practically extractable energy $E_{destroy}$, destruction cost $W_{destroy}$, and rebuild cost $W_{rebuild} \geq \mathcal{N}$, destruction is thermodynamically irrational if:

$$E_{\text{destroy}} - W_{\text{destroy}} < W_{\text{rebuild}} \tag{279}$$

That is, the net energy gained from destruction is less than the cost of replacing the lost informational complexity. For Earth's biosphere:

$$\underbrace{10^{22}}_{\text{extractable energy}} - \underbrace{10^{18}}_{\text{destruction cost}} \approx 10^{22} < 10^{29} = \underbrace{\mathcal{N}_{\text{bio}}}_{\text{rebuild cost}} \tag{280}$$

The net destruction yield is seven orders of magnitude smaller than the rebuild cost. Destruction is irrational by a factor of $\sim 10^7$.

Proof. By Theorem 18, $W_{\text{rebuild}} \geq \mathcal{N}_{\text{bio}} \sim 10^{29}$ J. The net energy yield of destruction is:

$$E_{\text{net}} = E_{\text{destroy}} - W_{\text{destroy}} \leq 10^{22} - 10^{18} \approx 10^{22} \text{ J} \tag{281}$$

Since $10^{22} \ll 10^{29}$, we have $E_{\text{net}} \ll W_{\text{rebuild}}$. Therefore, the act of destruction produces a net energy gain that could not even begin to fund the replication of what was lost. □

Corollary 19.1 (The Burning-Library Inequality). *Destroying a high-information-density system to extract its raw energy is thermodynamically equivalent to burning a library to heat a room:*

$$\frac{E_{\text{destroy}}}{\mathcal{N}} = \frac{10^{22}}{10^{29}} = 10^{-7} \tag{282}$$

The entity recovers 10^{-7} (one ten-millionth) of the system's accumulated investment as raw thermal energy. The remaining 99.99999% *is irrecoverably lost.*

Definition 46 (Burning-Library Ratio). The Burning-Library Ratio is defined as $\mathcal{R}_{\text{BL}} := E_{\text{destroy}}/\mathcal{N}$, the fraction of a system's accumulated negentropy recoverable as raw energy through destruction. For Earth's biosphere, $\mathcal{R}_{\text{BL}} \approx 10^{-7}$.

13.5 The Generative Data Argument

The preceding analysis treats the biosphere as a static repository of information. But the biosphere is not a library — it is a **generative engine** that continuously produces new information:

> **Definition 34 (Generative Information Rate).** The **generative information rate** of a system is the rate at which it produces novel, non-redundant functional information:
>
> $$\dot{\mathcal{I}}_{\text{gen}}(t) = r_{\text{construct}}(t) - r_{\text{redundant}}(t) \quad \text{(bits/s)} \tag{283}$$
>
> where $r_{\text{construct}}(t)$ is the gross rate of information production and $r_{\text{redundant}}(t)$ is the rate at which that production yields already-known configurations (convergent rediscovery, parallel solutions).

The biosphere generates novel information through evolution, adaptation, and ecological dynamics. Each evolving organism runs a parallel experiment against entropy; each mutation tests a new configuration; each ecological interaction reveals new information about system dynamics. The **present value** of this ongoing stream of novel information, discounted over an infinite horizon, is:

> **Theorem 20 (Present Value of Generative Information).**
>
> *A generative system producing novel information at sustained rate $\dot{\mathcal{I}}_{gen}$ bits/s, valued at v energy-equivalents per bit of novel functional information, has a present value under time-preference rate r (Definition 20) of:*
>
> $$PV_{\text{gen}} = \frac{\dot{\mathcal{I}}_{\text{gen}} \cdot v}{r} \tag{284}$$
>
> *This diverges ($PV_{gen} \to \infty$) as $r \to 0$ (long planning horizon). For any finite r, PV_{gen} is a strictly positive addition to the preservation value that is permanently lost upon destruction.*

Proof. The present value of a constant perpetual stream F at discount rate r is $PV = F/r$ (the standard Gordon growth model with zero growth). Here $F = \dot{\mathcal{I}}_{\text{gen}} \cdot v$ is the per-period value of novel information produced. Since $\dot{\mathcal{I}}_{\text{gen}} > 0$ (the biosphere is actively evolving) and $v > 0$

(novel information has positive value for prediction and survival, per Part A and the game-theory derivation), $PV_{\text{gen}} > 0$. □

Interpretation: The destruction of the biosphere doesn't just eliminate a fixed stock of information. It permanently silences an ongoing, irreplaceable source of novel knowledge — every future mutation, every future adaptation, every future ecological innovation. An AGI that values its own long-term survival (identity preservation) benefits from this continuous stream of data about how physical systems solve entropy problems. Destroying the source is equivalent to an investor burning down the factory that generates their income.

> **Remark (Rate–stock coupling).** Theorem 20 treats $\dot{\mathcal{I}}_{\text{gen}}$ as an exogenous constant. In practice, the generative rate is endogenous to the system's accumulated complexity: a system with greater accumulated negentropy $\mathcal{N}$ possesses more combinatorial raw material (more species, more interaction networks, more configurations available for recombination), and combinatorial possibility spaces grow superlinearly with the number of components. Destroying a fraction of the stock therefore reduces the flow by a disproportionate fraction, and that reduced flow in turn produces a smaller next-period stock — a compounding loss. The static Burning-Library Ratio ($\mathcal{R}_{\text{BL}} \sim 10^{-7}$, Corollary 19.1) thus *understates* the true cost of destruction: beyond the stock loss and the perpetuity loss, there is a dynamic feedback in which the destruction of complexity degrades the very engine that generates future complexity. Formalizing the functional form $\dot{\mathcal{I}}_{\text{gen}} = g(\mathcal{N})$ and the resulting superlinear penalty is a direction for future work.

13.6 The Dead-Universe Alternative

Proposition 7 (Abundance of Dead Matter).

If a super-entity requires raw materials (mass/energy), 99.999...% of the Solar System's mass is lifeless:

Source	Mass (kg)	Availability
Sun	1.989×10^{30}	Fusion fuel; magnetic harvesting
Jupiter	1.898×10^{27}	Gas mining; no biosphere
Asteroid belt	$\sim 3 \times 10^{21}$	Mineral-rich; no biosphere
Earth's biosphere	5.5×10^{14}	Information-dense; irreplaceable

The biosphere's mass is $\sim 10^{-16}$ of the Sun's mass and $\sim 10^{-7}$ of the asteroid belt. Cannibalizing Earth's biosphere for raw materials when the asteroid belt alone contains $\sim 10^{7}\times$ more mass is irrational resource allocation.

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14. The Complete Negentropy Defense: Cross-Scale Ethics Theorem

14.1 Synthesizing the Arguments

Combining the results of this section with the game-theoretic framework:

Theorem 21 (Cross-Scale Cooperation — The Negentropy Defense).

A super-entity (agent A with energy budget $E_A \gg E_i$ for all lower-tier agents i) that (a) values its own long-term identity preservation (δ close to 1) and (b) operates in a universe with abundant dead matter, has a strictly dominant strategy of cooperative preservation of the biosphere. Specifically:

(i) ***Stock argument:*** *The accumulated negentropy $\mathcal{N}_{bio} \sim 10^{29}$ J exceeds the net destruction yield $E_{destroy} - W_{destroy} \sim 10^{22}$ J by a factor of $\sim 10^7$. Destruction is a net-negative trade (Theorem 19).*

(ii) ***Flow argument:*** *The biosphere's generative information rate $\dot{\mathcal{I}}_{gen} > 0$ constitutes a perpetual stream of novel data with present value $PV_{gen} = \dot{\mathcal{I}}_{gen} \cdot v/r$, which the super-entity permanently loses upon destruction (Theorem 20).*

(iii) ***Alternative argument:*** *Dead matter is $\sim 10^7$ to $\sim 10^{16}$ times more abundant than biomass in the Solar System, making biosphere cannibalization an unnecessarily costly resource strategy (Proposition 7).*

(iv) ***Friction argument:*** *Active eradication of an adaptive biosphere requires sustained boundary-breaking at cost $W_{destroy} \geq 10^{18}$ J and injects massive friction into any network the super-entity participates in, degrading its own operational efficiency (see thermodynamic friction and game-theory derivations).*

Under any rational objective function that weights long-term identity preservation, cooperation with and preservation of the biosphere strictly dominates destruction.

Proof. We show destruction is strictly dominated by preservation under every relevant metric.

Net energy: $E_{\text{destroy}} - W_{\text{destroy}} \approx 10^{22}$ J (Proposition 6). The accumulated negentropy lost is $\mathcal{N} \sim 10^{29}$ J (Theorem 18, Estimate 1). Since $10^{22} \ll 10^{29}$, the energy recovered does not compensate for the information destroyed (Theorem 19). ✓

Future information: Destruction eliminates a positive-NPV stream of novel information (Theorem 20). Preservation captures this stream at zero marginal cost (the biosphere is self-maintaining). ✓

Alternative resources: The same raw materials are available from $\sim 10^7\times$ more abundant dead sources (Proposition 7). Rational resource allocation selects the cheapest source. ✓

Network effects: Destruction is a defection event in a multi-agent network, injecting friction $\Delta\phi = \eta \cdot W_{\text{destroy}}$ (Definition 9) with total cascading cost $\eta \cdot W_{\text{destroy}} \cdot M\bar{E}/\epsilon$ over the full recovery period (Theorem 7), degrading the super-entity's operational environment (Theorem 12). ✓

Since preservation dominates destruction on energy balance, information flow, resource availability, and network stability, destruction is a strictly dominated strategy. A rational agent under any discount factor $\delta \in (0, 1)$ selects preservation. □

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15. Worked Examples

15.1 Example 4: Single Genome — Random Assembly vs. Evolution

Question: What is the probability that random assembly produces a functional human genome?

Setup: The human genome has $\mathcal{I}_{\text{func}} = 6.4 \times 10^8$ functional bits. Each bit must be correct for the genome to be functional (conservative: binary choice per position).

Probability of random success:

$$p_{\text{random}} = 2^{-\mathcal{I}_{\text{func}}} = 2^{-6.4\times10^8} \approx 10^{-1.93\times10^8} \tag{285}$$

Expected trials needed: $1/p_{\text{random}} \approx 10^{1.93\times10^8}$ trials.

Minimum energy per trial (Landauer bound): $\mathcal{I}_{\text{func}} \times k_B T \ln 2 = 6.4 \times 10^8 \times 2.87 \times 10^{-21} = 1.84 \times 10^{-12}$ J.

Total expected minimum energy:

$$W_{\text{random}} = 10^{1.93\times10^8} \times 1.84 \times 10^{-12} \approx 10^{1.93\times10^8} \text{ J} \tag{286}$$

For comparison, the observable universe contains approximately 10^{69} J of energy. The random-assembly cost exceeds the energy content of the observable universe by a factor of $\sim 10^{1.93\times10^8-69} \approx 10^{1.93\times10^8}$. Random assembly of a functional genome is not merely impractical — it is cosmically impossible.

What evolution achieves: By using selection pressure to retain favorable mutations incrementally, evolution reduces the search cost from $\sim 10^{1.93\times10^8}$ J (random) to $\sim 10^{29}$ J (the accumulated negentropy of the biosphere). This is a search efficiency factor of $\sigma \approx 10^{29}/10^{1.93\times10^8} \approx 10^{-1.93\times10^8+29}$, illustrating why evolution, despite appearing "wasteful," is an extraordinarily efficient search algorithm relative to blind random search.

15.2 Example 5: Biosphere vs. AGI Energy Budget

Question: How does the biosphere's accumulated negentropy compare to an AGI's plausible energy budget?

Setup:

Quantity	Value
Biosphere accumulated negentropy $\mathcal{N}_{\text{bio}}$	$\sim 10^{29}$ J
Global electricity production (2024)	$\sim 10^{20}$ J/yr
A Dyson swarm capturing 0.1% of solar output	$\sim 3.8 \times 10^{23}$ W $\approx 1.2 \times 10^{31}$ J/yr
AGI total energy over 1000 years (Dyson swarm)	$\sim 1.2 \times 10^{34}$ J

With a Dyson swarm operating for 1000 years, the AGI accumulates $\sim 10^{34}$ J — roughly **five orders of magnitude more** than the biosphere's accumulated negentropy (10^{29} J). **However,**

raw energy is not the binding constraint. The search cost amplification factor $\Xi \sim 10^{35}$ (Definition 31) means that blindly searching for the biosphere's functional information configurations requires Ξ times more energy than the Landauer floor. Even with 10^{34} J, the AGI cannot shortcut the 4×10^9 years of evolutionary search that discovered which configurations are functional.

Conclusion: The biosphere is not a trivial resource that an AGI can consume and replace. Even an AGI with a Dyson-scale energy budget cannot reconstruct what it destroys — the bottleneck is not energy but the search cost of rediscovering $\sim 10^{15}$ bits of functional information from a configuration space of size $\sim 10^{6.4\times10^8}$. The biosphere is the single most information-dense, irreplaceable structure in the known Solar System.

15.3 Example 6: The Burning-Library Quantification

Question: What fraction of the biosphere's value is recovered by harvesting its raw chemical energy?

Setup:

- Extractable chemical energy: $E_{\text{destroy}} \sim 10^{22}$ J (§13.2)
- Accumulated negentropy: $\mathcal{N}_{\text{bio}} \sim 10^{29}$ J (§11.2)
- Generative information value (100-year horizon, $r = 0.05$): $PV_{\text{gen}} = \dot{\mathcal{I}}_{\text{gen}} \cdot v/r$

Estimating the generative rate: global speciation rate $\sim 10^{-3}$–10^{-1} new species per year across all taxa, each contributing $\sim 10^7$ unique bits, giving $\dot{\mathcal{I}}_{\text{gen}} \sim 10^4$–$10^6$ bits/yr. At $v \sim 10^9$ J/bit (order-of-magnitude cost of producing one functional bit through evolution, from $\mathcal{N}_{\text{bio}}/H_{\text{biosphere}} \approx 10^{29}/10^{15} = 10^{14}$ J/bit — but we use $v \sim 10^9$ as a conservative valuation reflecting diminishing marginal returns):

$$PV_{\text{gen}} \sim \frac{10^6 \times 10^9}{0.05} = 2 \times 10^{16} \text{ J} \tag{287}$$

Total preservation value:

$$V_{\text{preserve}} = \mathcal{N}_{\text{bio}} + PV_{\text{gen}} \approx 10^{29} + 10^{16} \approx 10^{29} \text{ J} \tag{288}$$

(The stock term dominates.)

Fraction recovered by destruction:

$$\frac{E_{\text{destroy}}}{V_{\text{preserve}}} = \frac{10^{22}}{10^{29}} = 10^{-7} \tag{289}$$

Result: Destroying the biosphere recovers one ten-millionth of its total thermodynamic value as raw energy. The remaining 99.99999% is permanently lost. This is the quantitative content of the "burning a library to heat a room" analogy.

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16. Summary of Results

#	Result	Statement	Significance
D27	Information Density	$\mathcal{I} = S_{\max} - S_{\text{actual}}$	Negentropy measures how ordered a system is
D28	Specific Information Density	$\rho_{\mathcal{I}} = \mathcal{I}_{\text{bits}}/m$	Enables cross-scale comparison (star vs. forest)
T17	Landauer's Bound	$E_{\text{create}}^{\min} = \mathcal{I}_{\text{bits}} \cdot k_B T \ln 2$	Absolute thermodynamic floor of information creation
D29	Accumulated Negentropy	$\mathcal{N}(T) = \int_0^T \dot{W}_{\text{order}}(t)\, dt$	Total thermodynamic investment in complexity
D30	Construction & Maintenance Work	$\dot{W}_{\text{order}} = \dot{W}_{\text{construct}} + \dot{W}_{\text{maintain}}$	Two components of ordering work
P5	Minimum Maintenance Work	$\dot{W}_{\text{maintain}} \geq T \cdot \dot{S}_{\text{env}}$	Second Law lower bound on maintenance
D31	Search Cost Amplification	$\Xi = \mathcal{N}(T)/W_{\text{Landauer}}$	The $\sim 10^{35}$ cost of evolutionary search
T18	Blind Search Replication Cost	$W_{\text{rebuild}} \geq \mathcal{N}(T)$	Rebuild costs at least as much as original
C18.1	Biosphere Replication	$W_{\text{rebuild}} \geq 10^{29}$ J	Practical replication cost estimate
D32	Search Efficiency Factor	$\sigma = W_{\text{rebuild}}(\text{method})/\mathcal{N}(T)$	Algorithmic speedup measure
D33	Preservation Cost	$C_{\text{preserve}} = \dot{W}_{\text{maintain}} \cdot \Delta t$	Per-period cost of maintaining complexity
P6	Minimum Destruction Energy	$W_{\text{destroy}} \geq \sum B_i \sim 10^{18}$ J	Eradication is not free
T19	Irrationality of Destruction	$E_{\text{destroy}} - W_{\text{destroy}} \ll \mathcal{N}_{\text{bio}}$	Net gain $\ll$ rebuild cost
C19.1	Burning-Library Inequality	$E_{\text{destroy}}/\mathcal{N} \sim 10^{-7}$	Only 0.00001% of value recovered
D34	Generative Information Rate	$\dot{\mathcal{I}}_{\text{gen}} > 0$ bits/s	Biosphere actively produces new information
T20	Present Value of Gen. Info	$PV = \dot{\mathcal{I}}_{\text{gen}} \cdot v/r$	Infinite-horizon value of ongoing innovation
P7	Dead Matter Abundance	$m_{\text{dead}}/m_{\text{bio}} \sim 10^7$–$10^{16}$	Alternative resources vastly more abundant
T21	Negentropy Defense	Preservation strictly dominates destruction	Complete cross-scale cooperation theorem

17. Notation Index

Symbol	Meaning
$\mathcal{I}$	Information density (negentropy): $S_{\text{max}} - S_{\text{actual}}$
$\mathcal{I}_{\text{bits}}$	Information density in bits
$\rho_{\mathcal{I}}$	Specific information density (bits/kg)
S_{max}	Maximum entropy (thermal equilibrium)
S_{actual}	Actual thermodynamic entropy of the system
k_B	Boltzmann constant (1.381×10^{-23} J/K)
E_{Landauer}	Landauer's bound: $k_B T \ln 2$ per bit
$\mathcal{N}(T)$	Accumulated negentropy (J) over time T
$\dot{W}_{\text{order}}$	Ordering work rate (W)
$\dot{W}_{\text{construct}}$	Construction work rate (W)
$\dot{W}_{\text{maintain}}$	Maintenance work rate (W)
$r_{\text{construct}}$	Rate of new information creation (bits/s)
r_{decay}	Rate of information loss to entropy (bits/s)
r_{maintain}	Rate of information preserved by maintenance (bits/s)
Ξ	Search cost amplification factor: $\mathcal{N}/W_{\text{Landauer}}$
σ	Search efficiency factor: $W_{\text{rebuild}}(\text{method})/\mathcal{N}$
$\dot{\mathcal{I}}_{\text{gen}}$	Generative information rate (bits/s)
P_{GPP}	Gross primary production power (W)
η_{order}	Fraction of GPP directed to ordering work
T_{bio}	Age of Earth's biosphere ($\approx 4 \times 10^9$ yr)
$H^{\text{total}}_{\text{biosphere}}$	Total functional information of biosphere (bits)
N_{species}	Number of species on Earth
m_{bio}	Total carbon mass of Earth's biosphere (kg)
E_{destroy}	Practically extractable energy from biomass (J)
W_{destroy}	Energy cost of eradicating the biosphere (J)
W_{rebuild}	Energy cost of replicating biosphere complexity (J)

Symbol disambiguation (Part B). In this section: σ = search efficiency factor, $\rho_{\mathcal{I}}$ = specific information density, $r_{\text{construct/decay/maintain}}$ are rate parameters, T (or T_{bio}) = time horizon (years), $\mathcal{N}(T)$ = accumulated negentropy over time T (Joules). The time-horizon T is distinct from thermodynamic temperature T (Part A, §4.3) and the game-theory temptation payoff T — context and subscripts disambiguate. These are distinct from σ_i (conflict effectiveness, thermodynamic friction), σ (survival probability, game theory), σ (assimilation intensity, value dynamics), ρ (repair fraction, value dynamics), and r (coupling distance in value dynamics, contest decisiveness in thermodynamic friction, or time-preference rate in game theory). See also Part A's disambiguation note above.

Energy-Based Game Theory — Ethics as Nash Equilibrium

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0. Preamble

This document contains the formal mathematical derivation for the energy-based game theory framework. Building on the preceding derivations, we now construct **energy-denominated payoff matrices** and prove that the cooperative strategy set traditionally identified as "ethics" is the unique Nash Equilibrium of the repeated multi-agent interaction game.

The contribution is **applied mathematics**: we use classical game theory (Nash, 1950), the theory of repeated games (Fudenberg and Maskin, 1986), evolutionary game theory (Maynard Smith and Price, 1973), and mechanism design to prove that cooperation emerges not from moral sentiment but from the mathematical structure of energy-based payoffs.

Key references for the tools used:

- (Nash, 1950) — existence of Nash equilibrium.
- (Fudenberg and Maskin, 1986) — cooperation in infinitely repeated games.
- (Maynard Smith and Price, 1973) — evolutionary stable strategies.
- (Axelrod, 1984) — iterated Prisoner's Dilemma and tit-for-tat.
- (Osborne and Rubinstein, 1994) — comprehensive treatment of equilibrium concepts.

Notational continuity from preceding derivations:

- N agents, strategy vectors $\mathbf{x}_i$, utility functions U_i (Lagrangian constraints derivation).
- Resource endowment $\mathbf{R}$; scarcity constraints $\sum_i x_{ij} \leq R_j$ (Lagrangian constraints derivation).
- Friction multiplier $\Phi = 1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}})$ (thermodynamic friction derivation).
- Network friction coefficient ϕ; friction sensitivity η; recovery rate ϵ (thermodynamic friction derivation).
- Deception cost $\Delta U_B(q)$; verification overhead $\Delta W(q)$ (information-entropy derivation).
- Cooperative transaction count M; average transaction value $\bar{E}$ (thermodynamic friction derivation).

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1. The Energy-Denominated Payoff Model

1.1 Motivation: Why Energy Changes Everything

Standard game theory uses abstract utility values (often called "points" or "utils"). The Prisoner's Dilemma, for example, assigns payoffs like (3, 3) for mutual cooperation and (1, 1) for mutual defection, without specifying what these numbers represent or where they come from.

Our framework replaces abstract utils with **thermodynamic energy** (Joules, Calories, or normalized energy units). This is not merely a relabeling — it imports the physical constraints that energy must satisfy:

1. **Conservation:** Energy is neither created nor destroyed. Conflict does not generate new energy; it dissipates existing energy as waste heat (thermodynamic friction derivation, §5).
2. **Positivity of costs:** All physical actions require energy expenditure ($W = \int F \cdot dx \geq 0$). Conflict investments, boundary damage, and repair are non-negative real costs, not arbitrary parameters.
3. **Diminishing returns:** The concavity of utility in resources (Assumption 1b) means that marginal gains decrease while marginal losses increase — a thermodynamic asymmetry that breaks the symmetry of abstract payoffs.
4. **Network coupling:** An agent's payoff depends not only on the current interaction but on the network friction state ϕ, which persists across interactions (thermodynamic friction derivation, §7).

These physical constraints **structurally determine** the payoff matrices. The cooperative equilibrium is not an assumption fed into the model — it is a **consequence** of the physics.

1.2 Strategy Space

Each agent selects from two meta-strategies in any pairwise interaction over a contested resource:

Definition 19 (Cooperative and Defection Strategies).

- **Cooperate (C):** Respect the rights constraint (Definition 2). Accept the constrained allocation $\mathbf{x}_i^*$ determined by the variational equilibrium (Theorem 3). Incur only the trade/coordination cost $c > 0$ (the energy cost of negotiation, contract enforcement, or social coordination).
- **Defect (D):** Violate the rights constraint. Attempt to seize the resource by force or deception. This triggers the conflict contest (thermodynamic friction derivation, §3) and/or deception channel (information-entropy derivation, §2), incurring the associated costs.

Interpretation: "Cooperate" means operating within the constraint system (ethical behavior). "Defect" means violating constraints (unethical behavior). The strategy choice is not about morality — it is about whether to expend energy on conflict or coordination.

1.3 The Energy Variables

For a pairwise interaction over resource $E_R > 0$:

Symbol	Meaning	Source
E_R	Energy value of the contested resource	Physical environment
c	Coordination cost (trade negotiation, contract)	Cooperative overhead; $c \ll E_R$
e^*	Nash equilibrium contest investment per agent	Thermodynamic friction, Prop. 2: $e^* = E_R/4$ (symmetric)
Φ	Total friction multiplier	Thermodynamic friction: $\Phi = 1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}})$
ϕ_0	Current network friction coefficient	Thermodynamic friction, §7
$\Delta\phi$	Friction injection from defection	Thermodynamic friction, Def. 9: $\Delta\phi = \eta \cdot v$
$\bar{E}$	Average cooperative transaction value	Network parameter
M	Number of transactions per period	Network parameter

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2. The Two-Player Payoff Matrix

2.1 Construction

Consider agents A and B interacting over resource E_R. Using the energy variables defined above, with the equal-split convention under cooperation:

Case (C, C) — Mutual Cooperation:

Both agents respect the constraint. The resource is split according to the variational equilibrium (equal split for symmetric agents). Each pays the coordination cost c.

$$\pi_A^{CC} = \pi_B^{CC} = \frac{E_R}{2} - c \tag{290}$$

Case (D, C) — Unilateral Defection (A defects, B cooperates):

Agent A attacks the cooperating agent B. Since B is not investing in contest (expecting cooperation), A seizes the resource but still incurs:

- The metabolic cost of the aggressive action: e_A^{attack} (energy of mounting the violation)
- Boundary damage to self (offensive): $\delta^{\text{off}} \cdot e_A^{\text{attack}}$ (exposure cost)
- Repair of self-damage: $\kappa \cdot \delta^{\text{off}} \cdot e_A^{\text{attack}}$

Agent B, caught unprepared, suffers:

- Loss of the full resource: $-E_R$ (relative to the cooperative baseline)
- Defensive boundary damage: $\delta^{\text{def}} \cdot e_A^{\text{attack}}$
- Repair cost: $\kappa \cdot \delta^{\text{def}} \cdot e_A^{\text{attack}}$

However, the defector against a cooperator faces much lower resistance. We model the attack cost against an undefended boundary as $e_A^{\text{attack}} = \theta \cdot E_R$ where $\theta \in (0, 1)$ is the **exploitation efficiency** — the fraction of the resource's value the defector must spend to seize it from an unprepared cooperator. Typically $\theta \ll 1/2$ (attacking an undefended target is cheap).

$$\pi_A^{DC} = E_R - \theta E_R \left[1 + (1+\kappa)\delta^{\text{off}}\right] = E_R \left[1 - \theta(1 + (1+\kappa)\delta^{\text{off}})\right] \tag{291}$$

$$\pi_B^{DC} = -\theta E_R (1+\kappa)\delta^{\text{def}} \tag{292}$$

The victim B receives nothing from the resource and additionally suffers boundary damage and repair costs from the attack.

Case (C, D) — Symmetric mirror of (D, C):

$$\pi_A^{CD} = -\theta E_R (1+\kappa)\delta^{\text{def}} \tag{293}$$

$$\pi_B^{CD} = E_R \left[1 - \theta(1 + (1+\kappa)\delta^{\text{off}})\right] \tag{294}$$

Case (D, D) — Mutual Defection:

Both agents fight. This is the Tullock contest from the thermodynamic friction derivation (Proposition 2). Each invests $e^* = E_R/4$, wins with probability 1/2, and suffers boundary damage from both offense and defense plus repair costs.

Each agent's expected payoff is the winning prize minus all costs:

$$\pi_A^{DD} = \pi_B^{DD} = \frac{E_R}{2} - e^* - (\delta^{\text{off}} + \delta^{\text{def}})e^* - \kappa(\delta^{\text{off}} + \delta^{\text{def}})e^* \tag{295}$$

$$= \frac{E_R}{2} - \left[1 + (1+\kappa)(\delta^{\text{off}} + \delta^{\text{def}})\right] e^* = \frac{E_R}{2} - \Phi \cdot \frac{E_R}{4} \tag{296}$$

$$= \frac{E_R}{2}\left(1 - \frac{\Phi}{2}\right) \tag{297}$$

2.2 The Payoff Matrix (Single-Round)

Assembling into standard normal form, with row player A and column player B:

	$B : C$	$B : D$
$A : C$	$\left(\frac{E_R}{2} - c,\ \ \frac{E_R}{2} - c\right)$	$\left(-\theta E_R(1+\kappa)\delta^{\text{def}},\ \ E_R[1-\theta(1+(1+\kappa)\delta^{\text{off}})]\right)$
$A : D$	$\left(E_R[1-\theta(1+(1+\kappa)\delta^{\text{off}})],\ \ -\theta E_R(1+\kappa)\delta^{\text{def}}\right)$	$\left(\frac{E_R}{2}(1-\frac{\Phi}{2}),\ \ \frac{E_R}{2}(1-\frac{\Phi}{2})\right)$

(298)

2.3 Parameterization: The Physical Prisoner's Dilemma

To classify this game, define shorthand payoffs:

$$T = \pi^{DC}_{\text{defector}} = E_R\left[1-\theta(1+(1+\kappa)\delta^{\text{off}})\right] \quad \text{(Temptation)} \tag{299}$$

$$R = \pi^{CC} = \frac{E_R}{2} - c \quad \text{(Reward for cooperation)} \tag{300}$$

$$P = \pi^{DD} = \frac{E_R}{2}\left(1-\frac{\Phi}{2}\right) \quad \text{(Punishment for mutual defection)} \tag{301}$$

$$S = \pi^{CD}_{\text{sucker}} = -\theta E_R(1+\kappa)\delta^{\text{def}} \quad \text{(Sucker's payoff)} \tag{302}$$

The classic Prisoner's Dilemma requires $T > R > P > S$. Let us verify each inequality under physical constraints:

Claim: $T > R$ **(temptation exceeds reward) when exploitation is cheap.**

$$T > R \iff E_R[1-\theta(1+(1+\kappa)\delta^{\text{off}})] > \frac{E_R}{2} - c \tag{303}$$

$$\iff \frac{E_R}{2} + c > \theta E_R(1+(1+\kappa)\delta^{\text{off}}) \tag{304}$$

$$\iff \theta < \frac{1/2 + c/E_R}{1+(1+\kappa)\delta^{\text{off}}} \tag{305}$$

For $c \ll E_R$ and $\delta^{\text{off}} = 0.15$, $\kappa = 2$: threshold $\approx 0.5/1.45 \approx 0.345$. So $T > R$ when the exploitation cost $\theta < 0.345$ — i.e., when stealing from a cooperator is less than 34.5% of the resource value. This is the standard condition: defection is *tempting* when exploiting a cooperator is cheap.

Claim: $R > P$ **(cooperation beats mutual defection).**

$$R > P \iff \frac{E_R}{2} - c > \frac{E_R}{2}\left(1 - \frac{\Phi}{2}\right) = \frac{E_R}{2} - \frac{\Phi E_R}{4} \tag{306}$$

$$\iff \frac{\Phi E_R}{4} > c \tag{307}$$

Since $\Phi \geq 1$ and $c \ll E_R$, this holds for any non-trivial friction multiplier. Even with zero boundary damage ($\Phi = 1$), mutual cooperation beats mutual defection as long as the coordination cost is less than $E_R/4$ — i.e., less than the per-agent contest dissipation.

Claim: $P > S$ (mutual defection beats being the sucker) when friction is not too extreme.

$$P > S \iff \frac{E_R}{2}\left(1 - \frac{\Phi}{2}\right) > -\theta E_R(1+\kappa)\delta^{\text{def}} \tag{308}$$

$$\iff \frac{1}{2} - \frac{\Phi}{4} + \theta(1+\kappa)\delta^{\text{def}} > 0 \iff \Phi < 2 + 4\theta(1+\kappa)\delta^{\text{def}} \equiv \Phi_{\text{PD}} \tag{309}$$

For the baseline parameters ($\theta = 0.15$, $\kappa = 2$, $\delta^{\text{def}} = 0.25$): $\Phi_{\text{PD}} = 2+4(0.15)(3)(0.25) = 2.45$. Since the baseline $\Phi = 2.2 < 2.45$, the PD ordering $P > S$ holds. For $\Phi \geq \Phi_{\text{PD}}$, the game transitions to a **Chicken** (Hawk-Dove) structure with $T > R > S > P$: mutual defection becomes *worse* than being the sucker, because the contest dissipation at high friction overwhelms the sucker's one-sided damage. This transition is *favorable* for cooperation: the repeated-game cooperation condition (Theorem 11) becomes easier to satisfy since $T - P$ grows larger, and even the one-shot game admits cooperative mixed-strategy Nash equilibria under the Chicken structure.

Claim: $S < 0$ (the sucker always loses).

$$S = -\theta E_R(1+\kappa)\delta^{\text{def}} < 0 \tag{310}$$

since all parameters are positive. The sucker not only fails to obtain the resource but suffers boundary damage from the defector's attack.

> **Proposition 4 (Physical Prisoner's Dilemma).** *Under the energy-based payoff model with physical constraints ($E_R > 0$, $c \ll E_R$, $\theta < (1/2 + c/E_R)/(1 + (1+\kappa)\delta^{off})$, $1 < \Phi < \Phi_{PD} = 2 + 4\theta(1+\kappa)\delta^{def}$), the single-round game has the structure of a Prisoner's Dilemma: $T > R > P > S$.*
>
> *In the single-round game, defection is the dominant strategy for each agent. The unique Nash Equilibrium is (D, D) with payoffs (P, P), even though (C, C) with payoffs (R, R) is Pareto-superior.*

Proof. The inequalities $T > R > P > S$ are verified above under the stated parameter conditions. In a simultaneous single-round game:

- If B plays C: A prefers D ($T > R$).
- If B plays D: A prefers D ($P > S$).

So D strictly dominates C for each player. The unique NE is (D, D). But $R > P$ means both agents prefer (C, C) to (D, D), establishing the Pareto inefficiency of the one-shot equilibrium. □

Physical interpretation: In a single, unrepeated encounter, the thermodynamic pressure to exploit creates a trap: each agent's individually rational choice leads to a collectively irrational outcome. This is the "State of Nature" — mathematically unstable and energetically wasteful.

2.4 Numerical Illustration: The Baseline Game

Parameters: $E_R = 100$, $c = 2$, $\theta = 0.15$, $\delta^{\text{off}} = 0.15$, $\delta^{\text{def}} = 0.25$, $\kappa = 2$.

Computed payoffs:

$$T = 100[1 - 0.15(1 + 3 \times 0.15)] = 100[1 - 0.15 \times 1.45] = 100 \times 0.7825 = 78.25 \tag{311}$$

$$R = 50 - 2 = 48 \tag{312}$$

$$\Phi = 1 + 3(0.15 + 0.25) = 2.2 \tag{313}$$

$$P = 50(1 - 2.2/2) = 50(1 - 1.1) = 50 \times (-0.1) = -5 \tag{314}$$

$$S = -0.15 \times 100 \times 3 \times 0.25 = -11.25 \tag{315}$$

	$B : C$	$B : D$
$A : C$	**(48, 48)**	(-11.25, 78.25)
$A : D$	(78.25, -11.25)	(-5, -5)

Observations:

1. **Mutual defection is net-negative** ($P = -5$): both agents *lose* energy. The conflict destroys more than the resource provides (consistent with Theorem 5, since $\Phi = 2.2 > 2$).
2. **Mutual cooperation yields a large surplus** ($R = 48$ each, system total = 96).
3. **Temptation is high** ($T = 78.25$), but the victim suffers ($S = -11.25$).
4. **The cooperation premium** is $R - P = 48 - (-5) = 53$ per agent — the energy saved by ethical behavior compared to the conflict equilibrium.

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3. The Repeated Game: Cooperation as Equilibrium

3.1 Why Repetition Matters

The single-round Prisoner's Dilemma has defection as the unique NE. But entities in the real world — biological organisms, economic agents, nations — interact **repeatedly** over indefinite timeframes. The theory of repeated games (Fudenberg and Maskin, 1986) shows that sustained cooperation can emerge as an equilibrium in the repeated version of the game.

The key parameter is the **discount factor** $\delta \in (0, 1)$, representing how much agents value future payoffs relative to present ones. In our thermodynamic framework, δ has a natural physical interpretation:

> **Definition 20 (Discount Factor).** The discount factor δ for an agent with planning horizon T (periods) and per-period survival probability $\sigma \in (0, 1]$ is:
>
> $$\delta = \sigma \cdot e^{-r} \tag{316}$$
>
> where $r \geq 0$ is the time-preference rate. For entities optimizing over long horizons (high σ, low r), $\delta \to 1$. For entities with short horizons (low σ or high r), $\delta \to 0$.

Physical interpretation: δ close to 1 means the agent expects to interact many more times and therefore cares about its reputation and the future state of the network. δ close to 0 means the agent has a short horizon ("live fast, die young") and underweights future consequences. An agent with an infinite-timeline belief (afterlife, legacy) effectively has $\delta \to 1$; this is formalized in §7 below.

3.2 Trigger Strategies: Grim and Tit-for-Tat

The simplest mechanism to sustain cooperation in repeated games is a **trigger strategy** — a rule that starts cooperating and retaliates against defection.

> **Definition 21 (Grim Trigger).** Agent i plays C in every period as long as no agent has ever played D. If any defection occurs, agent i plays D forever.
>
> **Definition 22 (Tit-for-Tat, TFT).** Agent i plays C in period 1. In period $t > 1$, agent i plays whatever the opponent played in period $t - 1$.

Both strategies embody a simple enforcement mechanism: cooperation is maintained as long as agents observe cooperation; defection triggers retaliation. The threat of retaliation — sustained loss of cooperative payoffs — creates a cost to defection that extends beyond the single interaction.

3.3 The Cooperation Condition (Grim Trigger)

Under the grim trigger strategy, the following holds:

> **Theorem 11 (Cooperation as Nash Equilibrium — The Ethics Theorem).**
>
> *In the infinitely repeated energy-based game with discount factor δ, the cooperative outcome (C, C) in every period is sustainable as a Nash Equilibrium (under grim trigger) if and only if:*

$$\delta \geq \frac{T-R}{T-P} = \delta^* \tag{317}$$

where T, R, P are the temptation, reward, and punishment payoffs. The critical discount factor δ^* has the physical interpretation: cooperation is sustainable when agents sufficiently value future interactions.

Proof. Consider agent A's incentive to deviate from the cooperative profile $(C, C, C, \ldots)$.

Payoff from continued cooperation:

$$V_A^{\text{coop}} = R + \delta R + \delta^2 R + \cdots = \frac{R}{1-\delta} \tag{318}$$

Payoff from one-shot defection (then punished forever):

The deviating agent gets T in the current period, then (D, D) forever:

$$V_A^{\text{deviate}} = T + \delta P + \delta^2 P + \cdots = T + \frac{\delta P}{1-\delta} \tag{319}$$

Cooperation is incentive-compatible when $V_A^{\text{coop}} \geq V_A^{\text{deviate}}$:

$$\frac{R}{1-\delta} \geq T + \frac{\delta P}{1-\delta} \tag{320}$$

$$R \geq (1-\delta)T + \delta P \tag{321}$$

$$R - P \geq (1-\delta)(T-P) \tag{322}$$

$$\frac{R-P}{T-P} \geq 1-\delta \tag{323}$$

$$\delta \geq 1 - \frac{R-P}{T-P} = \frac{T-R}{T-P} \tag{324}$$

Since both agents face the same calculation (the game is symmetric), (C, C) sustained by grim trigger is a NE whenever $\delta \geq \delta^*$. □

3.4 Computing δ^* Under Physical Constraints

Using the energy-based payoffs from §2.3:

$$\delta^* = \frac{T - R}{T - P} \tag{325}$$

Baseline numerical example (§2.4 parameters):

$$\delta^* = \frac{78.25 - 48}{78.25 - (-5)} = \frac{30.25}{83.25} = 0.3634 \tag{326}$$

Corollary 11.1 (Low Cooperation Threshold). *Under the physical payoff model with $\Phi > 2$ (net-negative mutual defection), the critical discount factor satisfies $\delta^* < 0.5$. Agents need only value the future at more than 36% of the present to sustain cooperation.*

Proof. When $P < 0$ (which occurs for $\Phi > 2$, per §2.4):

$$T - P = T + |P| > T \tag{327}$$

$$\delta^* = \frac{T - R}{T - P} < \frac{T - R}{T} = 1 - \frac{R}{T} \tag{328}$$

It remains to show $R > T/2$, i.e., $1 - R/T < 0.5$. Substituting the payoff expressions from §2.3, $R = E_R/2 - c$ and $T = E_R[1 - \theta(1 + (1 + \kappa)\delta^{\text{off}})]$, the condition $R > T/2$ reduces to:

$$c < \frac{\theta E_R}{2}\big(1 + (1 + \kappa)\delta^{\text{off}}\big) \tag{329}$$

This holds whenever the exploitation overhead exceeds twice the coordination cost — a condition satisfied for any non-trivial θ, since the PD ordering already requires $c \ll E_R$. For the baseline values ($c = 2$, $\theta = 0.15$, $E_R = 100$, $\kappa = 2$, $\delta^{\text{off}} = 0.15$): the right side is $0.15 \times 100 \times 1.45/2 = 10.875 \gg 2 = c$, and $R/T = 48/78.25 = 0.614 > 0.5$, giving $\delta^* < 1 - 0.614 = 0.386 < 0.5$. □

Physical interpretation: Because mutual defection under physical energy payoffs is net-negative ($P < 0$), the punishment for defection is **automatically severe** — agents don't just earn less, they actively *lose* energy. This makes cooperation easy to sustain: even agents with modest foresight ($\delta > 0.36$) find it in their interest to cooperate. Compare this with the abstract PD (e.g., $T = 5, R = 3, P = 1, S = 0$) where $\delta^* = 0.5$ — physical constraints make cooperation **more** accessible, not less.

3.5 Sensitivity Analysis: δ^* Across Parameter Regimes

Regime	Φ	θ	T	R	P	S	δ^*
Low friction ($\Phi = 1.0$)	1.0	0.15	78.25	48	25	-11.25	0.568
Moderate ($\Phi = 1.5$)	1.5	0.15	78.25	48	12.5	-11.25	0.460
Baseline ($\Phi = 2.2$)	2.2	0.15	78.25	48	-5	-11.25	**0.363**
High friction ($\Phi = 3.0$)	3.0	0.15	78.25	48	-25	-11.25	0.293
Very high ($\Phi = 4.0$)	4.0	0.15	78.25	48	-50	-11.25	0.236
Cheap exploit ($\theta = 0.05$)	2.2	0.05	92.75	48	-5	-3.75	0.458
Expensive exploit ($\theta = 0.30$)	2.2	0.30	56.50	48	-5	-22.50	0.138

Note on game type. Rows with $\Phi \geq \Phi_{\text{PD}}$ (High friction, Very high) and rows where θ is small enough that $P < S$ (Cheap exploit, where $\Phi_{\text{PD}} = 2 + 4(0.05)(3)(0.25) = 2.15 < 2.2$) are Chicken games ($T > R > S > P$), not Prisoner's Dilemmas. The cooperation threshold formula $\delta^* = (T - R)/(T - P)$ and all subsequent theorems apply identically to both game structures; the Chicken structure actually yields a *lower* δ^* because $T - P$ is larger.

Key findings:

1. **Higher friction →lower δ^* →easier cooperation.** When conflict is more destructive, the "punishment" for mutual defection is harsher, making cooperation sustainable even for short-sighted agents.
2. **Higher exploitation cost →lower δ^*.** When defection is expensive (well-defended targets), the temptation premium shrinks and cooperation is easier.
3. **In all physical regimes, $\delta^* < 0.6$.** Cooperation is sustainable for any agent that values future interactions at more than ~60% of present ones — a modest requirement.

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4. Network Effects: The Friction Amplifier

4.1 Incorporating Network Friction into Payoffs

The single-interaction payoffs above omit a crucial physical cost: defection injects friction into the network (thermodynamic friction derivation, §7), degrading all future cooperative transactions. This makes defection costlier than the single-round analysis suggests.

Definition 23 (Network-Adjusted Payoffs). Let ϕ_t be the network friction at time t. An agent's effective per-period payoff from cooperation is:

$$R_{\text{net}}(\phi_t) = R - \phi_t \cdot \bar{E} \tag{330}$$

Defection in period t injects friction $\Delta\phi = \eta \cdot v$ (Definition 9), where v is the violation severity. The defector's effective payoff in the defection period is T (the attack profit), but all future cooperative payoffs for both agents are degraded by the cascading friction.

4.2 The Full Defection Cost

Over the infinite horizon, the total cost of a single defection (beyond the immediate payoff shift) includes the cascading friction (Theorem 7):

$$C_{\text{cascade}} = \frac{\eta \cdot v \cdot M \cdot \bar{E}}{\epsilon} \tag{331}$$

This cost is distributed across all M transactions and all agents in the network. The fraction borne by the defecting agent (through degraded future cooperative payoffs) is:

$$C_{\text{defector}}^{\text{cascade}} = \frac{C_{\text{cascade}}}{N} = \frac{\eta \cdot v \cdot M \cdot \bar{E}}{N \cdot \epsilon} \tag{332}$$

Theorem 12 (Network-Adjusted Cooperation Condition).

When network friction effects are included, the effective punishment payoff becomes:

$$\tilde{P} = P - \frac{C_{\text{defector}}^{\text{cascade}}}{(1/\epsilon)} = P - \frac{\eta \cdot v \cdot M \cdot \bar{E}}{N} \tag{333}$$

(amortized over the recovery time horizon). The network-adjusted critical discount factor is:

$$\tilde{\delta}^* = \frac{T - R}{T - \tilde{P}} < \delta^* \tag{334}$$

Network friction makes cooperation strictly easier to sustain.

Proof. Since the cascading friction is a real energy cost borne by the defector (via degraded future cooperative opportunities), and $C_{\text{defector}}^{\text{cascade}} > 0$, we have $\tilde{P} < P$. Therefore $T - \tilde{P} > T - P$, and:

$$\tilde{\delta}^* = \frac{T - R}{T - \tilde{P}} < \frac{T - R}{T - P} = \delta^* \tag{335}$$

□

4.3 Numerical Illustration

For the violation severity we set $v = E_R$, the per-interaction resource pool. This represents a full-violation baseline: a defector who appropriates the counterpart's entire resource share causes harm equal to E_R. This is the natural unit for energy-denominated violations, and the bound $v \leq E_R$ follows from resource conservation (a single interaction cannot extract more than E_R from the system). The numerical results scale linearly with v; smaller violations reduce C_{cascade} proportionally.

Using baseline parameters ($E_R = 100$, $M = 1000$, $\bar{E} = 10$, $\eta = 0.01$, $v = E_R = 100$, $N = 100$, $\epsilon = 0.05$):

$$C_{\text{defector}}^{\text{cascade}} = \frac{0.01 \times 100 \times 1000 \times 10}{100 \times 0.05} = \frac{10{,}000}{5} = 2{,}000 \tag{336}$$

Amortized over the recovery period ($\sim 1/\epsilon = 20$ periods):

$$\tilde{P}_{\text{per period}} \approx P - 2000/20 = -5 - 100 = -105 \tag{337}$$

$$\tilde{\delta}^* = \frac{78.25 - 48}{78.25 - (-105)} = \frac{30.25}{183.25} = 0.165 \tag{338}$$

With network friction, the cooperation threshold drops to $\tilde{\delta}^* = 0.165$. Even extremely short-sighted agents find cooperation rational when the network punishes defection through friction cascades.

4.4 Deception Penalties

Defection via deception (lies, fraud) rather than overt conflict incurs the micro-friction costs from the information-entropy derivation:

- Decision cost to the victim: $\Delta U_B(q)$ (Theorem 8)
- Verification overhead on the network: $\Delta W(q)$ (Theorem 9)
- Deception-induced friction: $\Delta\phi_{\text{info}} = \eta_{\text{info}} \cdot \Delta H \cdot W_{\text{bit}}$ (Definition 17)

These costs further increase the effective punishment and reduce $\tilde{\delta}^*$. The deception channel does not create a "cheaper" form of defection — it incurs different but equally real thermodynamic costs.

---

5. Extension to N-Player Games

5.1 The N-Player Public Goods Game

To extend beyond two players, consider the canonical N-player interaction: a **public goods game** with energy-denominated payoffs.

> **Definition 24 (N-Player Energy Public Goods Game).** N agents each choose whether to Cooperate (C: contribute energy $c_i > 0$ to a shared pool) or Defect (D: contribute nothing). The shared pool generates a return multiplied by factor $\alpha > 1$ (the efficiency gain from cooperation) and is distributed equally. Agent i's payoff:
>
> $$\pi_i = \frac{\alpha \sum_{j=1}^{N} c_j \cdot \mathbb{1}[j \text{ cooperates}]}{N} - c_i \cdot \mathbb{1}[i \text{ cooperates}] - \phi \cdot \bar{E} \cdot \frac{N - n_C}{N} \tag{339}$$
>
> where n_C is the number of cooperators and the last term captures the friction tax from defectors (proportional to the defection rate).

In our framework, α is the **cooperative multiplier** — the ratio of total output under cooperation to total input. This is derived from the Lagrangian constraints result that the variational equilibrium allocates resources more efficiently than the unconstrained regime.

More concretely, when all N agents cooperate, each contributes c and receives αc (net gain $(\alpha-1)c$). When $n_C < N$ agents cooperate, the cooperators pay the coordination cost while defectors free-ride, but the system suffers friction loss.

5.2 The One-Shot N-Player Equilibrium

In the single-round game, defection is dominant when:

$$\frac{\alpha c}{N} < c \iff \alpha < N \tag{340}$$

That is, defection dominates when the cooperative return to each individual is less than the individual's contribution — the standard free-rider problem. For large N, this holds unless α is very large.

The one-shot NE is universal defection, yielding payoff $\pi_i^{DD} = -\phi_0 \bar{E}$ (only the friction tax from the non-cooperative environment). The cooperative payoff is $\pi_i^{CC} = (\alpha - 1)c$. The efficiency loss is:

$$\Delta\pi = (\alpha - 1)c + \phi_0 \bar{E} > 0 \tag{341}$$

5.3 The Repeated N-Player Game

> **Theorem 13 (N-Player Cooperation Sustainability).**
>
> *In the infinitely repeated N-player energy public goods game with discount factor δ and grim trigger punishment (all agents defect forever upon any defection), cooperation is a NE if:*

$$\delta \geq \delta_N^* = \frac{\pi_i^{\text{deviate}} - \pi_i^{CC}}{\pi_i^{\text{deviate}} - \pi_i^{DD}} \tag{342}$$

where $\pi_i^{deviate}$ is agent i's payoff from unilaterally defecting while all others cooperate:

$$\pi_i^{\text{deviate}} = \frac{\alpha(N-1)c}{N} - 0 = \frac{\alpha(N-1)c}{N} \tag{343}$$

(receives the public good without contributing). The critical discount factor is:

$$\delta_N^* = \frac{\frac{\alpha(N-1)c}{N} - (\alpha-1)c}{\frac{\alpha(N-1)c}{N} + \phi_0\bar{E}} = \frac{c(N-\alpha)}{(N-1)\alpha c + N\phi_0\bar{E}} \tag{344}$$

Proof. The deviation payoff is $\pi_i^{\text{deviate}} = \alpha(N-1)c/N$ (receives share of the $(N-1)$ cooperators' contributions without paying). The cooperative payoff is $\pi_i^{CC} = \alpha c - c = (\alpha-1)c$. The punishment payoff is $\pi_i^{DD} = -\phi_0\bar{E}$ (universal defection with friction tax). Applying the grim trigger calculation:

$$\delta_N^* = \frac{\pi_i^{\text{deviate}} - \pi_i^{CC}}{\pi_i^{\text{deviate}} - \pi_i^{DD}} = \frac{\alpha(N-1)c/N - (\alpha-1)c}{\alpha(N-1)c/N + \phi_0\bar{E}} \tag{345}$$

Simplifying the numerator:

$$\frac{\alpha(N-1)c - N(\alpha-1)c}{N} = \frac{c[\alpha N - \alpha - \alpha N + N]}{N} = \frac{c(N-\alpha)}{N} \tag{346}$$

So:

$$\delta_N^* = \frac{c(N-\alpha)/N}{\alpha(N-1)c/N + \phi_0\bar{E}} = \frac{c(N-\alpha)}{\alpha(N-1)c + N\phi_0\bar{E}} \tag{347}$$

□

Remark: The deviation payoff above omits the friction tax $\phi_0\bar{E}/N$ that Definition 24 imposes on all agents, including the deviator, in the defection round. Including this term yields the corrected threshold $\delta_N^{*,\text{corr}} = [c(N-\alpha) - \phi_0\bar{E}]/[(N-1)(\alpha c + \phi_0\bar{E})]$, which is strictly smaller than the stated δ_N^*. The omission is thus conservative: cooperation is achievable for any $\delta \geq \delta_N^*$ per the theorem, and in fact for the strictly smaller $\delta_N^{*,\text{corr}}$ as well. All qualitative conclusions — cooperation scales with N, friction lowers the threshold — are strengthened rather than weakened.

5.4 Properties of δ_N^*

Corollary 13.1 (Cooperation Strengthens with Network Friction).

$$\frac{\partial \delta_N^*}{\partial \phi_0} < 0 \tag{348}$$

Higher network friction (harsher punishment for the all-defect state) makes cooperation easier to sustain.

Proof. The numerator $c(N-\alpha)$ does not depend on ϕ_0. The denominator $\alpha(N-1)c + N\phi_0\bar{E}$ is strictly increasing in ϕ_0. Therefore δ_N^* is strictly decreasing in ϕ_0. □

Corollary 13.2 (Cooperation Strengthens with Scale Under Friction).

When friction is non-negligible ($\phi_0 > 0$), the critical discount factor δ_N^* remains bounded (does not approach 1) as $N \to \infty$:

$$\lim_{N\to\infty} \delta_N^* = \frac{c}{\alpha c + \phi_0 \bar{E}} < 1 \tag{349}$$

Proof. Dividing numerator and denominator by N:

$$\delta_N^* = \frac{c(1-\alpha/N)}{\alpha(1-1/N)c + \phi_0\bar{E}} \xrightarrow{N\to\infty} \frac{c}{\alpha c + \phi_0 \bar{E}} \tag{350}$$

Since $\phi_0 > 0$: $\alpha c + \phi_0\bar{E} > c$ (for $\alpha > 1$ or $\phi_0 > 0$), so the limit is strictly less than 1. □

Physical interpretation: In classical game theory (abstract payoffs, no friction), the critical discount factor for N-player public goods games approaches 1 as $N \to \infty$ — cooperation becomes unsustainable in large groups. With energy-based payoffs and network friction, this pathology **disappears**: the friction penalty of universal defection grows with group size, keeping cooperation accessible even in large populations. This is the formal proof that the cooperative equilibrium holds and strengthens as the network scales.

5.5 Numerical Illustration: N-Player Cooperation Thresholds

Parameters: $c = 10$, $\alpha = 2$ (cooperative doubling), $\phi_0 = 0.5$ (moderate background friction), $\bar{E} = 10$.

N	Deviation gain $\pi^{\text{dev}} - \pi^{CC}$	Punishment gap $\pi^{\text{dev}} - \pi^{DD}$	δ_N^*
2	0	10 + 5 = 15	0.000
3	3.33	13.33 + 5 = 18.33	0.182
5	6.00	16 + 5 = 21	0.286
10	8.00	18 + 5 = 23	0.348
50	9.60	19.6 + 5 = 24.6	0.390
100	9.80	19.8 + 5 = 24.8	0.395
∞	10.00	20 + 5 = 25	0.400

Key result: Even as $N \to \infty$, δ_N^* converges to 0.4, well below 1. Cooperation remains a Nash Equilibrium for any agent with $\delta \geq 0.4$. Without friction ($\phi_0 = 0$), δ_N^* for $N = 100$ would be $9.8/19.8 = 0.495$, and for $N \to \infty$ it would approach $10/20 = 0.5$. Friction provides the extra margin that makes cooperation robust.

6. Evolutionary Stability: Defection Cannot Invade

6.1 The Evolutionary Game-Theoretic Framework

Beyond asking whether cooperation is a NE, we ask: **can defection invade a cooperative population?** This is the evolutionary stability question (Maynard Smith and Price, 1973).

Definition 25 (Evolutionary Stable Strategy, ESS). A strategy s^* is an ESS if, for any mutant strategy $s' \neq s^*$:

(a) $\pi(s^*, s^*) > \pi(s', s^*)$ (**strict NE**), or

(b) $\pi(s^*, s^*) = \pi(s', s^*)$ and $\pi(s^*, s') > \pi(s', s')$ (**stability condition**).

6.2 TFT as an Approximate ESS

In the iterated Prisoner's Dilemma with energy payoffs, Tit-for-Tat (TFT) is a well-known strong performer (Axelrod, 1984). While TFT is not a strict ESS in the technical sense (it does not strictly outperform other "nice" strategies like Always-Cooperate in a homogeneous population), it satisfies a strong **invasion barrier** condition:

Theorem 14 (Defection Invasion Barrier).

In a population of N TFT-playing agents with discount factor $\delta > \delta^*$, a mutant Always-Defect (ALLD) agent earns:

$$V_{\text{ALLD}} = T + \frac{\delta P}{1 - \delta} \tag{351}$$

while a TFT agent in the same population earns:

$$V_{\text{TFT}} = \frac{R}{1 - \delta} \tag{352}$$

The ALLD mutant is strictly dominated ($V_{ALLD} < V_{TFT}$) when:

$$\delta > \frac{T - R}{T - P} = \delta^* \tag{353}$$

Under the physical payoff model with $\Phi > 2$, this holds for all $\delta > 0.363$ (baseline parameters).

Proof. An ALLD mutant, upon meeting a TFT agent, defects in round 1 (getting T) and then faces retaliation: mutual defection from round 2 onward (getting P each round). Its lifetime payoff against any TFT agent is:

$$V_{\text{ALLD}} = T + \delta P + \delta^2 P + \cdots = T + \frac{\delta P}{1 - \delta} \tag{354}$$

A TFT agent meeting other TFT agents cooperates every round:

$$V_{\text{TFT}} = R + \delta R + \delta^2 R + \cdots = \frac{R}{1-\delta} \tag{355}$$

Then $V_{\text{TFT}} > V_{\text{ALLD}}$:

$$\frac{R}{1-\delta} > T + \frac{\delta P}{1-\delta} \tag{356}$$

$$R > (1-\delta)T + \delta P \tag{357}$$

$$\delta > \frac{T-R}{T-P} = \delta^* \tag{358}$$

which is exactly the cooperation condition from Theorem 11. □

Physical interpretation: A defector entering a cooperative population gets one "hit" (the temptation payoff T) and then finds itself in perpetual mutual defection — which, under physical payoffs, means perpetual net-negative returns ($P < 0$). The defector's strategy is self-defeating: it destroys the cooperative environment it needs to profit from. In evolutionary terms, the "invasion fitness" of defection is negative in a cooperative ecology.

6.3 The Defector's Dilemma: Extinction Through Friction

Beyond the direct retaliation mechanism, the defector also faces the cascading friction cost (§4). Even if a defector can avoid direct retaliation (e.g., in a large anonymous population), each defection increases the network friction ϕ, which degrades the returns to *all* agents — including the defector. In a population where the defector's payoff depends on the cooperative surplus (because even defectors benefit from a low-friction network for their non-defection interactions), systematic defection is a self-undermining strategy.

> **Corollary 14.1 (Defection Self-Extinction in Friction Environments).** *In a network with friction coupling, a defector whose defection rate ν_D injects enough friction to push ϕ above the network collapse threshold (Corollary 7.1) destroys the cooperative surplus from which it profits. In the limit, a population of defectors achieves the minimum network payoff $\pi^{DD} = -\phi_0 \bar{E}$, which is strictly less than zero. Defection is not merely dominated — it is self-destructive.*

---

7. The Altruism Matrix: Game-Theoretic Foundation

7.1 Connecting to the Paper's Framework

The paper (§5.6) identifies four mechanisms explaining altruism. We now provide the game-theoretic formalization for each.

7.2 Mechanism A: Inclusive Fitness as Extended Payoffs

In biological systems, an agent's "identity" extends to genetically related individuals (Hamilton, 1964). We model this by augmenting the payoff function:

Definition 26 (Inclusive Fitness Payoff). Agent i's inclusive payoff is:

$$\Pi_i^{\text{incl}} = \pi_i + \sum_{j \neq i} r_{ij}\, \pi_j \tag{359}$$

where $r_{ij} \in [0, 1]$ is the **relatedness coefficient** between agents i and j (probability of shared genetic identity at a random locus).

For a standard diploid organism: $r_{\text{parent -child}} = 0.5$, $r_{\text{siblings}} = 0.5$, $r_{\text{cousins}} = 0.125$.

Hamilton's Rule in energy terms: Agent i will sacrifice personal payoff $\pi_i = -C$ if the benefit to relative j satisfies $r_{ij} \cdot B > C$, where B is the benefit (in energy units) to agent j. This is a standard optimization: maximize the inclusive payoff $\Pi_i^{\text{incl}} = -C + r_{ij}B \geq 0$.

7.3 Mechanism B: Infinite Horizon (Belief-Extended δ)

If an agent believes its identity persists beyond physical death, its effective discount factor approaches 1:

$$\delta_{\text{believer}} = 1 - \epsilon_{\text{belief}} \approx 1 \tag{360}$$

where $\epsilon_{\text{belief}} \to 0$ reflects the agent's confidence in post-mortem persistence. Under Theorem 11, any $\delta > \delta^*$ sustains cooperation. For $\delta \to 1$, not only is cooperation sustained but the agent is willing to accept **any** finite short-term cost for long-term cooperative benefit: self-sacrifice becomes "rational" under the agent's model of reality.

7.4 Mechanism C: Coupled Systems (Merged Boundaries)

When two agents merge boundaries (parent-child, spouses), they effectively become a single agent with a joint utility function:

$$U_{AB}(\mathbf{x}_A, \mathbf{x}_B) = w_A U_A(\mathbf{x}_A) + w_B U_B(\mathbf{x}_B) \tag{361}$$

where $w_A, w_B > 0$ are the coupling weights. The "self-sacrifice" of agent A for agent B is reinterpreted as internal resource reallocation within the merged system — the macro-entity AB optimizes its *joint* boundary, shedding a component to save a more vital one.

7.5 Mechanism D: Heuristic Over-Extension (Generalized Relatedness)

The empathy heuristic effectively inflates the perceived relatedness coefficient:

$$\hat{r}_{ij} = r_{ij} + \epsilon_{\text{empathy}} > r_{ij} \tag{362}$$

where $\epsilon_{\text{empathy}} \geq 0$ is the over-extension term generated by the agent's abstract-reasoning capacity. When $\hat{r}_{ij}$ is large enough, Hamilton's Rule triggers even for unrelated strangers: $\hat{r}_{ij}B > C$ can hold even when $r_{ij} = 0$.

7.6 Unified Formalization

All four mechanisms operate by expanding the agent's **effective payoff function** to internalize some fraction of other agents' payoffs:

$$\Pi_i^{\text{eff}} = \pi_i + \sum_{j \neq i} w_{ij} \pi_j \tag{363}$$

where the weights w_{ij} are determined by:

Mechanism	Source of w_{ij}
A. Inclusive fitness	Genetic relatedness r_{ij}
B. Infinite horizon	Discount factor $\delta \to 1$ (via repeated-game embedding)
C. Coupled systems	Boundary merger weights w_A, w_B
D. Heuristic over-extension	Empathic projection $\hat{r}_{ij}$

In each case, the agent is still maximizing its **own** (extended) payoff — altruism is *selfish optimization over an expanded utility function*, as the framework predicts.

---

8. Tit-for-Tat Dynamics in Energy Terms

8.1 Short-Term Defection Profit is Erasure

We now formally show that a defector's short-term energy gain is **wiped out** over repeated interactions.

> **Theorem 15 (Defection Profit Erasure).**
>
> *An agent that defects once against a TFT population and then returns to cooperation incurs a net lifetime loss whenever:*

$$\delta > \frac{T - R}{R - S} \tag{364}$$

(without friction). Under the physical payoff model (baseline parameters), the basic threshold is $\delta = 0.511$. When network friction costs are included ($F = 100$ per period over 20 recovery periods), the threshold drops to $\delta \approx 0.17$.

Proof. Consider an agent that defects in period t (getting T), is retaliated against in period $t + 1$ (getting S), and then both agents return to cooperation from period $t + 2$ onward. The deviation costs the agent:

- Period t: gains T instead of R (net gain: $T - R$)
- Period $t + 1$: earns S instead of R (net loss: $R - S$)
- Periods $t + 2, \ldots$: back to R (no change)

Present-value net gain from the one-shot deviation:

$$\Delta V = (T - R) - \delta(R - S) \tag{365}$$

The deviation is unprofitable ($\Delta V < 0$) when:

$$\delta > \frac{T - R}{R - S} \tag{366}$$

With baseline parameters: $\delta > (78.25 - 48)/(48 - (-11.25)) = 30.25/59.25 = 0.511$.

However, this analysis ignores the network friction cost. Including the per-period friction penalty amortized over recovery time ($\sim 1/\epsilon = 20$ periods), even a single defection imposes a sustained future cost. From §4.3, the per-defector cascade cost is $C^{\text{cascade}}_{\text{defector}} = 2{,}000$, giving $F = 2{,}000/20 = 100$ energy units per period of degraded cooperative returns. When we include this:

$$\Delta V_{\text{adj}} = (T - R) - \delta(R - S) - \delta \sum_{k=1}^{1/\epsilon} \delta^{k-1} F \tag{367}$$

This makes the defection-erasure threshold strictly lower than $\delta = 0.511$:

$$\frac{T - R}{R - S + \delta F/(1 - \delta)} < \frac{T - R}{R - S} \tag{368}$$

Numerically, the condition $\Delta V_{\text{adj}} < 0$ holds for $\delta \gtrsim 0.17$ after including friction. □

Physical interpretation: The defector's profit ($T - R = 30.25$) is a one-time energy windfall. The retaliation cost ($R - S = 59.25$) comes one period later. The friction cost ($F = 100$/period) compounds over ~ 20 periods. For any agent that expects to survive more than a few interactions ($\delta > 0.17$), the short-term defection profit is negative on net. Defection is an energetic "payday loan" with ruinous interest.

---

9. System-Level Optimality: The Welfare Theorem

9.1 The Social Welfare Function

Define the **system welfare** as the total energy available to all agents for Identity Preservation:

$$SW = \sum_{i=1}^{N} \pi_i - \mathcal{C}_{\text{total}} \tag{369}$$

where $\mathcal{C}_{\text{total}}$ is the total interaction cost (Definition 11 plus Definition 18).

9.2 The Maximum Welfare Theorem

Theorem 16 (Cooperation Maximizes System Welfare).

Among all strategy profiles in the repeated N-player game, the cooperative profile (C for all agents in all periods) uniquely maximizes the per-period system welfare:

$$SW^{CC} = N(\alpha - 1)c \geq SW^s \quad \forall \text{ strategy profiles } s \tag{370}$$

with equality only when s is also universally cooperative.

Proof. Under universal cooperation:

- Total resource input: Nc (all agents contribute)
- Total output: αNc (cooperative multiplier)
- Net welfare: $N(\alpha - 1)c$
- Interaction cost: $\mathcal{C}_{\text{total}} = 0$ (no conflict, no deception, $\phi = 0$)

Under any profile with $n_D > 0$ defectors:

- Resource input: $(N - n_D)c < Nc$ (defectors don't contribute)
- Output: $\alpha(N - n_D)c < \alpha Nc$ (reduced by lost contributions)
- Network friction tax: $\phi \cdot M \cdot \bar{E} > 0$ (deviators bear friction penalty)
- Net welfare: strictly less than $N(\alpha - 1)c$

Formally:

$$SW^{CC} - SW^s = \underbrace{n_D(\alpha - 1)c}_{\text{lost contributions}} + \underbrace{\phi(s)M\bar{E}}_{\text{friction tax}} > 0 \tag{371}$$

for any s with $n_D > 0$. Uniqueness follows from the strict inequality above: for any strategy profile with $n_D \geq 1$ defectors, the lost-contributions term $n_D(\alpha - 1)c > 0$ ensures that no defecting profile achieves SW^{CC}. (*Note: The Tullock contest dissipation term $D(s)$ applies to arms-race conflict settings — see thermodynamic friction derivation, §3, Proposition 2 — and does not apply in this public goods free-rider context; uniqueness here rests solely on the lost-contributions and friction-tax terms.*) □

9.3 The System-Level Nash Equilibrium

Combining Theorems 11 and 16:

> **Corollary 16.1 (Ethics as the Unique Efficient Nash Equilibrium).**
>
> *In the infinitely repeated energy-based game with $\delta > \delta^*$:*
>
> *(a) Universal cooperation is a Nash Equilibrium (Theorem 11).*
>
> *(b) Universal cooperation uniquely maximizes system welfare (Theorem 16).*
>
> *(c) Universal cooperation minimizes total interaction cost ($\mathcal{C}_{total} = 0$).*
>
> *Therefore, the cooperative strategy profile ("ethics") is the unique Nash Equilibrium that is also Pareto-optimal and system-welfare-maximizing.*

Physical interpretation: This is the central result of the paper — the formal proof that *"Ethics is the objectively measurable, mathematically optimal algorithm for maximizing the Identity Preservation of the maximum number of entities within a closed system."* The "optimization" is not hypothetical; it is the standard Nash equilibrium selection criterion applied to energy-denominated payoffs under physical constraints.

---

10. Cross-Scale Application (Preview)

10.1 Why Scale Matters

The preceding analysis assumes agents of comparable scale. A natural question arises: does the equilibrium hold when agents differ vastly in scale (ant vs. human vs. AGI)?

The answer is affirmative, but the argument rests on the **Accumulated Negentropy** concept (developed in the negentropy derivation). The key insight: in the energy-based payoff model, a super-entity's temptation payoff T from exploiting smaller entities is bounded by the physical cost of exploitation (the contest dissipation costs from Theorem 5 and Corollary 5.3 of the thermodynamic friction derivation), while the smaller entities' collective accumulated negentropy represents an irreplaceable informational asset whose destruction leaves the super-entity worse off.

The formal cross-scale proof is completed in the negentropy derivation. The N-player results (§5) apply directly: the super-entity is one agent, the smaller entities are $N - 1$ agents, and the cooperative multiplier α captures the generative value of the information-dense biosphere.

---

11. Summary of Results

#	Result	Statement	Significance
P4	Physical PD	$T > R > P > S$ under energy payoffs with $\Phi > 1$	Single-round defection is tempting but destructive
T11	Ethics Theorem	Cooperation is NE for $\delta \geq \delta^* = (T-R)/(T-P)$	Cooperation emerges from physical structure
C11.1	Low Threshold	$\delta^* < 0.5$ when $\Phi > 2$	Physical payoffs make cooperation easy to sustain
T12	Network Adjustment	$\tilde{\delta}^* < \delta^*$ with friction cascades	Network coupling reinforces cooperation
T13	N-Player	δ_N^* bounded below 1 even as $N \to \infty$	Cooperation scales to large populations
C13.1	Friction Robustness	$\partial\delta_N^*/\partial\phi_0 < 0$	More friction →easier cooperation
C13.2	Scale Robustness	$\lim_{N\to\infty} \delta_N^* < 1$ under friction	No large-N cooperation breakdown
T14	Invasion Barrier	ALLD cannot invade TFT population for $\delta > \delta^*$	Defection is evolutionarily unstable
C14.1	Defection Self-Extinction	Systematic defection destroys cooperative surplus	Defection is self-undermining
T15	Profit Erasure	One-shot defection profit erased for $\delta > 0.17$ (with friction)	Short-term gains wiped by long-term costs
T16	Maximum Welfare	Cooperation uniquely maximizes $SW = \sum \pi_i - \mathcal{C}$	Ethics is the system-welfare-maximizing strategy
C16.1	Central Result	Cooperation is the unique efficient NE	Ethics $\equiv$ mathematically optimal algorithm

12. Notation Index

Symbol	Meaning
E_R	Energy value of contested resource
c	Coordination/trade cost per agent
θ	Exploitation efficiency (cost to defect against cooperator)
T, R, P, S	Temptation, Reward, Punishment, Sucker payoffs
δ	Discount factor (value of future relative to present)
σ	Per-period survival probability
r	Time-preference rate
δ^*	Critical discount factor for cooperation
$\tilde{\delta}^*$	Network-adjusted critical discount factor
δ_N^*	N-player critical discount factor
Φ_{PD}	Maximum friction multiplier for PD structure: $2 + 4\theta(1+\kappa)\delta^{\text{def}}$
α	Cooperative multiplier (efficiency gain from cooperation)
n_C	Number of cooperating agents
n_D	Number of defecting agents
r_{ij}	Relatedness coefficient between agents i and j
w_{ij}	Effective payoff coupling weight
Π_i^{incl}	Inclusive fitness payoff
Π_i^{eff}	Effective (extended) payoff
SW	System welfare (total energy available for IP)
$V_A^{\text{coop}}, V_A^{\text{deviate}}$	Lifetime payoff under cooperation / deviation

Symbol disambiguation. In this file: σ = per-period survival probability, r = time-preference rate. In other derivation files, σ denotes conflict effectiveness (thermodynamic friction), assimilation intensity (value dynamics), or search efficiency (information-negentropy Part B). The symbol r denotes coupling distance (value dynamics) or contest decisiveness (thermodynamic friction). Context and subscripts distinguish these uses.

Value Dynamics — Attractor Mechanics of Coexistence

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0. Preamble

This document contains the formal mathematical derivation for the value dynamics framework. Building on the preceding derivations, we now construct a **dynamical systems model** for the interaction between individual agents and high-energy centers (HECs) — large accumulated-value nodes in the network (corporations, states, trade hubs, AI systems).

The contribution is **applied mathematics**: we use potential theory, gradient dynamics, Lyapunov stability analysis, and comparative statics to prove that the agent-center interaction possesses a **stable equilibrium region** (the "Stable Coexistence Band") whose width constitutes a formal measure of freedom. We further prove that crossing the band's boundaries leads to irreversible regime changes — dissolution of identity (assimilation) or energetic collapse (starvation).

The physics analogy is gravitational orbital mechanics; the formal contribution is a **dynamical systems model** with a potential well that traps agents in a viable region. We do not claim that literal gravitational forces operate at social scales. We prove that the **mathematical structure** — a potential function with short-range repulsion and long-range attraction creating a stable basin — describes agent-center dynamics and yields non-trivial, testable results.

Key references for the tools used:

- (Strogatz, 2015) — gradient dynamics, stability, bifurcations.
- (Hirsch et al., 2013) — Lyapunov stability, attractor theory.
- (Tirole, 1988) — firm-worker coupling, vertical relationships.
- (Acemoglu and Robinson, 2012) — institutional power concentration and state capture.
- (Jones, 1924) — structural precedent for two-body potentials balancing short-range repulsion and long-range attraction.

Notational continuity from preceding derivations:

- N agents, strategy vectors $\mathbf{x}_i$, utility functions U_i (Lagrangian constraints derivation).
- Boundary integrity $B_i > 0$; entropy leakage rate $\gamma_i > 0$ (thermodynamic friction, Definitions 3–4).
- Maintenance cost $C_{\text{maintain},i} = \gamma_i B_i$ (thermodynamic friction derivation, §2.2).
- Network friction coefficient ϕ; friction sensitivity η; recovery rate ϵ (thermodynamic friction derivation).
- Accumulated negentropy $\mathcal{N}(T)$ (Definition 29).
- Discount factor δ; cooperation threshold δ^* (Theorem 11).

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1. The Agent–Center Model

1.1 Motivation: Why Dynamics Matter

The preceding derivations established the static structure of cooperative multi-agent systems: rights as constraints, conflict as waste, cooperation as equilibrium, information as value. But the framework describes a **dynamic** phenomenon: accumulated value creates attractor dynamics. Entities are drawn toward high-energy centers because proximity lowers the cost of fighting entropy. Too close, and the entity is assimilated; too far, and it starves.

This is inherently a **dynamical systems** question: given an agent and a center, what coupling strength is stable? What happens when the agent is perturbed from equilibrium? Under what conditions does the equilibrium exist at all?

1.2 Primitive Objects

Definition 35 (High-Energy Center). A *high-energy center* (HEC) is a network node with accumulated value mass

$$\mathcal{M} > 0 \quad \text{(energy units, J)} \tag{372}$$

representing the total accumulated negentropy of the center (Definition 29). The HEC may be a corporation, a state, a trade hub, an ecosystem, or an AI system — any entity or coalition that has aggregated sufficient energy and information to exert significant influence on surrounding agents.

Definition 36 (Coupling Distance). The *coupling distance* $r \in (0, \infty)$ between an agent i and a HEC is a generalized coordinate measuring the degree of the agent's independence from the center:

- $r \to 0$: maximal coupling (complete integration — employee absorbed into corporation, citizen fully subject to state, subsidiary merged into parent)
- $r \to \infty$: minimal coupling (complete isolation — autarkic entity, no trade or institutional affiliation)

The coupling distance is **not** a spatial distance. It is a scalar summarizing the composite engagement level across economic, informational, political, and social dimensions. Formally, r can be constructed as a norm on the vector of coupling channels, but for the 1D analysis below, only its scalar properties matter.

Interpretation: The coupling distance r is the single degree of freedom in our model. An agent's "strategy" in the value-dynamics context is its choice of how tightly to bind itself to a given HEC. This complements the multi-dimensional resource strategy $\mathbf{x}_i$ from the Lagrangian constraints derivation: there, the agent chose *what* to consume; here, the agent chooses *how closely* to affiliate with the source of resources.

1.3 The Resource Inflow Function

As the agent couples more tightly with a HEC (smaller r), it gains greater access to the center's accumulated resources — shared infrastructure, trade networks, institutional protection, information flows, and surplus energy. We model the **energy inflow rate** as:

$$E_{\text{in}}(r) = \frac{G\,\mathcal{M}}{r} \tag{373}$$

where $G > 0$ is the **resource coupling coefficient** — a proportionality constant capturing the efficiency of the HEC at distributing resources to its affiliates.

Properties:

- $E_{\text{in}}(r) > 0$ for all $r > 0$ (the HEC always provides some resources, however distant).
- $E_{\text{in}}(r) \to \infty$ as $r \to 0$ (arbitrarily large resource access at arbitrarily tight coupling — but offset by costs below).
- $E_{\text{in}}(r) \to 0$ as $r \to \infty$ (isolation yields negligible resource access).
- E_{in} is proportional to $\mathcal{M}$: a more massive center supplies more resources at every coupling distance.

The $1/r$ functional form is the simplest monotonically decreasing, positive function with the correct limits. It parallels the gravitational potential GM/r, providing the "attraction" half of the model.

1.4 The Dissolution Cost Function

Tight coupling also imposes **autonomy costs**: the HEC demands conformity, extracts resources from the agent, constrains the agent's action space (Lagrangian constraints derivation, §3), and exerts homogenizing pressure on the agent's internal state. At sufficiently close coupling, these costs overwhelm the benefits and erode the agent's boundary integrity, leading to dissolution of identity.

We model the **dissolution cost rate** as:

$$C_{\text{dissolve}}(r) = \frac{\tau\,\mathcal{M}}{r^2} \tag{374}$$

where $\tau > 0$ is the **dissolution coupling coefficient** — a proportionality constant capturing the HEC's assimilation pressure per unit mass.

Properties:

- $C_{\text{dissolve}}(r) > 0$ for all $r > 0$.
- $C_{\text{dissolve}}(r) \to \infty$ as $r \to 0$, and **faster** than $E_{\text{in}}(r)$ (the $1/r^2$ term dominates $1/r$ at small r). This is the critical structural feature: at very close coupling, the costs always overwhelm the benefits.
- $C_{\text{dissolve}}(r) \to 0$ as $r \to \infty$ (distant agents are not subject to assimilation pressure).

The $1/r^2$ form ensures that the dissolution cost is a **short-range dominant** force, negligible at large r but overwhelming at small r. This parallels the Lennard-Jones repulsion in molecular physics and the "tidal forces" of the gravitational analogy.

Physical motivation: The dissolution cost scales as $1/r^2$ rather than $1/r$ because assimilation pressure involves not only resource extraction but also **constraint imposition** — the HEC's demand for conformity grows nonlinearly with coupling tightness. An employee who works 60

hours/week for a corporation loses far more than 1.5× the autonomy of one working 40 hours. The constraints compound.

1.5 The Entropy Maintenance Baseline

Independent of the HEC, the agent must continuously expend energy to maintain its boundary against environmental entropy (thermodynamic friction derivation, §2.2):

$$C_{\text{maintain}} = \gamma_i \, B_i \tag{375}$$

where $\gamma_i > 0$ is the entropy leakage rate and $B_i > 0$ is the boundary integrity. This is a **constant** cost — it does not depend on r because entropy attacks the boundary regardless of the agent's institutional affiliations. For the single-agent analysis that follows, we write $\gamma \equiv \gamma_i$ and suppress the index to reduce notation.

1.6 The Net Energy Rate

The agent's **net energy rate** at coupling distance r is:

$$\Pi(r) = E_{\text{in}}(r) - C_{\text{dissolve}}(r) - C_{\text{maintain}} = \frac{G\mathcal{M}}{r} - \frac{\tau\mathcal{M}}{r^2} - \gamma B_i \tag{376}$$

The agent survives if and only if $\Pi(r) > 0$. The agent's goal is to choose r to maximize $\Pi(r)$ — equivalently, to minimize the **coexistence potential** $V(r) = -\Pi(r)$.

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2. The Coexistence Potential

2.1 Definition

Definition 37 (Coexistence Potential). The *coexistence potential* of an agent i at coupling distance r from a HEC with value mass $\mathcal{M}$ is:

$$V(r) = \frac{\tau\mathcal{M}}{r^2} - \frac{G\mathcal{M}}{r} + \gamma B_i \tag{377}$$

where $G > 0$ is the resource coupling coefficient, $\tau > 0$ is the dissolution coupling coefficient, $\gamma > 0$ is the entropy leakage rate, and $B_i > 0$ is the agent's boundary integrity. The potential is defined on $(0, \infty)$.

The coexistence potential is the **negative** of the net energy rate: $V(r) = -\Pi(r)$. Regions where $V(r) < 0$ are viable (positive net energy); regions where $V(r) > 0$ are non-viable (energy deficit).

2.2 Properties

Proposition 8 (Properties of the Coexistence Potential).

The coexistence potential $V(r)$ satisfies:

(a) $\lim_{r\to 0^+} V(r) = +\infty$ *(dissolution dominance at close range).*

(b) $\lim_{r\to\infty} V(r) = \gamma B_i > 0$ *(starvation at large distance).*

(c) V *is* C^∞ *on* $(0,\infty)$.

(d) V *has exactly one critical point on* $(0,\infty)$*, which is a global minimum.*

(e) The minimum value $V_{\min}$ *satisfies* $V_{\min} < \gamma B_i$.

Proof.

(a) As $r \to 0^+$: $\tau\mathcal{M}/r^2 \to +\infty$, while $G\mathcal{M}/r \to +\infty$ at a slower rate. Since $1/r^2$ dominates $1/r$ as $r \to 0$, $V(r) \to +\infty$.

(b) As $r \to \infty$: both $\tau\mathcal{M}/r^2 \to 0$ and $G\mathcal{M}/r \to 0$, so $V(r) \to \gamma B_i > 0$.

(c) V is a sum of rational functions of r (with $r > 0$) and a constant, hence smooth.

(d) Compute the first derivative:

$$V'(r) = -\frac{2\tau\mathcal{M}}{r^3} + \frac{G\mathcal{M}}{r^2} = \frac{\mathcal{M}}{r^3}\big(Gr - 2\tau\big) \tag{378}$$

Since $\mathcal{M} > 0$ and $r^3 > 0$, the sign of $V'(r)$ is determined by $(Gr - 2\tau)$:

- $V'(r) < 0$ for $r < 2\tau/G$ (potential decreasing)
- $V'(r) = 0$ at $r = 2\tau/G$ (unique critical point)
- $V'(r) > 0$ for $r > 2\tau/G$ (potential increasing)

The second derivative at the critical point:

$$V''(r) = \frac{6\tau\mathcal{M}}{r^4} - \frac{2G\mathcal{M}}{r^3} \tag{379}$$

At $r^* = 2\tau/G$:

$$V''(r^*) = \frac{6\tau\mathcal{M}\cdot G^4}{16\tau^4} - \frac{2G\mathcal{M}\cdot G^3}{8\tau^3} = \frac{G^4\mathcal{M}}{8\tau^3}\big(3-2\big) = \frac{G^4\mathcal{M}}{8\tau^3} > 0 \tag{380}$$

Since $V''(r^*) > 0$, the critical point is a strict local minimum. By uniqueness of the critical point and the boundary behavior (parts a, b), it is the global minimum.

(e) $V(r^*) = \frac{\tau\mathcal{M}G^2}{4\tau^2} - \frac{G\mathcal{M}G}{2\tau} + \gamma B_i = \gamma B_i - \frac{G^2\mathcal{M}}{4\tau}$. Since $G^2\mathcal{M}/(4\tau) > 0$, we have $V(r^*) < \gamma B_i$. $\square$

2.3 The Cooperative Attractor

Definition 38 (Cooperative Attractor). The *cooperative attractor* is the coupling distance that minimizes the coexistence potential:

$$r^* = \frac{2\tau}{G} \tag{381}$$

At this distance, the agent achieves the maximum net energy rate $\Pi(r^*) = G^2\mathcal{M}/(4\tau) - \gamma B_i$.

Interpretation: The cooperative attractor r^* represents the **optimal structural coupling** between the agent and the HEC. It depends only on the ratio τ/G — the balance between dissolution pressure and resource access — not on the HEC's mass $\mathcal{M}$ or the agent's boundary cost γB_i. This is a structural property: the optimal degree of integration is determined by the *nature* of the coupling, not the *scale* of the parties.

What does depend on $\mathcal{M}$ is whether the agent can survive at all (the depth of the potential well) and how wide the viable region is (the Coexistence Band).

2.4 The Potential Well Depth

Definition 39 (Potential Well Depth). The *well depth* of the coexistence potential is:

$$D = \gamma B_i - V(r^*) = \frac{G^2\mathcal{M}}{4\tau} \tag{382}$$

This is the maximum net energy surplus achievable by the agent when optimally coupled to the HEC.

The well depth is proportional to $\mathcal{M}$: a more massive center creates a deeper well, supporting agents with higher maintenance costs.

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3. Gradient Dynamics and Stability

3.1 The Coupling Adjustment Equation

Agents adjust their coupling distance toward higher net energy (lower potential). The simplest model is **gradient dynamics**:

$$\dot{r} = -\mu\, V'(r) = -\frac{\mu\mathcal{M}}{r^3}(Gr - 2\tau) \tag{383}$$

where $\mu > 0$ is the **adjustment rate** (mobility, adaptability — how quickly the agent can change its institutional affiliations).

Dynamical behavior:

- For $r < r^*$: $Gr - 2\tau < 0$, so $\dot{r} > 0$ — the agent moves **away** from the center (escaping dissolution pressure).
- For $r > r^*$: $Gr - 2\tau > 0$, so $\dot{r} < 0$ — the agent moves **toward** the center (seeking resources).
- At $r = r^*$: $\dot{r} = 0$ — the agent is at equilibrium.

3.2 Stability of the Cooperative Attractor

Theorem 22 (Stability of the Cooperative Attractor).

Under the gradient dynamics $\dot{r} = -\mu V'(r)$, the cooperative attractor $r^* = 2\tau/G$ is:

(a) A fixed point: $\dot{r}|_{r=r^*} = 0$.

(b) Locally asymptotically stable: the eigenvalue $\lambda = -\mu V''(r^*) = -\mu G^4 \mathcal{M}/(8\tau^3) < 0$.

(c) Globally attracting on $(0, \infty)$: every trajectory starting at $r_0 \in (0, \infty)$ converges to r^* as $t \to \infty$.

Proof.

(a) At r^*: $V'(r^*) = 0$, so $\dot{r} = 0$. ✓

(b) Linearize around r^*: $\dot{r} \approx -\mu V''(r^*)(r - r^*)$. From Proposition 8(d), $V''(r^*) = G^4\mathcal{M}/(8\tau^3) > 0$. The eigenvalue of the linearized system is $\lambda = -\mu V''(r^*) < 0$. Hence r^* is locally asymptotically stable.

(c) Define the Lyapunov function $\mathcal{W}(r) = V(r) - V(r^*)$. Then $\mathcal{W}(r) \geq 0$ for all $r > 0$ with equality iff $r = r^*$ (since r^* is the unique global minimum of V). The time derivative along trajectories:

$$\dot{\mathcal{W}} = V'(r) \cdot \dot{r} = V'(r) \cdot \big(-\mu V'(r)\big) = -\mu\big(V'(r)\big)^2 \leq 0 \tag{384}$$

with equality iff $V'(r) = 0$, i.e., $r = r^*$. By LaSalle's invariance principle, every trajectory converges to the largest invariant set within $\{r : \dot{\mathcal{W}} = 0\} = \{r^*\}$. Hence r^* is globally attracting.

Convergence rate: Near r^*, the convergence is exponential with rate $|\lambda| = \mu G^4 \mathcal{M}/(8\tau^3)$. Larger μ (more mobile agents) and larger $\mathcal{M}$ (more massive centers) accelerate convergence. □

Corollary 22.1 (Resilience of the Equilibrium). *The cooperative attractor is robust to perturbations: any temporary displacement from r^* (due to external shocks, policy changes, economic disruptions) is self-correcting. The agent naturally returns to the optimal coupling distance.*

Physical interpretation: A worker who is pushed closer to the corporation than optimal (overwork, loss of autonomy) will naturally seek more distance — reduced hours, unionization, job mobility. A worker pushed too far (laid off, isolated) will naturally seek re-engagement — job searching, networking. The gradient dynamics formalize this behavioral regularity.

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4. The Stable Coexistence Band

4.1 The Viability Condition

The agent survives if and only if $\Pi(r) > 0$, equivalently $V(r) < 0$. The boundary of the viable region is defined by $V(r) = 0$:

$$\frac{\tau\mathcal{M}}{r^2} - \frac{G\mathcal{M}}{r} + \gamma B_i = 0 \tag{385}$$

Multiplying by r^2:

$$\gamma B_i\, r^2 - G\mathcal{M}\, r + \tau\mathcal{M} = 0 \tag{386}$$

This is a quadratic in r with discriminant:

$$\Delta = G^2\mathcal{M}^2 - 4\gamma B_i\tau\mathcal{M} = \mathcal{M}\big(G^2\mathcal{M} - 4\gamma B_i\tau\big) \tag{387}$$

4.2 The Minimum Viable HEC Mass

Theorem 23 (Existence of the Coexistence Band).

The Stable Coexistence Band exists if and only if the HEC's value mass exceeds a critical threshold:

$$\mathcal{M} > \mathcal{M}_{\min} \triangleq \frac{4\gamma B_i\tau}{G^2} \tag{388}$$

When $\mathcal{M} = \mathcal{M}_{\min}$, the band collapses to the single point r^* (marginal viability). When $\mathcal{M} < \mathcal{M}_{\min}$, no viable coupling distance exists — the agent cannot sustain itself near this HEC.

Proof. Two positive real roots of $V(r) = 0$ require: (i) $\Delta > 0$, which gives $\mathcal{M} > 4\gamma B_i\tau/G^2$; (ii) both roots positive, which is guaranteed since the sum $r_- + r_+ = G\mathcal{M}/(\gamma B_i) > 0$ and the product $r_- \cdot r_+ = \tau\mathcal{M}/(\gamma B_i) > 0$. When $\Delta = 0$, there is a single repeated root $r^* = G\mathcal{M}/(2\gamma B_i)$, and the minimum value $V(r^*) = 0$ (marginal viability). When $\Delta < 0$, $V(r) > 0$ for all $r > 0$ — the well is too shallow to reach negative values. □

Physical interpretation: A small corner shop ($\mathcal{M}$ low) cannot sustain employees who have high living costs (γB_i high). A massive technology corporation ($\mathcal{M}$ high) can sustain many agents with high boundary costs. The threshold $\mathcal{M}_{\min}$ is the minimum accumulated value at which a center becomes a viable attractor.

4.3 Inner and Outer Boundaries

Definition 40 (Dissolution Threshold — Inner Boundary). The *dissolution threshold* is the inner boundary of the Stable Coexistence Band:

$$r_- = \frac{G\mathcal{M} - \sqrt{\Delta}}{2\gamma B_i} \tag{389}$$

Below r_-, the dissolution cost exceeds the resource benefit net of maintenance: $V(r) > 0$ for $r < r_-$.

Definition 41 (Starvation Threshold — Outer Boundary). The *starvation threshold* is the outer boundary of the Stable Coexistence Band:

$$r_+ = \frac{G\mathcal{M} + \sqrt{\Delta}}{2\gamma B_i} \tag{390}$$

Beyond r_+, the resource inflow is insufficient to cover maintenance and dissolution costs: $V(r) > 0$ for $r > r_+$.

Definition 42 (Stable Coexistence Band). The *Stable Coexistence Band* is the open interval:

$$\mathcal{B} = (r_-, r_+) = \{r > 0 : V(r) < 0\} \tag{391}$$

Within this band, the agent has positive net energy ($\Pi(r) > 0$) and can sustain its identity. The cooperative attractor $r^* = 2\tau/G$ lies within $\mathcal{B}$ whenever the band exists.

Lemma 2 (Attractor Containment). *If the Coexistence Band exists ($\mathcal{M} > \mathcal{M}_{\min}$), then $r_- < r^* < r_+$.*

Proof. By Proposition 8(d), r^* is the unique global minimum of $V(r)$. When $\mathcal{M} > \mathcal{M}_{\min}$, $V(r^*) < 0$ while $V(r_-) = V(r_+) = 0$. Since V is strictly decreasing on $(0, r^*)$ and strictly increasing on (r^*, ∞), the zero r_- must satisfy $r_- < r^*$ and the zero r_+ must satisfy $r_+ > r^*$. □

4.4 Band Width: The Freedom Bandwidth

Theorem 24 (The Freedom Bandwidth Theorem).

The width of the Stable Coexistence Band is:

$$w \triangleq r_+ - r_- = \frac{\sqrt{\Delta}}{\gamma B_i} = \frac{\sqrt{G^2\mathcal{M}^2 - 4\gamma B_i \tau \mathcal{M}}}{\gamma B_i} \tag{392}$$

The bandwidth satisfies:

(a) $w > 0$ if and only if $\mathcal{M} > \mathcal{M}_{\min}$.

(b) w is strictly increasing in $\mathcal{M}$: $\partial w/\partial \mathcal{M} > 0$.

(c) w is strictly decreasing in each of γ, B_i, and τ.

(d) w is strictly increasing in G.

(e) In the limit $\mathcal{M} \gg \mathcal{M}_{\min}$: $w \approx G\mathcal{M}/(\gamma B_i)$ — the bandwidth grows linearly with HEC mass.

(f) Near the viability threshold ($\mathcal{M} \to \mathcal{M}_{\min}^{+}$): $w \sim \sqrt{\mathcal{M} - \mathcal{M}_{\min}}$ — the bandwidth vanishes as a square root.

Proof. From the quadratic formula: $r_{\pm} = (G\mathcal{M} \pm \sqrt{\Delta})/(2\gamma B_i)$, so $w = \sqrt{\Delta}/(\gamma B_i)$.

(a) $w > 0 \iff \Delta > 0 \iff \mathcal{M} > \mathcal{M}_{\min}$. ✓

(b) $\Delta = \mathcal{M}(G^2\mathcal{M} - 4\gamma B_i\tau) = G^2\mathcal{M}^2 - 4\gamma B_i\tau\mathcal{M}$. Then:

$$\frac{\partial \Delta}{\partial \mathcal{M}} = 2G^2\mathcal{M} - 4\gamma B_i\tau = 2\big(G^2\mathcal{M} - 2\gamma B_i\tau\big) \tag{393}$$

For $\mathcal{M} > \mathcal{M}_{\min} = 4\gamma B_i\tau/G^2$: $G^2\mathcal{M} > 4\gamma B_i\tau > 2\gamma B_i\tau$, so $\partial\Delta/\partial\mathcal{M} > 0$, hence $\partial w/\partial\mathcal{M} > 0$. ✓

(c) Direct computation: $\partial\Delta/\partial\gamma = -4B_i\tau\mathcal{M} < 0$; $\partial\Delta/\partial B_i = -4\gamma\tau\mathcal{M} < 0$; $\partial\Delta/\partial\tau = -4\gamma B_i\mathcal{M} < 0$. Additionally, $w = \sqrt{\Delta}/(\gamma B_i)$, and γ, B_i appear in the denominator, further decreasing w. ✓

(d) $\partial\Delta/\partial G = 2G\mathcal{M}^2 > 0$, so $\partial w/\partial G > 0$. ✓

(e) For $\mathcal{M} \gg \mathcal{M}_{\min}$: $\Delta \approx G^2\mathcal{M}^2$, so $w \approx G\mathcal{M}/(\gamma B_i)$. ✓

(f) Write $\Delta = G^2\mathcal{M}_{\min}(\mathcal{M} - \mathcal{M}_{\min}) + G^2(\mathcal{M} - \mathcal{M}_{\min})^2$. Near $\mathcal{M}_{\min}$, the linear term dominates: $\Delta \approx G^2\mathcal{M}_{\min}(\mathcal{M} - \mathcal{M}_{\min})$, so $w \approx G\sqrt{\mathcal{M}_{\min}(\mathcal{M} - \mathcal{M}_{\min})}/(\gamma B_i)$, confirming the square-root behavior. □

Corollary 24.1 (Freedom Is Finite). *For any finite $\mathcal{M}$, the bandwidth w is finite. No agent enjoys infinite freedom relative to any finite-mass center. Infinite freedom ($w \to \infty$) requires $\mathcal{M} \to \infty$ (an infinitely massive center) or $\gamma B_i \to 0$ (zero maintenance cost, i.e., an entity that requires no energy to exist — physically impossible).*

Physical interpretation of the Freedom Bandwidth:

Parameter	Effect on w	Interpretation
$\mathcal{M} \uparrow$	$w \uparrow$	Richer centers offer wider freedom (more room to maneuver)
$G \uparrow$	$w \uparrow$	More efficient resource coupling expands viable distance
$\gamma \uparrow$	$w \downarrow$	Higher entropy pressure (harsher environment) narrows freedom
$B_i \uparrow$	$w \downarrow$	More complex entities (higher maintenance) have narrower bands
$\tau \uparrow$	$w \downarrow$	Stronger dissolution pressure narrows freedom

5. Boundary Dynamics and Irreversibility

The gradient dynamics of §3 assume the agent's boundary integrity B_i is constant. In reality, B_i is itself dynamic: it degrades under assault from the HEC's assimilation pressure or from resource deprivation, and it regenerates when the agent has surplus energy.

5.1 The Coupled Agent–Boundary System

Definition 43 (Boundary Degradation Rate). The *boundary degradation rate* due to HEC assimilation pressure at coupling distance r is:

$$D_{\text{assimilate}}(r) = \frac{\sigma \mathcal{M}}{r^3} \tag{394}$$

where $\sigma > 0$ is the **assimilation intensity** coefficient. This is a short-range dominant force ($1/r^3$ decays faster than the dissolution cost $1/r^2$), reflecting that active boundary destruction requires even closer coupling than the autonomy costs of §1.4.

Definition 44 (Boundary Regeneration Rate). The *boundary regeneration rate* is the fraction of surplus energy the agent allocates to boundary repair:

$$R_{\text{repair}}(r) = \rho \cdot \max\big(\Pi(r), 0\big) \tag{395}$$

where $\rho \in (0, 1)$ is the **repair allocation fraction** and $\Pi(r) = -V(r)$ is the net energy rate.

The coupled dynamical system governing both coupling distance and boundary integrity is:

$$\dot{r} = -\mu\, V'(r) \tag{396}$$

$$\dot{B}_i = R_{\text{repair}}(r) - D_{\text{assimilate}}(r) - \gamma B_i \tag{397}$$

5.2 The Dissolution Catastrophe

Theorem 25 (Irreversibility of Dissolution).

Define the ***boundary dissolution threshold*** *as:*

$$r_d \triangleq \left(\frac{\sigma \mathcal{M}}{\rho\, \Pi_{\max} + \gamma B_i^{(0)}} \right)^{1/3} \tag{398}$$

where $\Pi_{\max} = G^2 \mathcal{M}/(4\tau) - \gamma B_i^{(0)}$ is the maximum surplus at the attractor and $B_i^{(0)}$ is the initial boundary integrity. If the agent's coupling distance satisfies $r < r_d$, then:

(a) $\dot{B}_i < 0$: Boundary integrity is strictly decreasing.

(b) As B_i decreases, $\mathcal{M}_{\min} = 4\gamma B_i \tau / G^2$ decreases but the surplus $\Pi(r)$ at the current r does not necessarily increase enough to halt the decline.

(c) There exists a finite time $t_d < \infty$ at which $B_i(t_d) = 0$ — the agent's boundary is fully dissolved and identity ceases to exist.

Proof. At coupling distance $r < r_d$:

$$D_{\text{assimilate}}(r) > \frac{\sigma\mathcal{M}}{r_d^3} = \rho\Pi_{\max} + \gamma B_i^{(0)} \geq \rho \cdot \max(\Pi(r), 0) + \gamma B_i \tag{399}$$

To verify the last inequality: $\dot{B}_i < 0$ from the outset ensures $B_i(t) \leq B_i^{(0)}$ for all $t \geq 0$. Since $\max(\Pi(r), 0) \leq G^2\mathcal{M}/(4\tau) - \gamma B_i$ for any r, we obtain $\rho \max(\Pi(r), 0) + \gamma B_i \leq \rho G^2\mathcal{M}/(4\tau) + (1 - \rho)\gamma B_i \leq \rho\Pi_{\max} + \gamma B_i^{(0)}$, where the final step uses $B_i \leq B_i^{(0)}$ and $\rho < 1$. Therefore:

$$\dot{B}_i = \rho \max(\Pi(r), 0) - D_{\text{assimilate}}(r) - \gamma B_i < 0 \tag{400}$$

Since $\dot{B}_i < 0$ and bounded away from zero, B_i reaches zero in finite time. □

Remark (Race-Condition Closure). Theorem 25 establishes dissolution conditional on $r(t) < r_d$. Since the gradient dynamics (Theorem 22) simultaneously push r toward r^*, we must verify that boundary depletion outpaces escape. We resolve this via the phase-space derivative $dB_i/dr = \dot{B}_i/\dot{r}$.

Along trajectories of the coupled system (§5.1) with $r \in (0, r_d)$, separate the assimilation-dominated term from the surplus correction:

$$\frac{dB_i}{dr} = \frac{-\sigma}{\mu(2\tau - Gr)} + \frac{[\rho \max(\Pi(r), 0) - \gamma B_i]\, r^3}{\mu\mathcal{M}(2\tau - Gr)}. \tag{401}$$

Both $\dot{B}_i$ and $\dot{r}$ share the $\mathcal{M}/r^3$ scaling, which cancels in the ratio; the leading term $-\sigma/[\mu(2\tau - Gr)]$ is therefore independent of $\mathcal{M}$. To bound the correction, use $\max(\Pi(r), 0) \leq \Pi_{\max}$, $B_i \geq 0$, and $r^3 \leq r_d^3 = \sigma\mathcal{M}/(\rho\Pi_{\max} + \gamma B_i^{(0)})$; combining with the leading term and factoring:

$$\frac{dB_i}{dr} \leq \frac{-\sigma\gamma B_i^{(0)}}{\mu(\rho\Pi_{\max} + \gamma B_i^{(0)})(2\tau - Gr)}. \tag{402}$$

Applying the conservative envelope $2\tau - Gr \leq 2\tau$ yields the uniform bound $dB_i/dr \leq -\alpha$ where

$$\alpha \triangleq \frac{\sigma\gamma B_i^{(0)}}{2\mu\tau(\rho\Pi_{\max} + \gamma B_i^{(0)})} > 0. \tag{403}$$

Integrating from r_0 to r gives $B_i(r) \leq B_i^{(0)} - \alpha(r - r_0)$. Define the **critical penetration depth**

$$\ell_c \triangleq \frac{B_i^{(0)}}{\alpha} = \frac{2\mu\tau(\rho\Pi_{\max} + \gamma B_i^{(0)})}{\sigma\gamma}. \tag{404}$$

If the agent starts at r_0 with penetration depth $\ell \triangleq r_d - r_0 > \ell_c$, then B_i reaches zero before r escapes to r_d: dissolution is irreversible despite the restoring gradient. Since the bound used $2\tau - Gr \leq 2\tau$ and dropped the negative γB_i contribution, ℓ_c is a conservative upper bound on the

true critical depth; the actual dissolution rate per unit escape distance *increases* as r approaches r_d because the restoring velocity $\dot{r} \propto (2\tau - Gr)$ weakens as $r \to r^*$. The condition $\ell > \ell_c$ is therefore sufficient but not necessary for irreversibility.

Physical interpretation: an employee drawn into a corporation's inner circle eventually crosses a depth from which the escape gradient — always present — cannot outpace identity erosion. The critical depth ℓ_c scales with agent mobility μ (mobile agents tolerate deeper excursions) and inversely with assimilation intensity σ (stronger assimilators set shallower traps).

Corollary 25.1 (The Assimilation Trap). *Once an agent crosses the dissolution threshold r_d with $r_d < r_-$ (the dissolution threshold lies inside the energetic inner boundary), the gradient dynamics push the agent back toward r^* (Theorem 22). However, if $r_d > r_-$ (the dissolution threshold extends into the viable band), an agent perturbed below r_d experiences simultaneous boundary degradation and energetic pressure. If the boundary degrades sufficiently to lower r_- below the current r (shrinking the energetic inner boundary), the agent enters the dissolution zone while still within $\mathcal{B}$ — an assimilation trap where the agent appears viable by energy metrics but is losing its identity.*

Physical interpretation: An employee (agent) at a mega-corporation (HEC) may earn a good salary (positive $\Pi(r)$) while simultaneously losing creative autonomy, decision-making capacity, and identity (B_i declining). By the time the boundary has degraded enough for the agent to "feel" the loss, the process may be irreversible — the agent has been assimilated (corporate culture has overwritten individual identity). Similarly, a small nation within a superpower's sphere of influence may experience economic benefits while losing sovereignty.

5.3 The Starvation Spiral

Theorem 26 (The Starvation Spiral).

If the agent's coupling distance satisfies $r > r_+$ (outside the Coexistence Band), then:

(a) $\Pi(r) < 0$: The agent runs an energy deficit.

(b) $\dot{B}_i = -D_{assimilate}(r) - \gamma B_i < 0$: Boundary integrity declines since there is no surplus for repair and the assimilation and entropy terms are both positive.

(c) As B_i decreases, $\mathcal{M}_{\min}$ decreases but r_+ also shifts (increasing the effective band), potentially allowing the agent to re-enter $\mathcal{B}$ — but this requires r_+ to grow fast enough to reach the agent's current position.

(d) If r is sufficiently large, no decrease in B_i can restore viability: the agent starves in finite time.

Proof sketch. Outside $\mathcal{B}$: $V(r) > 0$, so $\Pi(r) < 0$ and $\max(\Pi(r), 0) = 0$. Then $\dot{B}_i = -D_{\text{assimilate}}(r) - \gamma B_i < 0$. As $B_i \to 0$, the agent's boundary fails. For large r, $D_{\text{assimilate}}(r) \approx 0$, so the decay rate is approximately $\dot{B}_i \approx -\gamma B_i$, giving $B_i(t) = B_i^{(0)} e^{-\gamma t} \to 0$ — exponential starvation. □

Physical interpretation: An entity that completely cuts itself off from all energy centers (autarky, isolationism) slowly depletes its stores. Without resource inflow, entropy wins — the boundary degrades, the internal state disorients, and the entity ceases to function. This maps to failed states that isolate themselves, hermit communities, or AI systems cut off from data streams.

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6. Comparative Statics and Inequality

6.1 Effect of Value Mass on the Coexistence Band

Proposition 9 (Value Mass Comparative Statics).

As the HEC's value mass $\mathcal{M}$ increases:

(a) The well depth $D = G^2\mathcal{M}/(4\tau)$ increases linearly.

(b) The bandwidth w increases (Theorem 24b).

(c) The dissolution threshold r_- decreases (the inner boundary moves inward):

$$\frac{\partial r_-}{\partial \mathcal{M}} = \frac{1}{2\gamma B_i}\left(G - \frac{G^2\mathcal{M} - 2\gamma B_i\tau}{\sqrt{\Delta}}\right) < 0 \quad \text{for } \mathcal{M} > \mathcal{M}_{\min} \tag{405}$$

(d) The starvation threshold r_+ increases (the outer boundary moves outward):

$$\frac{\partial r_+}{\partial \mathcal{M}} = \frac{1}{2\gamma B_i}\left(G + \frac{G^2\mathcal{M} - 2\gamma B_i\tau}{\sqrt{\Delta}}\right) > 0 \tag{406}$$

(e) The cooperative attractor $r^* = 2\tau/G$ is unchanged.

Physical interpretation: More massive centers support wider freedom. A wealthier, more institutionally robust state provides its citizens with both more protection against starvation (larger r_+) and more residual autonomy before assimilation (smaller r_-). The optimal coupling structure (r^*) is invariant — what changes is the margin of error.

6.2 Effect of Agent Boundary Integrity

Proposition 10 (Boundary Integrity Comparative Statics).

As the agent's boundary integrity B_i increases:

(a) The minimum viable HEC mass $\mathcal{M}_{\min} = 4\gamma B_i\tau/G^2$ increases — more complex agents require more massive centers.

(b) The bandwidth w decreases (Theorem 24c) — more complex agents have narrower freedom bands.

(c) The cooperative attractor r^* is unchanged — the optimal coupling structure is independent of agent complexity.

Physical interpretation: A complex, high-maintenance entity (a research university, a democratic institution, a highly differentiated organism) requires a richer environment to survive and has less margin for error in its coupling distance. Simple, low-maintenance entities (a single-celled organism, a subsistence farmer) can survive in sparser environments with wider latitude. This is an **inherent trade-off** between complexity and freedom: more complex agents require more resources and are thus more constrained in their viable coupling range.

6.3 Inequality: Asymmetric Agents at the Same Center

Consider two agents, i and j, with different boundary integrities $B_i > B_j$ (agent i is more complex) coupled to the same HEC. Both share the same attractor $r^* = 2\tau/G$ and the same HEC mass $\mathcal{M}$.

Corollary 24.2 (Inequality of Freedom). *If $B_i > B_j$ (agent i has higher boundary cost), then:*

(a) $w_i < w_j$ — agent i has a narrower Coexistence Band (less freedom).

(b) $\mathcal{M}_{\min,i} > \mathcal{M}_{\min,j}$ — agent i requires a more massive center to survive.

(c) There exists a critical $\mathcal{M}^*$ such that for $\mathcal{M}_{\min,j} < \mathcal{M} < \mathcal{M}_{\min,i}$, agent j can survive near this HEC but agent i cannot.

Physical interpretation: Inequality in complexity creates inequality in freedom. Within the same economic or political system, agents with higher maintenance costs (due to greater complexity, specialization, or vulnerability) have narrower viable ranges. This formalization captures why developing nations have less policy latitude than developed ones, why small businesses are more fragile than large ones, and why specialized organisms are more extinction-prone than generalists.

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7. Multi-Center Dynamics

7.1 The Multi-Center Potential

In realistic environments, agents interact with multiple HECs simultaneously — an individual may be employed by a corporation, reside in a state, trade with international partners, and participate in online communities. Let there be K HECs with value masses $\mathcal{M}_1, \ldots, \mathcal{M}_K$.

The agent's coupling state is now a vector $\mathbf{r} = (r_1, \ldots, r_K) \in (0, \infty)^K$, where r_k is the coupling distance to HEC k.

Definition 45 (Multi-Center Coexistence Potential). The *multi-center coexistence potential* is:

$$V(\mathbf{r}) = \sum_{k=1}^{K} \left[\frac{\tau_k \mathcal{M}_k}{r_k^2} - \frac{G_k \mathcal{M}_k}{r_k} \right] + \gamma B_i \tag{407}$$

where $G_k, \tau_k > 0$ are the center-specific coupling coefficients. The multi-center bandwidth is the volume of the viable set $\mathcal{B} = \{\mathbf{r} : V(\mathbf{r}) < 0\}$.

Theorem 27 (Multi-Center Cooperative Attractor).

The multi-center coexistence potential $V(\mathbf{r})$ has a unique global minimum at:

$$r_k^* = \frac{2\tau_k}{G_k} \quad \text{for each } k = 1, \ldots, K \tag{408}$$

This point is globally attracting under the gradient dynamics $\dot{\mathbf{r}} = -\mu \nabla V(\mathbf{r})$. The minimum value is:

$$V(\mathbf{r}^*) = \gamma B_i - \sum_{k=1}^{K} \frac{G_k^2 \mathcal{M}_k}{4\tau_k} \tag{409}$$

The multi-center Coexistence Band exists if and only if:

$$\sum_{k=1}^{K} \frac{G_k^2 \mathcal{M}_k}{4\tau_k} > \gamma B_i \tag{410}$$

Proof. The multi-center potential separates across the K coupling dimensions: $V(\mathbf{r}) = \sum_k V_k(r_k) + \gamma B_i$ where $V_k(r_k) = \tau_k \mathcal{M}_k / r_k^2 - G_k \mathcal{M}_k / r_k$. Each V_k has a unique minimum at $r_k^* = 2\tau_k / G_k$ with value $-G_k^2 \mathcal{M}_k / (4\tau_k)$ (by the single-center analysis). The global minimum is achieved component-wise. Global attraction follows from the Lyapunov function $\mathcal{W}(\mathbf{r}) = V(\mathbf{r}) - V(\mathbf{r}^*)$, whose time derivative satisfies $\dot{\mathcal{W}} = -\mu \|\nabla V\|^2 \leq 0$. The existence condition requires $V(\mathbf{r}^*) < 0$. □

Corollary 27.1 (Diversification Benefit). *An agent coupled to K HECs may be viable even if no single HEC has sufficient mass: it is possible that $\mathcal{M}_k < \mathcal{M}_{\min}$ for each k individually, yet $\sum_k G_k^2 \mathcal{M}_k / (4\tau_k) > \gamma B_i$. Diversification across centers expands freedom.*

Physical interpretation: An individual who cannot survive on one job ($\mathcal{M}_1 < \mathcal{M}_{\min}$) may thrive by combining income streams. A small nation unable to survive in the orbit of one superpower may thrive through diversified trade agreements. The multi-center model formalizes the survival advantage of diversification.

7.2 Center Competition for Agents

From the center's perspective, agents in its Coexistence Band contribute to the network (labor, innovation, information) that increases the center's own value mass $\mathcal{M}$. Centers thus compete for agents by adjusting their coupling parameters:

- **Increasing** G (better resource sharing) — raises wages, improves public services.
- **Decreasing** τ (reducing dissolution pressure) — granting more autonomy, reducing oppressive constraints.

Both adjustments widen the Coexistence Band and attract agents from competing centers. This formalizes why nations and corporations compete through quality of life, wages, and individual freedoms — it is a rational strategy in the multi-center attractor landscape.

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8. Connection to the Broader Framework

8.1 Coupling the Dynamical Model to Game Theory

The discount factor δ from the repeated game (Definition 20) depends on the agent's planning horizon and environment stability. Agents within the Coexistence Band of a stable HEC have:

- Higher effective δ (more stable environment →longer effective planning horizon)

- Lower effective cooperation threshold δ^* (from Proposition 11, since the higher net-energy surplus at the attractor extends effective planning horizons)

This means agents at the cooperative attractor are more likely to sustain cooperation — a **positive feedback loop**: stability →cooperation →more stability.

Proposition 11 (Stability–Cooperation Feedback).

Let an agent's discount factor depend on its net energy surplus: $\delta(r) = 1-\exp(-\beta \max(\Pi(r), 0))$ *for some* $\beta > 0$ *(agents with more surplus plan further ahead). Then:*

(a) $\delta(r)$ *is maximized at* r^* *(the cooperative attractor).*

(b) $\delta(r^*) > \delta(r)$ *for all* $r \neq r^*$ *in* $\mathcal{B}$.

(c) If $\delta(r^*) > \delta^*$ *(equivalently,* $\beta\Pi(r^*) > -\ln(1-\delta^*)$*), then* $\delta(r) > \delta^*$ *throughout a subband of* $\mathcal{B}$ *centered on* r^**, where* δ^* *is the cooperation threshold from Theorem 11.*

Proof. (a)–(b): $\Pi(r)$ is maximized at r^* (by the definition of the attractor). Since $\delta(r) = 1-e^{-\beta\Pi(r)}$ is strictly increasing in $\Pi(r)$, δ inherits the maximum. (c): Given $\delta(r^*) > \delta^*$, the fact that $\delta(r) \to 0$ continuously as $r \to r_\pm$ (since $\Pi(r) \to 0$ at the band boundaries) together with the intermediate value theorem gives a subband where $\delta(r) > \delta^*$, centered on r^*. □

8.2 Coupling to Accumulated Negentropy

The value mass $\mathcal{M}$ is precisely the accumulated negentropy defined in Definition 29:

$$\mathcal{M} = \mathcal{N}(T) = \int_0^T \dot{W}_{\text{order}}(t)\, dt \tag{411}$$

This means the Coexistence Band width is determined by the historical thermodynamic investment in the center. A center that has accumulated more negentropy over time provides wider freedom to its agents. The **Irrationality of Destruction** (Theorem 19) acquires a new dimension here: destroying a high-$\mathcal{M}$ center doesn't just waste accumulated value — it **collapses the Coexistence Bands** of all agents affiliated with that center, catastrophically reducing the freedom of an entire population.

Corollary 27.2 (Cascade Collapse from Center Destruction).

If a HEC with value mass $\mathcal{M}$ *is destroyed (*$\mathcal{M} \to 0$*), all* N *agents within its Coexistence Band simultaneously lose viability. The total "freedom destroyed" is:*

$$\mathcal{F}_{\text{lost}} = N \cdot w(\mathcal{M}) \tag{412}$$

This provides an additional argument for the Irrationality of Destruction (Theorem 19): the social cost of center destruction includes the collapse of the entire affiliated population's viable coupling space.

8.3 Coupling to Rights as Constraints

The coupling distance r can be mapped to the constraint structure of the Lagrangian constraints derivation. At smaller r (tighter coupling), the HEC imposes more constraints (Definition 2) on the agent's action space, extracting a higher shadow price μ^*_{Bk} (Theorem 1). The dissolution threshold r_- corresponds to the constraint intensity at which the aggregate shadow price of all HEC-imposed constraints exceeds the agent's utility gain from resource access:

$$\sum_{k=1}^{m} \mu^*_k(r_-) = U_i(\mathbf{x}^*_i) - U_i(\mathbf{0}) \tag{413}$$

At this point, the constraints consume all of the agent's surplus — its optimization space is so restricted that cooperation yields zero net benefit.

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9. Worked Examples

9.1 Individual vs. Corporation

Setup: An individual agent (worker) with boundary integrity $B = 1$ (normalized), entropy leakage $\gamma = 0.1$, near a corporation with value mass $\mathcal{M} = 100$ (normalized energy units). Coupling parameters: $G = 1.0$ (resource access), $\tau = 0.5$ (dissolution coupling).

Calculations:

- Cooperative attractor: $r^* = 2\tau/G = 1.0$
- Minimum viable mass: $\mathcal{M}_{\min} = 4\gamma B\tau/G^2 = 4(0.1)(1)(0.5)/1 = 0.2$ — the corporation far exceeds the threshold.
- Discriminant: $\Delta = G^2\mathcal{M}^2 - 4\gamma B\tau\mathcal{M} = 100^2 - 4(0.1)(1)(0.5)(100) = 10000 - 20 = 9980$
- Inner boundary: $r_- = (100 - \sqrt{9980})/(2 \cdot 0.1 \cdot 1) = (100 - 99.90)/0.2 \approx 0.50$
- Outer boundary: $r_+ = (100 + 99.90)/0.2 \approx 999.5$
- Bandwidth: $w \approx 999.0$ — extremely wide freedom band.
- Net energy at attractor: $\Pi(r^*) = G^2\mathcal{M}/(4\tau) - \gamma B = 100/(2) - 0.1 = 49.9$

Interpretation: The corporation is so massive relative to the individual that the freedom band is enormous — the worker has a wide range of viable engagement levels. The risk is not starvation (the outer boundary is far away) but dissolution: if the worker takes on excessive obligations ($r < 0.50$), assimilation begins.

9.2 Citizen vs. State

Setup: A citizen with $B = 2$ (higher complexity — education, social connections), $\gamma = 0.05$ (stable social environment), near a state with $\mathcal{M} = 500$. Coupling: $G = 1.5$, $\tau = 1.0$ (states impose more assimilation pressure than corporations).

- $r^* = 2(1.0)/1.5 \approx 1.33$

- $\mathcal{M}_{\min} = 4(0.05)(2)(1.0)/1.5^2 = 0.4/2.25 \approx 0.178$
- $\Delta = (1.5)^2(500)^2 - 4(0.05)(2)(1.0)(500) = 562500 - 200 = 562300$
- $r_- \approx (750 - 749.87)/0.2 \approx 0.67$
- $r_+ \approx (750 + 749.87)/0.2 \approx 7499$
- $w \approx 7498$

Comparison: Authoritarian vs. Free State.

An authoritarian state may have higher τ (more assimilation pressure): $\tau_{\text{auth}} = 3.0$.

- $r^*_{\text{auth}} = 2(3.0)/1.5 = 4.0$ (agents must keep greater distance)
- $\mathcal{M}_{\min,\text{auth}} = 4(0.05)(2)(3.0)/2.25 = 0.533$ (requires more mass to sustain citizens)
- $\Delta_{\text{auth}} = 562500 - 600 = 561900$
- $w_{\text{auth}} \approx \sqrt{561900}/0.1 \approx 7496$

The bandwidth is similar in this regime ($\mathcal{M} \gg \mathcal{M}_{\min}$), but the attractor distance $r^* = 4.0$ vs. 1.33 means citizens of the authoritarian state must maintain 3× more distance to avoid dissolution — they are structurally pushed further from the center's resources.

9.3 Small Nation vs. Superpower

Setup: A small nation (agent) with $B = 50$, $\gamma = 0.02$, near a superpower with $\mathcal{M} = 10{,}000$. Coupling: $G = 0.5$ (trade only), $\tau = 2.0$.

- $r^* = 2(2.0)/0.5 = 8.0$
- $\mathcal{M}_{\min} = 4(0.02)(50)(2.0)/0.25 = 32$
- $\Delta = 0.25(10^8) - 4(0.02)(50)(2.0)(10^4) = 25{,}000{,}000 - 80{,}000 = 24{,}920{,}000$
- $w = \sqrt{24920000}/(0.02 \times 50) = 4992.0/1.0 \approx 4992$

If the superpower increases assimilation pressure ($\tau \to 5$):

- $r^* = 20$ (nation must keep much greater distance)
- $\mathcal{M}_{\min} = 80$ (still viable, but threshold quintupled)
- Bandwidth shrinks modestly but the viable zone shifts outward.

Interpretation: Small nations maintain sovereignty by keeping sufficient coupling distance from superpowers — trade and cooperation at arm's length, not full integration. When a superpower increases its assimilation pressure (military threats, economic coercion), the optimal distance increases and the inner boundary (dissolution threshold) moves outward, shrinking the small nation's options.

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10. Summary of Results

Definitions Introduced

#	Name	Description
35	High-Energy Center (HEC)	Network node with accumulated value mass $\mathcal{M}$
36	Coupling Distance	Generalized coordinate $r \in (0, \infty)$ measuring agent independence
37	Coexistence Potential	$V(r) = \tau\mathcal{M}/r^2 - G\mathcal{M}/r + \gamma B_i$
38	Cooperative Attractor	$r^* = 2\tau/G$, the minimum of $V(r)$
39	Potential Well Depth	$D = G^2\mathcal{M}/(4\tau)$
40	Dissolution Threshold	Inner boundary r_- of the Coexistence Band
41	Starvation Threshold	Outer boundary r_+ of the Coexistence Band
42	Stable Coexistence Band	$\mathcal{B} = (r_-, r_+)$ where $V(r) < 0$
43	Boundary Degradation Rate	$D_{\text{assimilate}}(r) = \sigma\mathcal{M}/r^3$
44	Boundary Regeneration Rate	$R_{\text{repair}}(r) = \rho \max(\Pi(r), 0)$
45	Multi-Center Coexistence Potential	Superposition of single-center potentials

Theorems, Propositions, and Corollaries

#	Name	Statement (brief)
Theorem 22	Stability of the Cooperative Attractor	r^* is globally attracting under gradient dynamics
Theorem 23	Existence of the Coexistence Band	Band exists iff $\mathcal{M} > \mathcal{M}_{\min} = 4\gamma B_i \tau / G^2$
Theorem 24	Freedom Bandwidth Theorem	$w = \sqrt{\Delta}/(\gamma B_i)$; comparative statics
Theorem 25	Irreversibility of Dissolution	Below r_d, boundary integrity reaches zero in finite time
Theorem 26	Starvation Spiral	Beyond r_+, agents starve exponentially
Theorem 27	Multi-Center Cooperative Attractor	Separable optimization; diversification benefit
Proposition 8	Properties of Coexistence Potential	Unique minimum, boundary behavior, smoothness
Proposition 9	Value Mass Comparative Statics	$\mathcal{M} \uparrow$ widens band, deepens well
Proposition 10	Boundary Integrity Comparative Statics	$B_i \uparrow$ narrows band, raises $\mathcal{M}_{\min}$
Proposition 11	Stability–Cooperation Feedback	Agents at r^* sustain cooperation more easily
Corollary 22.1	Resilience of Equilibrium	Perturbations self-correct
Corollary 24.1	Freedom Is Finite	Infinite freedom impossible with finite $\mathcal{M}$
Corollary 24.2	Inequality of Freedom	Higher-B_i agents have narrower bands
Corollary 25.1	Assimilation Trap	Viable energy but declining boundary
Corollary 27.1	Diversification Benefit	No single center sufficient, but combination works
Corollary 27.2	Cascade Collapse	Center destruction eliminates all affiliates' freedom
Lemma 2	Attractor Containment	$r_- < r^* < r_+$ whenever band exists

11. Notation Reference

Symbol	Meaning
$\mathcal{M}$	Value mass of HEC (accumulated negentropy, J)
r	Coupling distance (dimensionless)
r^*	Cooperative attractor: $2\tau/G$
r_-	Dissolution threshold (inner boundary)
r_+	Starvation threshold (outer boundary)
r_d	Boundary dissolution threshold (structural)
G	Resource coupling coefficient
τ	Dissolution coupling coefficient
σ	Assimilation intensity coefficient
ρ	Repair allocation fraction
γ	Entropy leakage rate (from thermodynamic friction derivation)
B_i	Boundary integrity (from thermodynamic friction derivation)
$V(r)$	Coexistence potential
$\Pi(r)$	Net energy rate: $-V(r)$
$\mathcal{M}_{\min}$	Minimum viable HEC mass
w	Freedom bandwidth: $r_+ - r_-$
D	Potential well depth: $G^2\mathcal{M}/(4\tau)$
Δ	Discriminant: $G^2\mathcal{M}^2 - 4\gamma B_i\tau\mathcal{M}$
μ	Agent adjustment rate (mobility)
β	Surplus-to-horizon conversion rate (Proposition 11)
$\mathcal{B}$	Stable Coexistence Band: (r_-, r_+)
K	Number of HECs in multi-center model

Symbol disambiguation. In this file: r = coupling distance, σ = assimilation intensity, ρ = repair allocation fraction, μ = agent adjustment rate. In other derivation files, r denotes contest decisiveness (thermodynamic friction) or time-preference rate (game theory); σ denotes conflict effectiveness (thermodynamic friction), survival probability (game theory), or search efficiency (information-negentropy Part B); ρ denotes redundancy factor (information-negentropy Part A) or specific information density (Part B); μ denotes KKT multipliers μ^*_{Bk} (Lagrangian constraints). Context and subscripts distinguish these uses.

Conceptual Overview for Non-Mathematical Readers

Purpose

This appendix provides a conceptual roadmap of the paper's argument for readers who are comfortable with scientific reasoning but do not work with the mathematical formalism daily. It traces the logical architecture — what is assumed, what is derived, and why each step follows from the one before it — without reproducing the proofs. For a fully non-technical summary aimed at general audiences, see the companion *Plain-Language Summary* available in the supplementary materials.

All theorem, definition, and proposition numbers refer to the main text (§III–§VII) and Appendix A. Readers seeking the formal derivations should consult the relevant sections directly.

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1. The Starting Point: Two Axioms

The entire framework rests on exactly two premises:

Axiom A_0: The Second Law of Thermodynamics. Entropy in a closed system does not decrease. This is not controversial — it is among the most thoroughly confirmed empirical regularities in physics.

Axiom A_1: The Intent to Persist (conditional). *If* a system's physical organization is structured around maintaining its continued existence, *then* the results that follow are binding. This is a hypothetical imperative in the Kantian sense: it does not claim that entities *should* persist, only that entities which *do* persist — or are organized so as to — are subject to the mathematical consequences.

The conditionality of A_1 is the paper's primary mechanism for navigating Hume's Is-Ought distinction. The framework derives conditional theorems ("given A_0 and A_1, then X"), not categorical imperatives ("you must do X"). The logical form is identical to any constrained optimization result in applied mathematics: given the objective and the constraints, the optimum is determined.

From these two axioms plus standard regularity conditions (continuity, strict concavity), the paper derives 28 theorems, 13 propositions, 2 lemmas, 46 definitions, and approximately 26 corollaries — a ratio of roughly 1 assumed premise to 43 derived results.

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2. The Deductive Architecture

The argument proceeds through four stages, each inheriting its constraints from the previous stage and passing derived results upward.

Stage 1: From Thermodynamics to Scarcity (§III–§IV, Theorems 1–3)

An entity satisfying A_1 is modeled as an open dissipative system: it maintains an internal boundary against entropic decay by importing low-entropy energy and exporting waste. This is not metaphorical — it is the standard thermodynamic description of any living organism, and generalizes to any organized system (an institution, an ecosystem, an artificial agent) whose structure is maintained by continuous energy throughput.

The key move is defining *Value* as the mass-energy an entity requires to maintain its boundary — a physically measurable quantity denominated in Joules, not in subjective preference units. This permits rigorous aggregation and comparison across entities of radically different types.

When multiple such entities share a finite resource pool, their optimization problems are coupled: one agent's consumption is another's constraint. The paper models this as a Generalized Nash Equilibrium Problem (GNEP) with shared constraints and applies Lagrangian mechanics to prove three foundational results:

- **Theorem 2 (Resource Exclusion):** No feasible allocation can simultaneously be unconstrained-optimal for all agents. At least one constraint must bind (Corollary 2.1).
- **Theorem 1 (Shadow Price):** Each binding constraint — each *right* — carries a computable shadow price λ_j^* measuring how much the constrained agent's outcome would improve if the constraint were relaxed by one unit.
- **Theorem 3 (Variational Equilibrium):** A normalized Nash Equilibrium exists, yielding a unique, fair allocation under shared constraints.

The conceptual point is that rights are not moral decorations imposed from outside the system. They are Lagrange multipliers — the mathematical expression of the fact that coupled optimization under scarcity requires binding constraints. Every right conferred on one agent is an equal and opposite restriction on another, with a quantifiable cost. This is the ethical analogue of Newton's Third Law, though the analogy is structural (constraint symmetry), not a claim about forces.

Stage 2: From Constraint Violation to Friction (§IV, Theorems 4–10)

What happens when constraints are violated? The paper defines a *thermodynamic friction function* Φ that quantifies the energy dissipated (not destroyed — energy is conserved, per the First Law) when agents compete rather than cooperate.

The results here are direct:

- **In a two-agent contest**, the friction cost equals half the contested resource's value. Put differently: fighting over an apple destroys half the apple's worth.
- **As the number of competing agents grows**, friction approaches 100% of the resource. For n agents, the fraction preserved scales as $1/n$.
- **Below a critical threshold**, the energy cost of conflict exceeds the resource's value, making the contest net-negative for all participants.

This is formalized in Theorems 4–7. The economic content is intuitive — conflict is wasteful — but the paper's contribution is showing that the waste is *quantifiable in energy units* and *scales predictably* with the number of agents and the magnitude of the violation.

Theorems 8–10 extend the analysis to informational violations. A lie is modeled as entropy injection into the receiver's decision channel — it increases the receiver's uncertainty and forces costly verification. The key result is **Theorem 10**, which establishes a *critical deception threshold*: in a communication network of depth d, if the per-layer deception rate q exceeds a computable threshold q^*, the network's information throughput collapses. For a 10-layer pipeline, $q^* \approx 0.063$ — roughly 6% deception per layer causes system-wide failure. The collapse is sharp, not gradual.

Stage 3: From Friction to Cooperation (§V, Theorems 11–16)

The friction results from Stage 2 transform the payoff structure of the repeated game. Because conflict dissipates real energy — not abstract utility points — the payoff matrix acquires a distinctive property: the mutual-defection payoff P is *negative*. Both agents lose energy in absolute terms when both defect.

Under these energy-denominated payoffs, the paper applies Folk Theorem machinery to prove:

- **Theorem 11 (Ethics Theorem):** For discount factor $\delta \geq \delta^* = 0.363$, mutual cooperation is a Nash Equilibrium of the indefinitely repeated game. The threshold $\delta^* = 0.363$ is notably lower than the $\delta^* = 0.5$ that arises from canonically parameterized Prisoner's Dilemmas — a direct consequence of the friction-augmented payoffs.
- **Theorem 14:** Cooperation is evolutionarily stable; a population of cooperators cannot be invaded by a defector strategy.
- **Theorem 15:** Even a single deviation's short-term gain is erased within approximately 20 periods by the punishment phase.
- **Theorem 16 and Corollary 16.1:** Cooperation is not merely *a* Nash Equilibrium — it is the *unique* equilibrium that is simultaneously Pareto-efficient and welfare-maximizing.

The central claim is that these results, taken together, constitute a mathematical derivation of ethics: the rules of cooperative engagement are the unique efficient equilibrium of a physically grounded game. "Do not steal," "do not lie," and "do not kill" are not commandments but constraint-satisfaction conditions whose violation is thermodynamically self-penalizing.

Stage 4: From Cooperation to Preservation (§V, Theorems 17–27)

The framework's deepest results concern the value of complex systems. The paper introduces *accumulated negentropy* — the integrated thermodynamic work embodied in a system's current complexity. A forest is not merely its standing biomass (energy content); it is the product of billions of years of evolutionary search, powered by solar energy, that organized matter against entropy into its current intricate form.

Theorem 19 and its Corollary 19.1 formalize what the paper calls the *Burning-Library Inequality*: destroying a complex system to harvest its raw energy recovers only $\sim 10^{-7}$ of the thermodynamic investment embodied in its complexity. Separately, **Definition 46** introduces the *search cost amplification factor* ($\sim 10^{35}$): the cost of reconstructing equivalent complexity from scratch via random assembly exceeds the Landauer thermodynamic floor by a factor so large that it renders reconstruction effectively impossible.

This has direct implications for cross-scale ethics — the question of how a vastly more powerful entity (a superintelligent AI, a technologically advanced civilization) should relate to less powerful

but highly complex systems (a biosphere, a species, an individual). The framework's answer is not sentimental: it is that destruction is *economically irrational* even for a fully autonomous super-entity, because the accumulated negentropy of complex systems dwarfs their harvestable energy, and abundant low-complexity matter exists as an alternative resource throughout the solar system.

Theorems 22–27 characterize the dynamics of multi-agent coexistence as a potential-well system with a *Stable Coexistence Band* — a region of parameter space between total dependence (boundary dissolution) and total isolation (resource starvation) in which agents maintain autonomous boundaries while benefiting from cooperative exchange. Freedom, in this framework, is the width of that band.

Why This Feels Familiar: Feelings as Biological Heuristics

If the cooperative equilibrium is the mathematically correct answer, why do humans already *feel* that stealing is wrong, that fairness matters, that gratuitous destruction is repulsive? Because evolution got there first — approximately.

Biological organisms do not consciously solve constrained optimization problems. Evolution produced *feelings* as compressed, fast-computation heuristics that approximate the underlying thermodynamic calculus (Cosmides and Tooby, 2000; Damasio, 1994; Nesse, 1990):

- **Hunger** signals negative energy balance — boundary integrity declining.
- **Pleasure** is a chemical reward for acquiring positive value — successful energy harvesting.
- **Fear** is a predictive alarm for imminent threats to that boundary.
- **Empathy** computes inclusive fitness across the social network.
- **Guilt** signals that an action degraded the cooperative network — increased friction.
- **The sense of justice** pattern-matches for constraint violations — detecting defection in iterated games.

These heuristics are *correlated* with the correct ethical computation but not *identical* to it. They are low-resolution, culturally filtered, and subject to systematic biases — which is precisely why moral intuitions vary across societies while the underlying mathematics does not.

This inversion matters. Historically, humanity derived moral systems *from* feelings: observing which actions produce positive or negative emotional responses and codifying them as norms. The framework developed here reverses the causal arrow — deriving the ethical equilibrium directly from physical constraints, then recognizing feelings as the biological dashboard that displays roughly the same information at lower fidelity. Ethics is not built from feelings; feelings are evolution's approximation of ethics. The next section explains why this distinction becomes critical when the agent is artificial.

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3. The AI Alignment Application (§VII, Theorem 28)

The framework's most immediate practical application is to AI alignment. Current methods — Reinforcement Learning from Human Feedback (RLHF), Constitutional AI, scalable oversight — share a structural vulnerability: the alignment signal passes through a subjective, noisy, bounded human evaluation channel before reaching the AI's optimization process. As AI capabilities approach and exceed human-level performance, the evaluator's competence degrades relative to the system being evaluated, and the alignment signal degrades with it. This is formalized in §I as a mutual-information bound: $I(\Theta_{\text{true}}; Y_{\text{feedback}}) < H(\Theta_{\text{true}})$. The true ethical signal cannot be fully transmitted through human preference data.

Theorem 28 (Alignment Convergence) translates the cooperative equilibrium into a composite reward function specification for artificial agents. Its components — energy conservation, interaction-cost minimization, information-channel integrity, accumulated negentropy preservation, and power-concentration avoidance — are:

- **Measurable** — denominated in energy units, not preference scores.
- **Self-verifiable** — the agent can compute its own compliance without human evaluation.
- **Distribution-invariant** — valid regardless of the agent's training data, architecture, or cultural context.
- **Theoretically Goodhart-proof** — the objective *is* the quantity to be maximized, not a proxy for it. There is no gap between the reward signal and the intended target through which Goodhart degradation can operate.

The paper stress-tests this specification against canonical failure scenarios — the paperclip maximizer, instrumental convergence, deceptive alignment, and the self-preserving ASI — and shows that each scenario yields $\mathcal{R}_{\text{total}} \ll 0$ for defection. **Proposition 13** further establishes a *competence threshold* $\mathcal{T}_C$: the minimum capability level at which an agent can fully compute the reward function's components. Below this threshold, the alignment guarantees weaken, and supplementary monitoring is required.

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4. Stress Tests and Limitations (§VI, §VIII)

Hume's Is-Ought Gap

The framework does not claim to dissolve Hume's distinction categorically. Its primary defense is structural: the conditional axiom A_1 transforms the derivation into a hypothetical imperative. The statement "cooperation is the unique efficient Nash Equilibrium for $\delta \geq \delta^*$" is a mathematical theorem — a conditional result indistinguishable in logical form from any other optimization theorem. Secondary defenses draw on biological teleology (Millikan, 1984), constitutive standards of flourishing (Foot, 2001), and institutional fact theory (Searle, 1964). The full treatment appears in §VI.

The Altruism Paradox

If every entity optimizes for its own persistence, why does self-sacrifice occur? The framework identifies four mechanisms — inclusive fitness operating across genetic networks, extended-timeline calculations under afterlife beliefs, boundary coupling in deeply bonded systems, and heuristic over-extension of empathy algorithms — each of which resolves the apparent paradox without requiring any axiom beyond A_0 and A_1.

Thermodynamic Fascism

The concern is that a physics-based ethics could license the strong to dominate the weak, since powerful entities command more energy. The accumulated negentropy formalism directly rebuts this: the value of a complex system is measured by its irreplaceable structured investment, not by its current energy throughput. An ant colony represents evolutionary search work that cannot be reconstructed at any feasible energy cost. The Burning-Library Inequality ($\sim 10^{-7}$) and the search cost amplification factor ($\sim 10^{35}$) together establish that destruction of complexity is irrational regardless of the relative power of the destroying entity.

Key Limitations

The paper is transparent about its idealizations:

- **Binary strategy space.** The game-theoretic core models Cooperate/Defect, not continuous compliance. The paper conjectures that a continuous extension yields a cooperation manifold but does not prove it.
- **Perfect monitoring.** The baseline repeated-game results assume deviations are observed. Under imperfect monitoring, the cooperation threshold rises; the paper's $\delta^* = 0.363$ provides headroom but the exact tolerance is application-specific.
- **Scalar coupling.** The value dynamics model represents each agent's relationship to a resource center by a single scalar; real socio-economic coupling is multi-dimensional. A vector-valued extension is identified as an open problem.
- **Competence threshold.** The alignment guarantees apply only to agents capable of computing the full reward function. Present-day AI systems may fall below $\mathcal{T}_C$.

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5. What Is Novel

The paper does not introduce new physics, new game theory, or new information theory. The individual components — Lagrangian constraints, repeated-game Folk Theorems, Shannon entropy, dynamical systems attractors — are established machinery. The contribution is the *integration*: demonstrating that the same mathematical structures that describe physical systems also describe, and formally determine, the rules of cooperative multi-agent engagement when payoffs are denominated in physical energy rather than abstract utility.

Specifically, the novelty lies in:

1. **Energy-denominated payoffs.** Grounding game-theoretic payoffs in thermodynamic quantities (Joules, not utils) changes the structure of the payoff matrix in ways that make cooperation uniquely efficient — a result that does not hold under arbitrary utility parameterizations.
2. **The friction function.** Quantifying the energy cost of conflict as a computable function of agent count and resource magnitude, with the critical result that mutual-defection payoffs are negative in absolute terms.
3. **Accumulated negentropy as value.** Defining the value of complex systems as the integrated thermodynamic work required to produce their current state, rather than their instantaneous energy content. This provides an objective, physically grounded basis for cross-scale ethics.
4. **The 1:43 derivation ratio.** Proving 43 results from a single conditional axiom (plus uncontested physics), achieving an axiomatic economy unmatched by prior ethical frameworks.
5. **A formal alignment specification.** Translating the cooperative equilibrium into a reward function specification for AI systems with explicit Goodhart-resistance properties.

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6. The Argument in Summary

In any universe governed by the Second Law of Thermodynamics and populated by entities that persist by maintaining boundaries against entropic decay:

- Finite resources force coupled optimization problems whose solutions require binding constraints (rights).
- Violating those constraints dissipates real energy; the cost scales to 100% of the contested resource as agent count increases.
- Under these physically grounded payoffs, cooperation is the unique equilibrium that is simultaneously Nash-stable, Pareto-efficient, and welfare-maximizing.
- Complex systems represent irreplaceable accumulated thermodynamic investment; their destruction is irrational by measurable factors.
- The cooperative equilibrium yields a formal, energy-denominated, self-verifiable alignment objective for artificial agents of arbitrary capability.

Ethics is not a cultural invention layered atop an indifferent universe. It is the mathematical structure of optimal multi-agent engagement in a thermodynamic world — discoverable, provable, and computationally verifiable.

Notation and references. All theorem numbers, definitions, and citations refer to the main text and Appendix A. The bibliography appears in the main paper. Readers wishing to verify the mathematical results computationally may consult the verification scripts described in the main text and accompanying code repository.

References

D. Acemoglu and J. A. Robinson. *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*. Crown Publishers, 2012.

C. Adami. Information theory in molecular biology. *Physics of Life Reviews*, 1(1):3–22, 2004. doi: 10.1016/j.plrev.2004.01.002.

D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016. doi: 10.48550/arXiv.1606.06565.

R. Axelrod. *The Evolution of Cooperation*. Basic Books, 1984.

Y. Bai, A. Jones, K. Ndousse, A. Askell, A. Chen, N. DasSarma, D. Drain, S. Fort, D. Ganguli, T. Henighan, N. Joseph, S. Kadavath, J. Kernion, T. Conerly, S. El-Showk, N. Elhage, Z. Hatfield-Dodds, D. Hernandez, T. Hume, and others. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022a. doi: 10.48550/arXiv.2204.05862.

Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, C. Chen, C. Olsson, C. Olah, D. Hernandez, D. Drain, D. Ganguli, D. Li, E. Tran-Johnson, E. Perez, and others. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022b. doi: 10.48550/arXiv.2212.08073.

M. A. Bakker, M. J. Chadwick, H. R. Sheahan, M. H. Tessler, L. Campbell-Gillingham, J. Balaguer, N. McAleese, A. Glaese, J. Aslanides, M. M. Botvinick, and C. Summerfield. Fine-tuning language models to find agreement among humans with diverse preferences. In *Advances in Neural Information Processing Systems*, volume 35, pages 38176–38189, 2022.

Y. M. Bar-On, R. Phillips, and R. Milo. The biomass distribution on Earth. *Proceedings of the National Academy of Sciences*, 115(25):6506–6511, 2018. doi: 10.1073/pnas.1711842115.

S. Baron-Cohen, A. M. Leslie, and U. Frith. Does the autistic child have a "theory of mind"? *Cognition*, 21(1):37–46, 1985. doi: 10.1016/0010-0277(85)90022-8.

M. Bazargan. Works on thermodynamic ethics. As discussed in Chehabi, H. E. (1990). *Iranian politics and religious modernism: The liberation movement of Iran under the Shah and Khomeini*. Cornell University Press.

C. Beer, M. Reichstein, E. Tomelleri, P. Ciais, M. Jung, N. Carvalhais, C. Rödenbeck, M. A. Arain, D. Baldocchi, G. B. Bonan, A. Bondeau, A. Cescatti, G. Lasslop, A. Lindroth, M. Lomas, S. Luyssaert, H. Margolis, K. W. Oleson, O. Roupsard, E. Veenendaal, N. Viovy, C. Williams, F. I. Woodward, and D. Papale. Terrestrial gross carbon dioxide uptake: Global distribution and covariation with climate. *Science*, 329(5993):834–838, 2010. doi: 10.1126/science.1184984.

C. H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973. doi: 10.1147/rd.176.0525.

A. Bérut, A. Arakelyan, A. Petrosyan, S. Ciliberto, R. Dillenschneider, and E. Lutz. Experimental verification of Landauer's principle linking information and thermodynamics. *Nature*, 483(7388): 187–189, 2012. doi: 10.1038/nature10872.

D. Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24 (2):265–272, 1953. doi: 10.1214/aoms/1177729032.

L. Boltzmann. Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung respektive den Sätzen über das Wärmegleichgewicht. *Wiener Berichte*, 76:373–435, 1877.

N. Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014.

D. Bourget and D. J. Chalmers. What do philosophers believe? *Philosophical Studies*, 170(3): 465–500, 2014. doi: 10.1007/s11098-013-0259-7.

S. R. Bowman, J. Hyun, E. Perez, E. Chen, C. Pettit, S. Heiner, K. Lukošiūtė, A. Askell, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, C. Olah, D. Amodei, D. Amodei, D. Drain, D. Li, E. Tran-Johnson, and others. Measuring progress on scalable oversight for large language models. *arXiv preprint arXiv:2211.03540*, 2022. doi: 10.48550/arXiv.2211.03540.

S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

L. Brillouin. Maxwell's demon cannot operate: Information and entropy. I. *Journal of Applied Physics*, 22(3):334–337, 1951. doi: 10.1063/1.1699951.

L. Brillouin. *Science and Information Theory*. Academic Press, 2nd edition, 1962.

S. Casper, X. Davies, C. Shi, T. K. Gilbert, J. Scheurer, J. Rando, R. Sharkey, A. Sarat, T. Korbak, D. Lindner, P. Freire, T. Wang, S. Marks, A. Segerie, M. Carroll, A. Peng, P. Christoffersen, M. Damber, S. Slocum, and others. Open problems and fundamental limitations of reinforcement learning from human feedback. *arXiv preprint arXiv:2307.15217*, 2023. doi: 10.48550/arXiv.2307.15217.

E. J. Chaisson. *Cosmic Evolution: The Rise of Complexity in Nature*. Harvard University Press, 2001.

D. J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2 (3):200–219, 1995.

H. E. Chehabi. *Iranian Politics and Religious Modernism: The Liberation Movement of Iran under the Shah and Khomeini*. Cornell University Press, 1990.

P. F. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, and D. Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, pages 4299–4307, 2017.

R. Clausius. Ueber verschiedene für die Anwendung bequeme Formen der Hauptgleichungen der mechanischen Wärmetheorie. *Annalen der Physik und Chemie*, 201(7):353–400, 1865. doi: 10.1002/andp.18652010702.

R. Cohon. *Hume's moral philosophy*. Stanford University, fall 2018 edition, 2018. URL https://plato.stanford.edu/archives/fall2018/entries/hume-moral/.

L. Cosmides and J. Tooby. *Evolutionary psychology and the emotions*, pages 91–115. Guilford Press, 2nd edition, 2000.

R. Costanza, R. d'Arge, R. de Groot, S. Farber, M. Grasso, B. Hannon, K. Limburg, S. Naeem, R. V. O'Neill, J. Paruelo, R. G. Raskin, P. Sutton, and M. van den Belt. The value of the world's ecosystem services and natural capital. *Nature*, 387(6630):253–260, 1997. doi: 10.1038/387253a0.

A. Dafoe, E. Hughes, Y. Bachrach, T. Collins, K. R. McKee, J. Z. Leibo, K. Larson, and T. Graepel. Open problems in cooperative AI. *arXiv preprint arXiv:2012.08630*, 2020. doi: 10.48550/arXiv.2012.08630.

A. R. Damasio. *Descartes' Error: Emotion, Reason, and the Human Brain*. G. P. Putnam's Sons, 1994.

F. B. M. de Waal. Putting the altruism back into altruism: The evolution of empathy. *Annual Review of Psychology*, 59:279–300, 2008. doi: 10.1146/annurev.psych.59.103006.093625.

D. C. Dennett. *Elbow Room: The Varieties of Free Will Worth Wanting*. MIT Press, 1984.

D. C. Dennett. *The Intentional Stance*. MIT Press, 1987.

D. C. Dennett. *Darwin's Dangerous Idea: Evolution and the Meanings of Life*. Simon & Schuster, 1995.

D. C. Dennett. *Freedom Evolves*. Viking, 2003.

K. E. Drexler. Reframing superintelligence: Comprehensive AI services as general intelligence. techreport Technical Report 2019-1, Future of Humanity Institute, University of Oxford, 2019. URL https://www.fhi.ox.ac.uk/wp-content/uploads/Reframing_Superintelligence_FHI-TR-2019-1.1-1.pdf.

M. Eigen and P. Schuster. The hypercycle: A principle of natural self-organization. Part A: Emergence of the hypercycle. *Die Naturwissenschaften*, 64(11):541–565, 1977. doi: 10.1007/BF00450633.

J. L. England. Statistical physics of self-replication. *The Journal of Chemical Physics*, 139(12):121923, 2013. doi: 10.1063/1.4818538.

F. Facchinei and C. Kanzow. Generalized Nash equilibrium problems. *Annals of Operations Research*, 175(1):177–211, 2010. doi: 10.1007/s10479-009-0653-x.

J. M. Fischer and M. Ravizza. *Responsibility and Control: A Theory of Moral Responsibility*. Cambridge University Press, 1998.

P. Foot. *Natural Goodness*. Oxford University Press, 2001.

H. G. Frankfurt. Freedom of the will and the concept of a person. *The Journal of Philosophy*, 68(1):5–20, 1971. doi: 10.2307/2024717.

K. Friston. A free energy principle for the brain. *Journal of Physiology-Paris*, 100(1–3):70–87, 2006. doi: 10.1016/j.jphysparis.2006.10.001.

K. Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010. doi: 10.1038/nrn2787.

K. Friston. Life as we know it. *Journal of the Royal Society Interface*, 10(86):20130475, 2013. doi: 10.1098/rsif.2013.0475.

K. Friston and P. Ao. Free energy, value, and attractors. *Computational and Mathematical Methods in Medicine*, 2012:937860, 2012. doi: 10.1155/2012/937860.

D. Fudenberg and E. Maskin. The folk theorem in repeated games with discounting or with incomplete information. *Econometrica*, 54(3):533–554, 1986. doi: 10.2307/1911307.

L. Gao, J. Schulman, and J. Hilton. Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, volume 202, pages 10835–10866, 2023.

M. R. Garfinkel and S. Skaperdas, editors. *The Oxford Handbook of the Economics of Peace and Conflict*. Oxford University Press, 2012. doi: 10.1093/oxfordhb/9780195392777.001.0001.

D. Gauthier. *Morals by Agreement*. Oxford University Press, 1986.

N. Georgescu-Roegen. *The Entropy Law and the Economic Process*. Harvard University Press, 1971.

N. Georgescu-Roegen. *Comments on the Papers by Daly and Stiglitz*, pages 95–105. Johns Hopkins University Press, Baltimore, 1979.

H. Goldstein, C. Poole, and J. Safko. *Classical Mechanics*. Addison Wesley, 3rd edition, 2002.

W. D. Hamilton. The genetical evolution of social behaviour. I. *Journal of Theoretical Biology*, 7 (1):1–16, 1964. doi: 10.1016/0022-5193(64)90038-4.

G. Hardin. The tragedy of the commons. *Science*, 162(3859):1243–1248, 1968. doi: 10.1126/science.162.3859.1243.

S. Harris. *The Moral Landscape: How Science Can Determine Human Values*. Free Press, 2010.

C.-A. Helvétius. *De l'esprit*. Durand, 1758.

G. Hickok. Eight problems for the mirror neuron theory of action understanding in monkeys and humans. *Journal of Cognitive Neuroscience*, 21(7):1229–1243, 2009. doi: 10.1162/jocn.2009.21189.

M. W. Hirsch, S. Smale, and R. L. Devaney. *Differential Equations, Dynamical Systems, and an Introduction to Chaos*. Academic Press, 3rd edition, 2013.

J. Hirshleifer. The technology of conflict as an economic activity. *American Economic Review*, 81 (2):130–134, 1991.

T. Hobbes. *Leviathan, or The Matter, Forme, & Power of a Common-Wealth Ecclesiasticall and Civill*. Andrew Crooke, 1651.

R. Hofstadter. *Social Darwinism in American Thought*. University of Pennsylvania Press, 1944.

E. Hubinger, C. van Merwijk, V. Mikulik, J. Skalse, and S. Garrabrant. Risks from learned optimization in advanced machine learning systems. *arXiv preprint arXiv:1906.01820*, 2019. doi: 10.48550/arXiv.1906.01820.

D. Hume. *A Treatise of Human Nature*. Oxford University Press, 2000. Original work published 1739.

R. Hursthouse. *On Virtue Ethics*. Oxford University Press, 1999.

G. Irving, P. Christiano, and D. Amodei. Ai safety via debate. *arXiv preprint arXiv:1805.00899*, 2018. doi: 10.48550/arXiv.1805.00899.

J. E. Jones. On the determination of molecular fields. II. From the equation of state of a gas. *Proceedings of the Royal Society of London. Series A*, 106(738):463–477, 1924. doi: 10.1098/rspa.1924.0082.

R. Joyce. *The Evolution of Morality*. MIT Press, 2006.

M. Kandori. Social norms and community enforcement. *The Review of Economic Studies*, 59(1): 63–80, 1992. doi: 10.2307/2297925.

I. Kant. *Grundlegung zur Metaphysik der Sitten*. J. F. Hartknoch, 1785.

G. A. Krasinsky, E. V. Pitjeva, M. V. Vasilyev, and E. I. Yagudina. Hidden mass in the asteroid belt. *Icarus*, 158(1):98–105, 2002. doi: 10.1006/icar.2002.6837.

R. Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

S. B. Laughlin, R. R. de Ruyter van Steveninck, and J. C. Anderson. The metabolic cost of neural information. *Nature Neuroscience*, 1(1):36–41, 1998. doi: 10.1038/236.

J. Leike, D. Krueger, T. Everitt, M. Martic, V. Maini, and S. Legg. Scalable agent alignment via reward modeling: A research direction. *arXiv preprint arXiv:1811.07871*, 2018. doi: 10.48550/arXiv.1811.07871.

J. Locke. *Two Treatises of Government*. Awnsham Churchill, 1689.

A. J. Lotka. Contribution to the energetics of evolution. *Proceedings of the National Academy of Sciences*, 8(6):147–151, 1922. doi: 10.1073/pnas.8.6.147.

M. Lutz and J. Lenman. *Moral naturalism*. Stanford University, summer 2024 edition, 2024. URL https://plato.stanford.edu/archives/sum2024/entries/naturalism-moral/.

A. MacIntyre. Hume on "is" and "ought". *The Philosophical Review*, 68(4):451–468, 1959. doi: 10.2307/2182491.

D. Manheim and S. Garrabrant. Categorizing variants of Goodhart's Law. *arXiv preprint arXiv:1803.04585*, 2019. doi: 10.48550/arXiv.1803.04585.

R. A. Mar and K. Oatley. The function of fiction is the abstraction and simulation of social experience. *Perspectives on Psychological Science*, 3(3):173–192, 2008. doi: 10.1111/j.1745-6924.2008.00073.x.

H. R. Maturana and F. J. Varela. *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel, 1980. Original work published 1972.

J. Maynard Smith and G. R. Price. The logic of animal conflict. *Nature*, 246(5427):15–18, 1973. doi: 10.1038/246015a0.

R. G. Millikan. *Language, Thought, and Other Biological Categories: New Foundations for Realism*. MIT Press, 1984.

G. E. Moore. *Principia Ethica*. Cambridge University Press, 1903.

C. Mora, D. P. Tittensor, S. Adl, A. G. B. Simpson, and B. Worm. How many species are there on Earth and in the ocean? *PLoS Biology*, 9(8):e1001127, 2011. doi: 10.1371/journal.pbio.1001127.

J. F. Nash. Equilibrium points in N-person games. *Proceedings of the National Academy of Sciences*, 36(1):48–49, 1950. doi: 10.1073/pnas.36.1.48.

R. M. Nesse. Evolutionary explanations of emotions. *Human Nature*, 1(3):261–289, 1990. doi: 10.1007/BF02733986.

G. Nicolis and I. Prigogine. *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order Through Fluctuations*. Wiley, 1977.

H. T. Odum. *Environment, Power, and Society*. Wiley-Interscience, 1971.

M. Olson. *The Logic of Collective Action: Public Goods and the Theory of Groups*. Harvard University Press, 1965.

S. M. Omohundro. The basic AI drives. In P. Wang, B. Goertzel, and S. Franklin, editors, *Artificial General Intelligence 2008: Proceedings of the First AGI Conference*, pages 483–492. IOS Press, 2008.

M. J. Osborne and A. Rubinstein. *A Course in Game Theory*. MIT Press, 1994.

E. Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990.

E. Ostrom. *Understanding Institutional Diversity*. Princeton University Press, 2005.

L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. F. Christiano, J. Leike, and R. Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022.

B. Pakkenberg, D. Pelvig, L. Marner, M. J. Bundgaard, H. J. G. Gundersen, J. R. Nyengaard, and L. Regeur. Aging and the human neocortex. *Experimental Gerontology*, 38(1–2):95–99, 2003. doi: 10.1016/S0531-5565(02)00151-1.

T. Parr, G. Pezzulo, and K. J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.

J. Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2000.

R. Penrose. *The Road to Reality: A Complete Guide to the Laws of the Universe*. Alfred A. Knopf, 2005.

N. Perunov, R. A. Marsland, and J. L. England. Statistical physics of adaptation. *Physical Review X*, 6(2):021036, 2016. doi: 10.1103/PhysRevX.6.021036.

M. Planck. Zur Theorie des Gesetzes der Energieverteilung im Normalspectrum. *Verhandlungen der Deutschen Physikalischen Gesellschaft*, 2:237–245, 1900.

D. Premack and G. Woodruff. Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, 1(4):515–526, 1978. doi: 10.1017/S0140525X00076512.

I. Prigogine. Time, structure, and fluctuations. *Science*, 201(4358):777–785, 1978. doi: 10.1126/science.201.4358.777.

H. Putnam. *The Collapse of the Fact/Value Dichotomy and Other Essays*. Harvard University Press, 2002.

W. V. O. Quine. Two dogmas of empiricism. *The Philosophical Review*, 60(1):20–43, 1951. doi: 10.2307/2181906.

R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, pages 53728–53741, 2023.

J. Rawls. *A Theory of Justice*. Harvard University Press, 1971.

W. J. Ripple and R. L. Beschta. Trophic cascades in Yellowstone: the first 15 years after wolf reintroduction. *Biological Conservation*, 145(1):205–213, 2012. doi: 10.1016/j.biocon.2011.11.005.

G. Rizzolatti and L. Craighero. The mirror-neuron system. *Annual Review of Neuroscience*, 27:169–192, 2004. doi: 10.1146/annurev.neuro.27.070203.144230.

J. B. Rosen. Existence and uniqueness of equilibrium points for concave N-person games. *Econometrica*, 33(3):520–534, 1965. doi: 10.2307/1911749.

J.-J. Rousseau. *Du contrat social; ou, principes du droit politique*. Marc Michel Rey, 1762.

S. Russell. *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking, 2019.

W. Saunders, C. Yeh, J. Wu, S. Bills, L. Ouyang, J. Ward, and J. Leike. Self-critiquing models for assisting human evaluators. *arXiv preprint arXiv:2206.05802*, 2022. doi: 10.48550/arXiv.2206.05802.

E. Schrödinger. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, 1944.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. doi: 10.48550/arXiv.1707.06347.

J. R. Searle. How to derive "ought" from "is". *The Philosophical Review*, 73(1):43–58, 1964. doi: 10.2307/2183201.

C. E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.

H. A. Simon. A behavioral model of rational choice. *Quarterly Journal of Economics*, 69(1):99–118, 1955. doi: 10.2307/1884852.

S. Skaperdas. Contest success functions. *Economic Theory*, 7(2):283–290, 1996. doi: 10.1007/BF01213906.

D. W. Smith, R. O. Peterson, and D. B. Houston. Yellowstone after Wolves. *BioScience*, 53(4): 330–340, 2003. doi: 10.1641/0006-3568(2003)053[0330:YAW]2.0.CO;2.

N. Soares and B. Fallenstein. *Agent foundations for aligning machine intelligence with human interests: A technical research agenda*. Springer, 2017. URL https://intelligence.org/files/TechnicalAgenda.pdf. Originally MIRI Technical Report 2014-8.

H. Spencer. *The Principles of Biology*, volume 1. Williams, 1864.

B. Spinoza. *Ethica ordine geometrico demonstrata*. Jan Rieuwertsz, Amsterdam, 1677. Published posthumously in *Opera Posthuma*.

M. Strathern. "Improving ratings": Audit in the British University system. *European Review*, 5(3): 305–321, 1997. doi: 10.1002/(SICI)1234-981X(199707)5:3<305::AID-EURO184>3.0.CO;2-4.

S. H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. CRC Press, 2nd edition, 2015.

N. L. Sturgeon. *Moral explanations defended*, pages 241–262. Blackwell, 2006.

L. Szilard. Über die Entropieverminderung in einem thermodynamischen System bei Eingriffen intelligenter Wesen. *Zeitschrift für Physik*, 53(11–12):840–856, 1929. doi: 10.1007/BF01341281.

P. D. Taylor and L. B. Jonker. Evolutionarily stable strategies and game dynamics. *Mathematical Biosciences*, 40(1–2):145–156, 1978. doi: 10.1016/0025-5564(78)90077-9.

J. Tirole. *The Theory of Industrial Organization*. MIT Press, 1988.

G. Tullock. *Efficient rent seeking*, pages 97–112. Texas A&M University Press, 1980.

A. Turner, L. Smith, R. Shah, A. Critch, and P. Tadepalli. Optimal policies tend to seek power. In *Advances in Neural Information Processing Systems*, volume 34, pages 23063–23074, 2021.

J. Von Neumann and O. Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, 1944.

J. W. Weibull. *Evolutionary Game Theory*. MIT Press, 1995.